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भाकृअनुप-राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी

राजेन्द्रनगर, हैदराबाद-500 030, तेलंगाना, भारत

ICAR-National Academy of Agricultural Research Management

Rajendranagar, Hyderabad-500 030, Telangana, India

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Abbreviations

AAU	Assam Agricultural University
ABF	Agri-Biotech Foundation
ABI	Agri-Business Incubator
ABS	Access and Benefit Sharing
ACP	Awareness Creation Program
ADG	Additional Director General
AERA	Agricultural Economics Research Association
AESSRA	Agricultural Economics and Social Science Research Association
AGTRI	Advances in Genomics Tools for Rice Improvement
AI	Artificial Intelligence
AICRP	All-India Coordinated Research Project
AICTE	All India Council for Technical Education
a-IDEA	Association for Innovation Development of Entrepreneurship in Agriculture
AIEEA	All India Entrance Examination for Admission
AIMA	All India Management Association
ALTENA	Asian Long Term Experimental Network for Agriculture
AMARA	Association for Management of Agricultural Research and Agripreneurship
ANGRAU	Acharya N. G. Ranga Agricultural University
ANRCM	Academy of Natural Resource Conservation and Management
APAR	Annual Performance Appraisal System
APMC	Agricultural Produce Market Committee
ARGM	Annual Rice Group Meeting
ARS	Agricultural Research Service
ASCI	Administrative Staff College of India
ASC	Agricultural Science Congress
ASEAN	Association of South East Asian Nations
ASRB	Agricultural Scientists Recruitment Board
ASTI	Agricultural Science and Technology Indicators
ATARI	Agricultural Technology Application Research Institute
AUs	Agricultural Universities
BARC	Bangladesh Agricultural Research Council
BASIX	Bhartiya Samruddhi Investments and Consulting Services Ltd.
BASU	Bihar Animal Sciences University
BAU	Bihar Agricultural University
BCIL	Biotechnology Consortium India Limited
BIOFIN	Biodiversity Finance Initiative

BIRAC	Biotechnology Industry Research Assistance Council
CAAST	Centre of Advanced Agricultural Science and Technology
CAI	Centre for Agri-Innovation
CAT	Common Admission Test
CAU	Central Agricultural University
CBI	Central Bureau of Investigation
CCS NIAM	Ch. Charan Singh National Institute of Agricultural Marketing
CCSHAU	Chaudhary Charan Singh Haryana Agricultural University
CDC	Career Development Centre
CDVL	Centre for Distance and Virtual Learning
CGIAR	Consultative Group for International Agricultural Research
CHRD	Center for Human Resource Development
CIFA	Central Institute of Freshwater Aquaculture
CIFE	Central Institute of Fisheries Education
CIFT	Central Institute of Fisheries Technology
CIIE	Centre for Innovation Incubation and Entrepreneurship
CIL	Coromandel International Limited
CIMMYT	International Maize and Wheat Improvement Center
CIRAD	The French Agricultural Research Centre for International Development
CIWA	Central Institute for Women in Agriculture
CJSC	Central Joint Staff Council
CMAT	Common Management Admission Test
CMFRI	Central Marine Fisheries Research Institute
COLLAge	Centre for Open and Lifelong Learning in Agriculture
CPI	Consumer Price Index
CRD	Completely Randomized Design
CRIDA	Central Research Institute for Dry land Agriculture
CRIJAF	Central Research Institute for Jute and Allied Fibers
CSAUA&T	Chandra Shekhar Azad University of Agriculture & Technology
CSIR	Council of Scientific & Industrial Research
CTCRI	Central Tuber Crops Research Institute
CTRI	Central Tobacco Research Institute
DAATTCs	District Agricultural Advisory and Transfer of Technology Centre
DAC&FW	Department of Agricultural Cooperation and Farmers Welfare
DFID	Department for International Development
DARS	Dryland Agriculture Research Station
DBT	Department of Biotechnology
DC	Data Centre
DCB	Development Credit Bank



DDG	Deputy Director General
DGR	Directorate of Groundnut Research
DIEO	Division for International and Educational Outreach
DKMA	Directorate of Knowledge Management in Agriculture
DRC	Disaster Recovery Centre
DST	Department of Science and Technology
DT	Design Thinking
DU	Delhi University
DWRP	Developing Winning Research Proposal
EDP	Executive Development Programme
EI	Extension Education Institute
e-NAM	National Agriculture Market
ENCORE	Enhancing Coastal And Ocean Resource Efficiency
EoDR	Ease of Doing Research
EP	Emeritus Professor
ERP	Enterprise Resource Planning
ES	Emeritus Scientist
ET	Education Technology
FADC	Food and Agriculture Division Council
FAI	Fertilizer Association of India
FAO	Food & Agriculture Organization
FCRI	Fisheries College and Research Institute
FDA	Food and Drug Administration
FDC	Faculty Development Center
FDP	Faculty Development Programme
FET	Field Experience Training
FICCI	Federation of Indian Chamber of Commerce and Industry
FLIs	Farmer Led Innovations
FOCARS	Foundation Course for Agricultural Research Service
FPC	Farmer Producer Companies
FPOs	Farmer Producer Organizations
FSRC	Financial Services Regulatory Commission
FSSAI	Food Safety and Standards Authority of India
FTAPCCI	Federation of Telangana and Andhra Pradesh Chambers of Commerce and Industry
FTAs	Farm Tele Advisors
FTF-ITT	Feed the Future - India Triangular Training
FYM	Farm Yard Manure
GADVASU	Guru Angad Dev Veterinary And Animal Sciences University
GBPUA&T	Govind Ballabh Pant University of Agriculture and Technology

GDD	Growing Degree Days
GEF	Global Environment Facility
GHGs	Greenhouse gases
GHMC	Greater Hyderabad Municipal Corporation
GIS	Geographic Information System
GKVK	Gandhi Krishi Vigyana Kendra
Gol	Government of India
GPS	Global Positioning System
GST	Goods and Services Tax
HEIs	Higher Education Institutions
HICC	Hyderabad International Convention Centre
HRD	Human Resource Development
IARI	Indian Agricultural Research Institute
IAS	Indian Administrative Service
IASRI	Indian Agricultural Statistics Research Institute
ICAR	Indian Council of Agricultural Research
ICM	Institute of Cooperative Management
ICMR	Indian Council of Medical Research
ICRISAT	International Crops Research Institute for Semi-Arid Tropics
ICT	Information and Communication Technology
ICTA-A	Climate Action Transparency Project on Adaptation
ICZM	Integrated Coastal Zone Management
IDDC	International Dryland Development Commission
IDP	Institutional Development Plan
IFPRI	International Food Policy Research Institute
IFS	Indian Forest Service
IGFRI	Indian Grassland and Fodder Research Institute
IGH	International Guest House
IGKV	Indira Gandhi Krishi Vishwavidyalaya
IGNFA	Indira Gandhi National Forest Academy
IIIT	Indian Institute of Information Technology
IIM	Indian Institute of Management
IIMR	Indian Institute of Millets Research
IIOPR	Indian Institute of Oil Palm Research
IIOR	Indian Institute of Oilseeds Research
IIP	Indian Institute of Packaging
IIRR	Indian Institute of Rice Research
IISER	Indian Institute of Science Education and Research
IISS	International Institute for Strategic Studies



IIT	Indian Institute of Technology
IIITDM	Indian Institute of Information Technology Design and Manufacturing
IIVR	Indian Institute of Vegetable Research
IJSC	Institute Joint Staff Council
ILS	Institute of Live Sciences
IMC	Institute Management Committee
IoT	Internet of Things (IoT)
IPE	Institute of Public Enterprise
IPR	Intellectual property rights
IPTM	Intellectual Property & Technology Management
IRC	Institute Research Council
IRGS	Institute of Research and Graduate Studies
IRM	Insect Resistance Management
ISAP	Indian Society of Agribusiness Professionals
ISAPM	Indian Society of Animal Production and Management
ISB	Indian School of Business
ISO	International Organization for Standardization
IT	Information Technology
ITK	Indigenous Technical Knowledge
ITMC	Institute Technology Management Committee
ITMU	Institute Technology Management Unit
IWELA	Indian Women Excellence and Leadership Award
JAC	Joint Advisory Council
JAU	Junagadh Agricultural University
JNTU	Jawaharlal Nehru Technological University
JNU	Jawaharlal Nehru University
KAU	Kerala Agricultural University
KCCs	Kisan Call Centers
KJWA	Koronivia Joint Work on Agriculture
KPIs	Key Performance Indicators
KVKs	Krishi Vigyan Kendras
LMS	Learning Management System
LSD	Latin square design
MABIF	Madurai Agribusiness Incubation Forum
MAFSU	Maharashtra Animal and Fishery Sciences University
MANAGE	National Institute of Agricultural Extension Management
MDP	Management Development Programme
MIDH	Mission for Integrated Development of Horticulture
MLA	Member of Legislative Assembly

MoEFCC	Ministry of Environment, Forest and Climate Change
MOOC	Massive Open Online Courses
MORE	Marketing Officers' Resurgence & Excellence
MoU	Memorandum of Understanding
MPUA&T	Maharana Pratap University of Agriculture and Technology
MVIRDC	M. Visvesvaraya Industrial Research and Development Centre
NAARM	National Academy of Agricultural Research Management
NAAS	National Academy of Agricultural Sciences
NABARD	National Bank for Agricultural and Rural Development
NADMP	National Agriculture Disaster Management Plan
NAEP	National Assessment of Educational Progress
NAHEP	National Agricultural Higher Education Project
NAIF	National Agricultural Innovation Found
NAPCC	National Action Plan on Climate Change
NARC	Nepal Agricultural Research Council
NARES	National Agricultural Research and Education System
NARS	National Agricultural Research System
NASF	National Agricultural Science Fund
NASI	National Academy of Sciences, India
NBA	National Biodiversity Authority
NBA	National Board of Accreditation
NBFGR	National Bureau of Fish Genetic Resources
NBPGR	National Bureau of Plant Genetic Resources
NBSSLUP	National Bureau of Soil Survey and Land Use Planning
NCCT	National Council for Cooperative Training
NCSCM	National Centre for Sustainable Coastal Management
NDDI	Normalized Difference Drought Index
NDRI	National Dairy Research Institute
NDVI	Normalized Difference Vegetation Index
NDWI	Normalized Difference Water Index
NERC	North Eastern Regional Centre
NET	National Eligibility Test
NF	National Fellow
NFDB	National Fisheries Development Board
NGOS	Non-Government Organizations
NIAM	National Institute of Agricultural Marketing
NIAP	National Institute of Agricultural Economics and Policy Research
NIDHI PRAYAS	National Initiative for Developing and Harnessing Innovations - Promoting and Accelerating Young and Aspiring technology entrepreneurs
NIF	National Innovation Foundation



NIOT	National Institute of Ocean Technology
NIPHM	National Institute of Plant Health Management
NIRD &PR	National Institute of Rural Development and Panchayat Raj
NITI	National Institution for Transforming India
NMSA	National Mission on Sustainable Agriculture
NP	National Professor
NPTEL	National Programme on Technology Enhanced Learning
NRC	National Register of Citizens
NRCB	National Research Centre for Banana
NRCC	National Research Centre on Camel
NRCM	Natural Resource Conservation and Management
NRM	Natural Resource Management
NRRI	National Rice Research Institute
NSTEDB	National Science & Technology Entrepreneurship Development Board
NSTMIS	National Science and Technology Management Information System
NTC	National Training Course
OPAC	Online Public Access Catalog
OPD	Outpatient Department
OUAT	Odisha University of Agriculture and Technology
PARC	Pakistan Agricultural Research Council
PAU	Punjab Agricultural University
PCK	Pedagogical Content Knowledge
PCT	Patent Cooperation Treaty
PDKV	Panjabrao Deshmukh Krishi Vidyapeeth
PDP	Project Directorate on Poultry
PGDETM	Post Graduate Diploma in Education Technology Management
PGDMA	Post Graduate Diploma in Management-Agriculture
PGDM-ABM	Post Graduate Diploma in Agribusiness Management
PGDTMA	Post Graduate Diploma in Technology Management in Agriculture
PJTSAU	Professor Jayashankar Telangana State Agricultural University
PME	Priority Setting, Monitoring and Evaluation
PPV&FRA	Protection of Plant Varieties & Farmer Rights Authority
PRA	Participatory Rural Appraisal
PVNRTVU	PV Narasimha Rao Telangana Veterinary University
QRT	Quinquennial Review Team
RAC	Research Advisory committee
RAG	Regional Advisory Group
RAWE	Rural Agricultural Work Experience
ReMS	Rashtriya e-Market System

RKVY	Rashtriya Krishi Vikas Yojana
RMP	Risk Management Plan
RSMs	Retail sales Managers
RTE	Ready to Eat
RVSKVV	Rajmata Vijayaraja Scindia Krishi Vishwa Vidyalaya
SAARC	South Asian Association for Regional Cooperation
SAMETI	State Agriculture Management & Extension Training Institute
SAS	Statistical Analysis System
SAUs	State Agricultural Universities
SBSTA	Subsidiary Body for Scientific and Technological Advice
SDGs	Sustainable Development Goals
SHUATS	Sam Higginbottom University of Agriculture, Technology and Sciences
SKLTSHU	Sri Konda Laxman Telangana State Horticultural University
SKNAU	Sri Karan Narendra Agriculture University
SKUAST	Sher-e-Kashmir University of Agriculture Sciences and Technology
SLAs	Students' Learning Approaches
SLTP	State Level Technical Programme
SMD	Subject Matter Division
SNRM	Strengthening Natural Resource Management
SOC	Senior Officers Committee
SSS	Skilled Support Staff
STEM	Society for Technology Management
SVVU	Sri Venkateswara Veterinary University
TAAS	Trust for Advancement of Agricultural Sciences
TAFE	Tractors and Farm Equipment
TANUVAS	Tamil Nadu Veterinary and Animal Sciences University
TARCMES	Technological Approaches for Resource Conservation and Management for Environmental Sustainability
TBI	Technology Business Incubator
TEFR	Techno-Economic Feasibility Report
TEL	Technology Enhanced Learning
TELAge	Technology Enhanced Learning in Agricultural Education
TERI	The Energy and Resources Institute



TISS	Tata Institute of Social Sciences
TLSOE	Teaching Learning Strategies of Online Education
TNAU	Tamil Nadu Agricultural University
TNAU UIC	Tamil Nadu Agricultural University University Innovation Cluster
TNFU	Tamil Nadu Fisheries University
TNJFU	Tamil Nadu Dr. J. Jayalalithaa Fisheries University
ToT	Training of Trainers
TPACK	Technology Pedagogy and Content Knowledge
TQIMC	Training Quality Improvement Measures Committee
TRIPS	Trade-Related aspects of Intellectual Property Rights
UAS	University of Agricultural Sciences
UASR	University of Agricultural Sciences, Raichur
UBKV	Uttar Banga Krishi Viswavidyalaya
UK	United Kingdom
UNDP	United Nations Development Programme
UNFCCC	United Nations Framework Convention on Climate Change
UoH	University of Hyderabad
USA	United States of America
USAID	U.S. Agency for International Development
VCD	Video Compact Disc
VNMKV	Vasantrao Naik Marathwada Agricultural University
WDP	Wheat Development Programme
WSSV	White Spot Syndrome Virus
WTO	World Trade Organization
ZBNF	Zero Budget Natural Farming
ZTMC	Zonal Agro-Technology Management Center
ZTMU	Zonal Technology Management Unit

From Director's Desk



Dr. Ch. Srinivasa Rao
Director

ICAR - NAARM continues to be the nodal training and capacity building institute, with the newly assigned role of think tank for the ICAR. I am happy to present the progress and achievements

of the Academy during the year 2020 in the areas of capacity building, research, education, policy engagement and entrepreneurship development in agriculture and allied sciences. I am proud to say that our Academy contributed significantly to the National Agricultural Research, Extension and Education System, despite the challenges posed by the Covid-19 pandemic. We organized 61 capacity building programmes, mostly in online mode, attended by more than seven thousand participants. These include two FOCARS programmes, Conference/Workshops/Seminars, Leadership Development programmes, Faculty Development programmes, Off-campus programmes, MOOC programmes, Entrepreneurial skill development programme for students, a tailor-made Virtual Training Programme for the IFS Probationers (2019-21 batch) from Indira Gandhi National Forest Academy (IGNFA), Dehradun and a collaborative programme for agribusiness professionals of Coromandel International Ltd.

We organized the 110th FOCARS, the flagship programme of our Academy, for 135 ARS probationers just before the Covid Pandemic in 2020. The 111th FOCARS with 37 ARS Trainees in physical mode was organized from 5th October 2020. All SOP practices were followed during the

conduct of the programme and was successfully completed. Several off-campus workshops were also organized at various Agricultural Universities. The Academy's faculty participated in meetings of UNFCCC Climate Change Dialogues and series of Webinars on the subjects of contemporary issues. Under the aegis of AMARA Society, a National Conference on Geospatial Technologies in Agriculture was organized, which was inaugurated by Dr. (Ms.) Tessy Thomas, DG (Aeronautical Systems), DRDO. The programme of MoU signing ceremony between ICAR and ASCI was held, in which both the Director General of ICAR and ASCI took part in the event.

With the unprecedented circumstances due to Covid-19 pandemic, we kept ourselves moving along with the new normal. The lockdown did create unforeseen challenges in all spheres of life. I am very happy to share that the Academy accepted the challenges positively and seamlessly adapted to online mode for almost all the mandated activities. Further, the Academy continued to work with full strength, adhering to the guidelines and SOPs prescribed by ICAR/Govt of India from time to time. I wish to put on record the commitment shown by our staff in attending office during the lock down in 2020 is really commendable.

The education programmes progressed smoothly as planned. All the students of PGDABM (2018-20) have been placed successfully in reputed agribusiness companies. It is the 10th batch in a row recording 100 per cent placement with handsome pay packages. Keeping in view the ongoing COVID-19 pandemic situation and travel restrictions in the country, the admission process for the 12th batch of PGDM (Agri-Business Management) programme was conducted virtually for admission

of 54 students. Classes for these students were commenced in online mode initially and later on switched over to offline classes. The 1st term contact programme for 34 students of PGD-TMA 2020 batch was conducted during Sep 24-26, 2020 through virtual mode attended by 34 students. The Academy successfully completed the online contact sessions for PG Diploma in Educational Technology and Management (PGDETM), during May 20-29, 2020.

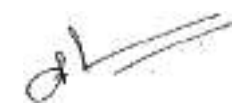
We have a total of 48 research projects, which comprised of 28 In-house projects, and 20 externally funded contract research and consultancy projects. Among them, we have projects of NAHEP, NASF, KRISHI, Farmers FIRST, Bio-informatics project of DBT, UNDP-BCIL and others. Under NAHEP, the Academy signed MoUs with six State Agricultural Universities. Further, Hon'ble Union Minister of Agriculture & Farmers' Welfare Shri Narendra Singh Tomar virtually launched ICAR's data centre 'Krishi Megh', and 'Data Recovery Centre' (DRC). The TBI of NAARM, a-IDEA, signed an MoU with NABARD and obtained 6.4 crore scaling-up grant to the entrepreneurial development project. The implementation of e-office cent per cent at the Academy was appreciable. The scientists got prestigious ICAR Awards including the coveted Rafi Ahmed Kidwai Award and Swami Sahajanand Saraswati Outstanding Extension Scientist Award. The Academy bagged 'Best Landscaping Institute Award' and 'Best Gardens Maintained Award' from the Department of Horticulture, Government of Telangana.

The Academy, following the stipulated COVID-19 guidelines and SOPs, celebrated many important international and national events

like International Women's Day, International Yoga Day, Swachhta Pledge, Mahila Kisan Diwas, Vigilance Awareness Week, 150th Birth Anniversary of Mahatma Gandhi, Rashtriya Ekta Diwas, Constitution Day, World Soil Day and Kisan Diwas. The Academy took several measures to combat the prevailing COVID-19 pandemic situation. These included frequent and regular sanitization, making use of 'Aarogyasetu' App, providing support to the Telangana State Government by providing Corona Quarantine Facility for accommodating transit international travelers from COVID-19 affected countries for isolation and quarantine with all necessities from Mar 17, 2020 till September 2020. Display boards were installed in the campus displaying *Do's and Don'ts* to prevent corona transmission.

The Academy is grateful to ICAR in general and Dr. Trilochan Mohapatra, Secretary-DARE and Director General, ICAR and Dr. R. C. Agrawal, Dy. Director General (Edn.) in particular for their unstinted guidance, support and encouragement. I acknowledge the contributions of Joint Director In-charge, all Heads of Divisions, various section in charges, Joint Director (Admn.) & Registrar, Chief Finance & Accounts Officer and all staff of ICAR-NAARM in strengthening the Academy.

I sincerely appreciate the efforts put in by Drs. I. Sekar, B.S. Sontakki, Ranjit Kumar, Tavva Srinivas, Alok Kumar, Sanjiv Kumar & BS Yashavanth who are instrumental in bringing out this report during this trying times of Covid. As Head of NAARM family, I place on record my appreciation to all for the strenuous work and sincere efforts in making the year 2020 successful and rewarding.



(Ch. Srinivasa Rao)

DIRECTOR

Date: April 15, 2021

Place: Hyderabad

Executive Summary

ICAR-National Academy of Agricultural Research Management is the nodal organization of ICAR for capacity building and engaged in enhancing the capacity of professionals at different hierarchical levels in the entire National Agricultural Research and Education Systems (NARES) in India. Apart from this, the Academy is also mandated to serve as a 'Think Tank' of ICAR. Despite the challenges posed by COVID-19 pandemic, during 2020, the Academy contributed significantly to the NARES and other stakeholders like Agricultural Universities, Krishi Vigyan Kendras (KVKs), Farmers' Producers Organizations (FPOs), Agri-startup ecosystem, etc. There were 61 capacity building programmes organized by the Academy in 2020 out of which, 33 programmes were need-based, 17 were workshops and 4 of them were off-campus programmes in addition to two flagship training programmes of Foundation Course of Agricultural Research Service (FOCARS) benefitting a total of 7,122 trainees of different institutes across the country.

The two Massive Open Online Courses (MOOC) organized during the year could attract 4,431 participants from various corners of the country. The FOCARS has been the flagship programme of the Academy for Agricultural Research Service (ARS) probationers. The 110th batch of FOCARS, in which a total of 135 newly recruited ARS Scientist-Probationers were trained during January 07-April 06, 2020. The programme was inaugurated by Dr. (Smt.) TAMILISAI Soundararajan, Hon'ble Governor, Telangana State. The 111th FOCARS were having 37 ARS Probationers which commenced from October 5, 2020, was inaugurated by Prof.

Rattan Lal, World Food Prize Winner-2020. The Academy also organized entrepreneurial skill development programme for students, a tailor-made virtual training programme for the 71 Indian Forest Service (IFS) Probationers (2019-21 batch) from Indira Gandhi National Forest Academy (IGNFA), Dehradun and a collaborative capability programme for 44 young agribusiness professionals of Coromandel International Limited.

The Academy has now evolved into a Think Tank of ICAR by providing advisory role and research policy advocacy in critical areas of importance in agriculture with changing needs. In this process, the Academy has developed four policy documents in critical areas such as climate change, agri-entrepreneurship development, agricultural education and faculty competence. These policy documents emerged after several rounds of brainstorming sessions and the stakeholders' consultations in the respective areas and key policy prescriptions were made for further strengthening of NARES.

The Post-Graduate Diploma in Management-Agribusiness Management (PGDM-ABM) is another flagship activity of the Academy. The AICTE - approved 2-year fully residential programme admitted 54 students for its 12th batch during the academic year 2020-21. With growing significance of agribusiness in the country, the demand of NAARM - trained students is also increasing. Therefore, all the students of 10th batch of PGDM-ABM got success in employment through campus placement with top agribusiness industries in India. Other academic programmes

of the Academy, viz. Post-Graduate Diploma in Technology Management in Agriculture (PGDTMA) and Post Graduate Diploma in Educational Technology Management (PGDETM) are one-year distance education programmes offered in online mode. Both the programmes are being offered in collaboration with University of Hyderabad (UoH). In these programmes, a total of 63 students took admission in the year 2020.

On the research front, the Academy had 48 research projects, which comprised of 28 In-house projects, and 20 externally funded contract research and consultancy projects. The sponsored projects are from various organizations including ICAR Schemes, NSTMIS, DST, DBT World Bank, NITI Aayog, IFPRI, NIPHM, ITC Pvt. Ltd., NASF, NDRI, UNDP-BCIL & NCSCM. The research projects span over six thematic areas viz. Agricultural Business Management for Inclusive Growth, Education Systems Management for Enriched Learning Quality and Environment, Enhancing Capacity for Leadership Governance, Extension Systems Management in Market-driven Environment, Information and Communication Management for Promoting Innovation and Governance and Mobilizing Science & Technology for Innovation in Sustainable Development. The faculty members of the Academy published 50 plus peer reviewed research papers in high impact journals. The other publications include books, book chapters, popular articles, training manuals, seminar proceedings, research reports etc. ICAR-NAARM is one of the consortia partners of Farmers FIRST Initiative launched by ICAR. The Academy was essentially involved in the capacity building activities and organized several workshops under the Farmers FIRST programme.

The Centre for Agri-Innovation (CAI) is located in the campus provides hand-holding support to the agri-startups through a-IDEA, the technology business incubator of the Academy. During the year, four incubatees have been inducted. Besides, 12 incubatees which were inducted during earlier years have been graduated in 2020. a-IDEA organized several Start-Up - FPO Immersion programmes and training programme for ABL units. It participated in BioAsia 2020, Asia's one of the largest biotechnology conference held at Hyderabad during Feb 17-19, 2020. It has launched Nidhi Prayas inviting innovators working on Idea/POC/Prototype in the field of agriculture and allied sectors to apply for "Nidhi Prayas" grant up to INR 10 Lakhs. It has organized series of webinars on various topics from sector specific to AMA (Ask Me Anything) sessions for startups, specifically focusing on finance and legal/compliances related issues.

Among others, the Technology Enhanced Learning in Agricultural Education (TELAge) is state-of-the-art Agricultural Education Lab, which was established in the Academy to develop digital courses, offer distance and online programmes, conduct virtual classes and support in-house programmes. With its technologically advanced infrastructure, the facility is now being used to offer MOOCs consultancy support to other organizations in developing and hosting digital content. Other neighbouring institutions like MANAGE, PJTSAU and NIPHM have been using this facility for generating their digital output. The facility is now renamed as Centre for Life Long Learning in Agricultural Education (COLLAge).

Several dignitaries graced different events of the Academy through online platform which include Prof Rattan Lal, World Food Prize Winner-2020; Padma Vibhushan Dr. RS Paroda, Former Secretary,

DARE & DG, ICAR. Dr. Trilochan Mohapatra, Secretary (DARE) & Director General (ICAR) also addressed the scientists- probationers of 110th and 111th batch of FOCARS. He urged the young scientists to come up with innovative ideas to undertake research for the benefit of stakeholders i.e. smallholder farmers.

During the year, the faculty and staffs of the Academy won four prestigious ICAR Awards including the coveted Rafi Ahmed Kidwai Award for Outstanding Research in Agricultural Sciences, Swami Sahajanand Saraswati Outstanding Extension Scientist Award along with two Cash

Awards in Administrative and Skilled Support Staff categories. One of the faculty has been conferred with Fellow of NAAS. Likewise, many of our faculty and staff have been recognized nationally and internationally for their outstanding work. The Academy bagged 'Best Landscaping Institute Award' (under central government institutions category) and 'Best Gardens Maintained Award' (under educational institute category) from the Department of Horticulture, Government of Telangana. The Academy has been striving to serve the farmers of the country which is recognized by winning the awards every year.

The ICAR-National Academy of Agricultural Research Management (NAARM) was established by the Indian Council of Agricultural Research in 1976 at Hyderabad. The major mandate of the Academy is to build capacity in agricultural research, education and extension education systems, and provide policy advocacy for the National Agricultural Research and Education System (NARES). To fulfill these mandates, Academy organizes various capacity building programmes for researchers, academicians, extension professionals, scholars, students and other stakeholders in NARES. The Academy strives to enhance individual and institutional capacity for innovation in NARES. Considering the strategic importance of agricultural research in food security and economic growth of the country, leadership, governance and innovation are emerging as prerequisite for the transformation of NARES into a more pluralistic innovation system. The Academy strives to generate a sense of fraternity and inculcate a scientific work culture amongst the agriculture scientists and faculty across the country. It plays a catalyzing role to usher planned transformation of the NARES in meeting its current and future challenges.

VISION

A global knowledge institution enabling National Agricultural Research and Education System (NARES) adapt to change through continuous innovation.

The Academy has leveraged its potential of excellent human resource management in the field of agricultural research into agribusiness

management, technology management and education technology management to develop and nurture students. The Academy offers higher education programme viz. Post Graduate Diploma in Management-Agribusiness Management (PGDM-ABM) having AICTE approval and accreditation from National Board of Accreditation (NBA), Diploma in Technology Management in Agriculture (DTMA) and Diploma in Educational Technology and Management (DETM) in distance mode in collaboration with University of Hyderabad (UoH). The Technology Business Incubator of the Academy, a-IDEA (Association for Innovation Development of Entrepreneurship in Agriculture) has earned laurels for its innovative approach and received national award for nurturing the Agri Startups.

MISSION

To enhance leadership, governance and innovation capacities of National Agricultural Research and Education System (NARES) through capacity strengthening, education, research, consultancy and policy support.

The Academy aims to emerge as an institution par excellence to facilitate and support a culture of dynamic management in agricultural research, education and frontline extensions, has been working with the mission of enhancing leadership, governance and innovation capacities of National Agricultural Research and Education System (NARES) through capacity strengthening, education, research, consultancy and policy support. The Academy is poised towards performing the role of a think tank for agriculture in advocating policies for strengthening National

Agricultural Research and Education system and for sustainable development of Indian agriculture.

1.1 Strategic framework

In tune with its Vision, Mission and Mandate, the Academy has developed a six-tier strategic framework during 12th Plan to steer different activities to achieve the set goals and targets. These are:

1. Agribusiness Management for Inclusive Growth

- Enhance capacities for food and agribusiness management education and research in NARES.
- Enhance capacities for entrepreneurship development in NARES and strategic management of agribusinesses.

2. Education System Management for Enriched Learning Quality and Environment

- Enhance capacities for agricultural education policy, planning and evaluation in institutions of NARES.
- Enhance capacities for faculty excellence and technology-enhanced learning for increasing learning effectiveness and continuous learning by deploying emerging media to create new and vibrant education environments.

3. Enhancing Capacities for Leadership and Governance

- Institutionalize a framework for leadership development at all levels of NARES - early, mid- career, senior professional and manager across all functions (research, education, extension).
- Institutionalize good governance in NARES by building capacities for effective

management of institutions, performance assessment, accountability, and developing strong organizational value systems and work culture that promote innovation.

4. Extension Systems Management in a Market driven Environment

- Enhance operational, adaptive and generational capacity of front-line agricultural extension systems to address emerging challenges.
- Enhance capacities for use of ICTs to provide customized knowledge, skills and solutions to farmers, farmer groups, rural communities and for social networking.

5. Information and Communication Management for Promoting Innovation and Governance

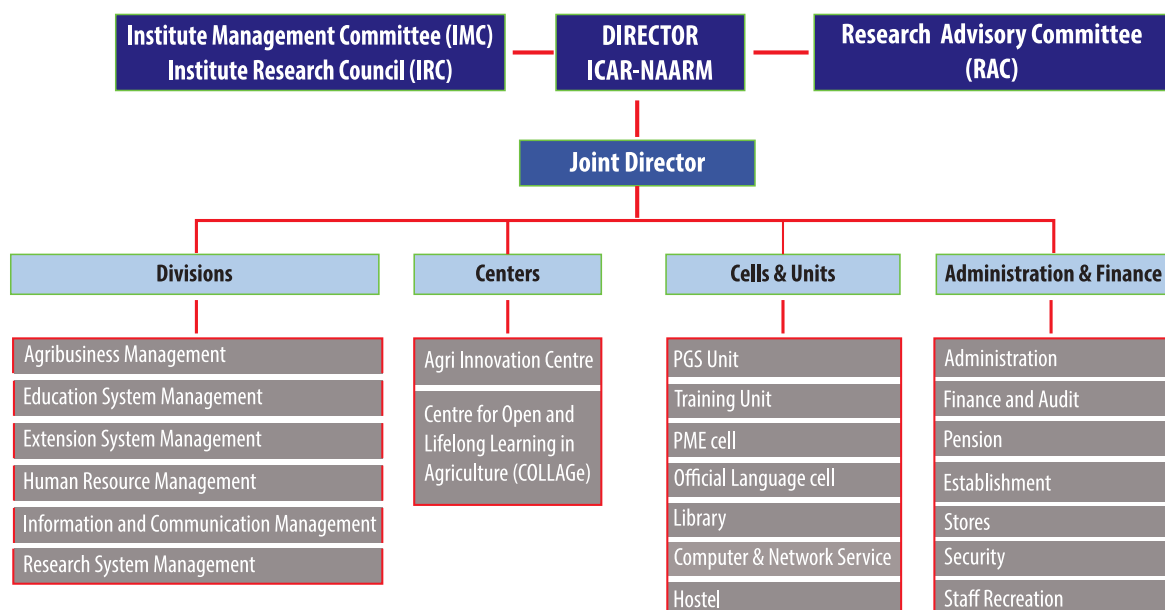
- Enhance capacities for integrating ICTs and Internet of Things (IoT) in institutional management and governance.
- Enhance capacities for managing data, information and knowledge in agriculture, scientific communication and institutional governance.

6. Mobilizing Science and Technology for Innovation and Sustainable Development

- Enhance capacities for research policy, strategy, priority setting, planning, management, monitoring and evaluation for a more pluralistic NARES, in the emerging contexts of food security, climate change and globalization.
- Enhance capacities for technology foresight and strategic management of intellectual property and commercialization of technologies



1.2 Organization and Management



1.2.1 Research Advisory Committee (RAC)

The Indian Council of Agricultural Research (ICAR) constituted the Research Advisory Committee (RAC) for three years from 14-05-2020 to 13-05-2023 as given below.

Prof. G. Padmanaban President NASI and Former Director IIS, Bangalore	Chairman
Dr. B. Venkateswarlu , Former Vice Chancellor, Vasanthrao Naik Marathwada Krishi Vidyapeeth, Parbhani, Maharashtra	Member
Dr. Pratibha Jolly , Science and Society Fellow, NASI Former Principal Miranda House, Delhi University	Member
Prof. V.K. Unni Public Policy Management Indian Institute of Management, Calcutta	Member
Dr. R.P. Singh Regional Manager-Operation & Resource, Breeding & Trait Development-Asia Pacific, Bayer BioScience Pvt. Ltd. Madhapur	Member
Dr. S.L. Goswami Former Director, National Academy of Agricultural Research Management, Rajendranagar, Hyderabad	Member
Dr. Ch. Srinivasa Rao Director, ICAR-NAARM, Hyderabad	Member
DDG (Agril. Education) , ICAR KAB-II, Pusa, New Delhi	Member
Two non-official members from rural/agricultural background nominated by the President, ICAR	Member
Dr. P. Krishnan , Principal Scientist ICAR-NAARM, Rajendranagar, Hyderabad	Member Secretary

1.2.2 Institute Management Committee (IMC)

The following officials were nominated as the members of the Institute Management Committee of NAARM for a period of 3 years (24-07-2020 to 23-07-2023).

Dr. Ch. Srinivasa Rao Director, ICAR-NAARM, Hyderabad	Chairman
Director of Agriculture Department of Agriculture, Government of Telangana	Member
Spl Commissioner and Director Agricultural Department Department of Agriculture Government of Telangana	Member
Director (Research) Prof. Jayashankar Telangana State Agricultural University, Rajendranagar, Hyderabad	Member
Shri Gone Shyamsunder Rao Seven Hills High School Mancherial, Telangana	Member
Shri Vallabhaneni Asha Kiran Vijayawada	Member
Dr. R. K. Mathur Director, ICAR-IIOPR, Andhra Pradesh	Member
Dr. V Tonapi , Director, ICAR-IIMR, Hyderabad	Member
Dr. Raghavendra Batta Director, National institute of animal Nutrition and Physiology, Bangalore	Member
Dr. SK Soam , I/c Joint Director, NAARM, Hyderabad	Member
Dr. M.K. Agnihotri ADG (HRD), ICAR, KAB-II, New Delhi	Member
F&AO , IIRR, Hyderabad	Member
Joint Director (Admn.) & Registrar NAARM, Hyderabad	Member Secretary

1.3 Linkages/Networking/Collaborations

NAARM has a strong strategic and functional network with 108 ICAR Research Institutes and 75 Agricultural Universities (AUs) and plays crucial role through capacity building of these institutions, agribusiness industry, scientists, and academia. Its network spans across various other National as well as International institutions.

1.3.1 National Institutes/ Organizations

1. All ICAR institutes and Agricultural Universities.
2. Various departments of the Government of India like, NITI (National Institution for Transforming India) Aayog, Department of Science and Technology (DST), Department of Biotechnology (DBT), National Science & Technology Entrepreneurship Development Board (NSTEB).
3. University of Hyderabad, Osmania University, Jawaharlal Nehru Technological University (JNTU), Indian Institute of Information Technology (IIIT), Council of Scientific & Industrial Research (CSIR), Indian Council of Medical Research (ICMR) and various institutions located at Hyderabad.
4. Ministry of Environment, Forest and Climate Change (MoEF & CC), National Board of Accreditation (NBA), Protection of Plant Varieties & Farmer Rights Authority (PPV&FRA), National Fisheries Development Board (NFDB) and State Biodiversity Boards.
5. Management Institutions located at Hyderabad viz. the Administrative Staff College of India (ASCI), Indian School of Business (ISB), National Institute of Agricultural Extension Management (MANAGE), National Institute of Rural Development and Panchayat Raj (NIRD & PR), and Institute of Public Enterprise

(IPE); and across the country such as Indian Institutes of Management (IIMs).

6. State Departments of Agriculture of various States (Telangana, Andhra Pradesh).
7. Private Sector Organizations (Agri Input Companies, Service Companies, etc.).
8. Non-Government Organizations (NGOs).
9. Other organizations viz. National Innovation Foundation (NIF), Federation of Indian Chamber of Commerce and Industry (FICCI), National Bank for Agricultural and Rural Development (NABARD) Centre for Innovation Incubation and Entrepreneurship (CIIE) etc.
10. Krishi Vigyan Kendras (KVKs).

1.3.2 International Organizations

1. CGIAR Institutions (CIMMYT, ICRISAT, IFPRI, etc.)
2. World Bank, FAO, UNDP and other United Nations Organizations
3. Department of International Development (DFID, UK).
4. SAARC Agricultural Centre (SAC).
5. NARS of South Asian Countries such as Bangladesh Agricultural Research Council (BARC), Nepal Agricultural Research Council (NARC), Pakistan Agricultural Research Council (PARC), and other NARS Institutions in Afghanistan, Sri Lanka and other ASEAN countries.
6. NARS in other African and South East Asian countries such as Kenya, Mali, Tanzania, Nigeria, Malawi, Liberia and Philippines.
7. Leading Land Grant and State Universities of USA and other Universities in Europe, Australia and other developing and developed countries.

Innovation, governance and knowledge form the fulcrum of the Academy's mission, which is driven by its Vision documents (2030 and 2050) as well as those of ICAR. Recognizing the enabling nature of role of emerging science and technologies across diverse sectors, it voices for their integration into agricultural value chain. Further, the evolving changes in agri-education sector shifting focus from 'teaching to learning outcome' signifies the requirement to develop newer institutional mechanisms and capacities of the system. The Academy also identifies the crucial need for a continuous stakeholder engagement to usher effective and efficient performance in the NARES with more transparency and accountability in place. Therefore, the activities of research, capacity building and education have been aligned to the six-tier strategic approach. Through a structured deployment of this strategy, the Academy envisages to play a catalytic role in the planned transformation process of the NARES. The subsequent sections present the annual progress and achievements in research, capacity building and the other activities of the Academy under the six strategic areas during the year 2020.

2.1 Agribusiness Management for Inclusive growth

With the shift in agricultural development strategies from a pure production oriented approach to a broader systems perspective, the importance of the agribusiness sector has grown significantly. Seen as an engine for growth, agribusiness and its related industries are receiving increased attention in policies and strategies that aim to promote investments in

agro-enterprises and agro based value chain development. This has prompted a need for deeper understanding of agribusinesses, agro-industries and agri-food value chains. The theme on Agribusiness Management for Inclusive Growth encompasses research, education, capacity building and consulting for developing future leaders in the field of agribusiness. It broadly aims to enhance capacities for food and agribusiness management education and research in NARES and to enhance capacities for entrepreneurship development in NARES and strategic management of agribusinesses.

Rapid urbanization, rising disposable income of households, expanding and diversifying food demand and globalization of agricultural economy have given impetus to research and capacity building in the area of Agribusiness Management. In order to meet the broader objectives of the theme, the Academy focuses on the following thrust areas for research, capacity building and education:

- **Agribusiness environment:** With a major focus on value chain analysis for agri-commodities and services; consumer's demands; price behavior and forecast analysis;
- **Public policy and impact:** With focus on market reforms like e-NAM, GST, etc.; financial inclusion and its spillover; spread and convergence of crop insurance;
- **Markets and institutional innovation:** Covering market integration of commodities; producer companies and FPOs; inputs and service delivery systems; diversification and

intensification of agriculture for high value commodities; business analytics;

- **Nurturing future leaders in agribusiness:** Through imparting quality education in agribusiness management.

2.1.1 New Market Reform and E-NAM in India: Effect on Market Choices and Price Realization for Smallholders

The study aims to examine the effect of policy changes in agricultural marketing in recent time known as New Farm Acts 2020, on business operations of e-NAM mandi, investment in post-harvest agriculture and farmers' price realization in India. The daily transaction data available on e-NAM portal and AGMARKNET for all the commodities and in all the mandis has been collected. The operations of e-NAM mandis were observed. Currently, 1000 APMC mandis are integrated into e-NAM platform, which are spread across 21 states and 3 union territories. The e-NAM has also expanded to six market places in Kerala state, where there is no APMC. Similarly, in Karnataka state, all APMC mandis are integrated into another online trading platform called ReMS (Rashtriya e-Market System). During this year, 2 ReMS markets have also been integrated with e-NAM to examine the inter-operability between two different platforms. The ongoing pandemic situation in the country also created an opportunity to recognize the Warehousing Development and Regulatory Authority WDRA-approved warehouses and FPOs as e-NAM market places. Thus, the farmers who have stocked their produce in these places, may sell their commodities from those places directly. The daily transaction data from e-NAM platform exhibited high volatility, particularly during October and November months. Besides, the value of

transaction during these two months were much higher than July to September months. It may be due to harvesting and marketing of *Kharif* season crops in October month. The study will analyse the transaction data from both the portals and also try to examine the trend in investment in post-harvest agriculture during rest of the project period. The field survey of farmers to understand their selling behavior will be undertaken after improvement in Covid situation in the country.



Fig. 2.1: Value of daily traded commodities reported on eNAM during July - November 2020

2.1.2 Mainstreaming the Small Ruminant Farmers into Efficient Value Chains in India

The study on mainstreaming the sheep farmers into efficient value chain in Telangana state was undertaken with the following objectives: (i) to study the socio-economic profile of the sheep farmers and carry out livelihood analysis of them, (ii) to study the production and marketing practices of sheep farmers, (iii) to identify the constraints in sheep value chains in the study area, and (iv) to suggest suitable strategies and policies to increase the income of sheep farmers through efficient value chains. Exploratory research design was selected as an appropriate design to carry out the study. Mahbubnagar and Nalgonda districts were selected for the primary survey based on the population status as per 19th Livestock Census, 2012. About 100 sheep farmers from each district were surveyed with the help of structured interview schedule. Besides, a sample



size of five value chain actors for each district were also interviewed to know about their activities, and the purchase and sale prices at each level in the market channel.

The average annual income of sheep farmers was Rs. 1,64,448 and 51.67 per cent of farmers had median annual income of Rs. 1,00,001 to Rs. 2,50,000. Among shepherds, 72.50 per cent farmers had sheep farming as the main occupation and rest 26.67 per cent had it as subsidiary occupation. 51.67% of them were small farmers (2.5-5 acre), followed by 25.83, 10.83, 10.00 and 1.67 per cent farmers belonged to marginal, landless, medium and large farmers. Majority (79.17%) of the respondents possessed medium flock size with the holding of 38-138 sheep, followed by 15.83% owning large flock size (>138 sheep) and 5.00 per cent had small flock size (<38 sheep). The overall experience of the farmers in sheep farming was about 19.37 years. Cost of production in rearing of sheep revealed that the average feed cost per animal in the study area was Rs. 45, health/ treatment cost was Rs. 161, labour cost was Rs. 1101, cost related to sheds, equipment and land lease was Rs. 238, and cost related to miscellaneous items was Rs. 41. Thus, the average cost of production of a sheep in the study area worked out to be Rs. 1,586 per annum. Production practices of sheep farmers in the study area showed that all the farmers in the study area were rearing the sheep in extensive system. The mean grazing time (hrs) and distance (km) in the study area was 9.41 hrs and 5.98 km per day respectively. Majority (82.50%) of sheep farmers didn't follow supplementary feeding, only 17.50 per cent of the sheep farmers were providing supplementary feeding. Home-grown ewes were the source of ewes for the majority (92.50%) of the sheep farmers. It was observed

that 92.50% of the respondents preferred to replace the ewes from their own flock and the rest (7.50%) purchased from outside in the study area, while 70.00% of the farmers replaced the breeding rams from their own flock, followed by 13.33% of them were found purchasing from outside, 9.17 per cent replacing from government schemes and 10.00% by exchange from others. Majority (70.00%) of the farmers marketed their sheep at 3-6 months of age, followed by 17.50 per cent at 9-12 months of age and 12.50 per cent at 7-8 months of age respectively.

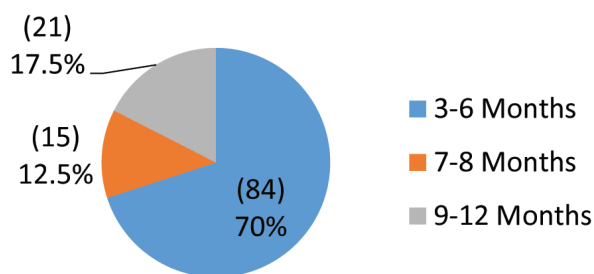


Fig. 2.2: Marketing age of sheep in Telangana

The average sale price of rams in the study area was Rs. 7056, ewes was Rs. 3053 and lambs was Rs. 3230. Majority (61.67%) of the sheep farmers sold their sheep biannually followed by 20.00 per cent of the sheep farmers sold their sheep annually and 18.33 per cent of the sheep farmers sold their sheep thrice a year. Marketing pattern revealed that nearly one-third (34.17%) of the farmers sold their sheep to the "local traders + outstation traders", followed by 29.17%, 15.83%, 5.83%, 5.00%, 4.16% and 3.33% selling their sheep to "local traders", "local traders + consumers", "outstation traders", "local traders + outstation traders + consumers", "local traders + butchers" and "outstation traders + consumers" respectively (Figure 2). Majority (91.67%) of the shepherds sold their sheep herds at their home followed by 5.83 per cent of the farmers in livestock market and 2.50 per cent of the farmers in haat. Majority

of the farmers (75.00%) expressed that price for marketable stock was fixed on the basis of physical appearance.

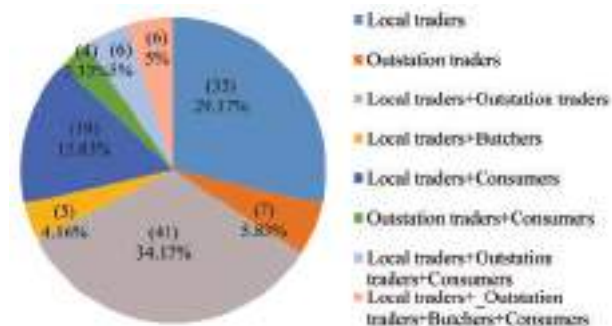


Fig. 2.3: Marketing pattern in the sale of sheep in Telangana

2.1.3 Techno-Economic Feasibility Studies for ICAR Agro-Processing Technologies/ Value Added Products

In a competitive world of scarce resources, new technologies fight for survival against developed technologies that promise immediate returns with comparatively little risk. Accelerated dissemination of innovative technologies requires efficient transfer mechanism to stakeholders. Prospective entrepreneur before adopting any technology would be interested to assess the technology feasibility. Lack of single window information to verify the various feasibility aspects along with other details of technologies is an important constraint which delays the technology adoption mechanism. It is in this context it is felt necessary to make available such required information in the form of a document i.e., techno-economic feasibility report for identified potential technologies which can help the entrepreneur in assessing the technology and also as a project report for obtaining support from financial institutions. The objectives of the study are to identify the Agro-Processing technologies/ value added products with commercial potential from ICAR Institutes; to estimate the project cost and assess the economic feasibility of identified

Agro-Processing technologies/value added products; and to document Techno-Economic Feasibility reports (TEFR) for the identified Agri-Processing technologies/value added products. Prioritization is in progress in consultation with Agr-innovate to identify the Agro-Processing Technologies from the 358 Agrinnovate identified ICAR technologies. Three products (Starch based Super absorbent Polymer; Cassava Resistant Starch (RS4); and Cassava Resistant Starch (RS5)) were identified that were listed by Agri-nnovate from CTCRI for the preparation of TEFR. Draft formats and parameters were prepared in consultation with the CTCRI Scientist concerned and the secondary data availability. Preliminary data on the inputs (Machinery, chemicals, labor, etc.) for a working model collected and estimated the economics.

2.1.4 Entrepreneurship Development through Farmer Led Innovations- A Case Study in Plantation Sector

The study was initiated to identify entrepreneurial Farmer Led Innovations (FLIs) in Plantation Sector. Among the 148 FLIs identified, an innovation index was developed to prioritize 28 FLIs for field testing. The project team and experts finalized about 28 parameters for each prioritized FLIs for field verification and validation and accordingly, questionnaire specific to each of the innovations along with the socio-economic details of the innovators was developed. A database on identified potential FLIs and a framework for prioritization of economically/ commercially potential FLI, profile of farmers-innovators along with details of innovations was developed.

The project team visited four innovators field/ location for field verification and validation of the innovations i.e., Tractor Operated Farm Waste Shredder for Coconut fronds in Tamil Nadu,



Mechanized Tractor Drawn Planter for Planting of Turmeric, Ginger and Turmeric Planter and Ginger/Turmeric Harvester in Maharashtra and Dehusking Machine for Arecanut. Questionnaire for hypothesis testing has been circulated among the farmer innovators to understand their entrepreneurial competencies, innovation proneness etc. The specifications for the App have been finalized and the same has been provided to C-DAC for App development.

2.1.5 Agri-Startups in India: Decoding the Determinants and Success Factors

The study has been undertaken with the objectives to investigate the motivations for endeavouring into agri startup; to examine the business processes followed by Agri-startups; to determine the factors influencing success of Agri startups; and to come up with cases of agri-startups. During the period, data from secondary sources were collected and studied. Additionally, eight agri-startups which are being incubated at NAARM technology business incubator i.e., aIDEA are also being interviewed.

It was found that agritech-startups account for 55% of the total agri-startups in India. Only 11% of the total agritech-startups have so far reached the scaling stage. The average age of agritech-startups which are in the scaling stage is found to be 4 years. These startups are trying to solve specific problems of agricultural value chain with close to a quarter into technology based devices (IoT, sensor, artificial intelligence based) which improves the efficiency of agricultural production and operation. The next big category is e-commerce and marketplace for agricultural produce in the agricultural value chain. Presence of startups can be seen beyond the metro cities and innovative startups evolving from smaller cities too. Karnataka is leading the race in terms

of total number of agritech-startups which is also having maximum number of startups altogether. The presence of startups in dairy, aquaculture and poultry sector is very minimal, majority being present in the crop and horticulture sector.

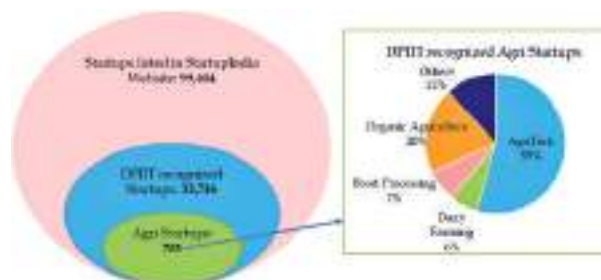


Fig. 2.4: Agri-startups and Distribution

More than 92% startups are in the crop and horticulture sector. Very minimal startups are into dairy/livestock sector which is 3% of total agritech-startups. These are particularly in fodder delivery, milk testing and IoT solutions for dairy farm operations. Same percentage of startups are in aquaculture with innovative aquaculture production processes.



Fig.2.5: Agritech-startups in different stages of agri value chain

2.1.6 Effect of COVID on Agri-Startups

With the launch of Start-up India initiative by the Prime Minister of India in 2015, almost all ministries and departments have taken several initiatives to promote young entrepreneurs and start-ups. Indian Council of Agricultural Research (ICAR), under Department of Agricultural Research and Education (DARE) has also established 25 Agri-

Business Incubators (ABIs) in first phase during 2017 and 25 more during second phase in 2020 through National Agricultural Innovation Fund (NAIF). Presently, these incubators are supporting more than 300 incubates in agriculture and allied sector. Apart from this, some institutions also got support from Department of Science and Technology (DST), Biotechnology Industry Research Assistance Council (BIRAC), Department of Agricultural Cooperation and Farmers Welfare (DAC&FW) of Government of India for creating incubation facilities. Thus, startup in agriculture sector is quite a new phenomenon. When the entire world including India is reeling under COVID-19 pandemic since January 2020, many of these agri-start-ups are feeling themselves into tailspin. Therefore, this quick study was conducted to understand the effect on agri start-ups due to the situations emerging from COVID-19 pandemic.

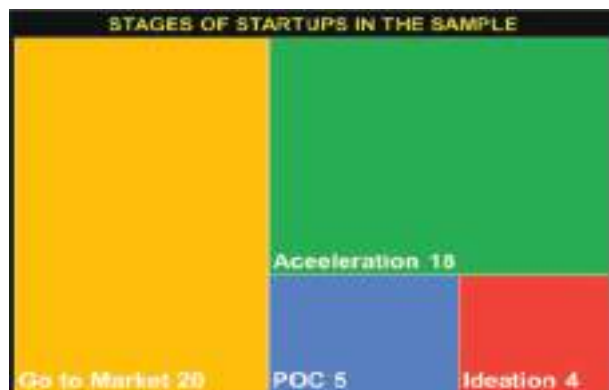


Fig. 2.6: Stages of Startups in the respondents

A study on the effect of Covid-19 on Agri-Start-ups in initial phase has been undertaken. To capture the implications of these extraordinary circumstances in the country on some of the Agri-Start-ups, a-IDEA, the technology business incubator of ICAR-NAARM conducted an online survey of about 47 agri-start-ups in different sub sectors. These start-ups were grouped in three category 1) Ideation (4 Nos) and POC stage start-ups (5 Nos) with a total of 9 start-ups; 2) Go-

To-Market stage start-ups (20 start-ups) and 3) Acceleration stage start-ups (19 start-ups). Fig. 2.6 shows the distribution of startups.



Fig. 2.7: Startup representing different sectors

Maximum number of start-ups are engaged in sustainable inputs in agriculture (22), followed by soil, water, weather technology using ICT, IoT, AI and ML (4). Other sectors which have been represented by start-ups are Agri-biotech, Innovative food technology, supply chain technology, animal husbandry, farm fresh retail, post-harvest technology, fisheries, organic farming, farm fresh retail in aquaculture. These start-ups have been working in different states of India. Many of them have business interests in multiple states. Most of the startup companies have started their revenue generation. In acceleration stage and Go-To-Market stage, about 10 per cent of the companies have not started revenues. This is a normal situation in startup ecosystem where almost 10% of the start-ups do not start revenue earning. This is due to the fact they are in a planning to come up with big launch or their product or service changes very fast due to the market situations and they are surviving mostly on the finances they have raised during this period. However, start-ups in ideation stage and POC stage, revenue generating start-ups are only two out of 9. This is again a normal situation in startup ecosystem.



Fig. 2.8: Startup generating revenues

Covid-19 is expected to affect start-ups business in different ways. The major constraints during the COVID 19 faced by the Go-To-Market stage and acceleration stage start-ups are of similar nature. Closure of production center/factory during the first lockdown period was the major roadblock. During lockdown-II, there might be some relaxation which was not known as survey was conducted before this. The issue of transportation was also highlighted which has been relaxed during lockdown-II.



Fig. 2.9: Constraints faced by Startups in Goto Market and Accelerator stage

During this period, “KISAN RATH” mobile app was launched by the government to ease part of the logistics constraints. Besides these, labour availability both in urban production centres and rural area for farm operations also became a serious issue. This is mainly due to reverse migration of labor from urban area to rural area, as the labours are willing to be near their family

members during this difficult time. Some start-ups in these two stages are also facing other constraints like burden of EMIs, unavailability of packaging material, increased receivables, hindered market accesses, etc.

Following initiatives need to be taken up for pulling out these start-ups:

- Ensuring liberal and easy funding in terms of equity need to be provided by the Government.
- Financial institutions may be instructed to help the start-ups to meet the working capital requirement in short and medium term for which respective incubator(s) may be trusted as counter guarantor.
- Tax exemption should be more clearly targeted towards start-ups.
- Incubators should be more professional and incubation center should be given more autonomy in terms of spending and earning.

2.1.7 Financial Accessibility and Occupational Risks of Vegetable Street Vendors in India

Street vendors are playing a vital role in the urban informal economy of India by providing employment and income, and goods and service to urban poor and marginal working class. In all the Indian cities, street vendors provide a low cost, decentralized and highly efficient system of distribution of a wide variety of goods of daily needs ranging from fruits and vegetables to clothes, utensils and many others. Majority of the street vendors are migrants from rural area. Around 10 million street vendors exist in India. Several vegetable vendors are engaged in last mile delivery of vegetables. Information about their livelihood and business were not available in the public

domain. Hence a study has been taken up which will help to understand the livelihood of street vendors selling fruits and vegetables and the risks faced by them. Attempts are made to find answers for the following research questions: i) Who are the street vendors and their socio-economic status and why they are engaged in vending activities?, ii) What challenges they are facing while working at street and sourcing the material?, iii) How they are contributing to the value chain of the commodities?, and iv) What are the policies that supports and affects their performances and risk exposure they face in their business?

2.1.8 Training Programme on 'Sustainable Management of Small-Scale Seed Enterprises'

To address the training needs of the small scale seed enterprises which have an important role in making available quality seed to farmers, a training programme on 'Sustainable Management of Small Scale Seed Enterprises' was conducted by ICAR-National Academy of Agricultural Research Management during October 05-09, 2020 in on-line mode. A total of 13 participants attended the training programme. Participants included representatives from Private Seed Industry (1), Krishi Vigyan Kendras (KVKs) (5), State Agriculture Universities (SAUs) (3) and Institutes of Indian Council of Agricultural Research (2) and others (2). Training participants represented seven states. Besides, the Guest Faculty from diverse areas in seed sector of the country viz., Sri Yaganti Venkateswara Rao, Former MD, Yaganti Seeds, Hyderabad, Dr. Soundareswaran, Professor (Seed Science and Technology), Tamil Nadu Agricultural University, Coimbatore and Dr. K.V. Sripathy, Scientist (SS), ICAR - Indian Institute of Seed Science, Mau also delivered lectures. The programme schedule comprised sessions of both

theoretical as well as hands-on exercises with 14 sessions each of 75 minutes' duration.

The training programme included sessions such as: Overview of Seed Business Scenario: Global and India; Types of Seed Enterprises; Formation of Small Scale Seed Enterprises; Understanding and Institutional Aspects of Seed Production and Marketing; Risk Analysis in Seed Production; Social and Environmental Aspects of Commercial Seed Business; Developing Business Plans for Small Seed Companies; Market Supply and estimation of Market Margins for a Seed Enterprise; Seed Processing and Storage Cost Estimation; Market Segmentation and Positioning; Breeders Seed Production & Supply to Companies/ Departments; Pricing Strategies and Mechanisms; Market Promotion; Seed Export Procedures; Seed Quality Regulation in India; Financial Viability Analysis and Market Distribution.

2.1.9 Business Plan Development and Accelerating FPOs/FPCs

ICAR-NAARM organized an online MDP on "Business Plan Development and Accelerating FPOs/FPCs" during Dec 14-19, 2020. In total, 34 participants attended the training, out of which 13 were from different ICAR institutes, 4 from State Agriculture Universities, 5 from KVKs, 6 represented FPOs, and 4 were individual or NGOs who were working with FPOs. One third of participants were women. The programme covered all the aspects of FPO, right from mobilization of farmers, FPO formation, and all aspects of business plan development for FPOs, like business model canvass (BMC), market research, marketing plan, etc. Apart from that other business management tools like management of supply chain, finances, networking with start-ups, digital and e-markets. We also invited several resource persons, including



NABARD representative, value chain financing company, Sammunati; supply chain company, Waycool. The founder of two successful federated FPOs- Sahaydri Farms and Sahaja Aharam also shared their experiences with the participants. Detailed legal and regulatory compliances and other operational challenges were discussed in detail. The participants were grouped in 6 different teams and they practiced and prepared their own business plan for one FPO. This was done by creating six groups which were working on the platform simultaneously.

2.1.10 MDP on 'Market Research & Value Chain Management of Agricultural Commodities'

An MDP on 'Market Research & Value Chain Management of Agricultural Commodities' was held at ICAR-NAARM, Hyderabad during 17-21 November, 2020 in online mode. The programme was conducted to orient the participants (i) to train participants on different aspects of market research such as market integration, market segmentation, consumer survey and so on; (ii) to expose them with various concepts of value chain management in different sectors of agriculture; and (iii) to equip them with the knowledge about various methodological approaches employed in the value chain management and research. The programme covered the following broad areas, viz. Concepts in market research; Business planning process; Market segmentation & product differentiation; Consumer survey; Concepts in

value chains; Value creation in the existing supply chains; Facilitating and strengthening of value chains; Descriptive analysis of market parameters; Operations and logistics management; Demand planning and forecasting; Enterprise Resource Planning (ERP); ICT in value chains; Commodity trading and Spot exchanges; Analytical tools applied in value chain management; Engaging Public Private Partnership along with people in VC; Developing business plans; Financing value chains and improving Business Environment of value chains. There were total 38 registered participants including 10 female participants. There were 7 non-ICAR participants which mainly included participants from KVKs and State Agricultural Universities. The programme consisted of a blend of lectures, case analysis, discussions and hand-on exercises.

2.1.11 Consultative Workshops on 'Academia-Industry-Government Linkages'

Two consultative workshops on Academia-Industry Government Linkages were organized as a part of World Bank sponsored National Agricultural Higher Education Project-Component 2A- one at Assam Agricultural University during Jan 27-28, 2020 and another at Punjab Agricultural University, Ludhiana during Jan 28-29, 2020. The aim of the workshops is to assist agricultural universities in strengthening their linkages with industry and structuring dialogue with state government to catalyze their participation in raising the quality and relevance



Fig. 2.10: Participants of Consultative Workshops on Academia-Industry-Government Linkages

of the agricultural higher education. In all, 133 experts from academia (Agricultural universities), industry and government participated in these workshops. Participants from North Eastern States, Punjab, Haryana, Himachal Pradesh and Jammu & Kashmir participated in the workshop.

2.1.12 Training Programme on Marketing Officers' Resurgence and Excellence for Agribusiness Professionals (MORE) of Coromandel International Limited

The objective of the program is to expose the participants with advances in agriculture technologies, soil health and nutrition management and to provide knowledge in Coromandel product portfolio and to bring attitudinal & behavioral changes to enhance Business management skills to make them ready to take higher responsibilities. The topics covered include Advances in agronomic practices of major crops, Climate change and balanced nutrition landscape, Farmer Producers Organizations (FPOs), Data analytics & Forecasting. Agribusiness

Management Skill development sessions on Business Model Canvas, Marketing Plan and Brand Management & Promotion strategies organized. The soft skills on Relationship Management, Negotiation and Conflict Management also covered during training programme.

2.1.13 Workshops on 'Development of Soft skills for Entrepreneurship among Agri-Graduates'

Collaborative Workshops were organised with 26 State Agricultural Universities on soft skills for entrepreneurship development during January-March 2020. These one-day awareness workshops were organised as a part of National Agricultural Higher Education Project (NAHEP)- Component 2, with an objective of creating awareness among agri-graduates on soft skills, innovativeness, and entrepreneurship. Faculty of 26 Universities were trained as master trainers by NAARM for organising these workshops targeting 3000 students.



Fig. 2.11: One-day workshop on Development of Soft skills for Entrepreneurship among Agri-graduates at PJTSAU, Hyderabad



Fig. 2.12: Workshop on Development of Soft skills for Entrepreneurship among Agri-graduates at PAU, Ludhiana



Fig. 2.13: Participants of Workshop on Soft skills for Entrepreneurship among Agri-graduates



2.1.14 Online Orientation Workshop and Training Program for ABI units

An online orientation workshop and training program for ABI units was conducted for three days during August 17-19, 2020 which was coordinated by Dr. K Srinivas and Dr. Manju Gerard of ICAR-NAARM, and Dr. Shiv Datt and Dr. Vikram Singh, from IP&TM Unit of ICAR. Main objective of the programme was to mainstreaming of newly created 25 ABIs into innovation and entrepreneurial ecosystem, and improving their skills to enhance efficiency and effectiveness of ABI. There were 58 registered participants of 25 ABI including PIs, CO-PIs and other contractual staff. The workshop was inaugurated by the Dr. T Mohapatra, Secretary, DARE and DG, ICAR and chaired by Dr. Ch Srinivasa Rao, Director NAARM. The programme was on virtual mode. All the participants gave their feedback positively after attending the programme and were ready to takeup the challenge. The valedictory session was addressed by Dr. Ch Srinivasa Rao, Director, ICAR-NAARM and Dr. Sanjeev Saxena, ADG (IP&TM), ICAR.

2.1.15 Post Graduate Diploma in Management-Agribusiness Management (PGDM-ABM)

The two-year fully residential Post Graduate Diploma in Management (Agriculture) (PGDMA) has been rechristened as Post Graduate Diploma in Management-Agribusiness Management (PGDM-ABM) during 2020. It has completed 11 years of its journey. The 10th batch (2018-20) comprising of 27 students underwent project work under the guidance of faculty member of the Academy as well as industry representatives. These projects dealt with the real-time problem's in the domains of value chain analysis of commodities,

market mapping, competitive analysis etc., helping to have a clear understanding of the current and future trends in the agribusiness sector. The batch graduated and got placed in esteemed organizations of agribusiness sector like Rallis India Limited, Samunnati, Waycool, PI industries, Basix SubK, etc. The 11th batch (2019-21) comprising of 48 students completed their summer training in several business organizations of repute. The students got hands-on training during their summer training. Due to COVID-19, many students had to resort to summer training from home and worked on problems identified by the organizations.



Fig. 2.14: PGDM-ABM 12th Batch Route to Final Admission

The 12th batch (2020-22) of the programme commenced at the Academy with 54 students (23 female and 31 male) taking admission. For the admission, the applicants were screened on the basis of CAT/CMAT scores and short-listed candidates were called for Analytical Writing Skill Test (AWST) and Personal interview (PI) on virtual platform amidst pandemic. The inauguration and

orientation programmes for the newly admitted students were organized virtually on August 6, 2020 while the new academic session started from August 10, 2020. The batch is represented by students from 13 states representing 27 different universities and 11 Agriculture and allied sciences disciplines at undergraduate level. Around 16% in the batch carry work experience.

The classes are being held online with all the students attending from their home. The students facing network issue at their home who were called back to campus have undergone COVID-19 test. All the necessary COVID guidelines are strictly followed in the campus. Apart from the academics, the students have shown their mettle in various national competitions organized by several Business Schools. PGDM-ABM students have secured 1st position in the event Brain-athon and 2nd position in the event Pitchers 2.0, Krishi Manthan held at IIM-A and 1st position in the event Krithya held in the e-conclave organized by SIIB, Pune. During the year, the students organized 5th edition of Sankalp, a student driven annual B-Fest started with an idea of 3 Cs- Collaborate, Compete and Connect.

2.2 Education System Management for Enriched Learning Quality and Environment

This theme is one of the core themes of the Academy, which essentially aims to enhance capacities for agricultural education policy, planning and evaluation in institutions of NARES and to enhance capacities for faculty excellence and technology-enhanced learning for increasing learning effectiveness and continuous learning by deploying emerging media to create new and vibrant education environments. In order to advance the objectives of this broad theme, the

Academy focused on the following thrust areas for its research, academic and capacity building activities. Lifelong Learning for Professional Development; Education beyond Higher Educational Institutions; Technology Enhanced Learning; Policy Advocacy on Education Systems Management.

2.2.1 Technology Enhanced Learning in Agricultural Education

The project on Technology Enhanced Learning (TEL) was conceptualized and implemented with a purpose to provide a futuristic, implementable, replicable and scalable model for a comprehensive technology enhanced and enabled education in agriculture. It involved developing and evolving state of art educational IT infrastructure in the form of TEL lab, ably backed up with content production, refinement and sharing strategies. Along the course of the project, several innovations, processes and technologies took shape and their effectiveness is seen in wider beneficiary base in the form of online learning, capacity building and digital course development. The established TEL lab has clearly come out with blue print as how to implement technology in education with multi-functional applications. This will go a long way in enhancing the quality of education in agriculture and also address various issues like reaching- the-unreached and provide access through offline education, online education, vocational education and distance education.

The TELAgE facility was innovatively used to offer Massive Open Online Courses through open source e-learning methodology. A total of 5 MOOCs were offered with success rate of 53 per cent and training over 2315 personnel. Certification of these programmes generated a revenue of about Rs. 16.57 lakhs. The protocols



were used to develop model digital courses, offer distance and online programmes, conduct virtual classes, and support some of the in house programmes. A total of 9 digital courses were developed. The methodology is blossomed into a revenue generation activity also by providing the consultancy-mode support to other agencies in developing and hosting the digital content. Programmes executed for institutes like MANAGE, NIRD&PR, NIPHM, ICAR-NDRI which generated a revenue of about Rs. 61.67 lakh. The project also promoted distance education through a hybrid model where the students can also get online resources and help, participate in online quizzes besides having resource material in

typical distance education mode. The TELAgE center also could provide virtual classes by which the contact sessions were conducted in two center simultaneously, thereby enhancing the participation rate in the distance programmes. One of the distance education programme was offered completely in online mode with 18 participants. A total of about 100 streaming hours were provided for virtual classes to support contact sessions of another distance education programme. The practices and knowledge generated were shared through capacity building programmes and policy workshops. Approximately 3000 personnel were exposed to the TEL protocols developed. The project outputs are summarized in table 2.1.

Table 2.1: Major Outputs of the Project on ‘Technology Enhanced Learning’

Outputs	Years					
	2015- 16	2016- 17	2017- 18	2018- 19	2019- 20	Total
No of video hours	184	396	314	124	262	1280
No of video modules	72	233	162	105	124	696
No of online courses	3	3	3	1	2	12
No of certificate programs	1	2	1	2	2	8
No. of beneficiaries trained through MOOC	98	529	516	482	690	2315
No. of beneficiaries trained through training / workshop	85	65	94	25	22	291
National visitors to lab	155	1015	734	302	319	2525
International visitors to lab	123	150	12	0	44	329
Average hits per day	372	1001	1082	827	989	854.2
Revenue generated through MOOC (Rs. in lakhs)	-	2.65	4.4	3.54	5.98	16.57
Revenue generated through Consultancy (Rs. in lakhs)	-	3.4	6.6	21	30.67	61.67
No. of other projects using TELAgE project outputs	-	2	1	1	2	5
Publications	0	2	1	1	2	6

2.2.2 Students’ Learning Approaches (SLAs) for enhanced learning outcomes in Agriculture Education: A Critical Analysis

An applied research project was taken up with the objectives: to analyse Students’ Learning Approaches (SLAs) among agriculture and

allied sciences for enhanced learning outcomes; to analyze the changes in Students’ Learning Approaches over time; to investigate the factors affecting Students’ Learning Approaches for effective learning outcomes; and to develop strategies for enhancing the quality of learning

outcomes in agriculture education. By using the instrument ASSIST (developed by Marton and Säljö (1976) and revised by Entwistle (1997) , the data were collected from 1894 students, representing 31 State Agricultural Universities. The students represented all streams of agricultural and allied sciences. The data collected for temporal dimensions from PGDMA students of NAARM was collected twice at the time interval of one year. Critical analysis on Students' Learning Approaches (SLAs) for enhanced learning outcomes in agriculture education indicated that majority of students are using strategic approach rather than deep / surface approach; more female students are using strategic approach than males; relatively more veterinary students are using deep approach and majority of PGDMA students are using strategic approach, followed by deep approach. There was moderate change in terms of approaches over time among PGDMA students. The student Learning Approaches should be considered in development and implementation of curriculum and also in designing assessment methodologies. One of the objectives has been integrated as an activity into National Agricultural Higher Education Project on 'Investments in Indian Council of Agricultural Research Leadership in Agricultural Higher Education'.

2.2.3 Ag-Academy: A Framework for scaling up e-learning platform

Agacademy is a cognitive edtech platform with a vision to create a National Agricultural Platform that enables educators to create content and engage with learners. Basically it enables educators to develop courses with a Mobile App and can be shared with different stake holders. Different mobile applications will be developed for educators as well as learners. Agacademy vision is to get the best minds from the country to

share their knowledge in an easy-to-understand form with the following objectives: To develop an artificial intelligence driven framework for scaling up of electronic learning platform for Indian agriculture; To develop the robust cloud based platform for hosting the courses developed by end users; Developing proper strategic, marketing and communication methodologies for popularizing the developed platform; to develop learning path construction for individual and customized learning of each and every end user; ensuring financially sustainable platform and assessing adaptation and impact of the platform among stakeholders.

Some of the significant achievements of the project was a Learning Management System at www.agacademy.net was installed and configured, progress has been made on developing frame work by initiating discussions with different stake holders, working and currently on Learning Path Construction Algorithms using artificial intelligence technologies has been identified Google Colaboratory as a platform for developing algorithms, The possibility of tying up with private company or a-idea incubated companies or empaneled firms of the academy, as a robust cloud based platform has been explored which requires huge IT resources.

2.2.4 Investments in Indian Council of Agricultural Research Leadership in Agricultural Higher Education

The objectives of the project are: i. to establish partnerships with globally recognized agricultural higher education institutions; ii. to promote the establishment of career development center at agricultural universities; iii. to improve curricula review processes and methods to consolidate and disseminate global best practices in agricultural education; and iv. to assist agricultural universities



in strengthening their linkages with industry and structuring dialogue with state government to catalyse their participation in raising the quality and relevance of agricultural higher education. A national workshop on academic excellence and resources generation and a training workshop on education management and academic leadership were organized. The recommendations of the workshops on 'Academia-Industry-Government Organization on Linkages' were drawn and a technical report on 'Global Best Practices in Higher Education' and a training manual for soft skills development were published. A policy brief on 'Re-imagining Higher Agricultural Education in India on the Face of Challenge from COVID-19 Pandemic - Strategies for Adapting to the New Normal' was published. Also a Hindi book titled '*Krishi-Udyamita Dvaara Atmanirbhar Bharat: Chhaatropayogee Gyaan Evam Kaushal Vikaas*' was published. A study on 'Curriculum Revision' involving BSMA members was carried out and developed an 'Inspired Teachers Database portal'. In addition, soft skills development training workshops were organized for 1997 students of 20 agricultural universities. Five Career Development Centers (CDCs) were established at 05 State Agricultural Universities and a Faculty Development Center at GBPUAT, Pantnagar. In total, 3280 participants benefited from 52 various programmes.

2.2.5 Development of Digital Content for MOOC on 'Rodents and Household Pest Management'

The main objective of the project was to enhance the core competencies of human resources in understanding the management aspects of rodents and household pests in day-to-day real time situation. As per the project requirement, 21 digital content modules have been developed to enhance the competency of field level pest control

operators. Each module covering 12 to 15 min duration was well edited along with animation and other presentation aids. Subject experts were identified by NIPHM and the methodology for content development was provided by NAARM to the invited experts.

2.2.6 Development of Digital Content for MOOC on 'Commercial Dairy Farming and Milk Processing and Value Addition'

The main aim of the project was to enhance the capacity of the dairy industry professionals and farmers/entrepreneurs for efficient and knowledge driven management of dairying and their products through e-content for implementation of MOOC courses by NDRI. As per the project requirement, 66 digital content modules are to be produced. Later NDRI revised the requirement of 60 modules on dairy management, milk processing, value addition and marketing. ICAR-NDRI and ICAR-NAARM together developed 60 digital content modules for offering MOOC on commercial dairy farming, which is first-of-its-kind initiatives. The MOOC launching programme was formally done by Dr. B.N. Tripathi, DDG (Animal Science) and Dr. R.C. Agrawal, National Director, NAHEP and DDG (Education) on 14th August, 2020. NDRI, Karnal and SRS, NDRI, Bangalore have identified experts for the content development.

2.2.7 Workshop on 'Prospects of Agricultural Education: Awareness among Higher Secondary School Administrators'

An awareness workshop on 'Prospects of Agricultural Education: Awareness among Higher Secondary School Administrators' was organized on Feb 24, 2020 at ICAR-NAARM. It was organised with an aim to share and communicate among Principals, Vice Principals, and Faculty on the prospects of agricultural

education and the need for quality manpower so that the 12th standard students would be aware of participating in the National Level entrance examination i.e. AIEEA-2020 (All India Entrance Examination for Admission) to be organised by ICAR in June 2020. About 35 participants including Principals, Vice Principals, and Faculty in Science from eight Higher Secondary Schools of Hyderabad participated in the programme. Mrs. Jaya Prada Bai B, DCEO- Hyderabad and Regional Joint Director inaugurated the programme and opined that these awareness programmes are very important for students to think beyond Engineering and Medicine stream, i.e. Agriculture and Allied Sciences. On this occasion, a flier of ICAR-AIEEA 2020 and a booklet on Career Opportunities in Agriculture and Allied Sectors were released. Dr. S.K. Soam, CPI NAHEP Project from ICAR-NAARM described the objectives and activities of project i.e. NAHEP and its contribution to quality and relevance of agricultural education. Dr. R Chandrashekar, Former Dean, SKLTSHU; Dr. D. Vishnuvardhan Reddy, Dean of Student Affairs of PJTSAU and Dr. ST Viroji Rao, Dean of Faculties of PVNR Telangana Veterinary University explained.



Fig. 2.15: Dignitaries and participants of workshop

2.2.8 Training Programme on 'Technology Enhanced Learning'

The main objectives of the training programme are (i) to enhance pedagogical methods with digital technology. (ii) to sensitize on technology

options for teaching and learning and (iii) to sensitize on digital learning management. The course was designed to apprise the teaching faculty of agricultural universities on use of technology for education and information retrieval. The programme encompassed concepts like pedagogical methods with digital technology, E-Learning, IPR for Digital Content and National Agricultural Education System, Basics of Teaching Learning & Teaching Methodologies, Digital Teaching and Creation Methods, Micro teaching, Learning Management, and Digital Information Retrieval.



Fig. 2.16: Director, NAARM with training participants

2.2.9 Virtual Training Workshop on 'Contemporary Education Technologies for Agricultural Education'

An online workshop was organised for the faculty of ANGRAU, Guntur with the objectives to apprise the scientist- trainees about current agricultural education scenario and the challenges faced by the agricultural education systems in the country; to enhance pedagogical methods with digital technology for enhanced digital education; and to sensitize on the usage of cloud based digital learning management systems. This training workshop on 'Contemporary Education Technologies for Agricultural Education' was conducted in online mode during June 08-12, 2020 for 20 faculties from six campuses of



ANGRAU, Guntur, Andhra Pradesh. This was the first of its initiative by the academy to cater to the capacity building demands through completely online mode in the area of online educational practices and protocols, duly considering the challenges faced due to Covid 19 scenario. Both the programme contents and delivery mode adopted for this online workshop amply instilled confidence among the participants to gear up for digital and online learning.

2.2.10 Training Programme on ‘Teaching Competency Enhancement’

The programme was organized during July 07-09, 2020 with 15 including 3 female participants. The objective of Online Training Programme on ‘Teaching Competency Enhancement’ was to orient the young faculty of SKLTSHU towards the basics of Teaching-Learning process, communication, presentation, technology in Teaching-Learning, developing winning research proposals, stress management etc. The Academy has switched over to new normal strategy of capacity building from on campus to online, duly considering the challenges faced due to Covid 19 scenario. The programme contents and delivery mode adopted for this online training sensitised the participants for enhancing the teaching competence.

2.2.11 Training Workshop on ‘Education Management and Academic Leadership’

The workshop was organised under National Agricultural Higher Education Project Component-2A in online mode during Sep 03-17, 2020 with 45 including 10 female participants. The primary objective of training workshop was to enhance the efficiency of Indian Agricultural Education System through Education Academic Leadership. The programme was aimed for the benefit of Senior

Administrators/ Heads of departments. Two members of External Advisory Panel of NAHEP from Texas A&M University and Cornell University also delivered lecture, apart from eminent experts from India. Forty-five enthusiastic administrators and senior faculty of agricultural universities from 18 states participated in the online training workshop. During the fifteen-day training programme, the participants were provided with theory and practical sessions on various topics related to education management and academic leadership by eminent personalities. An online application was developed to capture session wise reflections of the participants. The members of NAHEP External Advisory Panel i.e. Dr. Glenn C Shinn of Texas and A&M and Dr. K V Raman of Cornell University also interacted with the participants online.

2.2.12 Online training programme on E-Resources, E-learning and IoT

The programme was organized during Sep 05-09, 2020 with 35 including 14 female participants for the students of UAS Raichur. The objectives of the programme to the students of UAS, Raichur were (i) To apprise the trainees about agricultural scenario, and the agricultural research, education in the country. (ii) To enhance the knowledge on IOT in agriculture, IoT Policies and conduct Case Studies in Agriculture and also (iii) To sensitize on digital learning and digital resources. The training program had touched various contemporary aspects of education, technologies and digital resources that every student needs to be updated with. The feedback on the program-related aspects and the lectures, resource persons was sought through Google form at the end of the program. The analysis of the feedback suggested that the participants were very much satisfied with the program, as many parameters have

scored over four rating on a 5-point continuum. Overall, assessment parameters like course content, coordinator's skill and support, relevance, overall learning have all been rated above 4.5. The topics were desired by the participants on agriculture knowledge, roles and responsibilities of an agriculture student to have better carrier growth, useful computer software's for students. The session -wise feedback also showed highly positive response among the participants, with all the parameters like time allocation, coverage, usefulness and teaching aids were rated above 4.0.

2.2.13 Online Training Programme on 'Recent Innovations in Educational Technology'

An online training program on 'Recent Innovations in Educational Technology' was conducted during November 05-10, 2020 for 31 participants. The objectives of the programme were (i) To sensitize and update on the latest methods of teaching and learning; (ii) To provide practical experience on Technology Enhanced Learning in agricultural education; and (iii) To introduce innovative technology-based methodology in agricultural education. About 13 sessions were organized covering both theory and practicals. More emphasis was given on hands-on training to the participants on Digital Content Editing and LMS. Additional knowledge on application of Education Technology in Medical Science was imparted to the trainees in order to enable them to develop similar module in agriculture/veterinary sciences. The feedback of the trainees was the content of the training was very much apt for present Covid-19 situation and useful for enhancing efficiency and suiting the teaching requirement. Trainees urged to enhance the duration up to 12 days and

to include additional content on Blog Writing, Webpage development and tips for improvement in PPTs. During the valedictory programme, Dr. Ch. Srinivasa Rao, Director, NAARM appreciated the feedback provided by the participants and addressed the participants and encouraged to adopt the learning process in their educational environment.

2.2.14 Brainstorming Workshop on 'Networking of Agricultural Universities Alumni'

ICAR-National Academy of Agricultural Research Management, Hyderabad under National Agricultural Higher Education Project (Component 2A) organized a one-day Brainstorming Workshop on 'Networking of Agricultural Universities Alumni' at ICAR-NAARM, Hyderabad on 22nd January, 2020. This one-day workshop was organised with an objective to build a robust alumni platform for National Agricultural Research and Education System (NARES) and to map alumni through technological interventions and other strategies.



Fig. 2.17: Brainstorming Workshop on Networking of Agricultural Universities Alumni

The focus of the workshop was to build a robust alumni platform for National Agricultural Research and Education System (NARES) and to map the alumni of agricultural universities. Senior officials from Prof. Jayashankar Telangana State Agricultural University, Hyderabad, Sri



Konda Laxman Telangana State Horticultural University, Hyderabad and P.V. Narasimha Rao Telangana Veterinary University, Hyderabad attended the workshop. IT experts from various industries were also present during the workshop and demonstrated existing alumni mapping mechanisms including data harvesting through various social networks. In total, 25 participants participated in this workshop.

2.2.15 National Workshops on 'Career Development Centre'

Three National Workshops on Career Development Centre (CDC) were organized as part of the World Bank-funded National Agricultural Higher Education Project (NAHEP) component-2. The main objective of the workshop was to prepare the guidelines and protocols for the establishment of CDCs at five different locations of the country. The first Workshop was organized at Kerala Agricultural University (KAU), Thrissur on 7th February 2020 with 50 participants, out of which 16 were female participants representing agricultural universities and industry from the states of Karnataka, Kerala, Tamil Nadu, Telangana, Kerala and Uttar Pradesh.



Fig. 2.18: First National Workshop on Career Development Centre (CDC) at Kerala Agricultural University, Thrissur

The Second Workshop was organized at Sri Karan Narendra Agriculture University, Jobner-Jaipur on 12th February, 2020 with 47 male and 3 Female participants representing agricultural universities and industry from the states of Rajasthan,

Gujrat, Himachal Pradesh, Jammu & Kashmir, Uttarakhand and Punjab.



Fig. 2.19: Second Workshop Career Development Centre (CDC) at Sri Karan Narendra Agriculture University, Jobner-Jaipur

The Third workshop was organized at Indira Gandhi Krishi Vishwavidyalaya, Raipur on 14th February, 2020 with 51 participants, including 3 female participants representing agricultural universities and industry from the states of Odisha, Maharashtra, Chhattisgarh, West Bengal, Bihar and Madhya Pradesh. These Workshops were coordinated by Dr. N. Srinivasa Rao, Co-PI, NAHEP Component 2 & Principal Scientist and Dr. Alok Kumar, Co-PI, NAHEP Component 2 Principal Scientist, ICAR-NAARM, Hyderabad.



Fig. 2.20: Third National workshop on Career Development Centre (CDC) at Indira Gandhi Krishi Vishwavidyalaya, Raipur

2.2.16 Model Career Development Centers (CDCs) and Faculty Development Center (FDC) at Agricultural Universities established under NAHEP

Model Career Development Centers (CDCs) and Faculty Development Center (FDC) at Agricultural Universities were established under NAHEP Project, Component II. The NAHEP team of the academy facilitated the establishment of

Career Development Centers (CDCs) at Central Agricultural University (CAU), Imphal; Indira Gandhi Krishi Vishwavidyalaya (IGKV), Raipur; Sri Karan Narendra Agricultural University (SKNAU), Jobner; Sri Venkateswara Veterinary University (SVVU), Tirupati and Uttar Banga Krishi Viswavidyalaya (UBKV), Coochbehar. The major objective of the Career Development Center is to make the students more confident, and skilled and ready to support the country's mission of 'Atmanirbhar Bharat' through entrepreneurship. Faculty Development Centre (FDC) has been established at Govind Ballabh Pant University of Agriculture and Technology (GBPUAT), Pantnagar with the objective to strengthen teaching and enhance professional competency of the SAUs and bring in best global practices in the higher agricultural education system in the country.

For implementation of the scheme and constitution of CDC and FDC committee, the first virtual review meeting on status of establishment of CDCs / FDC with nodal officers / coordinators of CDCs/FDC was held on June 22, 2020 under the Chairmanship of Dr. S.K. Soam, Joint Director and CPI, NAHEP component 2. Dr. D. Thammi Raju, Co-PI briefly explained about the genesis, objectives and outputs of NAHEP component 2 and also discussed the focal areas of CDCs and FDC. The University officials presented the status and action plans of their CDCs & FDC. Drs. N. Srinivasa Rao and Alok Kumar coordinated the event.



Fig. 2.21: Participants of 1st Virtual Review Meeting on Status of Establishment of CDCs & FDC with Nodal Officers/Coordinators of CDCs/FDC

2.2.17 Capacity building of ICAR Nodal officers at SAUs through MDP at Various IIMs

ICAR-National Academy of Agricultural Research Management (NAARM) as part of the World Bank funded National Agricultural Higher Education Project (NAHEP) under component 2 'Investments in ICAR for Leadership in Agricultural Higher Education' had taken initiative for the capacity building of ICAR Nodal Officers / Senior University Officers from different State Agricultural Universities (SAU) through 4-5 days training at premier institutions such as Indian Institute of Managements (IIMs) in the area of leadership, innovation, etc. Around 15 ICAR Nodal officers from 13 SAUs trained in 03 batches at IIM, Kolkata and IIM, Ahmedabad. In the first batch, three participants attended training on 'Project Management' during 13-17 January, 2020 at IIM Kolkata. In second batch, 04 participants attended training on 'Interpersonal Effectiveness and Leadership Excellence' during 20-24 January, 2020 at IIM Kolkata. In third batch, 08 participants attended training on 'HR Auditing: Preparing the Ground for Strategic HRM' during 24-26 February, 2020 at IIM Ahmedabad.

2.2.18 Online Training on 'Pedagogy of Online Teaching: Philosophy and Methods' under FDC

Under the Faculty Development Centre (FDC) of Component 2 of NAHEP, GB Pant University of Agriculture & Technology (GBPUAT), Pantnagar conducted an Online Training on 'Pedagogy of Online Teaching: Philosophy and Methods' during 23-24 October 2020. Dr. Tej Partap, Vice Chancellor of GBPUAT, inaugurated the event. In his inaugural address, he stressed the need of training for teachers for very specific purposes, not only in teaching but also in research domains in research proposal writing, research proposal handling and also

soft skill training to be a good counselor, mentor, warden, etc. In addition, communication training will help in talking with different stakeholders especially farmers. He also pointed out to upgrade to the FDC as National College of Teachers Training. He appreciated the efforts of NAARM and NAHEP for giving FDC to GBPUAT.



Fig. 2.22: Participants of Online Training on 'Pedagogy of Online Teaching: Philosophy and Methods' under FDC

2.2.19 Online Training Workshop on 'Education Management and Academic Leadership'

The ICAR-NAARM has organised an Online Training Workshop on 'Education Management and Academic Leadership' during Sep 03 -17, 2020 with an objective to enhance the efficiency of Indian Agricultural Education System under NAHEP-Component 2. Forty-five enthusiastic administrators and senior faculty of agricultural universities including ten female participants from 18 states participated in the online training workshop. During the fifteen-days training programme, the participants were provided with theory and practical sessions on topics related to education management and academic leadership by eminent professionals. An online application was developed to capture session-wise reflections of the participants. The members of NAHEP External Advisory Panel i.e. Dr. Glenn C Shinn of Texas and A&M and Dr. K.V. Raman of Cornell University also interacted with the participants online.



Fig. 2.23: Director, NAARM with the Participants of Online Workshop on 'Education Management and Academic Leadership'

Dr. Ch. Srinivasa Rao, Director, ICAR NAARM, the Chief Guest of the valedictory programme highlighted the importance and status of agriculture education system in India. He emphasised that education would be the miracle to change, bringing relevancy to the system. Dr. Prabhat Kumar, National Coordinator of NAHEP Component 2, the Guest of Honour stressed on the importance of leadership and management in education system. He highlighted the need of learning and mentioned that it is a lifelong two-way process.

2.2.20 MOOC on 'Psychology of Learning for Stay-home Learning'

In its ever-continuing endeavors to meet the capacity building needs of NARES, the Academy launched a new fifteen days MOOC on Psychology of Learning during May 1-15,

2020. The programme is quickly conceptualized, synthesized and launched keeping in view the scenario demanding the utilization of time while staying at home. The programme aims to equip the teaching fraternity with understanding the psychology aspects of learning from student's perspective. In his inaugural address, Dr. Ch. Srinivasa Rao, Director briefed the importance of this course for teachers, faculty, trainers and others. He also indicated the success of earlier five MOOCs which is the precursor for the start of 6th MOOC course on Psychology of Learning.

The course was developed with a rich blend of videos, documents, presentations that are delivered through the NAARM e-Learning portal. The learning resources were developed using Academy's state of art facility, Technology Enhanced Learning Lab, which was established with the support of ICAR Education Division under Niche Area of Excellence. Besides the learning resources, the course facilitates learner activities and participation in online discussion forums followed by final online evaluation. The entire programme was offered in online mode and final certification was provided to the participants. The MOOC received a record 1658 registrations that represent entire country and covering wide range of subject domains, thereby benefiting the larger audience and aiding in the effective time utilization while at home.



Fig. 2.24: MOOC Video at NAARM e-Learning portal

The MOOC is offered through NAARM eLearning portal at <https://elearning.naarm.org.in>.

2.2.21 MOOC on Designing E-learning Content

The programme was organized during July 01-31, 2020 attracting 2773 including 1109 female participants. The objectives of the programme are (i) To develop systematic and quality instructions for presenting course content to meet learners' needs through technology in education and (ii) To emphasise on the Instructional Design Concepts, Models and Storyboarding and use the same in evolving E-Learning content design, with Rubrics and Assessments. This programme was offered through e-learning portal along with online registration. Innovative practices like peer evaluation, online evaluation and online assignment submission were used to evaluate the participants.

2.2.22 Post Graduate Diploma in Educational Technology Management (PGDETM)

The Post Graduate Diploma in Educational Technology Management, a year distant education programme is offered jointly by ICAR-NAARM and UoH Hyderabad. The 2nd Semester Contact Classes and Examinations for PGDETM-2019 batch were organised during Feb 08-11, 2020 for 26 candidates at ICAR-NAARM. The resource persons from NAARM and external faculty were invited to deliver sessions during the contact classes. The examination was conducted under the supervision of UoH and NAARM.



Fig. 2.25: PGDETM Examination during Feb 10-11, 2020 at TELAgE Lab, ICAR – NAARM



A total of 54 applications received for the 2nd batch of PGDETM admission. Out of 54 applications received, 40 were admitted after screening on payment of 1st semester course fee. The 2nd batch started from Jan 25, 2020 and the learning resource materials have been delivered to all the 40 candidates. The Online Contact Classes for PGDETM students were conducted from 20th to 29th May, 2020.

This was first of its kind initiative by the institute to conduct contact sessions through online mode amidst the challenges being faced due to COVID-19 pandemic situation. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM and Prof. Jeelani, Director, University of Hyderabad addressed the candidates of PGDETM. Fifty-two candidates attended the online programme from their respective places. The classes were from all the seven courses of the PGDETM Programme involving eight resource persons with 28 hours of contact sessions.



Fig. 2.26: Director, ICAR-NAARM during the contact classes for PGDETM

Followed by the Contact Classes, the Internal Assessment for PGDETM candidates (46 no.) was organised from June 01-10, 2020. The state of art facilities of TELAgE lab at NAARM has been used for organizing the Internal Assessment. The online contact classes for forty candidates of 2019 & 2020 batches were held and the students attended from their respective places. All the

seven courses of the PGDETM Programme were covered by various resource persons. Besides Contact Classes, Internal Assessment has been organized from December 01-10, 2020 for both the batches.

2.3 Enhancing Capacities for Leadership and Governance

Human Resource Management division has made noteworthy efforts to initiate and infuse a competency-based management research system through training, performance management, and teaching competency framework for achieving organizational goals. The division is involved in the development of competencies and capacities of scientists and research leaders in the ICAR and Agricultural Universities. Leadership and governance are the two important drivers to promote organizational effectiveness. Realizing this, the division is engaged in research and capacity building in these areas and associated areas of human resource management, competency management, teamwork, job and organizational stress and emotional intelligence. Through these endeavours, the division aims to develop individual and institutional capacities to deal with contemporary issues and achieve individual and organizational excellence. The strategic framework of the division envisages enhancing capacities for leadership and governance by institutionalizing a framework for leadership development at all levels of NARES- early, mid-career, senior professional and manager across all functions (research, education, extension), and institutionalizing good governance in NARES by building capacities for effective management of institutions, performance assessment, accountability, and developing strong organizational value systems and work culture that promote innovation.

To realize the above objectives, the following activities related to research and capacity building programs were carried out during the year 2020.

2.3.1 Emotional Intelligence and Organizational Effectiveness

The objectives of the project is to measure the emotional intelligence of scientific personnel and their organizational effectiveness in terms of team work, conflict resolution and leadership attributes and to suggest strategies to improve organizational effectiveness in NARES. Psychometric data was collected from 118 number of Senior and Principal Scientists of ICAR and SAU's regarding their status of emotional intelligence, team effectiveness, Thomas-Kilmann conflict modes, 180-degree leadership effectiveness attributes. The data was compiled and some general observations were arrived at. Furthermore data need to be collected in order to draw valid conclusions with statistical analysis to establish the relationship between the emotional intelligence and organizational behaviour of scientific manpower in NARES.

2.3.2 Team work and its effect on Organizational performance in NARES

The study aims to understand the teamwork skills and importance of teamwork among the stakeholders and its effects on organizational performance. In teamwork, one has to give value to maintenance and also give value to a job (Task) and people in team similarly one should have skills to the job and skills to maintain a good relationship with people in the team. Task Orientation: - Bothered about work & result oriented. Maintenance Orientation: - Bothered about people in the team and relationship Value: - Giving importance to something like work/people & make use of team synergy that teams

can produce when it is appropriate to use teams. Skills: - To enhance maintenance like checking on feelings of people or ability to do a job. The four categories i.e., Fully Functioning Team:- can be characterized as having members with a high-value commitment to both task & maintenance and with high skills in both areas. Lone Ranger: - Lack skills and do not have appropriate values towards task accomplishment. The Educably Retarded:- Have high values towards task accomplishment but low skills in task & maintenance and Trainer/Consultant: - Have adequate skills in both task and maintenance but tend to prize the maintenance functions so highly that little attention is paid to the task requirement resulting in lack of productivity. The depiction is mentioned below about task and maintenance orientation.

Out of 382 participants who participated in the team orientation behavior inventory (TOBI) study, 19.63% were found to be of Educably retarded profile, 58.9% were of Fully functioning team member profile, 5.5% were of Lone ranger profile and 15.97% were of Trainer consultant profile. A correlation analysis of publications with team work skills and team orientation behavior inventory (TOBI) indicated high correlation between publications with team work skills (0.87 at <0.0001 level of significance) and TOBI (0.78 at <0.0001 level of significance). A study with 2592 participants about team work skills indicated that 62.77% of participants required to develop better team work skills while 34% of participants have a moderate level of teamwork skills while mere 2.43% of participants have a high level of teamwork skills. During the investigation, ICAR team award winners were interviewed and 11 cases were prepared. The important factors affecting teamwork were support from superior authorities,

interdependence, communication, funding for the project, common understanding of the goal of the project, and ego problems among team members.

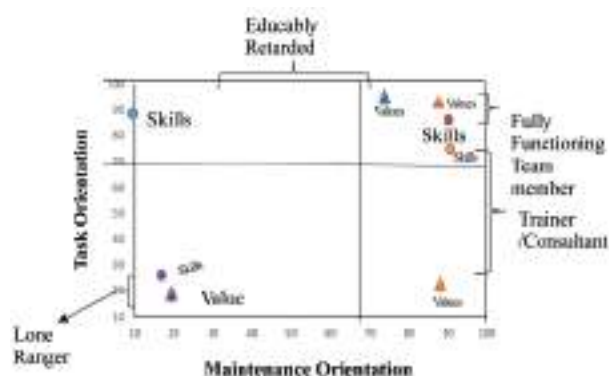


Fig. 2.27: Task Orientation Vs Maintenance Orientation

2.3.3 Competency Management System for Agricultural Research and Extension

Core competencies for functional and behavioural aspects were identified. Indicators for competencies identified for functional and behavioural domains were developed.

Case-I: Focused Group Discussion for developing competency dictionary for Scientists in Natural Resources Management.

Different levels of indicators for different roles of Scientists, Senior Scientists and Principal Scientists of Agronomy, Soil Science, Entomology,

Case-II: Focused Group Discussion for developing competency dictionary for Scientists in Agri-Business Management.

Different levels of indicators for different roles of Scientists, Senior Scientists and Principal Scientists of Agri-Business domain (Marketing, Operations, HRM, Finance, Business Analytics).

Some of the results emerged from the project are: Subject orientation got the first rank at scientist level indicating the development of core strength in the area of specialization; Research communication and research methodology domains given top priority at senior scientist and principal scientist

level; At pre-RMP level, project management and governance identified as priority domain.; Extension scientists gave top priority to research communication. Behavioral Indicators were developed for the following core competencies: Achievement Focus, Communication, Decision Making, Leadership, Strategic Awareness, Working relationships, and Creativity.

2.3.4 Identification of Best Practices Developed by the Leadership in Scientific and General Administration in NARES

National Agricultural Research and Education System (NARES) has been evolving over sometime in terms of leadership, administration, and research methodologies for improving the performance of the system. While some of the improvements were institutionalized in the system for long-lasting effect, some were not institutionalized due to various reasons and thus making performance improvements transient. One of the major reasons for this problem is personality-based best practices thus making the performance improvement of institutions transient. This project helps to make a comprehensive list of best practices followed by great Leaders, administrators, and researchers for improving the performance of the system and suggest ways and means for institutionalizing these best practices.

As mentioned earlier these practices were of personality based and thus have become transient after the exit of the person from the ICAR system as they are not institutionalized. During investigation, few best practices observed are 1. Reclamation of competent employees who have become litigants in courts and obstructing the functioning of organizations and making them participants in institutional development programmes; 2. Motivating the young scientists through periodical progress meeting at sectional/

divisional level; 3. Sharing of available resources at divisional level; 4. Motivating the youngsters for participation in training programmes in national and international levels; 5. Deputing young scientists for faculty upgradation programmes to institutes of eminence in their respective fields; 6. Promoting team building in all aspects of divisional/sectional activities; 7. Attending family functions of staff; 8. Sharing of views periodically at various levels; 9. Encouragement of faculty in various activities to achieve overall growth; 10. Conducting budget utilization meetings once in every week to utilize the complete budget given for the organization; 11. Calling all the people concerned for senior officer's committee meeting once in a month for resolving administration, financial and other issues of institute followed by action taken reports; 12. Conducting academic or research meetings once every month to resolve issues related to research/teaching/training of the institute; 13. Pursuing people with patience for publishing research papers in high impact factor journals/policy briefs /white paper/books/ book chapters/etc. and 14. giving targets to every achievement and pursuing relentlessly. Work is going on to collect more number of best practices from ICAR and State Agricultural Universities.

2.3.5 HR Interventions and their influence on Institutional Innovations

The study attempted to assess the moderation and mediation effects of key HR interventions on institutional innovation in ICAR. Data collected from 140 Respondents through interview schedule and Google based questionnaire were analysed to identify the link among HR interventions and institutional innovation. Data on research innovations in terms of scientific staff, research papers, research papers/scientist, IPs (patents and copyrights) and products during 2014-19 for 19 ICAR Institutes was analysed and indicated

that Animal Science Institutes have higher research papers /Scientists. HR Interventions (Participative Decision Making, Performance Appraisal, Training & Development, Reward Systems, Career Planning and Development, Information Sharing, Motivation, Team Spirit, Delegation, Employee's record and information, Remuneration, Talent Management, Manpower planning, Goal orientations, Job enrichment and job rotation, Coaching and Mentoring) identified for institutional innovation.

2.3.6 Digital Teaching Evaluation in Agricultural Education Using TPACK Model

The study used classical Technology Pedagogy and Content Knowledge (TPACK) model to evaluate the Digital Teaching Aptitude among Agricultural University teachers and identify the issues in digital teaching. The modified survey instrument was sent online to more than 1500 participants of the various training programmes of Academy, who had exposure to digital teaching techniques. Data from a total of 200 participants was collected during the survey. Preliminary assessment suggests that Pedagogical Content Knowledge (PCK) and Pedagogical Knowledge (PK) for the teachers are high as compared to other combinations of TPACK factors.

2.3.7 Gender-based ICT led Knowledge Management in Agriculture

A study was conducted on 'Gender based ICT led knowledge management in agriculture' and the salient findings of the study are the following: A higher percentage of men were found to be informed with media available in agriculture and animal husbandry compared to women. Male farmers had higher 'networking' compared to women, while women had highly localized interactions. Majority of men and women

respondents used TV, whereas the majority of the farm women had to depend on others for phone usage. They preferred TV followed by phone and Newspaper for agriculture information. Women had higher crop-specific knowledge gaps compared to men, especially in plant protection. Television, mobile and newspaper are the most preferred and accessed ICTs for agriculture information by both male and female farmers. The media preference for agriculture, is highly correlated with the access and use of media. Therefore, exposure to new media is essential to develop preference. Age, educational status, land holding and farming experience of farmers (males and female) did not show significant relationship with media preference and use for agriculture. ICTs access and use showed significant gender differentiation, implying thereby that farm women still have limited access and use of ICTs. Access to Wi-Fi indicated gender difference due to the differential awareness and access to the technology. The access to the new media and training on their use is essential for using ICT extensively in the field of agriculture; A mobile app 'm-Prioritize' was developed for prioritization of field/farm level problems/constraints. This app primarily meant for extension/research professionals, which was pilot tested on a sample of about 100 professionals and was found to be user-friendly.

2.3.8 Online Training Programme on Stress Management

The programme was organized during July 07-10, 2020 with 22 participants. The objectives of this training program include to develop an in-depth understanding of personal and psychological causes of stress/anxiety and its effects-physical, social and organizational behavior, to understand the mechanism of occurrence of

various stress related disorders and to develop constructive means for coping up with distress and achieve productive and enriched life-style. This 4-day program includes psychological and medical aspects of stress and its management, interpersonal and time management strategies and the role of emotional and spiritual intelligence in understanding and managing stress in a holistic manner.

2.3.9 New Entrants Assimilation Training for Young Agribusiness Professionals (NEAT Phase-1) of Coromandel International Limited

The first phase of Collaborative Capability Building Program (NEAT) of 5 days was organized by the Academy online for 44 professionals of CIL on Fridays & Saturdays during Sep 11-26, 2020. The objective of the program is to provide the basic understanding of contemporary issues related to agribusiness industry, crop production and crop protection practices and to upgrade the knowledge and skill in sales and marketing domain.



Fig. 2.28: Director, NAARM interaction with NEAT participants

The topics covered include Agriculture and Indian economy, Agronomy aspects of major field crops, Soil health based nutrient marketing, Basics in plant health management and plant protection

measures, etc. The behavioral aspects such as personality development, positive thinking and emotional intelligence were also covered in this programme.

2.3.10 Online MDP on Leadership Development (a pre-RMP Programme)

The Online MDP was organized during Dec 08-19, 2020 with 43 including 6 female participants. The programme provided an opportunity for the pre-RMPs, who are in the “middle zone”, to develop smooth and gradual transformation to step into future leadership position. The major objectives of the programme are: To help participants understand their own personality dimensions; To explore the leadership challenges and core competencies needed in present day context; To develop understanding and skills in research and technology management, Information and knowledge management and administration and finance management; and to adapt ideas of diverse skills for sharpening capacities to effectively execute various roles in the organization. The programme covered the areas of Leadership and personality development; Research and technology management; Information and knowledge management; and Administration, finance, vigilance management & lessons from epics.



Fig. 2.29: Participants of the Management Development Program on Leadership Development

2.3.11 Motivation, Positive Thinking and Communication Skills for Technical

Officers of ICAR

The programme was organized during Dec 17-22, 2020 with 23 including 8 female participants. Modern management skills, effective personnel management system, adoption of modern technology for work simplification, automation have become essential components for providing constructive services by the technical officers in the ICAR System. These officers play a pivotal role in extending research support in the organization. This training programme was intended to improve skills and efficiency of technical officers in discharging functions and providing constructive technical support in meeting the organizational goals and targets. The programme consisted of a blend of lectures, and case analysis. Sessions on motivation, positive thinking, team work, time management, online communication, in mainstream media, data visualization etc. The programme was conducted in online mode.



Fig. 2.30: Participants of the Training Programme on ‘Motivation, Positive Thinking and Communication Skills for Technical Officers of ICAR’

2.3.12 Training Workshop for Vigilance Officers of ICAR Institutes

The programme was organized during Aug 05-07, 2020 inviting 33 participants. The objectives of the workshop were enhancing competencies of the Vigilance Officers to provide support for vigilance matters in the Institute. The entire



programme was organized through online. Eminent resource persons from various reputed institutions delivered lectures besides interaction sessions with Director (DARE) and Chief Vigilance Officer, ICAR. The participants were exposed to different vigilance procedures to be followed and preventive vigilance. The participants were sensitized about various aspects of vigilance i.e., duties and responsibilities of vigilance officers, disciplinary procedures, recruitment procedures, RTI, handling of court cases, conduct rules, purchase and work procedures etc. Director, ICAR-NAARM and Shri. Shaleen Agrawal, Director (DARE) & CVO (DARE & ICAR) addressed the participants.

2.4 Extension Systems Management in Market-driven Environment

This is one of the core themes of the Academy, which essentially aims to enhance operational, adaptive and generational capacity of front-line agricultural extension systems to address emerging challenges and to enhance their capacities for use of ICTs to provide customized knowledge, skills and solutions to farmers, farmer groups, rural communities and for social networking.

The Academy focuses on the following specific areas for its research, academic and capacity building, and policy advocacy activities. i. Extension Policy, Planning and Management ii. Extension Information Systems. The core research areas envisaged under this theme are assessment of research role and performance of extension scientists in ICAR institutes; evaluation and enhancement of new frontline extension initiatives of ICAR; ecosystem analysis of functional effectiveness and convergence and effectiveness of Krishi Vigyan Kendras; development of quality resources for capacity building based on empirical data and case studies and provide policy inputs

for enhancing the efficiency and effectiveness of frontline extension systems of ICAR and SAUs. The identified research themes are being addressed by undertaking institutional and externally funded research projects in interdisciplinary approach.

2.4.1 Management and Impact Assessment of Farmer FIRST Projects

The chief objectives of the project are: (i) to organize need based training workshop region wise to the Pls/Co-Pls of FFP implementing organizations; (ii) to populate the technologies implementable by FFP implementing centres and assess the attributes, spill-over and technology readiness for scaling up (iii) to document success stories and farmers' innovations under FFP, and (iv) to develop good practices for institutionalization of FFP approach. Farmer FIRST team of ICAR-NAARM, provided hand holding support in extension tools and analysis of data to FFP institutes. During this period, the team performed monitoring in ICAR-RC for NEH Region (Mrungar, Umtham villages of Umiyam), ICAR-CIRB (Dhansu village of Hisar) and CCSHAU (Bure village of Hisar). Sixteen FFP institutes were selected/identified for FET of 110th FOCARS to evaluate the sectoral changes in the FFP identified clusters of villages, due to their FFP interventions. Under Farmer FIRST Programme, the FFP team of ICAR-NAARM compiled and reviewed the progress of 8 FFP institutes under ATARIs-Bengaluru and Hyderabad. They were also involved in the documentation of feasible technologies of selected FFPs (42 centers-84 technologies) and the outcome based technological intervention of 42 centers (2100 farmer partners). The success pathways of innovative farmers documented, which had helped 11 farmers in getting "Innovative Farmers Award" on ICAR-NAARM Foundation day. The information pertaining to inspiring farmers (12 farmers) is compiled and documented.



Fig. 2.31: FFP Project team interacting with farmers

2.4.2 Research role analysis of extension scientists in ICAR

This study was undertaken to assess the research role of extension scientists in ICAR. It was taken up in the light of general criticism that extension scientists in ICAR are not doing enough and quality research. Specific objectives are: (i) To identify the research roles of Extension Scientists of ICAR, (ii) To analyze the existing research roles of Extension scientists of ICAR and (iii) To identify gaps and suggest interventions for bridging the research role. Ex-post facto research design with online survey approach was used. Pretested online questionnaire was used to collect data from extension scientists in ICAR (total number 192 during study period). Prior consent was ensured from the extension scientists to be respondents in the online survey. Over a period of 3 months, responses were received from 110 respondents making it about 62% response rate to online survey.

Majority (57%) of the respondents were in the age group of 35-55 years, 59% are male, 43% in Principal Scientist and 39% in Scientist cadres, 87% are recruited through ARS and 34% had prior working experience, Workload was perceived as heavy by 49%, moderate by 39% and very heavy by 12%, Majority (80%) perceived 'research' as the most important role compared to other

roles like field extension, training, teaching, coordination, etc., During the study period, majority of the extension scientists in ICAR had 1-2 inhouse and 1 externally funded project (as PI and Co-PI), Analysis of self-reported data on publications during last five years indicated that on an average each extension scientist published 3 research articles in peer reviewed journals, 2 conference papers and 2 popular articles per year, Priority areas of research identified by respondents are : Pluralism and convergence, Innovative approaches/Institutional innovations like FPOs, Climate change, Doubling/enhancing farmers' income, ICTs/IOTs/Digital solutions, Market-led extension, Mainstreaming youth and women in agriculture, Entrepreneurship development, Major constraints reported are: Lack of resources, Lack of knowledge and skills in undertaking rigorous methodologies, Lack of recognition of efforts, Lack of focus due to multi-tasking, Inadequate budget allocation for undertaking social sciences research, Over-involvement in other support/service activities (role conflict), Important suggestions given by respondents are training on advanced research methodologies in extension research, creating a functional network of extension scientists in ICAR and Initiate network projects on extension models, clarity on role of extension scientist in the system to strengthen extension research, provision of contract staff like Young Professionals for data collection, tabulation, and fieldwork.

2.4.3 Online Training Programme on Communication and Management Skills for Extension Professionals

The Academy organized a 21-days Online Training Programme on 'Communication and



Management Skills for Extension Professionals' from 1-21 October, 2020. The programme was attended by 53 participants (38 males and 15 females) representing KVK, Agricultural University and other institutions from 15 States and 2 Union Territories. This training programme was organized with the aim of enabling the participants to benchmark and enhance their communication and management skills for rendering effective extension and advisory services. MS Teams and Zoom Platforms were used for imparting training in Online mode. Progressive and experiential learning strategies were followed to allow participants to progress in learning from known to unknown and immediately applying the learnt KAS through structured exercises (individual and team). Learning log was used to track the summative and formative evaluations respectively. Conceptual understanding and practical exercises, structured experiences and individual and group exercises were implemented towards achieving the programme objectives. The programme consisted of 46 training sessions on 34 topics related to theme and spread over 16 working days, with 6 training hours each day.

This training programme enabled the participants to practice the learnt knowledge and skills to come out with innovative learning outputs like i) video-recording of short speeches and ii) writing of short communications, success stories, news/feature stories on the themes of Role of Women in Agriculture, Role of Agriculture in promoting Sustainable Food, Nutritional and Livelihood Security and Atmanirbhar Bharat Abhiyan. Additionally, an innovative Directory of Participants and Digital Group Photo of the participants were also brought out by the participants under the guidance of Programme

Directors. Participants were organized into small groups for fostering cooperative and peer learning.



Fig. 2.32: Online Training Programme on 'Communication and Management skills for Extension Personnel'

2.4.4 Online Training Programme on 'Innovative practices in Extension Research and Evaluation'

The programme was organized during Sep 08-28, 2020 with 59 participants. The objectives of training are to equip participants with the latest knowledge, skills and innovative practices needed for conducting impact-oriented and quality extension research and evaluation; to orient the participants with the advanced methodology used in social science research and to facilitate networking among the participants for inter-institutional research in extension. This programme was organized to develop the critical abilities of extension professionals to conduct quality research and evaluation. The training programme was conducted in two modules. In the starter module, topics on developing skills required for extension researchers like innovative methodology, use of software, sampling framework, etc. were provided with hands-on exercises to the participants. In the path-finder module, innovative practices on advanced research methods, writing scientific articles, etc. were covered. The participants

demonstrated their learnings through group exercise of concept note writing in prospective researchable areas in extension.



Fig. 2.33: Online Training Programme on 'Innovative practices in Extension Research and Evaluation'

2.4.5 Webinar series on 'Emerging Trends in Extension and Social Sciences Research'

The main objective of the Webinar Series was to orient the students, research scholars, academicians and scientists towards emerging research areas and introduce new research methodologies. The Webinar was conducted during June 10-16, 2020 and the main contents covered were: Behaviour change research; Psychology of disaster preparedness; Documentation and Validation of ITK; Conjoint analysis; Modelling farmers support for agro-tourism development; Measuring entrepreneurial traits; Constructing composite indices; Impact Evaluation and Network approach to small group research. The beneficiaries are scientists, Assistant & Associate Professors from state agricultural universities, students, and international delegates. A total 245 participated in the webinar series including 68 females. This webinar series was extremely useful for the youngsters to carry out effective research in the present scenario apprising them the emerging research areas and introducing new research methodologies.



Fig. 2.34: International webinar series on "Emerging Trends in Extension and Social Sciences Research"

2.4.6 Virtual Training on Participatory Rural Appraisal (PRA) and Communication Strategies (PRACom-2020)

ICAR-NAARM organized a tailor-made virtual training programme on "PRACom-2020: Participatory Rural Appraisal (PRA) and Communication Strategies" during 09-11 September, 2020 for the IFS Probationers (2019-21 batch), from Indira Gandhi National Forest Academy (IGNFA), Dehradun. The participants comprised of 71 IFS Probationers including two officers belonging to Royal Bhutan Forest Service. This training programme, third in the series since 2018, is being organized as a part of Memorandum of Understanding (MoU) signed between ICAR-NAARM and IGNFA.

During the inaugural address, Dr. Trilochan Mohapatra, Secretary, DARE and DG, ICAR called



upon all the IFS Probationers to make the best use of the network of agricultural researchers and extension functionaries spread across the country to further their conservation and sustainable development agenda, by forging functional partnership. Dr. Ch. Srinivasa rao, Director, ICAR-NAARM provided an overview about the course and Shri. Bharat Jothi IFS, Director, IGNSA, presided over the inaugural session.



Fig. 2.35: A Glimpse of the Inaugural Session of PRACoM-2020 for IFS Probationers from IGNSA

During this training module, the IFS probationers were introduced to various tools of PRA techniques and provided hands on exercises to relate how the tools can be used for real-life decision-making situations. The participants were provided an overview on basic etiquettes on communication and imparted necessary skills for leveraging digital tools for meeting various professional communication situations. The participants simulated a situation involving discussion / interaction between various experts or media and prepared a video, which was appraised by communication experts. The training also included a hands-on exercise on using analytical hierarchy process (AHP) analyzer, an online application developed by NAARM, for prioritization of administrative decisions. The probationers were sensitized on the need to work with researchers and citizens as partners, so as to harness the benefits of synergy, by presenting relevant case studies.

2.5 Mobilizing Science and Technology for Innovation and Sustainable Development

Catalysing agricultural science and technology for innovation is important for good governance in NARES, and sustainable development in agriculture. Conducting research therefore becomes vital for use of science based interventional tools for effective research management and technology management in agriculture. Our efforts are towards infusing innovative ideas and bringing in a new milieu in the entire agricultural research system with the use of advancement in information technologies for online management system and digital framework in order to increase the productivity in research, reduce the drudgery, avoid unnecessary overlapping, and enhance the overall efficiency in research. Special focus is accorded to agricultural technology management with the application of management fundamentals for efficient management of research outputs and for reaping competitive advantage in the organization. In order to advance the objectives under the broad theme, the division of Research Systems Management focusses on broad areas such as:

- Research Management and Policy Advocacy
- Innovation and Technology Management
- Research in the context of food security, environmental sustainability and globalization.

2.5.1 Profiling the Resources Devoted to Agricultural R&D and Development of National Framework for Evidence-based S&T Governance in India

Research in India and developing countries is predominantly public-funded and the generation of desired outputs and outcomes often depends upon effective resource use, among others.

However, the governments as well as researchers have been grappling with ways to gauge the efficiency and effectiveness of research organizations. Human resource and finance are the major tangible inputs (drivers) for research. But research productivity is also influenced by intangible inputs like research facilitation, leadership, governance, culture, etc. Most global and national ranking frameworks capture only the quantitative and easily measurable tangible inputs. However, they do not reveal adequately the 'problem areas' specific to an institute that a research administrator at the institute level can attempt to intervene through appropriate management decisions, the results of which could be perceptible in the short or medium term.

Addressing this crucial gap in the performance assessment methods, a methodological framework for assessing 'Ease of Doing Research' (EoDR) in research organisations was developed based on an extensive review of various national and international research performance assessment frameworks, series of consultations among the experts and subject matter experts. EoDr. refers to the perceived ease with which the researchers can carry out their research work in respective institutions. Technically, the Ease of Doing Research Index (EoDRI) is computed by aggregating the distance to frontier scores of 22 indicators (grouped under 5 parameters). EoDr. framework will be a handy and effective decision support tool for the research managers to transform their institutions. The framework takes a major departure from existing frameworks and focuses on the *intangible inputs* and *process* dimensions of the research ecosystem, which could be significant enablers of the tangible inputs.

As a part of collaborative research undertaken by NAARM with Department of Science and

Technology, New Delhi International Food Policy Research Institute (IFPRI), Washington the data pertaining to (i) human resources deployed (ii) financial resources used (iii) research focus and (iv) research outputs were collected from all ICAR institutions and Agricultural Universities for the period from 2014-15 till 2017-18. An interactive Dashboard to draw ready inferences from the datasets was also developed. The GoI has released the S&T Indicators tables and the R&D Statistics 2019-20, wherein the data collected by NAARM for the agricultural sector have been integrated. Through this project, NAARM was part of a national initiative for collecting S&T data for the farm sector and thus, aid in evidence-based policy making in India. The Academy also developed a prototype for the Dashboard jointly with one of the IT firms empaneled with NAARM, to aid in data-driven decision making in the domain of agricultural research.

The Frameworks for Ease of Doing Research and Evidence-based Research Governance were presented in the USAID-IFPRI co-hosted side event, *Agricultural Science and Technology Indicators (ASTI): A Critical Resource for Agricultural Innovation System (AIS) Capacity Development*, on 20th November, 2020, on the margins of the FAO Tropical Agriculture Platform (TAP) Partners Assembly. A roadmap for regional cooperation and networks to strengthen capacity in national agricultural research systems and to enhance the overall relevance and use of evidence was prepared.

2.5.2 Agricultural Research and Development Infrastructure in Andhra Pradesh and Telangana

A focused study was undertaken as the potential for agricultural development being high in the



two states of Andhra Pradesh and Telangana with the following objectives: (i) To assess the status and gaps of infrastructure and resources available for agriculture research in the study states, (ii) To analyze the productivity and efficiency of available resources and infrastructure, and (iii) To assess the impact of agricultural research infrastructure on farmers. Some of the salient observations and key findings are given below. The absence of laboratory facilities in the state of Andhra Pradesh may be provided with the central budget supported as part of the special status to be granted to it. Some of these facilities include biotechnology labs, pesticide residue testing lab, etc. It was encouraged for setting up of two fisheries research labs as part of the SVVU to supplement the existing fisheries research station at Undi in locations like Nellore and Machilipatnam. A state of the art fruit and vegetable processing research station under ICAR need to be set up at Tirupati to encourage technologies that can be emulated by farmers and small/ medium entrepreneurs.

Filling up of the vacant posts both in the state (SAUs) and central sector institutions (ICAR) in both the states must be the top priority. For instance, in the case of the horticulture university of Telangana, while the office buildings are ready but to operate the university with academic programmes, there is an acute shortage of teaching/ research staff to an extent of 50 per cent. Similarly, budgetary support for adequate operational expenses of the institutions also need to be ensured. Some specific recommendations are: Promotion of FPOs, the establishment of horticulture resource Centres, strengthening research stations, PHT centre in horticulture colleges & research stations, business incubation centres, etc. Establishing new gardens/ rejuvenating old gardens, Centre of Excellence, collection & sorting centres, rural markets, ripening chambers, reefer vans, etc.

Establishing tissue culture units for Banana and other crops.

Promoting certified nurseries and also training the nurserymen in modern and hi-tech nursery management are required to be given. Some other recommendations are hereunder: Integration of MIDH and RKVY components for the sector's development in the Andhra Pradesh; Improving the logistics for better management of the post-harvest processing and marketing of the produce; Some more infrastructure gaps like integrated pack houses for mango, creation of processing facilities for mango and sweet orange; Improvement in cold chain facilities is another concern in Telangana for horticulture development; AICRP centre on Chilies and Banana need to be started in Telangana; Strengthening horticulture extension to ensure efficient transfer of technologies and also reduce/ avoid the non-core and non-technical workload of scientists and departmental officers; Utilizing the surface water area to the maximum potential because of the new irrigation projects like Kaleshwaram etc.; Establishment of more wholesale and adequate retail marketing facilities and ensuring the cold chain; Exploring the domestic markets in the country like West Bengal and North-East states for freshwater fish markets; Need for creating infrastructure for retail marketing outlets, cold chains, etc. in both the states; Need to create fishing harbours, landing centres, and jetties with NFDB and RIDF funds; Demarcating land for brackish water shrimp farming by getting permission from coastal aquaculture authority.

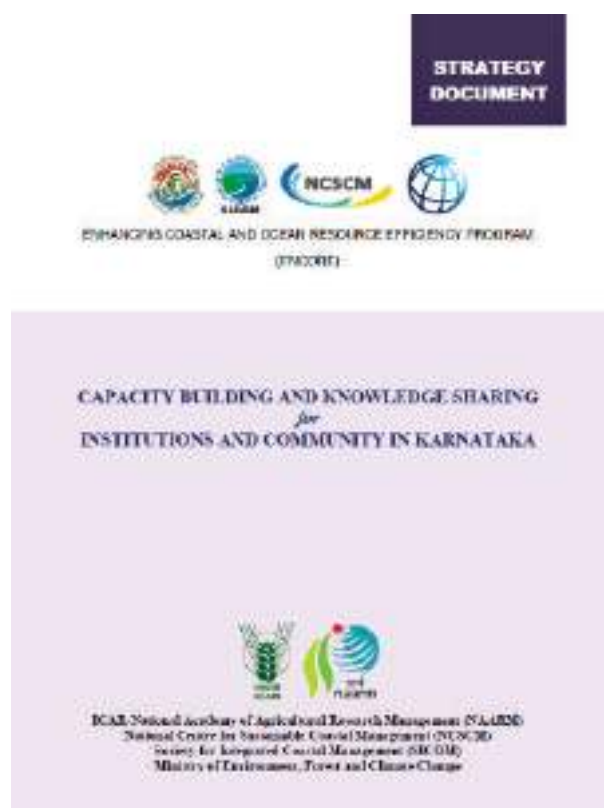
2.5.3 Strengthening National Programs through Policy Advocacy

Natural Resource Management, Biodiversity Conservation and Sustainable Resource Use are

the domains with significant national focus across various central and state ministries/ departments.

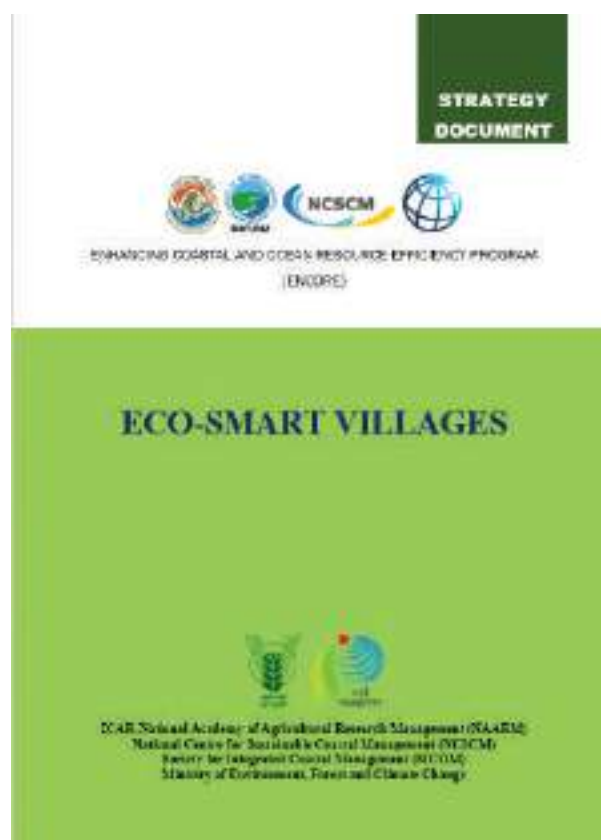
2.5.3.1 DPR and ToR for Capacity building and Knowledge Sharing of Individuals and Institutions in Karnataka

The Detailed Project Report (DPR) and Terms of Reference (ToR) for Capacity building and knowledge sharing of individuals and institutions in Karnataka has been prepared comprising of detailed modules for 02 Awareness Creation Programs (ACP) for coastal community; 08 Skill Development Programs (SDP) for the fishers and 02 Management Development Programs (MDP) for the state officers. Further a comprehensive framework for developing *MyCoast*- A one stop solution for coastal dwellers and a Single Window Platform for real-time fish marketing and e-auctioning were also included for implementation in Karnataka under ICZM and further scaling up for the entire country.



2.5.3.2 DPR and ToR for Development of Model Coastal Eco-Smart Villages

The Detailed Project Report (DPR) and Terms of Reference (ToR) for *Development of Model Coastal Eco-smart Villages* was prepared based on extensive review, expert consultation and field visits. Under the project a framework was developed for the interventions to be implemented with respect to plantation activities, soil and water conservation, energy conservation and few specific centralized solutions, so as to develop the identified five villages viz., Kantiyajal (Gujarat), Agonda (Goa), Apsara Konda (Karnataka), Akkampettai cluster (Puducherry) and Baghmunda (Odisha)) for developing as model eco-smart coastal villages in collaboration with five different States/UT.





2.5.4 Framework for Mainstreaming Biodiversity Conservation in Agricultural Development Schemes in India



Strengthening Natural Resource Management (SNRM)
A Framework for Mainstreaming Biodiversity
Conservation in the Development Schemes of
Agriculture Sector in India



The framework for mainstreaming biodiversity conservation in the development schemes of agricultural sector in India was prepared based on detailed assessment of the implementation status of biodiversity relevant public schemes of agriculture sector. Agrobiodiversity relevant components are spread across various development schemes and it was observed that diverse agro-ecosystems impact the extent of relevance of different development schemes spatially. Further, the extent of implementation of different development schemes also depended on the structure of the scheme components.

The report recommended reforms in identified public schemes to ensure enhanced biodiversity outcomes for agro-biodiversity conservation through public finance and identified key entry

points and called for creation of a separate Mission on Agro-biodiversity focused on Plant Genetic Resources as a Sub-mission of National Mission on Sustainable Agriculture (NMSA) under National Action Plan on Climate change (NAPCC).

The salient recommendations were presented in Eurasia-Pacific Virtual Dialogue of BIOFIN on Finance for Nature on 12 Nov, 2020 attended by representatives from various countries and senior officials from UNDP and National Biodiversity Authority (NBA), MoEFCC, GoI. Subsequently, it was presented in the GoI-BIOFIN Steering Committee meeting chaired by Additional Secretary, MoEFCC on 20 Nov 2020 for needful action.



Fig. 2.36: Screen grab of Eurasia BIOFIN Dialogue and GoI-BIOFIN Steering Committee Meeting

2.5.5 Impact Assessment of Research and Development Programs on Food and Nutritional Security

The project envisions to profile key research and developmental programs on food and nutritional security with an aim to develop methodological framework for stratification of their success levels.

Profiling of various programs with an emphasis on programs related to food grains has been done and status chart has been prepared for the programs and programs have been preliminarily classified on common parameters viz., budget, target group, impact etc. Further through expert consultation the draft classification will be improved to identify key programs for stratifying their success levels.

2.5.6 Commercialization of Agricultural Technology Scheme - National Agriculture Innovation Fund (NAIF) (Component-I) (IPTM - ZTMC)

NAARM-ZTMU is involved in capacity building in IP management and tech transfer/commercialization domain by providing trainings and also serving as “help desk” for Agri Business Incubators/Startups/SAUs. NAARM-ZTMU helps other ICAR institutes in conducting patent search as prior art for patent filings. Acting as institutional mechanism to protect/manage innovations/intellectual properties (IPs) generated within institute, Unit has filed three copyright applications for software/literary work developed at NAARM. ZTMU at NAARM is actively involved in research understanding the trends of emerging technologies in agriculture. Following are the details of research work carried out; 1. Patent Landscape analysis to understand status

of applications of Internet of Things in protected growth conditions, specifically in crop plants. 2. Analysis of IP Management and Technology Commercialization Practices in ICAR. 3. Publication entitled ‘Technology management through ICAR-Agri Business Incubators: Status and Strategies. 4. Publication entitled ‘Women led agri-startups: Courage to Innovate for Change’ 5. Study on Indian startups using IoT in Agricultural Sector.

2.5.7 Sensitization Workshop on ‘Implementation of Access and Benefit Sharing (ABS) Regulations in Agriculture Research’

The Biological Diversity Act, 2002 and the Access and Benefit Sharing (ABS) Guidelines, 2014 has significant implications for the scientists working on various bio-resources including microbes. The law has significant relevance to the agricultural scientists in ICAR and AUs who are handling biological resources. Recognizing this, NAARM had organized the 2nd Awareness-cum-Sensitization Workshop on ‘Implementation of ABS regulation in Agricultural Research’ during 8-9 June, 2020 through online mode in collaboration with Tamil Nadu Agriculture University (TNAU), as a part of UNDP - Biotech Consortium India Limited (BCIL) sponsored project. Dr. Varaprasad, Former Director, Indian Institute of Oilseed Research; Dr. KK. Lal, Director, ICAR-NBFGR, Lucknow



Fig. 2.37: Glimpse of the session - Resource persons interacting with participants



and experts from UNDP, ICAR institutions and National Biodiversity Authority (NBA), Chennai, served as the resource persons. Dr. Ruchi Pant, Head, NRM and Biodiversity emphasized the global significance of the subject in her inaugural address. The aim of the workshop was to enhance capacities of academic and research institutions in the public and private sector, assessing genetic resources to pursue research and transfer technology for better understanding the legal provisions and guidelines related to ABS.

There were over 50 participants comprising of researchers from NARES, especially those from crop sciences background participated in the workshop. The training included practical exercise on ABS Process through group discussion (i.e. crops, horticulture and biotechnology), which facilitated the participants in better understanding of ABS implementation, process, status and challenges with reference to agricultural research in India. The workshop also introduced the participants to the different types of ABS applications at the National Biodiversity Authority (NBA), which further complimented the legal understanding of the Biological Diversity Act through deliberations on the ABS handbook for researchers under preparation in UNDP.

2.5.8 Online MDP on 'Priority Setting, Monitoring and Evaluation (PME) of Agricultural Research Projects'

The 8th Management Development Programme (MDP) on 'Priority setting, Monitoring and Evaluation (PME) of Agricultural Research Projects' for the In-charges of PME Cell of ICAR institutions was organized by the ICAR-National Academy of Agricultural Research Management (NAARM) during 12-17 October, 2020. The In-charges and members of PME Cells from 32 ICAR institutions, representing 8 subject matter

divisions, nominated by the Director General of ICAR, participated in the programme.



Fig. 2.38: Screenshot of interaction of resource persons with the participants

The online MDP comprised modules on principles of priority setting, monitoring, evaluation and impact assessment of agricultural research projects. The participants were sensitized on the various metrics and tools used for measuring individual and institutional research performance, global and national frameworks for measuring research efficiency and ICAR guidelines pertaining to the various functions of PME Cell like managing research publications, consultancy services, contract research, IP and technology management. The training included interactive sessions, hands-on exercises, case studies and experience sharing. The participants were sensitized on various online tools and applications for research review, project proposal preparation, checking plagiarism and choosing appropriate journals for research communication. Dr. PK Joshi, Former Director, NAARM & NIAP graced the valedictory function. The participants were urged to institutionalize the learnings in their day-to-day activities and were assured all support from NAARM in their endeavours.

2.5.9 Virtual Training on 'Design Thinking in Research Project Formulation and Implementation'

An Online Training on *Design Thinking in Research Project Formulation and Implementation* was

organized by the ICAR-National Academy of Agricultural Research Management (NAARM), Hyderabad during 24-29 August, 2020. About 45 agricultural researchers, including 13 females, representing 24 agricultural research disciplines from 15 ICAR institutions and 12 Agricultural Universities, shortlisted from over 150 applicants from across India, participated in the programme.



Fig. 2.39: Snapshot of the interaction of Resource Persons with the Participants

The participants were introduced to the concept of Design Thinking (DT) and its use in various aspects related to research management, with relevant illustrations and case studies. Innovative exercises were designed to instil DT perspectives among the participants in research problem identification, prioritization of research solutions and research concept development. The participants were provided DT practice for prototyping and visualization of project activities, output, outcome and impact. Interactive online breakout group sessions were conducted, wherein the participants had hands-on exercises on the use of various tailor-made online DT tools for research review, project proposal preparation, monitoring and evaluation.

2.5.10 Training Workshops on Developing Winning Research Proposals (DWRP)

NAARM organized two off-campus Training Workshops on 'Developing Winning Research Proposals (DWRP)' for the faculty members of

Maharana Pratap University of Agriculture and Technology (MPUAT), Rajasthan at College of Agriculture, Udaipur during 03-05 Jan 2020 and for those of Kamdhenu University, Gujarat at College of Dairy Sciences, Amreli during 29-31 Jan 2020. A total of 25 faculty members from MPUAT (07 women) and 23 faculty members from Kamdhenu University (04 women) participated in the training workshops.



Fig. 2.40: Participants of the Training Workshop on DWRP at MPUAT, Udaipur



Fig. 2.41: Participants of the Training Workshop on DWRP at Kamdhenu University, Amreli

The program aimed at imparting the skills for writing research proposals that can win funds from donors focusing on the needs of the stakeholders and sensitizing the participants on the various components of a research proposal. The core content of the program included elements of research concept writing; preparation of project logframes and making winning research proposals. Most exercises were through hands-on mode and the participants were introduced



to the online applications developed by NAARM viz., AHP Analyser, Research Concept Writer and Project Logframe Writer.

2.5.11 Awareness cum Sensitization Workshop on 'Access and Benefit Sharing'

India is a front runner in terms of implementation of biodiversity conservation. The Biological Diversity Act, 2002 and the Access and Benefit Sharing (ABS) Guidelines, 2014 have significant implications for the Scientists working on various bio-resources, including microbes. The law has significant relevance to the agricultural scientists in ICAR and AUs who are handling biological resources. Recognizing this, NAARM organized an Awareness-cum-Sensitization Workshop on 'Implementation of ABS regulation in agricultural research' during 22-23 January, 2020 in collaboration with UNDP, GEF and Biotechnology Consortium India Limited (BCIL), drawing experts from all relevant organizations including National Biodiversity Authority (NBA), Chennai. The aim of the workshop was to enhance capacities of academic and research institutions in the public and private sector, assessing genetic resources to pursue research and transfer technology for better understanding the legal provisions and guidelines related to ABS.

The Workshop provided an overview on ABS implementation, process, status and challenges with reference to agricultural research in India. The sessions were planned to be interactive and relevant Case Studies from NARES were discussed. The event was attended by 25 senior researchers from ICAR and State Agricultural Universities associated with research and technology management at the institute level like PME Cell / ITMU specifically working on crop sciences. The workshop also introduced the different

types of ABS applications at NBA, which further enabled the participants to understand the legal provisions of the Biological Diversity Act.



Fig. 2.42: Participants of the Sensitization Workshop on 'Implementation of ABS Regulations in Agriculture Research'

2.5.12 Webinar on 'Breaking Myths of Science Communication'

A Webinar on Breaking Myths on Science Communication was organized on May 4, 2020 jointly with College of Dairy Science, Amreli, Kamdhenu University under National Agricultural Higher Education Project (NAHEP). About 45 faculty members of the university participated in the program. Dr. RC. Agarwal, DDG (Education), ICAR in his presidential address complemented the University and NAARM for jointly taking this initiative during the lockdown and sensitizing the faculty members on the very relevant topic. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM in his special address highlighted the needs and issues in science communication. He also emphasized the need to remove the barriers in the minds of young researchers and acquire the basic skills for science communication. Dr. P. Krishnan, Principal Scientist, NAARM and the Lead Speaker dwelt on the various myths in science communication in terms of structure of research papers, authorship, choice of journals, publication process, etc.



Fig. 2.43: Interaction during the Webinar on 'Breaking Myths of Science Communication'

2.5.13 Online EDP for Master Trainers on 'Access and Benefit Sharing' (ABS)

NAARM organized an Online EDP on 'ABS Regulations and Nagoya Protocol' during 15-17, July 2020 through Video Conferencing in collaboration with UNDP and Biotech Consortium India Limited (BCIL), New Delhi. Over 30 scientists representing various regions and crops from NARES were selected from among the participants of the earlier training on "ABS implementation in India" conducted by the Academy.



Fig. 2.44: Director, ICAR-NAARM interacting with participants

Dr. VB. Mathur, Chairman, National Biodiversity Authority (NBA), Chennai; Dr. Ruchi Pant, Head, NRM and Biodiversity, UNDP and Dr. Vibha Ahuja, Chief General Manager, BCIL, during their

interaction with the participants, assured all support in their efforts to sensitize the concept among their peers in the region. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM, during his inaugural address, informed that ABS concepts will be integrated in relevant capacity building programs of the Academy. He invited the participants to contact NAARM for any support for designing/organizing the ABS sensitization programs.

Dr. Alejandro Lago, Project Manager, UNDP-GEF Global ABS Project provided an overview of ABS, CBD and Nagoya Protocol. The ABS Clearing House Mechanism under the Nagoya Protocol was explained by Dr. Matthew Daas, Programme Officer, CBD, Montreal. Dr. Vibha Ahuja from BCIL highlighted the ABS potential of Indian Biotech research sector. The ABS regulations in the context of IPR, TRIPS and WTO were elucidated by Dr. SK. Soam, Joint Director, NAARM. Dr. K.S. Varaprasad, Senior Consultant, UNDP anchored an interactive session with the participants along with the Course Directors. The participants were provided hands-on demonstration on using the NBA's online application for ABS filing.

2.5.14 Virtual workshop-cum-training on 'Intellectual Property Rights in Agricultural Research and Education in India'

A virtual workshop-cum-training on 'Intellectual Property Rights in Agricultural Research and Education in India' was organised for 14 days, which inaugurated by Dr. Trilochan Mohapatra, Secretary (DARE) and DG (ICAR). The aim of the programme was to imparting requisite knowledge on domain technologies and the regulatory issues associated for successful acquisition and commercialization of IPRs. The training provided a complete overview and knowledge generation, protection and management of Intellectual Property Rights in agriculture and support all their endeavors in becoming a successful IPR. enlightened individual.



The workshop and training programme was organized by Education division and IP&TM Unit of ICAR under the umbrella of National Higher Agricultural Education Programme.



Fig. 2.45: Virtual workshop on 'IPR in Agricultural Research and Education in India'

2.5.15 Foundation Course for Agricultural Research Service (FOCARS)

The Foundation Course for the Agriculture Research Service (FOCARS) is the flagship program of the Academy wherein the newly recruited agricultural researchers acquire necessary knowledge, develop relevant skills and build right perception and attitude to meet the contemporary challenges faced by the sector. During the year 2020, two batches of FOCARS (110 & 111) were trained. In 110th batch, a total of 135 ARS Probationers including 56 female probationers representing 28 ARS disciplines and 22 States have undergone the Foundation Course. Dr. (Smt.) TAMILISAI Soundararajan, Hon'ble Governor, Telangana State inaugurated the 110th FOCARS at the Academy on January 07, 2020. In her inaugural address, Smt. Soundararajan highlighted the role of research in strengthening agricultural production and efforts made towards achieving the goal of doubling the farmers' income set by the Hon'ble Prime Minister of India by the Year - 2022. The Hon'ble Governor urged the young probationers to take agriculture

sector to the new heights and bring second Green Revolution in the country.

The 111th batch of the FOCARS comprised of 37 ARS Probationers including 17 women, representing 15 different disciplines and 17 States / 2 Union Territories from across India including the North-Eastern States, Jammu and Ladakh. It was conducted in-person under a unique circumstance and the pandemic-imposed challenges were successfully overcome. The digital and virtual modes of interactions were effectively leveraged to connect the probationers with globally renowned experts from various walks of life during the specially designed slots for *Knowledge Quests* and *evening interactions*. Many eminent researchers, serving and retired VCs, DDGs, Directors and Senior Officers from Development Departments interacted with the participants. The Course was inaugurated by Prof. Rattan Lal, World Food Laureate & Director, Carbon Management and Sequestration Center, The Ohio State University, U.S.A. Dr. Trilochan Mohapatra, Secretary (DARE) & Director General (ICAR) while interacting with the scientist-probationers highlighted the need for building their core competencies in preparing competitive project proposals, science communication, nurturing agri-entrepreneurs, developing start-ups and using frontier tools like IoT, AI and block chain technology in the agricultural value chain during their foundation course at NAARM.



Fig. 2.46: Dr. (Smt.) TAMILISAI Soundararajan, Hon'ble Governor, Telangana State addressing 110th FOCARS during inauguration programme at NAARM



Fig. 2.47: Dr. (Smt.) TAMILISAI Soundararajan, Hon'ble Governor, Telangana State with 110th FOCARS Probationers and Faculty at NAARM



Fig. 2.48: Snapshot of Virtual Inauguration of 111th FOCARS by Prof. Rattan Lal, World Food Laureate



Fig. 2.49: Dr. Trilochan Mohapatra, Secretary (DARE) & Director General (ICAR) with 111th FOCARS Probationers and Faculty at NAARM

2.5.16 Diploma in Technology Management in Agriculture (DTMA)

NAARM offers one-year Diploma Programme in Technology Management in Agriculture (DTMA) since 2011 in distance education mode in collaboration with the University of Hyderabad (UoH) as a stepping stone towards better management of agricultural technologies within the changing landscapes of IP regime. The course is specially designed to cater to the needs of building a strong IP portfolio along with efficient technology management, diffusion and commercialization skills among the scientists, teachers, students and

professionals engaged in agriculture and related industries. The number of students admitted in 2020 was 23 with 4 female and 19 male students. These students represent 11 domain states. Most of the students are in-service candidates representing various Governmental and Non-Governmental Organizations such as Agricultural Universities, ICAR (CMFRI, NRCM, NRCC), Du Pont, IDBI Bank, Metro cash & carry India Private Ltd, Thermo fisher Scientific India Private Ltd, Global Green Company Ltd, MAFSU, Weather Risk Management Services Pvt Ltd, State Departments.

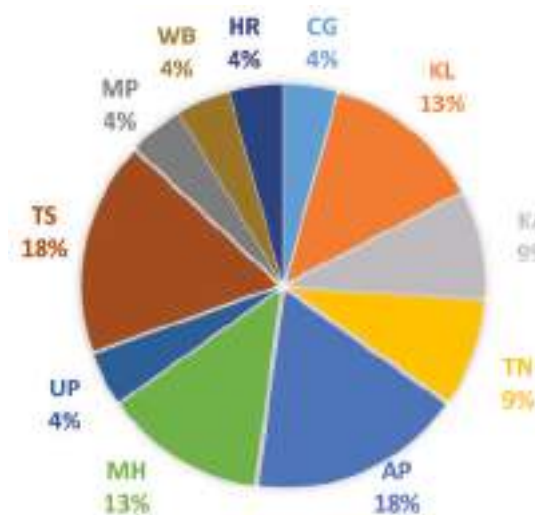


Fig. 2.50: States Representation in 9th Batch of DTMA

Two contact sessions of 3 days duration were conducted at the Academy in January for 8th batch and in September for the 9th batch covering the entire course. Resource persons from NAARM, Industry and other organizations contributed towards the lectures.



Fig. 2.51: Participants of 2nd Term Contact Session of PGDTMA



2.6 Information and Communication Management for Promoting Innovation and Governance

Enhancing the research, capacity building and consultancy in the areas of IT based decision support systems; GIS in agricultural management; ICT applications for village knowledge centers, Development of digital multimedia and e-learning resources for agriculture; Geospatial knowledge discovery and management; ICTs in agri-supply chains; ICT policy for NARS; Emerging information technology and systems; ICTs in participatory technology development and transfer; Designing e-learning systems and Distance training are the objectives to carryout activities related to research and capacity building programmes. In order to advance the objectives of the broad theme, the Academy focused in the following broad areas for its research, academic and capacity building activities. Application of GIS, GPS and remote sensing technologies; Development of information systems and knowledge management; Statistical computing and data analytics.

2.6.1 KRISHI: ICAR Research Data Management for Knowledge Management

The guidelines for managing the research data was issued by the Council in 2014 and the implementation across institutes was taken in a project mode with ICAR-IASRI as the lead center and ICAR-NAARM, ICAR-IARI, ICAR-NBSSLUP and ICAR-DKMA as the core institutes involving in developing processes, tools and oversee the implementation. The objectives of the project are to develop repositories of proven technologies and publications, to develop unit level repositories of experimental and observational data, to develop geo-portal for visualizing and analysis of spatial data and to develop agricultural knowledge portal to provide access and manage knowledge

repositories. The project aims at improving internal data management through use of standards across institutes/IT applications, improve data accessibility to researchers both within and outside ICAR and enhance the visibility of the ICAR through centralized repositories to show case technology, publications, mobile applications, ICT initiatives etc. The Academy is mainly involved in conceptualizing repositories for publications, technology, and experimental data.

The AICRP information system was made ready for AICRP on Castor, Sunflower and Safflower. The information system is useful in creating the trials under different crop at the beginning of the season. The randomized layout for each locations is made available to concerned locations as well as to the coordinating units. The datasets can be downloaded and uploaded by the individual centers and can be analyzed statistically and the reports are made available in the form of Excel Files. A varietal information system was conceptualized and a detailed database structure was developed. The information system contains about 32 fields covering different aspects of varieties from the contributed breeders to pedigree, registration status etc. The system is under testing and after testing, it will be made available to all institutes.

The AICRP Information system is being implemented since 2017 in Maize. A comprehensive data analysis was undertaken to understand the quality of the datasets submitted by different locations. The yield traits calculated at each location in 67 multi-location trials were considered. Assuming the genotypes tested as random, the heritability(h^2) of yield traits at individual locations were calculated based on the usual formula. The square-root of the heritability (h) is usually used to assess the quality of the data. Based on the analysis, it was found that more than

70% of trials are considered to be of good. The proposed methodology after validation will be used in the future during the analysis stage.

The AICRP information system 2.0 has been initiated during the year 2020. The new version is expected to provide better visualization of results, rigorous quality checks and better analytical capabilities to improve decision making. Based on the feedback from different AICRPs, a new feature to add past trials has been created. Using this feature, the previous year trials can be created and uploaded to the centralized database. To ease the trial creations, excel templates for different types of experiments have been created. After filling up the details in the templates, the same can be uploaded to create the trials. Further, provisions for uploading all locations data using a single or multiple excel files is being developed.

2.6.2 Establishment of Skill Development in Bioinformatics Infrastructure Facility (SKILL-BIF) on Agricultural Biotechnology

The SKILL-BIF centers use lecture materials, video clippings, demonstrations, tutorials and online facilities for teaching. It also periodically organizes hands-on trainings on various aspects of application of bioinformatics tools and molecular techniques in agricultural research and education. In this project an experiment was conducted on In-silico study of abiotic stress related transcription factor binding sites in the promoter regions of banana germin-like protein genes. The aim of the study to identify the transcription factor binding sites involved in abiotic stress in banana. In the study, 96 Transcription factor binding sites were recognized from promoter regions of 31 MuGLP genes. The identified TFBs were mostly belongs to WRKY family, fewer from MYB but not found the NAC family TFbs on the MuGLP.

Banana (*Musa*) belongs to the family of Musaceae, which is a widely consumed fruit in India. It has high nutritional value and grows in all the suitable climatic conditions. In the production of banana there are various stresses including biotic and abiotic declines resulting in huge yield losses. There are some Germin-like proteins (GLPs) that are involved in tolerance of abiotic stresses. There are many research works related to it, but the abiotic stress associated transcription factor (TFs) binding sites are still unknown in MuGLP genes of banana. Numerous TFs, mostly the WRKY, NAC and MYB are related to abiotic stress tolerance. We have studied the promotor regions from MuGLPgenes for the presence of WRKY, NAC and MYB in it. Phylogenetic analysis was done for MuGLP mRNA169 Promotor regions using MEGA V7.0 software. TFs binding sites was examined by PlantPAN 2.0 and remaining investigation was done with Genevestigator. The present findings on TFs binding sites in MuGL Pgenes may be beneficial for further characterization of abiotic stresses in banana is given in figure 2.59.



Accession No.	Length (bp)	TF Binding Site	TF Name
1	100	ATGTTT	WRKY
2	100	ATGTTT	WRKY
3	100	ATGTTT	WRKY
4	100	ATGTTT	WRKY
5	100	ATGTTT	WRKY
6	100	ATGTTT	WRKY
7	100	ATGTTT	WRKY
8	100	ATGTTT	WRKY
9	100	ATGTTT	WRKY
10	100	ATGTTT	WRKY
11	100	ATGTTT	WRKY
12	100	ATGTTT	WRKY
13	100	ATGTTT	WRKY
14	100	ATGTTT	WRKY
15	100	ATGTTT	WRKY
16	100	ATGTTT	WRKY
17	100	ATGTTT	WRKY
18	100	ATGTTT	WRKY
19	100	ATGTTT	WRKY
20	100	ATGTTT	WRKY
21	100	ATGTTT	WRKY
22	100	ATGTTT	WRKY
23	100	ATGTTT	WRKY
24	100	ATGTTT	WRKY
25	100	ATGTTT	WRKY
26	100	ATGTTT	WRKY
27	100	ATGTTT	WRKY
28	100	ATGTTT	WRKY
29	100	ATGTTT	WRKY
30	100	ATGTTT	WRKY
31	100	ATGTTT	WRKY

Figure 2.52: Transcription factor binding sites from 31 MuGLP promoter region



2.6.3 Decision Support System of Maize Inbred Germplasm

A web based decision support system on maize germplasm was designed and developed. The database on maize germplasm was created with the data on traits viz. time of anthesis, time of silk emergence, density of spikelets, number of kernel rows, number of kernels per row, grain type, plant height, ear placement height, anthocyanin colouration at base of glume, anthocyanin colouration of anthers, anthocyanin colouration of silks, grain colour, kernel row arrangement and 1000-kernel weight against each inbred lines. Different modules such as searching of database, queries, reports have been created. The system has been strengthened with module on GDD for anthesis calculation.



Fig. 2.53: Maize germplasm decision support database

2.6.4 Mega Environment Analysis Tools for AICRP Trials

A new project on 'Mega Environment Analysis Tools for AICRP Trials' was started from June 2020, to understand the effect of Weather on Genotypic x Environment Interaction in multi-environment trials. The specific objectives of the project include to incorporate effect of resources (weather and soil) on yield at each location in G x E interactions in Multi-Environment Interactions; to predict

yield at any given location based on AICRP Multi-Location Trials and to prepare analytical tools for the proposed methodology. The target crops are Maize and Sorghum. For the purpose of study, the data collected through AICRP sorghum and AICRP maize were used. The bi-plots of two-way mean tables provide better understanding of the interactions. The trial data are compiled as per the trial type wise (such as IVTs, AVTs etc.) and trait-wise (yield or other traits).

2.6.5 Training needs Assessment in Data Analytics for Scientists and Faculty

A new study on 'Training needs Assessment in Data Analytics' for Scientists and Faculty has been taken up with the specific objectives of benchmarking the knowledge level of scientists in data analytics and identifying the skill gaps in data analytics among scientists. It was observed that from the various training programmes conducted so far, the knowledge level of scientists in data analytics is not very high. Skill gaps in various data analytical skills is being analysed from the responses and feedbacks received from the participants of the training programme.

2.6.6 Geo-Spatial Knowledge System for Monitoring Climate Change in Agro Climatic Zone-X: Southern Plateau and Hills Region

Surface water levels monitoring is a prerequisite for studying hydrological and ecological developments. Remote sensing (RS) data sets provide spatial and temporal data which aid in mapping the water bodies at the regional and global scale. The temporal changes in the extent of surface water bodies in Telangana state were monitored through index methods, using indices like Normalized Difference Vegetation Index (NDVI), Normalized Difference Water Index (NDWI) and Modified NDWI (mNDWI), and

machine learning algorithm like random forest in Landsat 8 scenes. Google Earth Engine (GEE), a free and web-based cloud computing platform was used for processing earth observation data, based on the time series Landsat images and compared with real-time groundwater levels. The results showed a significant increase ($p < 0.01$) in both surface water area and groundwater levels in Telangana state, especially after 2015, which could be due to the water conservation projects implemented by the Government of Telangana since 2015. A schematic showing the methodology for surface water extraction is presented in Fig 2.61. The results of the algorithm developed with water indices for estimating the extent of water bodies within the region of interest on Google Earth Engine (GEE) code editor before and after execution is shown in Fig 2.62.

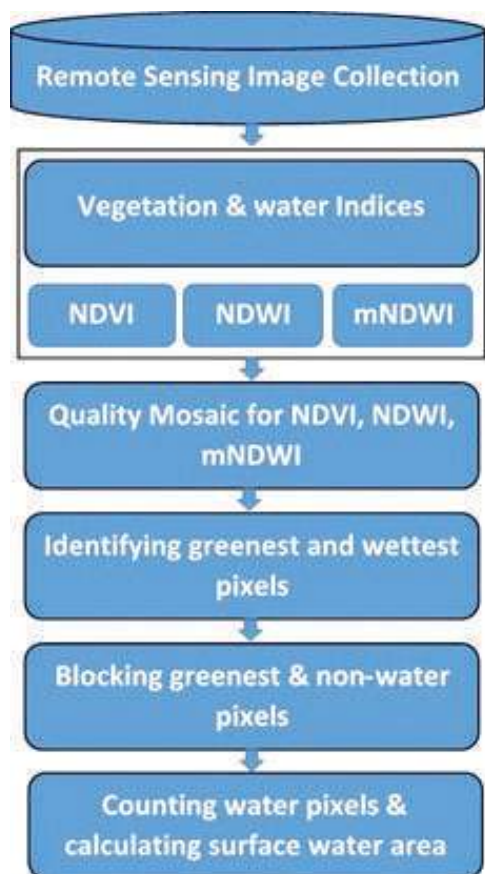


Fig. 2.54: Schematic for the surface water calculation performed in the study

The extent of surface water cover ranged from 1158.83 km² (2015) to 2189.76 km² (2013) for both the indices during the study period. The results of the present study indicated that, there was not significant ($p < 0.05$) variation in the extent of surface water cover estimated by two methods (NDWI and mNDWI) in the Telangana state (Fig 2.63). The extent of surface water estimated using RF algorithm showed a similar trend (Fig 2.64) as estimated by the algorithms (NDWI and mNDWI) and there was no significant difference ($p < 0.05$). The study showed that the extent of surface water had a significant positive correlation ($r = 0.85$; $p < 0.05$) with rainfall. The surface water area estimate was highest in the year 2013 (2189.76 km²) in the Telangana state, when the annual rainfall was 1348.7 mm and lowest during 2015 (1158.83 km²), when the rainfall was also the lowest i.e. 857.3 mm (Fig. 2.65).

The comparison of the study results with Central Ground Water Board (CGWB) ground water level data using Inverse Distance Weighting (IDW) interpolation method showed a positive correlation between ground water level and estimated surface water area ($R^2 = 0.86$). From the year 2015 onwards, the extent of surface water was observed to increase, due to higher rainfall, except in 2017, possibly because of drought spell. The ground water level showed an increasing trend from the year 2015 (Fig 2.66), which could be due to the Mission Kakatiya, a specialized water conservation project, launched by the Government of Telangana in 2015, aimed at rejuvenating 47,000 tanks and lakes in the state by accelerating the development of minor irrigation infrastructure by involving the community in irrigation management.



Fig. 2.55: Results of NDWI algorithm before and after execution

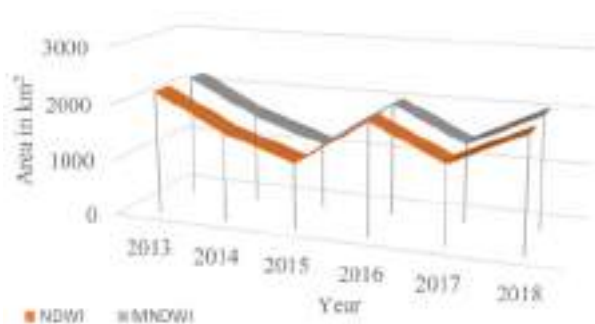


Fig. 2.56: Trend between NDWI and mNDWI

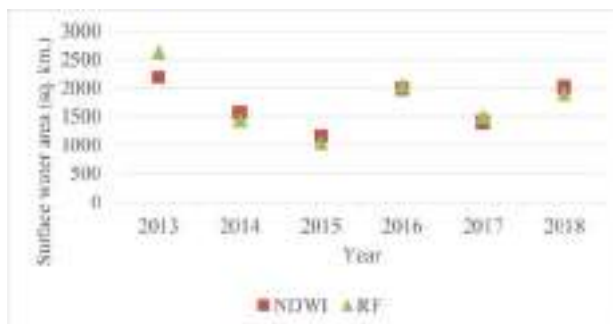


Fig. 2.57: Trend between NDWI and RF algorithms

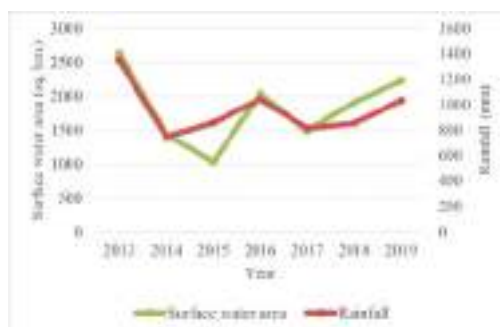


Fig. 2.58: Comparison of surface water area with rainfall data

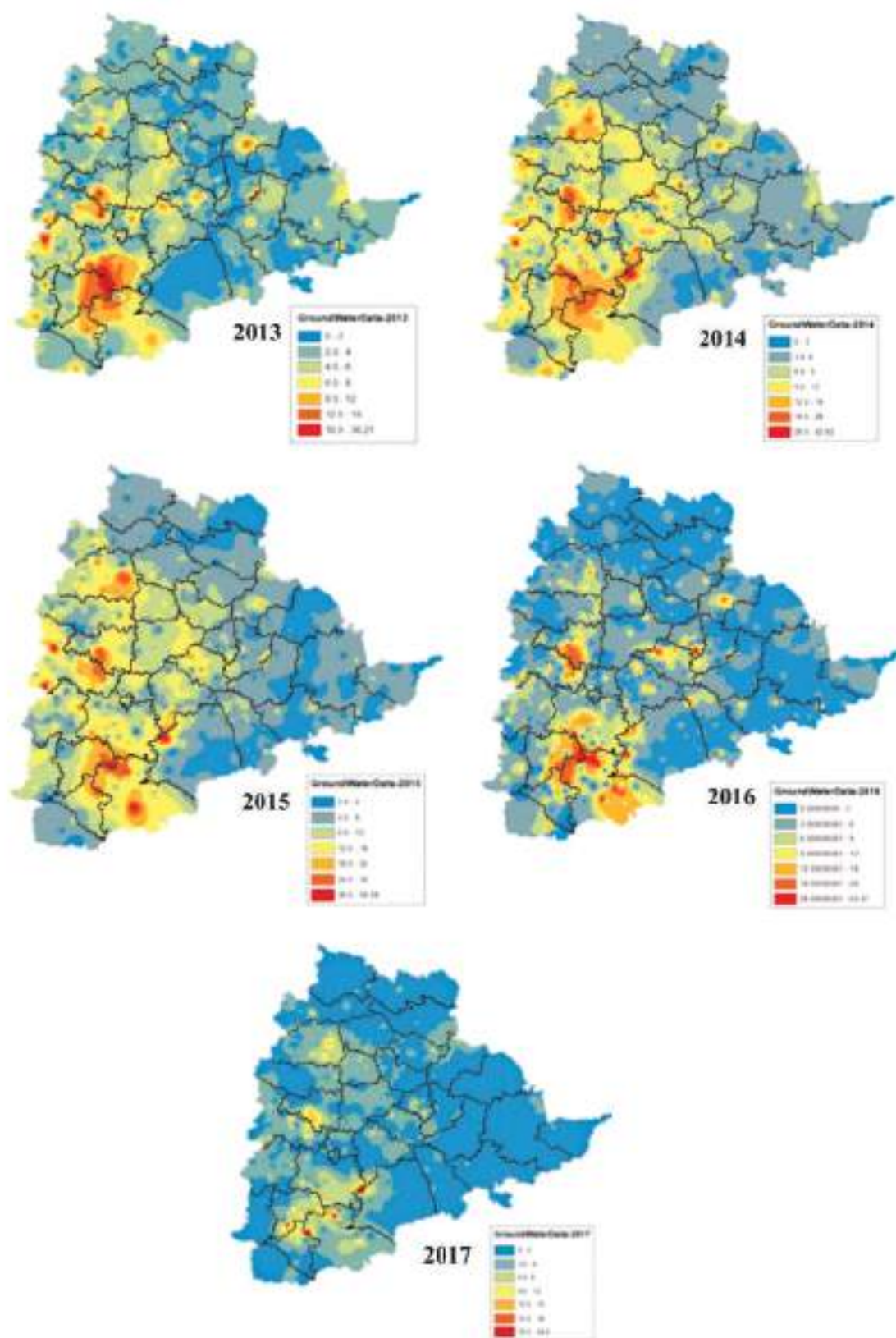


Fig. 2.59: Trend of ground water level in Telangana state during 2013-2017

2.6.7 Text Analytics of the Queries Raised at Kisan Call Centres in South India

In order to harness the potential of ICT in Agriculture, Ministry of Agriculture, GOI launched the “Kisan Call Centres (KCCs)” scheme on January 21, 2004. Under this scheme, farmers can call to Kisan Call Centre and enquire about various queries/problems related to agriculture and allied sectors. The Farm Tele Advisors (FTAs) answer the call and simultaneously record the query details which has made available a large amount of text data. This study has employed text analytics techniques to study the queries raised by farmers in Kisan Call Centres of South India. The results from the exploratory study of the text corpus indicated that a majority of queries (>60%) are related to weather. Apart from weather, queries on market rate of different commodities and pest & nutrient management practices also found to be appearing very frequently. Among the bigrams, ‘market rate’ appeared the highest number of times. The Topic Modeling using the

Latent Dirichlet Allocation technique suggested that the 1440 documents consisting district-wise & month-wise queries can be broadly categorised into 12 topics. Among them, the topic on ‘pest and nutrient management in paddy, chilli and cotton crops’ is found to be prominent with 495 documents. The cluster analysis resulted in division of 116 districts into 5 homogenous clusters based on the queries raised. The results are useful in appropriate planning communication techniques to send advisories to the farmers.

2.6.8 Focused Group Discussion on ‘Development of Guidelines for Collaborative Research Projects on Digital Solutions in Agriculture’

A Focused Group Discussion (FGD) on Development of Guidelines for Collaborative Research Projects on Digital Solutions in Agriculture was organized at the Academy on February 19, 2020 as part of National Agricultural Higher Education Project (NAHEP) with the objective to establish partnerships with the

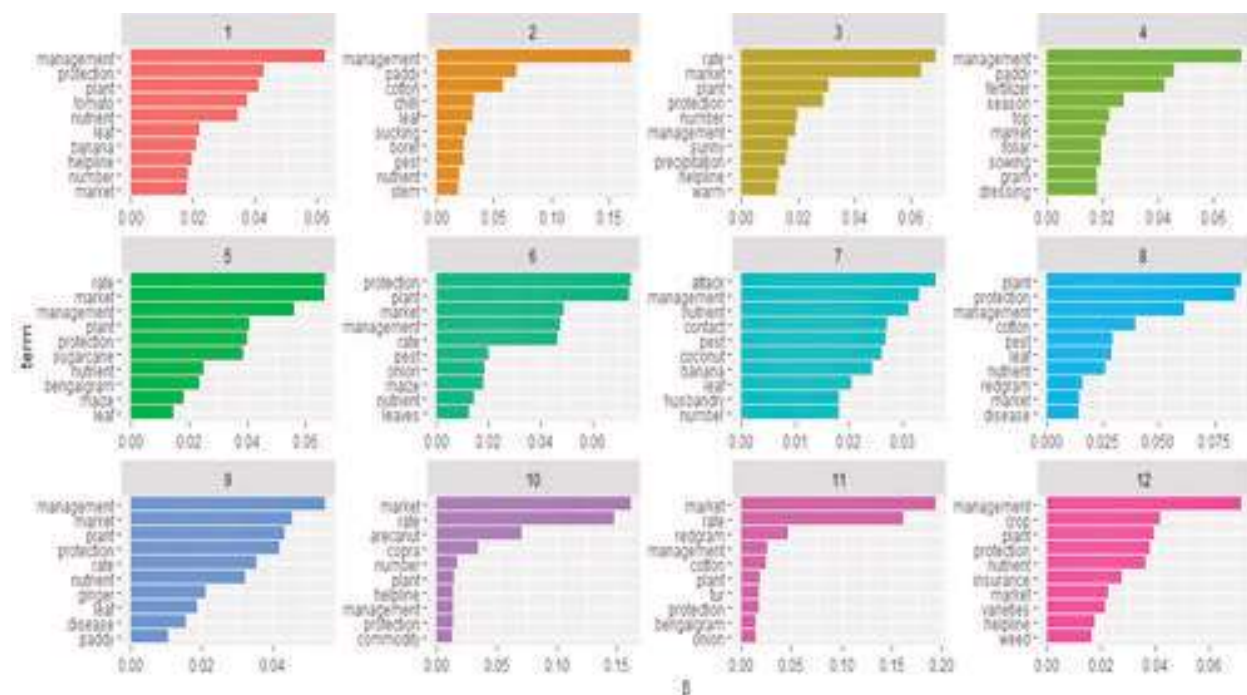


Fig. 2.60: Probabilities of Top 10 Words in each Topic

globally recognized Agricultural Higher Education Institutions (HEIs). The FGD was held with participation from 15 senior officials from IASRI, State Agricultural Universities and IT Industries. As a planned agenda for this meeting, the guidelines and project proposal format were discussed and finalized for development of digital solutions in agriculture through partnership between academic/research institutions and private IT firms/Startups.



Fig. 2.61: Focused Group discussion on 'Development of Guidelines for Collaborative Research Projects on Digital Solutions in Agriculture'

2.6.9 ICAR Summer School on 'ICT Applications in Agricultural Education and Extension'

The ICAR sponsored summer school was organized during Feb 19- Mar 10, 2020 with 24 participants. The objectives of the programme were: To sensitize entry and middle level NARS professionals about the concept of ICT and its use in Agricultural Education and Extension; To enhance the competency of NARS professionals in using various tools and techniques of ICT use in Agricultural Education and Extension; and to impart appropriate skills for ICT use and its applications in Agricultural Education and Extension.

The participants were exposed to organizing webinars through *zoom* and were made to write

blogs in groups. Around 5 blogs were written by them. Each participant was registered on at least 05 new social media professional platforms like LinkedIn, Research gate, YouTube etc. Online sessions were organized with practicals on Internet of Things (IOT) with hands on experience from industry personnel as well as institutes like National Institute of Plant Health Management (NIPHM) for Practical classes on drone.



Fig. 2.62: Use of drones in Agriculture by ICAR Summer School participants of ICT Applications in Agricultural Education & Extension



Fig. 2.63: Director, NAARM giving away certificates to Summer School participants of ICT Applications in Agricultural Education & Extension

2.6.10 Training Programme on 'Analysis of Experimental Data'

The online training programme was organized during Aug 05-11, 2020 with 161 participants. The objectives of the programme was to introduce R software for data analysis and to analyse data collected from different types of experiments using appropriate statistical techniques. The



programme was organized in on-line mode due to COVID-19 pandemic. There were more than 240 participants applied through on-line and the training programme was designed to restrict the on-line lectures to 3-4 hours a day spread equally in the forenoon and in the afternoon. Participants were encouraged to raise doubts/questions through chat during the lectures as well as at the end of each session. The questions were answered by the resource persons. A separate optional session was conducted during Sunday to those who are not able to follow the instructions. After the end of each session, one self-exercise sheet was provided and the participants were asked to find out solutions. These were discussed next day before each session and thus, allowing the participants to work through and learn. There were several advantages as well as disadvantages of offering the course through on-line. One biggest advantage of offering the course through on-line was we could train six times more participants as compared to off-line mode due to lab restrictions. Nearly 73% of the participants are attending for the first time in an on-line course spanning more than one day. Around 63% have said that their decision to attend this was influenced due to on-line mode. About 25% have mentioned that they are uncomfortable in on-line training mode. Some of the disadvantages of on-line mode are discomfort, interactions with other participants and distractions due to office work.

2.6.11 Training Programme on 'Analysis of Experimental Data using Statistical Analysis System' (SAS)

A six-day online training programme on "Analysis of Experimental Data using Statistical Analysis System (SAS)" was conducted at ICAR-National Academy of Agricultural Research Management, Hyderabad during Nov 09-17, 2020. A total of 110

participants including 37 women participants from ICAR institutes, SAUs and other Central/State Govt. Institutes attended the training programme. The training programme comprised of both theoretical and practical aspects of analyzing data obtained through different agricultural experiments such as basic and advanced Design of Experiments, Correlation and Regression, Cluster Analysis and Principal Component Analysis. The hands-on practical sessions were conducted by using real-life data in SAS University Edition environment. The training programme was conducted using MS Teams (online) using web-based software.



Fig. 2.64: A Snapshot of Virtual Training Programme on 'Analysis of Experimental Data using Statistical Analysis System' (SAS)

2.6.12 Online Training Programme on 'Design of Experiments'

ICAR-NAARM conducted an online training programme on 'Design of Experiments' for researchers of M/s UPL Ltd. during Nov 02-07, 2020. This programme was designed considering the requirement of the research staff of the company. The objectives of the training are to impart knowledge and skill in laying out different research designs, collecting data and analysis of the data using Bluesky statistics, a free and open source software. The participants, 25 in numbers, comprised of researchers working in the areas of field experimentation and quality testing. The

modules covered were hands on training with Bluesky Statistics, Tests of significance, Design and analysis of experimental designs viz. CRD, RBD, LSD, Factorial Experiments, Split and split-split plot, Correlation and Regression.



Fig. 2.65: Participants of the Online Training Programme on 'Design of Experiments'

2.6.13 Online National Seminar on 'Big Data Analytics in Agriculture'

The online seminar was organized during Dec 10-11, 2020 with 83 participants. The programme focused mainly on policy discussions on data science and big data applications with special reference to agriculture. An online programme, which included Big Data Applications in Crop Improvement, Big Data Tools and Techniques, Drone Based Sensing for Agriculture, Cloud Computing, Spatio-temporal Big-data for Agriculture and scope for potential Interventions and Modernizing the application of Advanced Analytics in Agriculture, Building and Delivering insights using SAS Via AI / ML Platform and Big Data Initiatives in ICAR were delivered by Academicians and Software experts. Experience sharing by various faculty was done. A panel discussion was also held on identifying potential thematic areas, opportunities, bottlenecks and policy issues in Animal sciences, Fisheries, Natural Resources Management, Artificial intelligence in Agriculture, Computing, Training

and other Policy issues with experts from different domains.



Fig. 2.66: Participants of the National Seminar on 'Big Data Analytics in Agriculture'

2.6.14 Geospatial Analysis using QGIS & R

This training programme was meant for Scientist/ Faculty/Technical Officers working in the area of Natural Resource Management from ICAR Institutes, State Agricultural Universities and KVK's. It was conducted during 27th February to 3rd March 2020 with 15 participants. The training class schedule and topics covered mainly on introduction to geospatial technologies in agriculture, relational database and geodatabase concepts for GIS, introduction to GIS, GPS and remote sensing, exploring data from open source platforms for geodata, spatial database design and implementation using quantum GIS open source software, spatial analysis like clipping, joining, buffering, interpolation, krigging etc., image processing and land use land cover using R studio.



Fig. 2.67: Training Programme on Geospatial Analysis using QGIS and R



2.6.15 Online Training Programme on 'Advanced Bio-informatics Tools and its Applications in Agriculture'

The Virtual Training Programme on 'Advanced Bio-informatics Tools and its Applications in Agriculture' under SKILL-BIF, DBT Project was organized by the ICAR-National Academy of Agricultural Research and Management, Hyderabad from Sep 14-19, 2020. About 73 Agricultural Researchers, including 37 women, representing 19 Agricultural Research disciplines from 20 ICAR Institutes and 19 Agricultural Universities virtually participated in the programme. Another online Training Programme on 'Advanced Bio-informatics Tools and its Applications in Agriculture' was organized from 7th to 11th December, 2020. There were 39 Agricultural Researchers, including 12 women, representing 17 Agricultural Research disciplines from 20 ICAR Institutions and 8 Agricultural Universities participated in the programme. This training programme was conceived with an idea to provide an insight to the participants about the enormous potential and application and management of tools of Bio-informatics in the domain of agriculture. The participants were provided with theory and hands-on-practical sessions on various software in Bio-informatics and tools on different topics related to emerging requirements in agriculture and allied sectors.



Fig. 2.68: Virtual Training Programme on 'Advanced Bioinformatics Tools and its Applications in Agriculture'

2.6.16 Training programme on 'Koha for Library staff of ICAR'

The six-day Capacity Building Training program on 'Koha for Library Staff of ICAR' was organized by the ICAR – National Academy of Agricultural Research Management (ICAR – NAARM), Hyderabad from Feb 20-25, 2020 for 19 participants with the objective to make the participants understand the application of Koha in library automation, to provide hands on training on various modules of Koha related to acquisition, cataloguing, OPAC etc., and to provide the knowledge of broad aspects of Koha installation and customization as per institute requirements. During the course of the training programme, the trainees were exposed to introduction of Linux; installation of Ubuntu Operating System, installation and configuration of Koha software; theoretical and practical aspects on various modules of Koha such as Administration, Acquisition, Patrons, Cataloguing, Circulation, Serials, OPAC, Reports, Tools etc.



Fig. 2.69: Participants with the Faculty of NAARM in the Training programme on 'Koha for Library staff of ICAR'

2.7 Research Projects (Completed & On-going)

2.7.1 Completed Research Projects during 2020

S. No.	PME Code	Project Title	Team	Duration	
				From	To
In-House					
1	AGED/NAARM/SIL /2017/012/00132	Team Work and its effect on organizational performance in NARES	RVS Rao Alok Kumar KH Rao	April 17	March 20
2	AGED/NAARM/SIL/ 2018/014/00153	Emotional Intelligence and Organizational Effectiveness	P. Ramesh RVS Rao	April 18	April 20
3	AGED/NAARM/SIL/ 2017/011/00131	Structural Changes in Agricultural Marketing in India: Effect on Inclusiveness, Efficiency and Profitability	Ranjit Kumar Sanjiv Kumar PC Meena B Ganesh Kumar N Sivaramane	April 17	March 20
4	AGED/NAARM/SIL/ 2016/009/00119	Gender based ICT led information management in agriculture	N. Sandhya Shenoy	July 16	March 20
5	AGED/NAARM/SIL/ 2018/027/00147	Decision Support System of Maize Inbred Germplasm	N Sunil, IIMR JC Shekar, IIMR N. Srinivasa Rao	July 17	July 20
6	AGED/NAARM/SIL/ 2017/014/00134	Geo-Spatial Knowledge System for monitoring climate change in agro Climatic zone-X : Southern plateau and hills region	P D Sreekanth B S Yashavanth P Venkatesan	April 17	April 20
7	AGED/NAARM/SIL/ 2018/023/00143	Text Analytics of the queries raised at Kisan Call Centres in South India	BS Yashavanth PD Sreekanth Jabir Ali (MANAGE)	April 18	March 20
8	AGED/NAARM/SIL/ 2018/024/00144	Assessment of technological interventions in teaching-learning Process	S. Senthil Vinayagam G.R.K. Murthy D. Thammi Raju	April 18	March 20
Extra Mural					
1	AGED/NAARM/CCP/ 2019/004/00158	Adoption of Natural Farming and its effect on crop yield and farmers livelihood in India	Ranjit Kumar_ Sanjiv Kumar B S Yashavanth, PC Meena, P Ramesh AK Indoria Sumanta Kundu Manjunath	Jan 19	March 20



S. No.	PME Code	Project Title	Team	Duration	
				From	To
2	AGED/NAARM/SCL/2018/015/00154	Economic analysis of Eucalyptus and Subabul based agroforestry ecosystem in Telangana and Andhra Pradesh	P Venkatesan Sivaramane	Sep 18	Aug 20
3	AGED/NAARM/SOL/2019/006/00160	Competency analysis among faculty of agricultural universities in technology enhanced learning	G.R.K. Murthy Surya Rathore	April 19	March 20
4	AGED/NAARM/SOL/2018/026/00146	Development of digital content for Agri-Warehousing management	S. Senthil Vinayagam G.R.K. Murthy	Nov 19	Dec 20
5	AGED/NAARM/SOL/2014/009/00100	Technology enhanced learning in agriculture education	G.R.K. Murthy D. Thammi Raju S. Senthil Vinayagam	Jan 14	March 20
6	AGED/NAARM/SOL/2017/003/00123	Agricultural research and development infrastructure in select Indian States	K. Kareemulla S. Ravichandran	July 17	March 20
7	AGED/NAARM/SIL/2019/009/00164	Orientation cum awareness and implementation of Access and Benefit Sharing (ABS) Guidelines in ICAR	K. Kareemulla B. Ganesh Kumar P. Krishnan	Nov 19	June 20
8	AGED/NAARM/SOL/2018/013/00152	Profiling the resources devoted to agricultural R&D and development of national framework for evidence-based S&T governance in India	Ch Srinivasa Rao P. Krishnan B. Ganesh Kumar M. Balakrishnan S.K. Soam B.S. Yashavanth	Oct 18	March 20
9	AGED/NAARM/SCP/2018/001/00140	Strengthening and integrating agricultural science & technology indicators (ASTI) in Indian NARES	Ch Srinivasa Rao P. Krishnan B. Ganesh Kumar S.K. Soam	Oct 17	March 20
10	AGED/NAARM/SIL/2019/009/00163	Developing a DPR & ToR for (i) Capacity building / knowledge sharing for institutions and community in Karnataka (ii) Eco-smart Villages	Ch Srinivasa Rao P. Krishnan S.K. Soam B.S. Sontakki I. Sekar G.R.K. Murthy	Sep 19	May 20

(Characters in Project Codes: S-Intra Institutional, C-Inter Institutional, I-Institute Funded, O- Externally Funded, C- Consultancy, L-Institute is Leader, P-Institute is Partner)

2.7.2 Ongoing Research Projects

In-House Projects

S. No.	PME Code	Project Title	Team	Duration	
				From	To
1	AGED/NAARM/SIL/2017/013/00133	Competency management system for agricultural research and extension	KH Rao, Alok Kumar RVS Rao, P Ramesh	Apr. 17	March 22
2	AGED/NAARM/SIL/2018/010/00149	HR Interventions and their influence on Institutional innovations	Alok Kumar RVS Rao P Ramesh	Oct 18	June 21
3	AGED/NAARM/SIL/2020/05/00171	Happiness and psychological well-being of scientific personnel in NARES	P. Ramesh RVS Rao BS Yaswanth	June 20	May 22
4	AGED/NAARM/SIL/2020/04/00170	Identification of best practices developed by the leadership in scientific and general administration in NARES	R V S Rao, P Ramesh Alok Kumar, K Kareemulla Senthil Kumar S Ravichandran D Thammiraju Ch Srinivasarao	June 20	Feb 23
5	AGED/NAARM/CCL/2019/007/00161	Mainstreaming the small ruminant farmers into efficient value chains in India	B Ganesh Kumar Sanjiv Kumar D Thammi Raju PC Meena, Tavva Srinivas A Dhandapani	Oct 18	Sep 21
6	AGED/NAARM/SIL/2018/030/00150	Farmer Producer Companies in India: A study on their management practices and business potential	N Sivaramane Ranjit Kumar, PC Meena Bharat S Sontakki K Srinivas, P Venkatesan	Oct 18	Sep 21
7	AGED/NAARM/SIL/2019/012/00166	Agri Startups in India: Decoding the determinants and success factors	Sanjiv Kumar K Srinivas Ranjit Kumar	June 19	May 21
8	AGED/NAARM/SIL/2020/01/00167	New Market Reform and E-NAM in India: Effect on market Participation and price gain for smallholders	Ranjit Kumar Sanjiv Kumar A Dhandapani K Srinivas, P C Meena K.H. Rao	June 20	May 22
9	AGED/NAARM/SIL/2020/02/00168	Studies and documentation of Techno-Economic feasibility of value added products from millets in India	Tavva Srinivas B. Ganesh Kumar GRK Murthy, K.Kareemulla Sudha Mysore	June 20	May 22
10	AGED/NAARM/SIL/2020/03/00169	Financial accessibility and occupational risks of vegetable street vendors in India	P C Meena Sanjiv Kumar	June 20	May 22



S. No.	PME Code	Project Title	Team	Duration	
				From	To
11	AGED/NAARM/SIL/2017/007/00127	Research role analysis of extension scientists in ICAR	VKJ Rao Bharat S Sontakki Surya Rathore, Alok Kumar	April 17	Dec 20
12	AGED/NAARM/SIL/2020/06/00172	Assessment of training (online and offline/physical) needs in data analytics	S. Ravichandran Surya Rathore	June 20	May 21
13	AGED/NAARM/SIL/2020/07/00173	Mega environment analysis tools for AICRP Trials	A. Dhandapani	June 20	May 22
14	AGED/NAARM/SIL/2018/021/00141	Ag-Academy: A framework for scaling up e-learning Platform	VV Sumanth Kumar P Krishnan, SK Soam N Srinivas Rao	April 18	March 21
15	AGED/NAARM/SIL/2018/020/00142	Digital teaching evaluation in agricultural education using TPACK model	GRK Murthy S Senthil Vinayagam M Balakrishnan	Oct 18	Sep 21
16	AGED/NAARM/SIL/2018/031/00151	Students' Learning Approaches (SLAs) for enhanced learning outcomes in agriculture education: A critical analysis	D Thammi Raju P Ramesh Bharat S Sontakki P Krishnan	Oct 18	Sep 21
17	AGED/NAARM/SIL/2017/009/00129	Developing a framework for enhancing agricultural research management efficiency	Ch Srinivasa Rao P Krishnan, I Sekar SK Soam, B Ganesh Kumar K. Kareemulla VV Sumanth Kumar	April 17	March 21
18	AGED/NAARM/SIL/2019/008/00162	Economic analysis of food prices and policies for the development of Indian agriculture	MB Dastagiri B Ganesh Kumar A Dhandapani	June 19	July 21
19	AGED/NAARM/SIL/2020/08/00174	Impact assessment of research and development programs on food and nutritional security	Umesh Hudedamani Ramesh Naik P Krishnan, Tavva Srinivas B S Yashvanth Vijaykumar	June 20	May 21
20	AGED/NAARM/SIL/2020/09/00175	Dynamics of agriculture and food systems in different Agro-ecosystems of India	Ramesh Naik Umesh Hudedamani P Krishnan, BS Yashavant N Subash, Monoranjana Mohanty Ch Srinivasa Rao	June 20	May 21

2.7.3 Externally Funded Contract Research (Sponsored/Collaborative/Grant-in-Aid Projects)

Sl. No.	PME Code	Title of Project	Team	Duration		Budget (in Lakhs)	Sponsored by
				From	To		
1	AGED/NAARM/ COP/ 2017/001/00121	Management and Impact Assessment of Farmer FIRST Project	P Venkatesan Bharat S Sontakki, N Sivaramane	Mar 17	Mar 21	65.00	ICAR, New Delhi
2	AGED/NAARM/ COP/ 2019/001/00155	Investments in Indian Council of Agricultural Research leadership in agricultural higher education (NAHEP)	SK Soam, D Thammi Raju N Srinivasa Rao, Alok Kumar VV Sumanth Kumar Surya Rathore, Sanjiv Kumar M Balakrishnan, SS Vinayagam P Krishnan, BS Yashavanth	Feb 19	Mar 21	1704.17	ICAR-NAHEP, New Delhi
3	AGED/NAARM/SOL/ 2015/007/00110	KRISHI: ICAR Research data repository for knowledge management	A Dhandapani SK Soam	Nov 15	Mar 21	76.00	ICAR, New Delhi
4	AGED/NAARM/SOL/ 2019/005/00159	Establishment of skill development bioinformatics infrastructure facility (Skill-BIF) for agricultural biotechnology (DBT)	M Balakrishnan SK Soam P Supriya	Dec 18	Mar 21	44.46	DBT, New Delhi
5	AGED/NAARM/CCL/ 2019/011/00165	Development of digital content for MOOC on commercial dairy farming, milk processing & value addition	S Senthil Vinayagam GRK Murthy	Apr. 19	Mar 21	23.30	NDRI, Karnal
6	AGED/NAARM/CCL/ 2019/010/00164	Rodents and household pest management under MOOC platform	S Senthil Vinayagam GRK Murthy	Apr. 19	Mar 21	7.40	NIPHM, Hyderabad
7	AGED/NAARM/SOL/ 2019/002/00156	Entrepreneurship development through farmer led innovations – A case study in plantation sector	S Senthil Vinayagam GRK Murthy, B GaneshKumar P Venkatesan	Apr. 19	Mar 22	84.66	NASF, ICAR, New Delhi
8	AGED/NAARM/SOL/ 2008/003/00067	IP management and transfer/ commercialization of agricultural technologies scheme	Umesh Hudedamani K Srinivas	Apr. 14	Mar 21	19.15	ICAR, New Delhi
9	AGED/NAARM/CCL/ 2019/007/00161	Strengthening natural Resource management (SNRM): A framework for mainstreaming biodiversity conservation in the development schemes of agricultural sector in India	Ch. Srinivasa Rao P. Krishnan SK Soam I. Sekar	Aug 19	Oct 21	29.50	UNDP-BCIL
10	AGED/NAARM/SOL/ 2016/002/00112	Establishment of agri-business Incubation (ABI) centres under XII plan scheme for NAIF	K Srinivas, Manju Gerard Umesh Hudedamani Ramesh Naik, Sanjiv Kumar	Jan 16	Mar 21	79.15	ICAR, New Delhi



2.8 Educational Programmes

2.8.1 Project Reports Submitted by 10th batch (2018-20) PGDMA Students

Sl. No.	Roll No.	Student Name	Title of the Project	Guide Name
1	PGDMA-1801	Aiswarya Sukumaran	Value chain analysis of mustard and business development of Spec-X Pro	Yashavanth BS
2	PGDMA-1802	Anand Singh	Investment pattern in different agripreneurship in India and world	R V S Rao
3	PGDMA-1803	Amal Maria Jacob	To identify areas of intervention for institutional building of FPOs	Manju Gerard
4	PGDMA-1804	Amarnath. C.K	Understanding the performance of ConTrak in the Malabar Milma Region and suggesting strategies	Bharat S Sontakki
5	PGDMA-1805	Atul Kumar	Assessment of warehousing operation of DORB in given study area	A Dhandapani
6	PGDMA-1806	Dastageerapasha	Competitive assessment of ADM purchase process (Soybean)	I Sekar
7	PGDMA-1807	Durgesh Chavan	Identification of gap between estimated and actual yield of cotton in Maharashtra	S K Soam
8	PGDMA-1808	Gayatri Gopan	Developing a database management tool to facilitate market linkage between FPOs and institutional buyers	Ranjit Kumar
9	PGDMA-1809	G Srikanth	Identification of fertilizer consumption pattern in Andhra Pradesh	Sanjiv Kumar
10	PGDMA-1810	Gollapalli Prasanth Mourya	Impact study – Chilli farm value chain development through Integrated Agri Extension Platform (IAEP) in Andhra Pradesh	B Ganesh Kumar
11	PGDMA-1811	Harshita Srivastava	Feasibility study to develop catalytic infrastructure for agri value chain.	Sanjiv Kumar
12	PGDMA-1812	I C Kariappa	Comparison of various agricultural loan schemes of nationalized banks vis-a-vis federal bank	K Kareemulla
13	PGDMA-1814	K Sai Prajeeth Reddy	Market research on tender coconut in India	N Sandhya Shenoy
14	PGDMA-1815	Lingala Thrishali	To study the effectiveness of crop advisors in Telangana in paddy crop and to study the perception of Geogreen product among farmers	K H Rao
15	PGDMA-1816	Madagoni Vamshi Krishna	Consumption pattern of different agrochemical brands in Telangana	Sivaramane N
16	PGDMA-1817	Mitali	Envisage ways to use letter of credit to get cheaper finance suitable to the model of company	Sreekanth P D
17	PGDMA-1818	Naveen Yadav N	Digitization of FPO.	Alok Kumar
18	PGDMA-1819	Nivedita Yogiraj Patil	Implementation of ISO 22000:2018	P Venkatesan
19	PGDMA-1820	Poorna Shree G	Development of FPO portal & Agri elevate portal	S Ravichandran

Sl. No.	Roll No.	Student Name	Title of the Project	Guide Name
20	PGDMA-1821	Ramya Nargund	Market research on tender coconut in southern India	S Senthil Vinayagam
21	PGDMA-1822	Rashmi Pandey	Benchmarking the catch masala variants	V V Sumanth Kumar
22	PGDMA-1823	Sadish Kumar	Establishing AMLA for FPOs in Bihar	M BalaKrishnan
23	PGDMA-1824	Siddharth P	Digitization of FPO.	P C Meena
24	PGDMA-1825	Shalini Singh	Identification of new products based on value chain analysis	Srinivas Kondapi
25	PGDMA-1826	Shetty Sharath Kumar	Role of technology and banking correspondent (BC) in JLG lending	Surya Rathore
26	PGDMA-1827	Shivani Balasaheb Doifode	Market potential study of commercial production and marketing of Compressed Bio Gas (CBG) by FPOs	M.B Dastagiri
27	PGDMA-1828	Sumeet Singh	Mapping operational flow and devising just-do-it approach for increasing process efficiency	Ch Srinivasa Rao

2.8.2 Summer Internship of 11th batch (2019-21) PGDM-ABM Students

Sl. No	Roll No.	Student Name	Title of the Project	Guide Name
1	PGDMA-1901	Aayush Awasthi	Price positioning of 3 way herbicide for sugarcane	Ch Srinivasa Rao
2	PGDMA-1902	Abhishek Singh	Market research and brand expansion strategies	M Balakrishnan
3	PGDMA-1904	Aiswarya Santhosh	Implementation of TAFE sponsored free farm equipment rental scheme to mitigate impact of COVID-19 on small & marginal farmers	S K Soam
4	PGDMA-1905	Ajithkumar	Perception of end users towards Nimitz	Alok Kumar
5	PGDMA-1906	Akansha Gupta	Change in mass media consumption pattern among Indian farmers during COVID-19 scenario	P Venkatesan
6	PGDMA-1907	Aman Kumar Sharma	Role of ICT in digital farming practices- an in depth study.	M B Dastagiri
7	PGDMA-1908	Annett Skinner	Diagnostic study of agri value chains in Assam	S Senthil Vinayagam
8	PGDMA-1909	Anshu Singh	Project I: Competitive analysis and market potential assessment for Agri-Tech based services as a retail product for farmers as well as developing business universe for expansion of operations. Project II: Market positioning of IFFCO kisan in Cattle feed and Aqua feed range.	K H Rao
9	PGDMA-1910	Arvind Mehara	Project I: Competitive analysis and market potential assessment in for Agri-Tech based services as a retail product for farmers. Project II: Market positioning of spice range as well as developing business universe for expansion of operations	Surya Rathore



Sl. No	Roll No.	Student Name	Title of the Project	Guide Name
10	PGDMA-1911	Ashish Pandey	Research and analysis of Samunnati's clientele and making recommendations to increase business volume with them.	B Ganesh Kumar
11	PGDMA-1912	Aurodeep Kamal	To study Grower's satisfaction levels & purchase intention behavior of galileo and galileo way in Punjab & Haryana	S K Soam
12	PGDMA-1913	Badush Salim	To study the consumer behaviour towards the fertilizer segment and farm equipments segment in UAE	N Sivaramane
13	PGDMA-1914	Byri Siva	Storage finance products for FPOs : Warehouse receipt Finance	Ch Srinivasa Rao
14	PGDMA-1915	Celine Chacko	Mapping of wheat in UP and Bihar	Sanjiv Kumar
15	PGDMA-1916	Chaithra P M	Evaluation and analysis of crops grown in Sirguppa, Haveri, Chikballapur districts of Karnataka and business calendar preparation for Falca.	M Ramesh Naik
16	PGDMA-1917	Chatakonda Kusumanjali	Implementation of TAFE sponsored free farm equipment rental scheme to mitigate impact of COVID-19 on Small & marginal farmers	P C Meena
17	PGDMA-1918	Chelumalla Ranjith	Mapping of research paddy potential in Telangana	Tavva Srinivas
18	PGDMA-1919	Devadi Hari Prasad	Evaluation of crop value chains and business opportunities in medicinal plants	K Kareemulla
19	PGDMA-1920	Divyanshu Yadav	Market and competitive research for good earth alliance	B S Yashavanth
20	PGDMA-1921	G Sai Vijita	Effectiveness of rallis current digital initiatives benchmarking with competition and way forward in uncertain and changing environment	Ranjit Kumar
21	PGDMA-1922	Ganesha T B	Customer satisfaction towards free tractor rental Scheme and end to end validation of orders	Alok Kumar
22	PGDMA-1923	Gourishetty Nikhil	To plan the layout and design of waycool distribution centre	P D Sreekanth
23	PGDMA-1924	Gowtham A N	Wallmart project- Baseline cum assessment study in coffee and turmeric	P Krishnan
24	PGDMA-1925	Gunjan Sharma	Adoption level of pexalon in Punjab and Haryana region of North India.	B Ganesh Kumar
25	PGDMA-1926	Hardika Mehta	Evaluation of crop value chains and business opportunities in medicinal plants	N Sivaramane
26	PGDMA-1927	Harshana M T	COVID assessment through gendered approach	I Sekar
27	PGDMA-1928	K Shravya	A study on geographical ICT based e-agriculture Support module in Uttar Pradesh for hybrid paddy grown area	M B Dastagiri
28	PGDMA-1929	Karlapudi Venkatesh	Analysing scope of agri-inputs for Samunnati FPOs.	R V S Rao
29	PGDMA-1930	Kokate Utkarsh Kiran	How to use digitization for enhancing business	Sanjiv Kumar

Sl. No	Roll No.	Student Name	Title of the Project	Guide Name
30	PGDMA-1931	Kola Haritha	Project- I: Competitive analysis and market potential assessment in for Agri-Tech based services as a retail product for farmers. Project-II: Market Positioning of Animal feed as well as developing business universe for expansion of operations.	VV Sumanth Kumar
31	PGDMA-1932	Koushik S	Wallmart project- Baseline cum assessment study. Sustainable livelihoods and market access program.	B S Sontakki
32	PGDMA-1933	Kshitiz Saxena	A study on dealer's satisfaction towards products and services provided by Dharti Agro Chemicals Pvt. Ltd.	P Venkatesan
33	PGDMA-1935	Nora Mohanty	Mobile app services and business adoption strategies for Good Earth Alliance Digital Pvt Ltd	S Ravichandran
34	PGDMA-1936	Prakshyati Satpathy	Covid assessment through a gendered approach	Manju Gerard
35	PGDMA-1937	Pranjali Siwal	Assessment study on COVID impact through gender approach.	K Kareemulla
36	PGDMA-1938	Priyanka	Project I: Research and analysis of Samunnati's clientele and making recommendations to improve the business volumes with the assigned company. Project II: Impact of pandemic induced lockdown on Agri value chains.	P D Sreekanth
37	PGDMA-1939	Renuka S	Understand the future trends and acceptance of UAV (Drones) in different crops by Indian farmers	S Ravichandran
38	PGDMA-1940	Rishav	Farmers perception on Bio-stimulants and its emerging trends	A Dhandapani
39	PGDMA-1941	Sachin Kattimani	RACP-Rajasthan Agriculture Competitiveness Project	K H Rao
40	PGDMA-1942	Shivanand Masali	Business opportunities & revenue models for Good Earth Alliance Digital	VV Sumanth Kumar
41	PGDMA-1943	Shubham Rawat	Landscape analysis of full stack agritech apps and suggest offerings for the FPO Gateway.	N Srinivasa Rao
42	PGDMA-1944	Siddanagouda Patil	Development of farmers development centre in Haveri district of Karnataka	P C Meena
43	PGDMA-1945	Sree Bala V P	Implementation of TAFE sponsored free farm equipment rental scheme to mitigate impact of covid-19 on small & marginal farmers	B S Yashavanth
44	PGDMA-1946	Supriya T D	Intervention of farm mechanization in chilli and soybean in Andhra Pradesh and Madhya Pradesh	Umesh Hudedamani
45	PGDMA-1947	Sushant Chandra	Customer satisfaction towards free tractor rental Scheme and end to end validation of orders.	Ranjit Kumar
46	PGDMA-1948	Vedavyas V Pandurangi	Supply chain mapping of pulses and spices	Manju Gerard
47	PGDMA-1949	Vishakha Wankhade	Study of orange value chain and VAAPCO SWOT analysis	Surya Rathore



Sl. No	Roll No.	Student Name	Title of the Project	Guide Name
48	PGDMA-1950	Yenugula Dorayya Naidu	Competitive analysis of fall army segment in Karnataka	M Balakrishnan

2.8.3 Students (M.Sc. & Ph. D) Guided by NAARM Faculty as Major Guide/Supervisor

Sl. No.	Name of the Faculty	Student	Title of the Thesis	Program MSc/PhD	University/ Institute
1.	M. Balakrishnan	D Karunakaran	Development of algorithm for prediction of fish yield in reservoirs using artificial neural network	Ph.D	Bharathiar University, Coimbatore
2.	Bharat S Sontakki	Bhuvana Rao N	Ecosystem analysis of effectiveness of krishi vigyan kendras in southern india	PhD	PJTSAU, Hyderabad
3.	S Senthil Vinayagam	Katta Akhila	E-Resources effectiveness in Agri-business education – A case study	PGDETM	ICAR NAARM - UoH
4.	D Thammi Raju	Dinesh U Lokhande	Use of social media in veterinary education and service	PGDETM	ICAR NAARM - UoH
5.	D Thammi Raju	M Balakrushna	Role of social media as a means of education for school going children in Kosgi town of Narayanapet District	PGDETM	ICAR NAARM - UoH
6.	D Thammi Raju	M Sowmya	Perceptions of school teachers on new educational policy- 2020: An analysis	PGDETM	ICAR NAARM - UoH
7.	Alok Kumar	Aswathy Joseph	Dimensions of active learning strategies in higher education	PGDETM	ICAR NAARM - UoH

3.1 Policy Research and Advocacy

The Academy has undertaken policy research on various issues and has been networking with various experts across the NARES. A brief on the Documents developed during 2020 are summarized hereunder:

3.1.1 Ease of Doing Research (EoDR): A Methodological Framework for Agricultural Research Organizations

Research in India and developing countries is predominantly public funded and the generation of desired output and outcomes depend upon effective resource use, among others. However, the governments as well as the researchers have been grappling with ways to gauge the efficiency and effectiveness of research organizations. Human resource and finance which are the major tangible inputs (drivers) for research. But research productivity is also influenced by intangible inputs like research facilitation, leadership, governance, culture, etc.

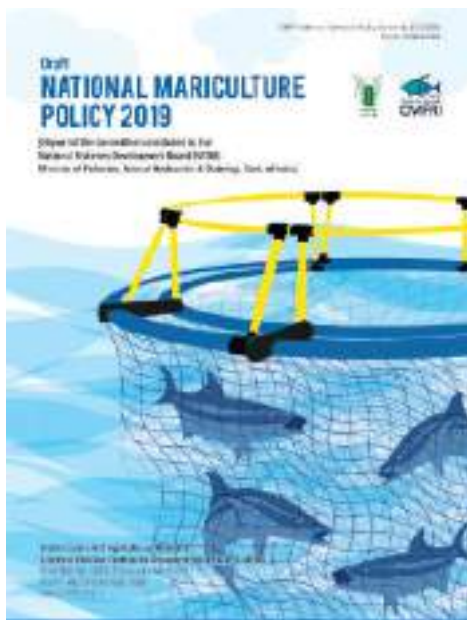


Most global and national ranking frameworks capture only the quantitative and easily measurable tangible inputs. However, they do not reveal adequately the 'problem areas' specific to an institute that a research administrator at the institute level can attempt to intervene through appropriate management decisions, the results of which could be perceptible in the short or medium term.

Addressing this crucial gap in the performance assessment methods, a methodological framework for assessing 'Ease of Doing Research' (EoDR) in research organisations was developed based on an extensive review and expert consultations. It refers to the perceived ease with which the researchers can carry out their research work in respective institutions. Technically, the Ease of Doing Research Index (EoDRI) is computed by aggregating the distance to frontier scores of 22 indicators (grouped under 5 parameters). EoDRI will be a handy and effective decision support tool for the research managers to transform their institutions.

The EoDr. framework takes a major departure from existing frameworks by focusing on the intangible inputs and process dimensions of the research ecosystem, which could be significant enablers of the tangible inputs. The EoDr. framework would aid the institutions under NARES in their self-discovery and renewal, by providing material evidence to research administrators and managers at different levels to take targeted and outcome-oriented interventions. The framework places individual researchers at the centre of attention and promises to make the research environment more exciting and enabling.

3.1.2 Draft National Mariculture Policy 2019 (NMP2019)

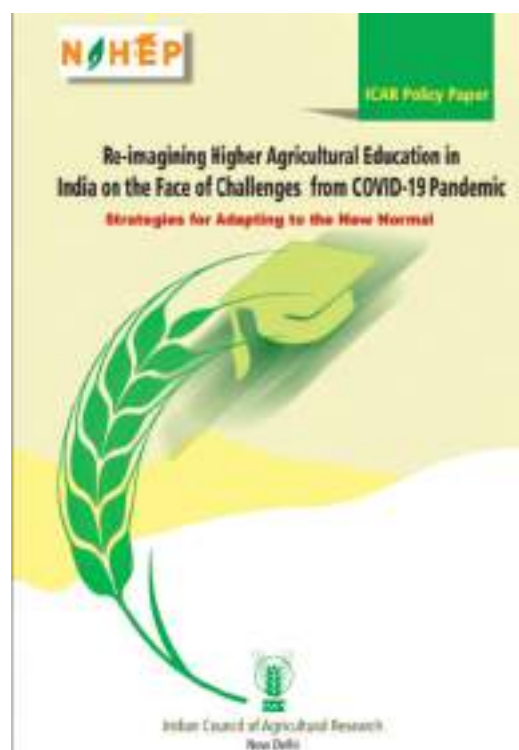


NAARM was part of the Drafting Committee constituted by National Board of Fisheries Development, Hyderabad vide its order NFDB/HRD/ Mariculture Policy dated 06.06.2018 - under the Chairmanship of Director, CMFRI. The Policy was made based on extensive review and consultations.

The National Mariculture Policy 2019 has been framed to ensure a structured growth of mariculture for meeting the growing demand of fish in the country. The overall strategy of National Mariculture Policy 2019 (NMP 2019) was to increase seafood production in a sustainable and responsible manner, ensure socio-economic development, enhance food, health and nutritional security and safeguard gender, social equity and environment. The policy has factored in the interests of all stakeholders especially the small and marginal fishers, while keeping in view the need of the growth of the sector in an ecologically sustainable manner.

The Draft National Policy has been published by ICAR and CMFRI on behalf of National Fisheries Development Board, Ministry of Fisheries, Animal Husbandry and Dairying, Govt. It is hoped that this Policy will also be able to tap major opportunities that are expected to emerge from the Blue Economy initiative of the country.

3.1.3 Re-imagining Higher Agricultural Education in India on the Face of Challenge from COVID-19 Pandemic - Strategies for Adapting to the New Normal



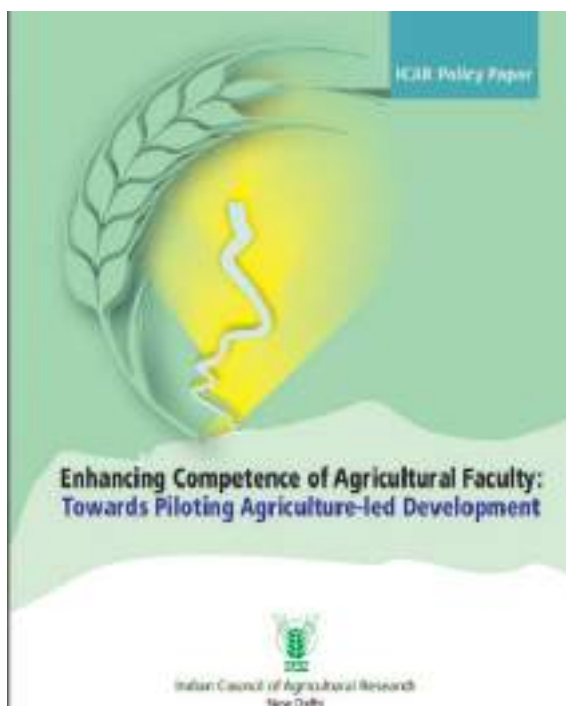
The COVID-19 pandemic caused significant disruption to the global education systems. The current pandemic paved the way for many disruptive innovations in the use of technology for education, which otherwise takes a much longer time. This Policy Paper maps the perception of university administrators, faculty members and students on the potential impacts of the crisis on agricultural higher education, which provide adequate pointers for planning for the short-



term and preparing for the long-term future – in our endeavour to convert the crisis into an opportunity.

The Policy brief provides strategies for the Universities to build on their experiences, capitalizing on this momentum and adapt their academic content and delivery to the New Normal. It also provides broad contours for future actions have been effectively captured.

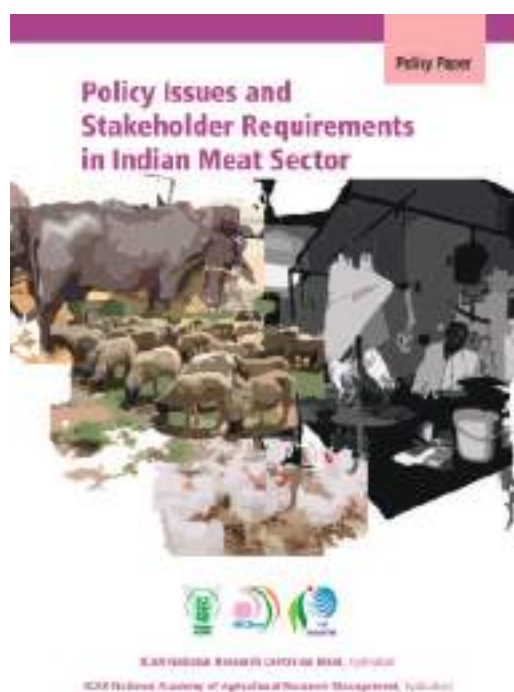
3.1.4 Enhancing Competence of Agricultural Faculty: Towards Piloting Agriculture-led Development



The human resource developed by the National Agricultural Education System has played a pivotal role in transforming agricultural scenario in the country by achieving self-sufficiency in food production. While there are various factors which influence the quality of higher education like funding support, physical infrastructure, quality of human resources, etc., the competence of faculty members plays the most significant role in defining the quality of the graduates from

the institutions. A policy brief was developed on the subject based on extensive review and consultations, which called for critical review of the current status, identify issues and advancing policy perspectives to address the gaps. The policy paper attempts to identify the issues and suggest strategies for enhancing the competence of the faculty members serving in the agricultural universities.

3.1.4 Policy Issues and Stakeholder Requirements in Indian Meat Sector



Meat sector activities are highly complex and regulated by several Central and State Governmental agencies. In spite of burgeoning demand for animal sourced foods and major cash inflows through export of buffalo meat, this sector is the most neglected with poor infrastructure, highly unorganized, and operated and managed by certain section of the society. Hence, identification of issues in meat sector, reasons for

continuous failure of the concerned agencies to modernize the municipal abattoirs and regulate the live animal transport, marketing, retailing and inability to streamline the supply-chain are needed to provide inputs for appropriate policy changes.

NAARM developed a Policy Brief jointly with NRC-Meat and NAAS, Hyderabad Chapter to address the core issues related to meat value chain including meat animal and poultry production, transportation, live animal marketing, slaughtering, meat inspection, processing, export, retailing. This document discusses the existing policies, acts, directive principles and regulatory issues related to meat sector activities and provide the framework to ensure sustainable development of livestock and poultry sector.

The policy document was prepared based on discussions and wide interactions among various meat and poultry industry stakeholders, meat exporters, poultry farmers, policy makers, regulators, personnel from Municipal administration and State Animal Husbandry Departments, butchers, retailers, meat processors and consumers.

The document suggests some effective interventions like salvaging male buffalo calves, effective culling of spent female buffaloes, increased role of veterinarian in ensuring animal health and food hygiene along the supply chain, training of meat inspectors, creation of disease free zones, modernization of existing livestock markets, tagging of animals at livestock markets to ensure traceability, designing of transportation vehicles to ensure better animal welfare and creation of more avenues for entrepreneurship and start-up business ventures in meat processing and marketing.

3.1.5. National Framework for Mainstreaming Biodiversity Conservation in Agricultural Development Schemes in India



Strengthening Natural Resource Management (SNRM)
A Framework for Mainstreaming Biodiversity
Conservation in the Development Schemes of
Agriculture Sector in India



The framework for mainstreaming biodiversity conservation in the development schemes of the agricultural sector in India was prepared based on a detailed assessment of the implementation status of biodiversity relevant public schemes of the agriculture sector. Over 40 success stories have been documented from these states with significant contribution to biodiversity conservation.

The report recommended reforms in identified public schemes to ensure enhanced biodiversity outcomes for agrobiodiversity conservation and identified key entry points. It also calls for the creation of a separate Mission on Agrobiodiversity focused on Plant Genetic Resources as a Sub-mission of National Mission on Sustainable Agriculture (NMSA) under National Action Plan on Climate change (NAPCC).

4.1 Centre for Agri-Innovation (CAI)

The Centre for Agri-Innovation of the Academy functions through the technology business incubators from ICAR-NAIF (National Agriculture Innovation Fund), Department of Science & Technology (DST), National Bank for Agriculture and Rural Development (NABARD), Government of India (GoI) supported Business incubator a-IDEA (Association for Innovation Development of Entrepreneurship in Agriculture) which also supports activities of BioNest, an Agri- Bio-incubator for Agri-Biotechnology Startups. BioNest has received financial support from BIRAC, DBT, GoI, which runs in partnership with Agri-Biotech Foundation (ABF), Rajendranagar, Hyderabad.

a-IDEA fosters innovation and entrepreneurship

in agriculture in India. Since 2014, it has been supporting startups across 14 domains of agriculture and allied sectors such as dairy, fisheries and animal husbandry. It has helped entrepreneurs to sensitize, ideate, incubate and accelerate their innovative early-stage startups that are scalable to become competitive food and agri-business ventures through capacity building, mentoring, networking and advisory support. These startups supported by a-IDEA are in turn disrupting the value chains of agriculture across farm gate to food plate. These innovations are touching upon the lives of farmers, as well as rural and urban consumers.

The a-IDEA is presently focusing on different sub-sectors of agriculture innovations across the value chain in agriculture:



Fig. 4.1: Sector wise Focus by Technology Business Incubator, a-IDEA

4.1.1 Incubatee Startups during 2020

Sl. No.	Name of Incubatee	Year of Induction	Industry Sector	Company Profile
1.	Gramworkx	2020	Soil, Water & Weather tech	IoT and AI-enabled smart farm resource management tool, which helps the farmers guide, optimizes, and monitor utilization of water. https://gramworkx.com/#/
2.	MandiGate	2020	Supply Chain	It is a commodity marketplace for agricultural produce where farmers/bulk sellers/licensee agents can list their products and buyers (industrialists, wholesale dealers, etc.) can review and place their bulk orders through an online portal or mobile app.
3.	Temperatech	2020	Post-Harvest	Developed completely of grid solar-powered cold storage solution to increase farmer incomes
4.	Greenpod Labs	2020	Post-Harvest	Active packing material by using nanotechnology to prevent post-harvest losses of produce

4.1.2 Graduated Startups in 2020

Sl. No.	Name of the Startup	Year of Induction	Industry Sector	Company Profile
1.	Oorvi Pvt. Ltd	April 2019	Waste Management	Floral Waste Management to Value Added Products http://oorvi.org/
2.	ChemLife Innovations Pvt. Ltd	April 2019	Waste Management	Conversion of Paddy Stubble into Value Added Products http://www.chemlifeinnovations.com/
3.	JSAmey Biotech Pvt Ltd	September 2018	Waste Management	Food Waste Management from the Hospitality Industry
4.	Urbankisaan Hydrotech Pvt Ltd	September 2018	Urban/Vertical Farming	Vertical Hydroponic Urban Farming
5.	Fishy Farmer Pvt. Ltd.	April 2019	Aquaponics	Commercial Production of Exotic Greens and Vegetables Using Aquaponics
6.	AgVerse Technology Pvt. Ltd	April 2019	Animal Husbandry	Artificial Insemination Gun for Early & Easy Heat Detection & Conceiving
7.	Aadhishakti Pvt. Ltd	April 2019	Farm Mechanization	ICT based Custom Hiring Centres for Farm Machineries & Equipment's to Farmers
8.	Pekhansa Pvt. Ltd	April 2019	Innovative Food Technology	Pani Puri Vending Machine
9.	Transity Digital Solutions, Pvt. Ltd	January 2018	Supply Chain Technology	SaaS Aggregator Platform for Transportation of Agriculture Produce http://transity.co/
10.	Skycraft Aerospace Pvt. Ltd.	January 2018	Precision/Smart Agricultures	Addresses Commercial and Enterprise Drone Needs http://www.skycrafts.com/



Sl. No.	Name of the Startup	Year of Induction	Industry Sector	Company Profile
	Robic Rufarm India Pvt. Ltd.	July 2018	ICT in Agriculture	Real-Time Pond Monitoring and Advisory Services for Aquaculture https://robicrufarm.com/
	Bloom Foods India	August 2018	Innovative Food Technology	Ready to Eat (RTE) Foods with a Special Focus on Traditional Non-Veg Pickles. https://bloomfoods.co.in/

4.1.3 Agri Udaan: Food and Agribusiness Accelerator

Agri Udaan is a focused program targeted towards startups that are trying to scale up their innovative startups. There are very few accelerators in India focused on Food and Agri-Business. With this background, a-IDEA, a technology business incubator at NAARM when it started its operations in 2015, realized the importance of such programs to support Agritech-startups that are facing challenges in scaling up. In this endeavor, a-IDEA launched the Agri Udaan, the Food & Agribusiness Accelerator. Agri Udaan is the flagship accelerator program of a-IDEA, NAARM. It brings on board a range of diverse partners to create an inclusive and collaborative ecosystem focusing on catalyzing scale-up stage Food and Agribusiness startups through rigorous mentoring, industry networking and investor pitching. It is a unique platform for scale-up stage innovators, entrepreneurs and startups in Food and Agribusiness sectors to showcase their products/services and to receive valuable inputs from Mentors, Incubators, R&D Institutions, Agribusiness Industry and Investors.

a-IDEA, NAARM launched India's first Food & Agribusiness Accelerator in partnership with CIIE, IIM-A in 2015. In this program, out of ~192 startups which have applied, 20 startups were mentored and eight startups were supported. Out of these eight startups, three startups got funding to the tune of ₹3 Cr. Further in 2017, the

2nd series of Food and Agribusiness Accelerator "Agri Udaan 2.0" was launched with the support of Department of Science and Technology in association with Caspian, Yes Bank, Marico, NeML. Nearly 579 startups applied, out of which 40 startups were mentored, 10 startups cohort was shortlisted and supported further, out of which 4 startups got funding to the tune of 6.25 Cr INR. In 2019, a-IDEA, NAARM with the support of DST launched its **Agri Udaan 3.0** in association with NABARD, Caspian, AgriInnovate, Omnivore, and other prominent partners.

Agri Udaan 3.0 launched in Hyderabad on 31st August 2019, under which roadshows were organised in Bengaluru, Pune, Patna, Lucknow, Gurugram, Bhopal, and Guwahati. Providing a truly Pan-India presence, Agri Udaan seeks to bring out innovations from across the country while also aiming to provide support to the startups across the country. The Top-10 Startup cohort of Agri Udaan 3.0, selected from a pool of 415 startups, pitched in front of 35+ investors across the spectrum of Food and Agribusiness. The aim of these events is to mobilize investors and have the startups pitch in front of them with the hope of facilitating investment raised for the startups. The Demo Days also serve as the first point of contact for the startups to liaison with investors across Food and Agribusiness industry that can result in investment over time.

4.1.4 Agri Udaan Cohort Startups of Agri Udaan 3.0

Sl. No.	Name of the Startup	Year of Induction	Industry Sector	Company Profile
1.	Hydrogreens	Nov 2019	Animal Husbandry	Micro Climate Control System for Growing Fodder https://hydrogreenfodder.com/
2.	Poultrymon	Nov 2019	Animal Husbandry	Smart Monitoring System for Poultry Farms and Hatcheries http://www.mlitsol.com/
3.	Amvicube	Nov 2019	Agritech	Non-Destructive Testing Instrument for Paddy http://www.amvicube.com/
4.	SNRAS Systems	Nov 2019	Fisheries	Recirculatory Aquaculture System https://www.snrasystems.com/
5.	Auraphyll	Nov 2019	Agri Biotech	Bio-stimulants and Elicitors https://www.auraphyll.com/
6.	Whrrl	Nov 2019	Agri Fintech	Blockchain Backed Lending Ecosystem for Warehouse Finance Receipts http://www.whr.loans/
7.	Bariflo Labs	Nov 2019	Fisheries	Hypolimnetic Aeration Device for deep Aeration, Feeding, and Screening for Shrimps https://bariflolabs.com/
8.	Han Agrocure	Nov 2019	Supply Chain Technology	Medicinal Mushroom Cultivation https://hanagrocure.com/
9.	Farmer Fresh Zone	Nov 2019	Supply Chain Technology	Online Marketplace for Organic, Safe to Eat Vegetables and Fruits https://www.farmersfz.com/
10.	Krishi Tantra	Nov 2019	Soil, Water and Weather Technology	Soil, Water & Weather Technology Testing https://krishitantra.com/

COHORT STARTUPS OF AGRI UDAAN 3.0



Fig. 4.2

4.1.5 Events Conducted by NAARM Business Incubator a-IDEA during 2020

The technology business incubator at NAARM,

a-IDEA conducted various events in different parts of the country so as to reach to as many agri entrepreneurs as possible.



Events of a-IDEA during 2020

S. No.	Name of the Event	Location	Date
1.	16 th Call for BIRAC BIG Grant	Hyderabad	1 st January 2020
2.	Call for Incubation	Hyderabad	4 th January 2020
3.	Startup-FPO Immersion Program, MABIF	Madurai, Tamil Nadu	10 th January 2020
4.	Startup-FPO Immersion Program, Reliance Foundation	Radhanpur, Gujarat	16 th January 2020
5.	Pitch Perfect	a-IDEA, NAARM	21 st January 2020
6.	Call for Incubation	Hyderabad	6 th February 2020
7.	Pitch Perfect	a-IDEA, NAARM	23 rd February 2020
8.	Startup-FPO Immersion Program, MABIF, BASIX	Durgapur, West Bengal	29 th February 2020
9.	Startup-FPO Immersion Program, NABARD	Kurnool, Andhra Pradesh	3 rd March 2020
10.	Agri Udaan 3.0, 1st Demo Day	Mumbai	5 th March 2020
11.	Agri Udaan 3.0, 2nd Demo Day	Bengaluru	7 th March 2020
12.	Call for Incubation	Hyderabad	9 th March 2020
13.	Call for NIDHI PRAYAS Grant	Hyderabad	29 th June 2020
14.	17 th Call for BIRAC BIG Grant	Hyderabad	1 st August 2020
15.	Inauguration Ceremony of a-IDEA's Scaling Up Operation by NABARD Chairman	Hyderabad	27 th August 2020
16.	Call for Incubation	Hyderabad	23 rd November 2020

4.1.6 Turnover and Employment by Startups of a-IDEA

An analysis of all the startups reveals that the startups have the potential to generate the overall value in the economy which would add

to the national income. Since 2014, there has been a continuous increase in the turnover of the company which shows that these companies are contributing to the sector through their well laid out plan and consistent efforts.



These startups are also responsible to generate employment for the educated youth in agriculture directly and also indirectly. An estimate of employment generated by the startups of NAARM

business incubator, a-IDEA was 877 in 2019-20 which is continuously increasing with increase of business of these startups and entry of new startups.

4.1.7 Webinars Conducted by a-IDEA during 2020

Sl. No.	Title of Webinar	Date of event
1	Sustainable Income for farmers in post COVID scenario using financial tools	18 th April 2020
2	AMA Session with Angel Investor Mr. Naga Prakasam	23 rd April 2020
3	Aquaculture sector during and Post COVID times	20 th May 2020
4	Tackling Dairy sector amid COVID times	23 rd May 2020
5	Legal Issues during and after COVID-19 in Startups sector	26 th May 2020
6	Approaches of Public funded research organizations in Agri-technology generation and its transfer in New Normal situation	28 th May 2020
7	Scope of Medicinal & Aromatic crops Sector in Present & Future Scenario	8 th June 2020
8	Startups in Animal Husbandry: present and future	10 th June 2020
9	Understanding BIRAC BIG Scheme Guidelines & Eligibility in Agri and Agri-Biotech Sector	13 th August 2020
10	BIRAC BIG Grantee talk	20 th August 2020
11	BIRAC BIG for Agritech & Allied Sector Startups: Sky is the limit	24 th August 2020
12	BIRAC BIG AMA Session	31 st August 2020
13	Ask Me Anything Live Webinar Session with a-IDEA Team-1	1 st September 2020
14	Ask Me Anything Live Webinar Session with a-IDEA Team-2	3 rd September 2020
15	BIG Awareness Session	10 th September 2020

Glimpses of the Events at a-IDEA



1st Startup-FPO Immersion Program, MABIF



2nd Startup-FPO Immersion Program, Reliance Foundation, Gujarat



3rd Startup-FPO Immersion Program BASIX, West



4th Startup-FPO Immersion Program NABARD,



Agri Udaan 3.0 Demo Day 1, Bengaluru



Agri Udaan 3.0 Demo Day 2, Mumbai



Translational Workshop at TNAU



BIRAC's 8th Foundation day



BIO ASIA 2020

4.1.8 Scaling Up Boost of a-IDEA from NABARD

a-IDEA, received financial support from NABARD for the next five years to scale up the operations. Shri Govinda Rajulu Chintala, Chairman of the NABARD has operationalized the programme on 27th August 2020. A-IDEA has served unleashing innovation and entrepreneurship in agriculture, leading it to become the country's foremost Agritech incubator with the support of ICAR, DST and BIRAC since 2014. Now NABARD has joined

the group of supporters for scaling up of a-IDEA to continue its journey of excellence to achieve dynamic and sustainable agri-entrepreneurial development. This would lead to long term benefits for the farming ecosystem in general. Shri G R Chintala, Chairman of NABARD, initiated the programme in the presence of Shri Y K Rao, CGM TSRO, Shri Sudhir Kumar Jannawar, CGM, APRO, Dr. Ch. Srinivasa Rao, Director ICAR-NAARM and President a-IDEA, and all the prominent dignitaries.



Fig. 4.3: Inauguration Ceremony of a-IDEA's Scaling Up Operation by NABARD Chairman



Figure 4.4: 1st PMRC Meeting of the Joint Programme of NABARD and a-IDEA, NAARM

Glimpses of the Programmes Promotions through Webinars



a-IDEA, NAARM and Robin Gufarm are organizing a webinar session on

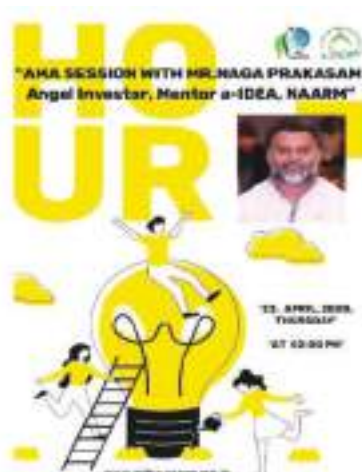
AQUACULTURE SECTOR PRE AND POST COVID TIMES

To discuss and address the existing challenges in Aquaculture sector during covid times

SPEAKERS

- Mr. Rishi Mehta, Founder, Robin Gufarm
- Dr. Neelkanth Thakur, ICAR, NARS, CITA
- Mr. Harishankar Rao, Senior Scientist, Chennai Production
- Mr. Anil Kumar, Founder, AGRISYS
- Dr. G. Venkatesh, ICAR, NARS, NARS (Mumbai)

30TH MAY 2020 | 4:00 PM IST
www.aidea.naarm.org.in



AMA SESSION WITH MR. NAGA PRAKASH

Angel Investor, Mentor a-IDEA, NAARM

13. APRIL 2020, THURSDAY
AT 6:00 PM

www.aidea.naarm.org.in



LEGAL ADVISORY TO STARTUPS DURING AND POST COVID TIMES

28TH MAY 2020 | 02:00 PM
www.aidea.naarm.org.in



TACKLING DAIRY SECTOR AMID COVID TIMES

In conversation with


Kuldip Sharma
 Consultant, Think Dairy


Dr. Rongit Kumar
 Head, Agri Business Dev, ICAR-NAARM


Mr. Babbar
 Founder, Sahitya


Dr. AK Singh
 In Scientist, ICAR-NRI

23RD MAY 2020 | 4:00 PM IST
www.aidea.naarm.org.in



APPROACHES OF PUBLIC FUNDED RESEARCH ORGANIZATIONS IN AGRI-TECHNOLOGY GENERATION AND ITS TRANSFER IN NEW NORMAL SITUATION

In conversation with


Dr. Seetha Mysore
 IIS, Bangalore
 Speaker


Dr. Sangeet Saxena
 IIS, Bangalore
 Speaker


Dr. D. Srinivas Rao
 IIS, Bangalore
 Speaker


Mr. Anand Kulkarni
 IIS, Bangalore
 Speaker


Dr. Anurag Kumar
 IIS, Bangalore
 Speaker

28TH MAY 2020 | 4:00 PM IST
www.aidea.naarm.org.in



Scope of Medicinal & Aromatic Plants in Present & Future Scenario

In conversation with


Dr. Ajit Katre
 Former Acting Director, CIMAP


Dr. Anil Dange
 Scientist, AWR PCA Lab


Ms. Manal Sharma
 Founder, Mytic Auro


Mr. Arundhan Prasad
 Founder, Bioethanol

5th JUNE 2020 | 4:00 PM IST
www.aidea.naarm.org.in



Startups in Animal Husbandry: Present and Future

In conversation with


Dr. Mahesh Chander
 ICAR, NRI


Mr. Vasant Kamath
 HydroGrowth


Mr. Priyank Saxena
 Agripreneur

10th JUNE 2020 | 4:00 PM IST
www.aidea.naarm.org.in

Glimpses of the Programmes Promotions through Webinars



α-IDEA, Technology Business Incubator of NAARM
Presents
NIDHI PRAYAS GRANT

Who Can Apply?

- Innovators working on innovative idea/POC/Prototype in Agri and Allied sectors

Focus Areas:

- ICT & IoT in Agriculture
- Precision/Smart Agriculture
- Innovative Food Technology
- Supply Chain Technology
- Post-Harvest Technology
- Soil, Water & Weather Tech.
- Urban/ Vertical Farming
- Agri Biotech
- Agri-Biotech
- Farm Mechanization
- Sustainable Inputs
- Animal Husbandry
- Fisheries
- Farm Fresh Retail

avail grant-in-aid upto ₹10 Lakh

Applications are open till 18th July 2020

Apply at:
<https://www.tbs.com/alpha-idea/nidhi-prayas/apply>
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α-IDEA, Technology Business Incubator of NAARM
BIOTECHNOLOGY IGNITION GRANT
Grant-in-aid up to **50 Lakhs INR**

WHO CAN APPLY?

- Entrepreneurs or startups working on innovative idea/POC/Prototype
- A registered company or an individual
- Researchers or Academicians

FOCUS AREAS:

- Agri-Biotech, Industrial Biotech, Food & Nutrition
- Agrotrans, Agri Waste Management and Allied sectors

17th BIG call open from 1st August to 15th September 2020

Contact us for further details:
Email: harishharish@naarm.in or events.alpha@naarm.in
Phone: +91 90100 30195/040-24581421/326
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α-IDEA, Technology Business Incubator of NAARM
BIOTECHNOLOGY IGNITION GRANT
grant-in-aid up to **50 Lakhs INR** **BIG** FOR INNOVATORS

Who Can Apply?

- Entrepreneurs or startups working on innovative idea/POC/Prototype
- A registered company or an individual
- Researchers or Academicians

Focus Areas:
Agri-Biotech, Industrial Biotech, Food & Nutrition, Agrotrans, Agri Waste Management and Allied sectors

17th Call for BIG Proposals 1st Aug - 15th Sept. 2020

Contact:
✉ events.alpha@naarm.in or harishharish@naarm.in
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Pitch Perfect
26th November 2020
If you are an entrepreneur or startup with an Innovative Idea/POC/ Prototype

Inviting Applications for INCUBATION PROGRAM

Apply on or before 23rd Nov, 2020

Apply Even: <http://www.alpha-naarm.org.in/naarm2020>

Why α-IDEA?

- Access to L&I & OT&A
- Provide linkage for free testing product & business
- Hands-on mentoring & networking
- Investment Advisor & Grant Support

Ms. Harish Harish - Executive Director & Co-Founder, α-IDEA, Technology Business Incubator of NAARM
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4.1.9 Participation of a-IDEA, NAARM in Key Activities

Sl. No.	Name of the Event	Location	Date
1.	India Fund Fest	Chandigarh	7 th February, 2020
2.	National Summit on Women and Education Empowerment - Indian Women Excellence and Leadership Award (IWELA) 3.0	New Delhi	17 th January, 2020
3.	Global Startup Competition	Bangalore	31 st January, 2020
4.	Bio Asia 2020	HICC, Hyderabad	17-19 February, 2020
5.	Translational Workshop	TNAU UIC (University Innovation Cluster) Coimbatore	15 th March, 2020
6.	BIRAC's 8 th Foundation day	Online	20 th March, 2020
7.	Sustainable income for farmers in post COVID-19 era using financing tools	Online	18 th April, 2020
8.	Faculty Development Program	Online	14-18 September, 2020
9.	BIG Conclave	Online	

4.1.10 Accolades for a-IDEA, Incubatees

- ❖ Whrrl has been shortlisted as one of "India's most innovative Startup" in DCB bank Innovation Carnival
- ❖ Farmer Fresh Zone received Best Innovator award from Destination Kerala
- ❖ Agnext represented in the panel of National Conclave on Artificial Intelligence by CII.
- ❖ Krishi-Tantra raised an investment from Omnivore and Nab ventures
- ❖ Krishi-Tantra, Whrrl, and Jeevabhumi (Tracex) were selected in AIP programme of Ag Hub of PJTSAU
- ❖ Agverse and City greens raised grant of Rs 25 lakhs each from RAB/ Raftar scheme of DOA&FW
- ❖ Holywaste has become official floral waste management partner with Department of Municipal and district administration of Govt. of Telangana
- ❖ Green Pod Labs has been shortlisted in Rockstart Accelerator
- ❖ BharatRohan story has been featured in Business Time as prominent startup working with farmer

- ❖ Temparatech has won a special prize at EDF Pulse India Awards
- ❖ Vihari Knukullu, Founder of Urban Kisan has been recognised as one among the 50 Next Generation leaders to watch by Project Management Institute
- ❖ The Better India has featured Hydrogreens in its article

4.2 Centre for Open and Lifelong Learning in Agriculture (COLLAge)

The Academy has been involved in strengthening the digital and online learning on many fronts. As a part of Technology enhanced Learning, the Academy has established the Centre of Lifelong Learning in Agricultural Education (COLLAge) which is involved in handholding, production and organizing online learning programmes.

Activities Undertaken by COLLAge during 2000

4.2.1 Organizing MOOCs

Considering the Covid factor and the need of online learning, the centre has offered two MOOCs during the lockdown period as below:

- a) MOOC on Psychology of Learning during 1-15 May 2020. It received 1652 registrations and 930 certifications. The intended audience are the teaching faculty of various educational organizations. The program also helped faculty utilize their time effectively during lockdown period to hone their teaching skills using proven psychology principles.
- b) MOOC on Designing E-Learning Content during 1-31 July 2020. It received a record number of 2773 registrations and 1342 certifications. The program targeted the teaching faculty on understanding the underlying principles of designing e-learning content for online learning.

Besides catering to the knowledge needs of vast clientele, the centre was successful in making MOOC as a sustainable activity by way of revenue generation through nominal certification charges. Approximately Rs 15.5 Lakhs was generated as revenue from two MOOCs.

4.2.2 Launch of Digital Courses

In one of its first-of-its-kind initiatives, the Academy partnered with ICAR-NDRI in developing digital content for offering MOOC on Commercial Dairy Farming with 23 modules, which was formally launched on 14th August 2020 through

virtual launch programme cohosted by NDRI and NAARM under NAHEP.

4.2.3 Digital Infrastructure Development

The existing infrastructure for COLLAGe has been strengthened with better illumination scheme for better digital quality of content.

4.2.4 Content Development in Consultancy Mode

Content development for NDRI and PJTSAU were done. A total of 60 modules for NDRI and 15 for PJTSAU were finalized. Two more modules are in processing.

4.2.5 Support of Distance Education Programme

About 40 modules of the Academy's distance education programme-PG Diploma in Education Technology Management (PGDETM) were refined for better quality and access. The same were published through e-learning portal (elearning.naarm.org.in). Besides this, the centre also offered to support to conduct online examinations for the students of the distance education programme. The COLLAGe proved highly effective for the students during the Covid Pandemic and ensured a smooth transition and conduct of the academic programme.



Fig. 4.5: Digital Content Development on Plagiarism for MOOC Content for PJTSAU

5.1 Training Programmes attended by ICAR-NAARM Employees

Sl. No.	Name	Programme	Duration (in days)	Organized by
1.	Alok Kumar	FDP on 'Principles of Management'	84 days	IIT Kharagpur & NPTEL (Online)
2.	Alok Kumar	FDP on 'The Essence of Leadership'	42 days	IIM, Bangalore & NPTEL (Online)
3.	BS Yashavanth	e-Faculty Training Programme on Machine Learning Technologies for Social Media Analytics	5 days	IIT, Guwahati (Online)
4.	D Thammi Raju	EDtech I Indian Education Congress	1 day	Franchise India Holding Ltd (Online)
5.	D Thammi Raju	Edu Tech Summit 2020	1 day	Franchise India (Online)
6.	D Thammi Raju	Brainstorming Session on Transforming Higher Agricultural Education in India	1 day	NAAS (Online)
7.	G R Ramakrishna Murthy	Evidence Based Teaching and Learning Strategies	3 days	IISER, Bhopal
8.	G R Ramakrishna Murthy	Training Teachers for Artificial Intelligence in Schools	4 days	Machine Intelligence Research Labs, USA and Bombay Teachers' Training College, Mumbai
9.	KH Rao	Case teaching and Case writing Workshop	8 days	AIMA, New Delhi (On-line)
10.	Manju Gerard	UNDP Awareness cum Sensitization Workshop on Implementation of Access and Benefit Sharing Regulations in Agriculture Research	3 days	ICAR-NAARM (On-line)
11.	Manju Gerard	MDP on "Creative Thinking, Problem Solving and Decision Making"	3 days	IIM Kozhikode (Online)
12.	Manju Gerard	Certificate Course "Leadership & Organization Development for Women Scientists/ Technologists"	5 days	Centre of Organizational Development, Hyderabad (Online)
13.	Manju Gerard	Certificate Course on IP Portfolio Management & Commercialization"	10 days	STEM, Hyderabad (Online)

Sl. No.	Name	Programme	Duration (in days)	Organized by
14.	P Supriya	International Training Program on "Agriculture 4.0 : Precision and Automated Ag Technologies"	7 days	Mahatma Phule Krishi Vidyapeeth, Rahuri (Online)
15.	P Supriya	Certificate Course on "Creativity & Innovation Management in Research"	6 days	Engineering Staff College of India, Hyderabad (Online)
16.	P Supriya	International Faculty Development Program on "Bioinformatics, Big Data and Drug Discovery – Prospects & Challenges"	14 days	G.B. Pant University of Agriculture and Technology, Pantnagar (Online)
17.	Ranjit Kumar	Executive Program on Policy Analysis and Program Evaluation	5 days	TISS, Mumbai
18.	Ranjit Kumar	Certificate Course on Successful Negotiation: Essential Strategies & Skills	7 days	Coursera by University of Michigan (Online)
19.	S Ravichandran	Quantitative Methods in Social Sciences	21 days	ICAR-National Institute of Agricultural Economics and Policy and Research, New Delhi (Online)
20.	S Ravichandran	Webinar on Agriculture During COVID-19 Economic Package and Reforms	1 day	ICAR-National Institute of Agricultural Economics and Policy and Research, New Delhi (Online)
21.	S Ravichandran	Webinar on Decision Tree	1 day	M/S Statcraft Ltd. (Online)
22.	S Ravichandran	Webinar on Designing a Curriculum for Online and Hybrid Classes	1 day	M/S Harvard Business Publishing (Online)
23.	Sanjiv Kumar	Certificate Course on Desktop I: Fundamentals Desktop II: Intermediate	15 days	Tableau Software (Online)
24.	Sanjiv Kumar	Certificate Course on The Data Scientist's Tool box	7 days	Johns Hopkins University through Coursera (Online)
25.	Umesh Hudedamani	Certificate Course on Immersion Program for Managers of New Incubation Centers	2 days	Venture Center, Pune (Online)
26.	Umesh Hudedamani	Certificate Course on Professional Patent Landscaping Workshop	1 day	Turnip Innovations, Mumbai (Online)
27.	Umesh Hudedamani	Certificate Course on IP Rights in Agricultural Research and Education in India	14 days	NAHEP and IP&TM, New Delhi (Online)



5.2 Capacity Building Programs Organized by ICAR-NAARM

Sl. No.	Title of the Programme	Duration	Programme Coordinators	No. of Participants	Programme Type
1.	Training Programme on 'Developing Winning Research Proposals (DWRP)' under Institutional Development Plan (NAHEP, ICAR, New Delhi) at MPUAT, Udaipur	3-5 Jan, 2020	SK Soam, P Krishnan	25	Off-Campus Programme
2.	110 th Foundation Course for Agricultural Research Service	7 Jan-4 Apr, 2020	RVS Rao, N Srinivasa Rao, T Srinivas	135	FOCARS
3.	Workshop under Extramural Project on 'Competency Analysis among faculty of AUs in TEL'	11 Jan, 2020	S Senthil Vinayagam, GRK Murthy	22	Workshop
4.	Brainstorming Workshop on Networking of Agricultural Universities Alumni (NAHEP Component 2A)	22 Jan, 2020	VV Sumanth Kumar, D Thammiraju, M Balakrishnan	35	Workshop
5.	Awareness cum Sensitization Workshop Implementation of Access and Benefit Sharing Regulations in Agriculture Research:	22-23 Jan, 2020	K Kareemulla, B Ganesh Kumar, P Krishnan	21	Workshop
6.	Technology Enhanced Learning (under TELAgE Project)	25-31 Jan, 2020	VV Sumanth Kumar, GRK Murthy	22	Need Based Programme
7.	Consultative Workshop on Academia Industry Government Linkages for Quality Agricultural Education (Under NAHEP) in collaboration with Assam Agricultural University (AAU), Khanapara, Guwahati	27-28 Jan, 2020	Sanjiv Kumar	63	Workshop
8.	Capacity Building Programme for CJSC Members of ICAR Hqs/ Institutes	27-31 Jan, 2020	C Muralidharan, P Ramesh, ZH Khilji	89	Need Based Programme
9.	Consultative Workshop on 'Academia Industry Government Linkages for Quality Agricultural Education' under NAHEP in collaboration with Punjab Agricultural University (PAU), Ludhiana	28-29 Jan, 2020	S Senthil Vinayagam	70	Workshop

Sl. No.	Title of the Programme	Duration	Programme Coordinators	No. of Participants	Programme Type
10.	Training Programme on 'Developing Winning Research Project for Faculty Members of Kamdhenu University, Amreli	29-31 Jan, 2020	SK Soam, P Krishnan	23	Need Based Programme
11.	Training Programme on 'Survey Data Analysis using R'	3-7 Feb, 2020	N Sivaramane, A Dhandapani	10	Need Based Programme
12.	Training Programme on 'Administrative & Finance Management For Section Officers AAO/AFAO/JAO/ Assistants of ICAR HQs / Institutes' at ICAR-CIFT, Cochin	5-11 Feb, 2020	PG Kohad, N Vijayalakshmi	36	Off Campus Programme
13.	Regional Workshop on 'Career Development Centre' in collaboration with KAU, Thrissur (Under NAHEP)	7 Feb, 2020	Alok Kumar, N Srinivasa Rao	50	Workshop
14.	Faculty Development Training on "Enhancing Efficiency and Effectiveness to Develop Entrepreneurship Ecosystem" at PDKV, Akola	11-20 Feb, 2020	KH Rao, Ranjit Kumar, K Srinivas	30	Off Campus Programme
15.	Regional Workshop on Career Development Centre in collaboration with SKNAU, Jobner (Under NAHEP)	12 Feb, 2020	Alok Kumar, N Srinivasa Rao	50	Workshop
16.	Regional Workshop on Career Development Centre in collaboration with IGKV, Raipur (Under NAHEP)	14 Feb, 2020	Alok Kumar, N Srinivasa Rao	50	Workshop
17.	Summer School on "ICT Applications in Agricultural Education & Extension"	19 Feb- 10 Mar, 2020	Surya Rathore, M Balakrishnan, Sanjiv Kumar	24	Summer School
18.	Training Programme on KOHA for Library Staff of ICAR	20-25 Feb, 2020	N Srinivasa Rao, VV Sumanth Kumar	19	Need Based Programme
19.	National Conference on 'GIS and Remote Sensing Applications in Agriculture'	21-23 Feb, 2020	PD Sreekanth, M Balakrishnan	89	Conference
20.	Enhancing Efficiency and Behavioural Skills of Stenographers Grade-III, PA, PS, PPS and Sr. PPS of ICAR at NAARM	24-29 Feb, 2020	PG Kohad, M Dinesh	35	Need Based Programme



Sl. No.	Title of the Programme	Duration	Programme Coordinators	No. of Participants	Programme Type
21.	Workshop on “Prospects of Agricultural Education: Awareness among Higher secondary School Administrators” (Under NAHEP)	24 Feb, 2020	BS Yashavanth	13	Workshop
22.	E-resources, E- learning and IoT for Students and Staff of UAS Raichur	25 Feb, 2020	GRK Murthy	60	Off Campus Programme
23.	Training Programme on ‘Geospatial Analysis using QGIS & R’	27 Feb- 3 Mar, 2020	PD Sreekanth, BS Yashavanth	15	Need Based Programme
24.	Training on ‘Entrepreneurial Skill Development for SVVU students	6-12 Mar, 2020	Ranjit Kumar, Sanjiv Kumar	27	Need Based Programme
25.	Exposure-cum-Educational Visit for Skilled Support Staff of ICAR-NAARM to Krishi Vigyan Kendras (KVK's)	13-16 Mar, 2020	Laxman M Ahire, M Shekhar Reddy	23	Need Based Programme
26.	MOOC on “Psychology of Learning”	1-15 May, 2020	P Ramesh, GRK Murthy, S Senthil Vinayagam	1658	MOOC
27.	MDP on Implementation of Access and Benefit Sharing Regulations in Agriculture Research: Awareness cum Sensitization Workshop	8-9 June, 2020	K Kareemulla, B Ganesh Kumar, P Krishnan	56	Workshop (Online Mode)
28.	Training course on “Contemporary Education Technologies for Agricultural Education” for the faculty of ANGRAU, Guntur	8-12 June, 2020	VV Sumanth Kumar, GRK Murthy, S Senthil Vinayagam	20	Need Based Programme (Online Mode)
29.	Massive Open Online Course (MOOC) on Designing E Learning Content	1-31 July, 2020	GRK Murthy, S Senthil Vinayagam	2773	MOOC
30.	Training Programme on “Teaching Competency Enhancement Programme for Faculty of SKLTSHU(Sponsored by SKLTSHU, Mojerla, Telangana State)	7 July, 2020	D Thammi Raju, VV Sumanth Kumar, S Senthil Vinayagam	15	Need Based Programme (Online Mode)
31.	Training Programme on “Stress Management”	7-10 July, 2020	P Ramesh, RVS Rao	22	Need Based Programme (Online Mode)

Sl. No.	Title of the Programme	Duration	Programme Coordinators	No. of Participants	Programme Type
32.	Implementation of Access and Benefit Sharing Regulations in Agriculture Research: Awareness cum Sensitization Workshop	7-10 July, 2020	K Kareemulla, B Ganesh Kumar, P Krishnan	78	Workshop (Online Mode)
33.	Master Trainers on Access and Benefit Sharing Regulations in India and the Nagoya Protocol	15-17 July, 2020	K Kareemulla, B Ganesh Kumar, P Krishnan	28	Workshop (Online Mode)
34.	Workshop for Vigilance Officers of ICAR Institutes	5-7 Aug, 2020	S Senthil Vinayagam	33	Workshop (Online Mode)
35.	Training Programme on "Analysis of Experimental Data Using R"	5-11 Aug, 2020	A Dhandapani, S Ravichandran	161	Need Based Programme (Online Mode)
36.	Orientation Workshop and Training Program for ABI units	17-19 Aug, 2020	K Srinivas, Manju gerard	58	Workshop (Online Mode)
37.	Access and Benefit Sharing (ABS) in the Context of Agriculture and Allied Sectors for the State Biodiversity Boards	21 Aug, 2020	K Kareemulla, B Ganesh Kumar, P Krishnan	108	Workshop (Online Mode)
38.	Training Programme on "E-Resources, E-Learning and IoT" for Staff of UAS, Raichur	24 Aug, 2020	GRK Murthy, S Senthil Vinayagam, M Balakrishnan	30	Need Based Programme (Online Mode)
39.	Training Programme on "Design Thinking in Research Project Formulation and Implementation"	25-29 Aug, 2020	SK Soam, P Krishnan	46	Need Based Programme (Online Mode)
40.	MDP on "Intellectual Property Valuation and Technology Management"	1-5 Sep, 2020	Manju Gerard, K Srinivas	31	Need Based Programme (Online Mode)
41.	Workshop on Education Management and Academic Leadership (sponsored under NAHEP)	3-17 Sep, 2020	D Thammi Raju, S Senthil Vinayagam	45	Workshop (Online Mode)
42.	Innovative Practices in Extension Research and Evaluation (21 days)	8-28 Sep, 2020	P Venkatesan, BS Sontakki, N Sivaramane, Surya Rathore	59	Need Based Programme (Online Mode)
43.	Training program on "PRACom-2020: Participatory Rural Appraisal (PRA) and Communication Strategies" (IGNFA) Dehradun	9-11 Sep, 2020	SK Soam, BS Sontakki, P Krishnan	71	Need Based Programme (Online Mode)



Sl. No.	Title of the Programme	Duration	Programme Coordinators	No. of Participants	Programme Type
44.	Collaborative Capability Building Programme on New Entrants Assimilation Training for Sales and Marketing Managers of Coromandal International Limited.	11-26 Sep, 2020	KH Rao, Ranjit Kumar, N Sivaramane	44	Need Based Programme (Online Mode)
45.	Training on 'Advanced Bioinformatics Tools and its Applications in Agriculture'	14-19 Sep, 2020	M Balakrishnan, SK Soam	73	Need Based Programme (Online Mode)
46.	Training Programme on 'Communication and Management Skills for Extension Professionals'	1-21 Oct, 2020	BS Sontakki, P Venkatesan, Surya Rathore	53	Need Based Programme (Online Mode)
47.	Training Programme on 'Sustainable Management of Small Scale Seed Enterprises'	5-9 Oct, 2020	T Srinivas, K Kareemulla	13	Need Based Programme (Online Mode)
48.	Training Programme on "E-Resources, E-Learning and IoT" for Students of UAS, Raichur	5-9 Oct, 2020	GRK Murthy, S Senthil Vinayagam, M Balakrishnan	35	Need Based Programme (Online Mode)
49.	111 th FOCARS	5 Oct, 2020-2 Jan, 2021	M Balakrishnan, VV Sumanth Kumar, P Krishnan	37	FOCARS
50.	MDP on 'Priority Setting, Monitoring and Evaluation of Agricultural Research Projects'	12-17 Oct, 2020	P Krishnan, T Srinivas	35	Need Based Programme (Online Mode)
51.	Training Programme on "Design of Experiments" for Officials of UPL Advanta Limited	2-7 Nov-2020	S Ravichandran, A Dhandapani, N Sivaramane	26	Need Based Programme (Online Mode)
52.	Training Programme on "Recent Innovations in Educational Technology"	5-10 Nov, 2020	S Senthil Vinayagam, GRK Murthy	31	Need Based Programme (Online Mode)
53.	Training Programme on "Analysis of Experimental Data using SAS"	9-17 Nov, 2020	S Ravichandran, A Dhandapani, BS Yashavanth	110	Need Based Programme (Online Mode)
54.	Collaborative Capability Building Programme on 'MORE (Marketing Officers' Resurgence & Excellence)' for Coromandal International Limited	16-20 Nov, 2020	KH Rao, Ranjit Kumar, N Sivaramane	50	Need Based Programme (Online Mode)

Sl. No.	Title of the Programme	Duration	Programme Coordinators	No. of Participants	Programme Type
55.	MDP on “Market Research & Value Chain Management of Agricultural Commodities”	17-21 Nov, 2020	B Ganesh Kumar, Sanjeev Kumar, PC Meena	39	Need Based Programme (Online Mode)
56.	Training Programme on “Administrative and Financial Management” for Administrative and Finance Officers of ICAR”	23-27 Nov, 2020	BD Phansal, ZH Khilji	43	Need Based Programme (Online Mode)
57.	Training on ‘Advanced Bioinformatics tools and its Applications in Agriculture’	7-11 Dec, 2020	M Balakrishnan, SK Soam, P Supriya	39	Need Based Programme (Online Mode)
58.	MDP on ‘Leadership Development’ (a pre-RMP programme)	8-19 Dec, 2020	RVS Rao, Alok Kumar, KH Rao	41	Pre-RMP programme (Online Mode)
59.	National Seminar on “Big Data Analytics in Agriculture”	10-11 Dec, 2020	S Ravichandran, A Dhandapani	83	National Seminar
60.	MDP on ‘Business Plan Development and Accelerating FPOs/FPCs’	14-19 Dec, 2020	Ranjit Kumar, K Srinivas	35	Need Based Programme (Online Mode)
61.	Training Programme on ‘Motivation, Positive Thinking and Communication Skills’ for Technical Officers of ICAR (T-5 & above) (Online Mode)	17-22 Dec, 2020	BS Yashavanth, Laxman M Ahire	23	Need Based Programme (Online Mode)

5.3 Participation in Seminar/Symposium/ Conferences/Workshops/Meetings

	Name	Programme	Venue	Period
January	BS Yashavanth	International Conference on ‘Importance of Statistics in Global Emerging Scenario’	Savitribai Phule Pune University, Pune	Jan 2-4, 2020
	M Balakrishnan	Consultative Workshop on ‘Finalization of Course Contents for PGDETM’	ICAR-NAARM	Jan 11, 2020
	D Thammi Raju	Brainstorming workshop on ‘Networking of Agricultural Universities Alumni’	ICAR-NAARM	Jan 22, 2020
	Ch Srinivasa Rao	Brainstorming Session on ‘Challenges and Opportunities in Meat Sector	ICAR-NRC Meat, Hyderabad,	Jan 28, 2020
	Ch Srinivasa Rao	Meeting of Board of Management of ANGRAU	Lam, Guntur, Andhra Pradesh,	Jan 30, 2020



	Name	Programme	Venue	Period
February	D Thammi Raju	Digital Project Proposals Evaluation Committee Meeting under NAHEP Component 2	Online	Feb 6, 2020
	S Senthil Vinayagam	National workshop on Career Development Centre (CDC)	Kerala Agricultural University, Thrissur	Feb 7, 2020
	Ch Srinivasa Rao	3 rd Meeting of Executive Council of National Council for Cooperative Training (NCCT), Ministry of Agriculture & Farmers' Welfare, Govt of India.	Krishi Bhawan, New Delhi	Feb 17, 2020
	Ch Srinivasa Rao	ICAR Senior Officers Committee Meeting	Krishi Bhawan, New Delhi	Feb 18, 2020
	D Thammi Raju	Focused Group Discussion on 'Development of Guidelines for Collaborative Research Projects on Digital Solutions in Agriculture'	ICAR-NAARM, Hyderabad.	Feb 19, 2020
	D Thammi Raju	National Conference on Geospatial Technology in Agriculture organized by Association for Management of Agricultural Research and Agripreneurship (AMARA)	ICAR-NAARM, Hyderabad.	Feb 20-21, 2020
	I Sekar	National Conference on Geospatial Technology in Agriculture organized by Association for Management of Agricultural Research and Agripreneurship	ICAR-NAARM, Hyderabad.	Feb 20-21, 2020
	Ch Srinivasa Rao	111 th Meeting of Executive Council of NAAS	NAAS Secretariat, New Delhi	Feb 21, 2020
	M Balakrishnan	International Conference on 'Banana- 2020'	ICAR-NRCB, Trichy, Tamil Nadu	Feb 22-25, 2020
	D Thammi Raju	Workshop on 'Prospectus of Agricultural Education: Awareness among Higher Secondary School Administrators'	ICAR-NAARM, Hyderabad.	Feb 24, 2020
	S Ravichandran	Meeting on 'Research Infrastructure in Horticulture in Telangana'	Dr.Konda Laxman Horticulture University, Mulugu, Telangana	Feb 24, 2020
	Ch Srinivasa Rao	24 th Meeting of Food and Agriculture Division Council (FADC) of Bureau of Indian Standards, Govt of India.	Krishi Bhawan, New Delhi	Feb 26, 2020
	Ch Srinivasa Rao	91 st Annual General Meeting of ICAR Society	NASC Complex, New Delhi	Feb 27, 2020

	Name	Programme	Venue	Period
March	M Ramesh Naik & H Umesh	PUSA Krishi Vigyan Mela	IARI, New Delhi	Mar 1-3, 2020
	D Thammi Raju	NAHEP Review Meeting on Investments in ICAR Leadership in Agricultural Higher Education	ICAR-NAARM	Mar 5, 2020
	T Srinivas	Meeting of the Expert Committee Report presentation to the Director General, ICAR on Strengthening ICAR-NAARM as Think Tank of ICAR	DG Office, Krishi Bhavan, New Delhi	Mar 13, 2020
	Ch Srinivasa Rao	Think Tank Committee meeting with DG, ICAR for final report presentation	DG's Committee Room, Krishi Bhawan, New Delhi	Mar 13, 2020
	S Ravichandran	Infrastructure in Fisheries in Telangana	Department of Fisheries, Government of Telangana	Mar 13, 2020
	S Senthil Vinayagam	Institute Management Committee	NRC on Meat, Hyderabad	Mar 16, 2020
	Ch Srinivasa Rao	Meeting with DDG (Edn), ICAR (through video conferencing)	Meeting with DDG (Edn) (Online)	Mar 18, 2020
April	Ch Srinivasa Rao	ICAR Directors' Meeting	Online	Apr 10, 2020
	Ch Srinivasa Rao	ICAR SOC Meeting	Online	Apr 22, 2020
May	Ch Srinivasa Rao	CGIAR Review Meeting	Online	May 4, 2020
	Ch Srinivasa Rao	ICAR SOC Meeting	Online	May 6, 2020
	Ch Srinivasa Rao	Meeting with DG, ICAR	Online	May 7, 2020
	H Umesh	Virtual Review Meeting of Project on "Mainstreaming Biodiversity Conservation in Agriculture Development".	Online	May 8, 2020
	Ch Srinivasa Rao	Discussion on BIOFIN Project	Online	May 8, 2020
	Manju Gerard	"Rebooting the Economy with the New Normal"	Startup India	May 12, 2020
	Ch Srinivasa Rao	Special Invitee for the Technical Session II - Crop Production of the Virtual 55 th Annual Rice Group Meeting (ARGM) of IIRR, Hyderabad.	Online	May 12, 2020
	Ch Srinivasa Rao	Meeting Chaired by Hon'ble Secretary, DARE and DG, ICAR along with ADGs, SMD Heads, DDGs, Directors of NAARM, IASRI and NIAEPr. to discuss on Revamping of ICAR Website.	Online	May 13, 2020
	Manju Gerard	Webinar "How to Search to Know the Uniqueness of Your Idea"- Questel	Online	May 14, 2020



	Name	Programme	Venue	Period
May	Alok Kumar KH Rao	"Sustainable Agriculture: Where Are We Headed After 2020?" - CropIN	Online	May 26, 2020
	Alok Kumar	Communication Skills for a Charismatic Leader- NPTEL	Online	May 27, 2020
	Ch Srinivasa Rao	12 th Meeting of Academic Council of PJTSAU, Hyderabad	Online	May 27, 2020
	Alok Kumar KH Rao	"Approaches of Public Funded Research Organizations in Agri-Technology Generation and its Transfer in New Normal Situation" a-IDEA and NAARM-TBI	Online	May 28, 2020
	Ch Srinivasa Rao	UNFCCC Delegation Meeting.	Online	May 28, 2020
	Ch Srinivasa Rao	Panelist for the Webinar on Approaches of Public funded Research Organizations in Agri-technology generation and its transfer in New Normal situation, organized by a-Idea, NAARM.	Online	May 28, 2020
	Ch Srinivasa Rao	21 st Meeting of Board of Management of PJTSAU, Hyderabad.	Online	May 29, 2020
June	Ch Srinivasa Rao	June Momentum for Climate Change - UNFCCC Global Climate Change Negotiations -Agriculture Session as Government of India Delegation Member.	Online	Jun 1-10, 2020
	Ch Srinivasa Rao	Webinar Panel Discussion on "Covid-19: Impacts and New Normal in Agriculture" by NAAS.	Online	Jun 5, 2020
	S Senthil Vinayagam	State Level Technical Program (SLTP) of Agricultural Engineering of ANGRAU	Online	Jun 5-6, 2020
	GRK Murthy	SLTP of Agricultural Engineering program of ANGRAU	Online	Jun 5-6, 2020
	Ch Srinivasa Rao	ICAR SOC Meeting	Online	Jun 9, 2020
	Ch Srinivasa Rao	Chief Guest for the Valedictory Session of the Online Course on "SciCom for Smart Scholars" organized by ICAR-CIFE, Mumbai & NAHEP-CAAST Project.	Online	Jun 11, 2020
	Ch Srinivasa Rao	Meeting chaired by Member Secretary, NDMA with the Officers of Department of Agriculture Cooperation and Farmer's Welfare regarding approval of National Agriculture Disaster Management Plan (NADMP), organized by NDMA New Delhi	Online	Jun 11, 2020
	I Sekar	Virtual meeting and scrutiny for PGDMA 2020-22 Programme at ICAR-NAARM, Hyderabad	Online	Jun 15-20, 2020

	Name	Programme	Venue	Period
June	D Thammi Raju	PGDMA Interviews for 2020-22 Batch applicants	Online	Jun 15-20, 2020
	Ch Srinivasa Rao	Meeting on "Impact of National Agricultural Research System" being organized by NAAS	Online	Jun 15, 2020
	Ch Srinivasa Rao	293 rd Meeting of Board of Management of ANGRAU	Online	Jun 18, 2020
	Ch Srinivasa Rao	112 th Meeting of the Executive Council of National Academy of Agricultural Sciences	Online	Jun 19, 2020
	D Thammi Raju	1 st Virtual Review Meeting on Status of establishment of CDCs / FDC with nodal officers / Coordinators of CDCs/FDC under National Agricultural Higher Education Project (NAHEP)		Jun 22, 2020
	Alok Kumar	Webinar on Entrepreneurial Motivation-NPTEL	Online	Jun 22, 2020
	Ch Srinivasa Rao	Presentation of newly elected Fellows, 2020 of NAAS as part of the Annual General Body meeting (Section: Natural Resource Management and Plant Protection)	Online	Jun 24, 2020
	Ch Srinivasa Rao	Meeting of Associates of NAAS Hyderabad Chapter	Online	Jun 24, 2020
July	Ch Srinivasa Rao	Meeting on Implementation of E-Office and Rules and Bye Laws - chaired by AS(D) & Secretary, ICAR with Directors of ICAR Institutes.	Online	Jun 24, 2020
	D Thammi Raju	Virtual Signing Ceremony of Tripartite Memorandums of Understanding for establishment of CDCs& FDCs under Component - 2 of NAHEP	Online	Jul 2, 2020
	Ch Srinivasa Rao	Inauguration of CDC under NAHEP and signing MoU	Online	Jul 2, 2020
	S Senthil Vinayagam	TQIMC Meeting: Review of Training Programmes of NIRD, Hyderabad	Online	Jul 3, 2020
	Ch Srinivasa Rao	Inauguration of ABS Program and Presidential Address in the Awareness cum Sensitization Workshop on 'Implementation of Access and Benefit Sharing Regulations in Agriculture Research'	Online	Jul 6, 2020
	Ch Srinivasa Rao	Virtual Launch of 2020 Global Food Policy Report Building Inclusive Food System Co-organized by IFPRI South Asia, ICAR and Trust for Advancement of Agricultural Sciences (TAAS)	Online	Jul 6, 2020



	Name	Programme	Venue	Period
	Ranjit Kumar	Socio-economic lecture series organised by Kerala Agricultural University, Thrissur	Online	Jul 6, 2020
	Ch Srinivasa Rao	Meeting to review the final version of Coffee Table Book and website of ICAR Journals under the Chairmanship of DG, ICAR	Online	Jul 8, 2020
	Ch Srinivasa Rao	Panelist in Webinar on "Agriculture Startups - Opportunities and Challenges" organised by Agrovision Foundation	Online	Jul 10, 2020
	D Thammi Raju	Review Meeting on Disaster Recovery Centre Under NAHEP chaired by DDG (Education), ICAR	Online	Jul 11, 2020
	D Thammi Raju	Joint Advisory Meeting of PGD Courses	Online	Jul 13, 2020
	I Sekar	Joint Advisory Committee Meeting of Post Graduate Diploma Courses with UOH	Online	Jul 13, 2020
	Ch Srinivasa Rao	ICAR Senior Officers Committee Meeting	Online	Jul 14, 2020
	Ch Srinivasa Rao	Inauguration of Online EDP for Master Trainers on Access and Benefit Sharing (ABS) Regulations in India and Nagoya Protocol	Online	Jul 15, 2020
	D Thammi Raju	NAHEP Meeting on Procurement of Goods	Online	Jul 16, 2020
	Ch Srinivasa Rao	92 nd Foundation Day & Award Ceremony of ICAR	Online	Jul 16, 2020
July	Ch Srinivasa Rao	Review meeting of all the conveners of NAAS Regional Chapters	Online	Jul 17, 2020
	Ch Srinivasa Rao	4 th Meeting of Executive Council of National Council for Cooperative Training	Online	Jul 17, 2020
	Ch Srinivasa Rao	Meeting of the 104 th Academic Council, ANGRAU, Lam, Guntur	Online	Jul 17, 2020
	Ch Srinivasa Rao	Felicitation ceremony of Dr. Rattan Lal, World Food Laureate, Distinguished University Professor, Ohio State University, USA followed by Lecture on "Soil Science in Sustainable Food System beyond COVID-19"	Online	Jul 21, 2020
	Ch Srinivasa Rao	Inaugural session of The virtual Annual Zonal Workshop of KVKs under Zone-10 (AP, Telangana, Tamil Nadu and Puducherry) hosted by ICAR-ATARI, Hyderabad	Online	Jul 23, 2020
	Ch Srinivasa Rao	Panelist in discussion on "Technology backstopping and strengthening of KVKs to combat the challenges posed by COVID 19 pandemic" in the virtual Annual Zonal Workshop of KVKs under Zone-10 hosted by ICAR-ATARI, Hyderabad	Online	Jul 23, 2020

	Name	Programme	Venue	Period
July	Ch Srinivasa Rao	Meeting on 'Engaging Retired Scientists for strengthening Agricultural Sector'	Online	Jul 23, 2020
	Ch Srinivasa Rao	Meeting on Management of Websites and Social Media under the Chairmanship of DG, ICAR with all SOC Members	Online	Jul 28, 2020
	Ch Srinivasa Rao	Expert for the Virtual Brainstorming Workshop to develop a framework for propelling ICAR-CIWA as a Global Institute for Women in Agriculture	Online	Jul 31, 2020
August	Ch Srinivasa Rao	Meeting on Transfer Policy/Guidelines for Scientists of ICAR under the Chairmanship of DG, ICAR	Online	Aug 1, 2020
	Manju Gerard	Webinar on " Need for IP Protection: The Dilemma Between Publishing & Patenting" organised by IKP, Hyderabad	Online	Aug 5, 2020
	Ch Srinivasa Rao	SOC Meeting of ICAR & Boosting India's Agricultural Export: Action Plan of ICAR under the Chairmanship of Secretary, DARE & DG, ICAR to discuss the consolidated action plan in a VC meeting of the SOC members	Online	Aug 5, 2020
	Ch Srinivasa Rao	Meeting with the Chairman of the Task Force on Agriculture Research Compendium to finalize the Chapter Scheme and the Authors	Online	Aug 7, 2020
	Ch Srinivasa Rao	113th Meeting of the Executive Council of National Academy of Agricultural Sciences	Online	Aug 10, 2020
	Ch Srinivasa Rao	Inauguration of Krishi Megh, ICAR Data Centre (DC), ICAR Disaster Recovery Centre, Accreditation Portal, and Alumni Network Portal by the Hon'ble Agriculture Minister, GOI.	Online	Aug 11, 2020
	P Supriya	National Webinar on Advances in Genomics Tools for Rice Improvement (AGTRI-2020) organised by Bihar Agricultural University, Sabour	Online	Aug 12, 2020
	Ch Srinivasa Rao	Board of Management Meeting of PJTSAU	Online	Aug 13, 2020
	Ch Srinivasa Rao	27th General Body Meeting of NAAS	Online	Aug 13, 2020
	Ch Srinivasa Rao	Launch of MOOC Course Content on Commercial Dairy Farming – A Joint Initiative of ICAR-NDRI and ICAR-NAARM (Online) (by (DDG (AS), DDG (Edn), Director, NDRI and Director, NAARM)	Online	Aug 14, 2020
	D Thammi Raju	National Core Committee in the Field of Veterinary Extension Education, as a Member of BSMA, Constituted by ICAR	Online	Aug 18, 2020



	Name	Programme	Venue	Period
August	S Senthil Vinayagam	Institute Research Council Meeting-2020 of ICAR-IIMR, Hyderabad	Online	Aug 19, 2020
	Ch Srinivasa Rao	Launch of MOOCs on “Information Handling Skills for Teaching, Learning and Research” under NAHEP (IG) Subproject on National Knowledge Management Centre for Agricultural Education and Research as Guest of Honour being organised by PJTSAU, Hyderabad	Online	Aug 19, 2020
	S Senthil Vinayagam	National Conference on Transformation of Agricultural Extension – Strategies for Effective Reformation-2020 organised by ANGARU, Bapatla	Online	Aug 20-21, 2020
	Ch Srinivasa Rao	Brainstorming Session on Strategies for enhancing Soil Organic Carbon for Food Security and Climate Action (NAAS Hyderabad Chapter activity)	Online	Aug 21, 2020
	H Umesh	Consultative meeting on prospective plan for Telangana State organised by Telangana Horticulture Institute Red Hills, Hyderabad	Online	Aug 24, 2020
	Ch Srinivasa Rao	Participated as a Chief Guest in the Online Short Certificate Training course on E-resources, E-Learning and IoT. NAHEP-IG Project organised by UAS, Raichur in collaboration with NAARM	Online	Aug 24, 2020
	Ch Srinivasa Rao	Meeting with the Chairman of the Task Force on Agriculture Research Compendium to finalise the Chapter Scheme and the Authors	Online	Aug 24, 2020
	Manju Gerard	Webinar, “Legal Framework for Protection of Plant Varieties in India: Challenges and Opportunities” organised by ICRISAT, Hyderabad	Online	Aug 26, 2020
	Ch Srinivasa Rao	Fourth Convocation of PJTSAU	Online	Aug 27, 2020
	Ch Srinivasa Rao	Inauguration of NABARD & NAAARM a-IDEA incubation facility at Centre for Agri Innovation by Shri Govinda Rajulu Chintala, Chairman, National Bank for Agricultural and Rural Development (NABARD)	Online	Aug 27, 2020

	Name	Programme	Venue	Period
September	Ch Srinivasa Rao	ICAR-NAARM Foundation Day Celebrations and Foundation Day lecture by Dr. R.S. Paroda, Former Director General, ICAR on "Reorienting agricultural research for innovation & development"	Online	Sep 1, 2020
	Ch Srinivasa Rao	ICAR Senior Officers Committee Meeting	Online	Sep 3, 2020
	I Sekar	Participated as External examiner for Public Defence and final Viva- voce examination for PhD student of Tamil Nadu Agricultural University	Online	Sep 5, 2020
	Ch Srinivasa Rao	NAAS Meeting under the Chairmanship of Dr. Panjab Singh on Agriculture Research Compendium	Online	Sep 7, 2020
	S Ravichandran	Final cum Concluding workshop of the DST-NSTMIS project "Agricultural Research and Development Infrastructure in Andhra Pradesh and Telangana	Online	Sep 8, 2020
	Ch Srinivasa Rao	Virtual Meeting of Regional Advisory Group (RAG) – First Meeting of Stakeholders for FY 2020-21 of National Bank for Agriculture & Rural Development (NABARD), Telangana Regional Office	Online	Sep 10, 2020
	Ch Srinivasa Rao	NAAS Sectional Committee Meeting of NRM for election of Fellowship and selection of Associateship 2021	Online	Sep 10, 2020
	Ch Srinivasa Rao	First meeting with Society of Agribusiness Professionals ISAP as the agency for the third party evaluation for all the schemes	Online	Sep 12, 2020
	T Srinivas	First meeting with Indian Society of Agribusiness Professionals (ISAP) identified by ICAR as the agency for the third party evaluation for all the schemes which is mandated by Finance Ministry	Online	Sep 12, 2020
	I Sekar	Meeting with Indian Society of Agribusiness Professionals (ISAP) as the agency for the third party evaluation for all schemes as mandated by Finance Ministry	Online	Sep 12, 2020
	Ranjit Kumar	Virtual Workshop on 'AgroEco2050 Foresight Study on Natural Farming: Andhra Pradesh' organised by RySS, CIRAD and FAO	Online	Sep 15, 2020
	Ch Srinivasa Rao	294 th meeting of Board of Management of ANGRAU	Online	Sep 15, 2020

	Name	Programme	Venue	Period
September	Ch Srinivasa Rao	Meeting & celebrations of 150 th Birth Anniversary of Mahatma Gandhi on 2 nd October 2020 under the Chairmanship of DG, ICAR	Online	Sep 17, 2020
	Ch Srinivasa Rao	Participated as Chairman of Technical Session on “Patent Cooperation Treaty” in the virtual workshop-cum-training on “Intellectual Property Rights in Agricultural Research & Education in India”	Online	Sep 18, 2020
	Ch Srinivasa Rao	Panelist in Virtual Brainstorming Session on “One World-One Health” organised by NAAS	Online	Sep 19, 2020
	Ch Srinivasa Rao	IMC Meeting of NAARM	Online	Sep 19, 2020
	Ch Srinivasa Rao	UNFCCC Meeting of SB chairs on agenda items related to Koronivia Joint Work on Agriculture with BASIC Group	Online	Sep 23, 2020
	Ch Srinivasa Rao	Steering Committee meeting on implementation of ICAR Research Data Repository for Knowledge Management (KRISHI) under the Chairmanship of DDG (NRM), ICAR	Online	Sep 24, 2020
	Ch Srinivasa Rao	Participation as Chairman of Technical Session on “IP Valuation & creation of agri-startups for business development” in the fourteen days virtual workshop-cum-training on “Intellectual Property Rights in Agricultural Research & Education in India”	Online	Sep 25, 2020
	T Srinivas	Webinar on “Farm Bills 2020”	Online	Sep 26, 2020
	Alok Kumar	National Webinar on “Farm Bills 2020: Understanding the Implications”	CAAST, IARI (ICAR-NAHEP)	Sep 26, 2020
	Ch Srinivasa Rao	NAAS Sectional Committee Meeting - NRM	Online	Sep 28, 2020
	Ch Srinivasa Rao	Pensioners Meet organized by ICAR-NAARM	Online	Sep 29, 2020
	Ch Srinivasa Rao	“Government initiatives and strategies for Nutrisensitive Agriculture”	Online	Sep 29, 2020
	Ch Srinivasa Rao	Webinar on “New Agriculture Acts – How will they benefit the farmers compared to earlier Acts?” with Her Excellency Dr. Tamlisai Soundararajan, Hon’ble Governor of Telangana	Online	Sep 30, 2020
	Ch Srinivasa Rao	Inaugural session on Global Summit on Responsible AI for Social Empowerment (RAISE 2020) by Hon’ble Prime Minister of India	Online	Oct 05, 2020

	Name	Programme	Venue	Period
October	Ch Srinivasa Rao	ICAR Senior Officers Committee Meeting	Online	Oct 05, 2020
	Ch Srinivasa Rao	48 th EEI Management Committee Meeting	Online	Oct 07, 2020
	Ch Srinivasa Rao	XXV Meeting of ICAR Regional Committee No. II comprising the states of West Bengal, Odisha, Telangana, Andhra Pradesh and the Union Territory of Andaman & Nicobar Islands.	Online	Oct 08, 2020
	Ch Srinivasa Rao	23 rd Board of Management Meeting of PJTSAU through Video Conference	Online	Oct 09, 2020
	Ch Srinivasa Rao	Virtual Workshop and Annual Review Meeting of ABIs/ ZTMCs/ITMUs of Crop Science Division ICAR under NAIF Scheme	Online	Oct 09-10, 2020
	Ch Srinivasa Rao	NAHEP External Advisory Committee Meeting	Online	Oct 13, 2020
	D Thammi Raju	1st Meeting of External Advisory Panel, constituted under NAHEP	Online	Oct 13, 2020
	T Srinivas	Webinar on "Agriculture and Food Policy for Five Trillion Dollar Economy"	Online	Oct 14, 2020
	Ch Srinivasa Rao	Brainstorming Session on "Agriculture and Food Policy for the Five Trillion Dollar Economy" organised by NAAS	Online	Oct 14, 2020
	Ranjit Kumar	Virtual Expert Workshop on Foresight Study "AgroEco 2050" organised by RySS-CIRAD-FAO	Online	Oct 15, 2020
	Ch Srinivasa Rao	खाद्य और कृषि संगठन (एफओ) की 75 वीं वर्षगांठ और विश्व खाद्य दिवस के अवसर पर स्मारक सिक्का - माननीय प्रधानमंत्री के साथ वीडियो सम्मेलन आधारित बातचीत	Online	Oct 16, 2020
	I Sekar	Virtual program of World Food Day & 75 th year of Foundation of FAO	Online	Oct 16, 2020
	Ch Srinivasa Rao	Brainstorming Session on 'Transforming Higher Agricultural Education in India' organized by NAAS under the convenorship of Dr. R.B. Singh, Former President, NAAS.	Online	Oct 20, 2020
	Ch Srinivasa Rao	Exploration and Brainstorming Meeting on Building Synergies & Strengthening Partnership between MoEFCC-IGNFA & ICAR-NAARM	Online	Oct 20, 2020
	Ch Srinivasa Rao	Guest of Honour during the inaugural function of Ten days NAHEP online Training Program on "Climate Risk Assessment and its Management through Agrometeorological Approaches" organised by Dryland Agriculture Research Station (DARS), SKUAST-K, Srinagar.	Online	Oct 21, 2020



	Name	Programme	Venue	Period
October	H Umesh	Brain storming session on preparation of Comprehensive Horticulture Development Plan for Telangana organised by Telangana Horticulture Institute Red Hills, Hyderabad.	Online	Oct 22, 2020
	Ch Srinivasa Rao	Meeting with Dr. Rattan Lal, World Food Laureate, & Distinguished University Professor, Ohio State University, USA to discuss on Soil Conservation and Management: Challenges and strategies	Online	Oct 27, 2020
November	S Senthil Vinayagam	National Conferences on Vigilance and Anti-Corruption organised by CBI, New Delhi	Online	Oct 27-29, 2020
	Ch Srinivasa Rao	ICAR SOC Meeting	Online	Nov 05, 2020
	Ranjit Kumar	Sub-committee meeting on Agri-processing & Agri Infra Development in Bihar Bihar Industries Association, Patna	Online	Nov 07, 2020
	Ch Srinivasa Rao	114th Meeting of the Executive Council of National Academy of Agricultural Sciences	Online	Nov 12, 2020
	T Srinivas	Webinar on "Proposed Changes on APAR" by the APAR committee	Online	Nov 18, 2020
	Ch Srinivasa Rao	Presentation on proposed changes in the existing APARs in the meeting of the Committee to review APAR Online	Online	Nov 18, 2020
	Ch Srinivasa Rao	UNFCCC Climate Change Dialogues -Preparatory Meeting	Online	Nov 19, 2020
	H Umesh	Virtual Workshop and Review Meeting of National Agriculture Innovation Fund (NAIF) Scheme by IP&TM Unit, New Delhi	Online	Nov 23-24, 2020
	Ch Srinivasa Rao	Virtual Workshop and Annual Review Meeting of ZTMCs/ ITMUs/ ABIs under ICAR Institutes of NRM; Agril. Engg. and Agril. Edn. Divisions of ICAR	Online	Nov 23-24, 2020
	S Senthil Vinayagam	Video Conference on 'Inclusion of 2 days Preventive Vigilance modules for the Induction and Mid-Career training programme' by the Central Vigilance Commissioner	Online	Nov 24, 2020
	P Supriya	International e-conference on 'Advances and Future Outlook in Biotechnology and Crop Improvement for Sustainable Productivity' organised by University of Horticultural Sciences, Bagalkot	Online	Nov 24-27, 2020
	P Supriya	National Workshop on Intellectual Property Management in Agriculture organised by ICAR-Indian Institute of Agricultural Biotechnology, Ranchi	Online	Nov 28, 2020

	Name	Programme	Venue	Period
December	Ranjit Kumar	International Webinar on 'Scaling Pre-Monsoon Dry Sowing with Natural Farming' organised by RySS, Guntur	Online	Dec 03, 2020
	Ch Srinivasa Rao	Annual Conference of VCs of SAUs/CAUs and Directors of ICAR Institutes	Online	Dec 5, 2020
	Ch Srinivasa Rao	24 th Meeting of Board of Management of PJTSAU	Online	Dec 7, 2020
	Ch Srinivasa Rao	UNFCCC Feedback Session of "Koronivia workshop on socioeconomic and food security dimensions of climate change in the agricultural sector" by UNFCCC Secretariat.	Online	Dec 7, 2020
	D Thammi Raju	Future Prospects for a Paid Internships in Veterinary Science in Denmark	Online	Dec 8, 2020
	Ch Srinivasa Rao	Panelist in a Session on "Reinventing Rice Production System to achieve the SDGs" in the 1st Indian Rice Congress-2020 organised by Association of Rice Research Workers, Cuttack, Odisha in collaboration with ICAR, ICAR-NRRI, ICAR-IIRR, ICAR-IARI, OUAT, Society	Online	Dec 8, 2020
	I Sekar	Participated as a member-nominee of Review Committee of National Agricultural Science Fund Scheme of ICAR	Online	Dec 9, 2020
	Ranjit Kumar	PJTSAU – PG Faculty Board Meeting	Online	Dec 10, 2020
	Ch Srinivasa Rao	Participated as a Member of Selection-cum-Standing Committee in the National Review Workshop of ICAR- National Professor (NP), National Fellow (NF), Emeritus Scientist (ES) and Emeritus Professor (EP) programmes	Online	Dec 10-11 & 22, 2020
	Ch Srinivasa Rao	ICAR Senior Officer Committee Meeting	Online	Dec 11, 2020
	Ranjit Kumar	Virtual Policy Seminar on 'Food and Agricultural Trade in the New Policy Environment' organised by IFPRI	Online	Dec 17, 2020
	Ch Srinivasa Rao	NAAS Task Force Meeting to review the progress of the 'Compendium on Impact of Agricultural Research'	Online	Dec 17, 2020
	Ch Srinivasa Rao	Meeting of the ICAR Committee for revision and filing of APAR online.	Online	Dec 17, 2020
	T Srinivas	Brainstorming Session on "Recent Agricultural Market Reforms in India"	Online	Dec 18, 2020
	Ranjit Kumar	Brainstorming Session on 'Recent Agricultural Market Reforms' organised by NAAS & NIAP through video conferencing	Online	Dec 18, 2020



	Name	Programme	Venue	Period
December	I Sekar	Brainstorming Session on 'Recent Agricultural Market Reforms in India' organised by NAAS & NIAP	Online	Dec 18, 2020
	Ch Srinivasa Rao	Meeting of Programme Committee of NAAS	Online	Dec 22, 2020
	Ch Srinivasa Rao	First Meeting of ICT Steering Committee under Chairmanship of Secretary, DARE & DG, ICAR	Online	Dec 24, 2020
	Ch Srinivasa Rao	Virtual event of "Hon'ble Prime Minister of India addressing the farmers and releasing PM Kisan money to farmers"	Online	Dec 25, 2020
	I Sekar	The event online of the address by the Hon'ble PM of India to the farming community and release of PM Kisan money to farmers	Online	Dec 25, 2020
	Ch Srinivasa Rao	115 th Meeting of the Executive Council of National Academy of Agricultural Sciences through video conferencing	Online	Dec 26, 2020
	Ch Srinivasa Rao	105 th Academic Council Meeting of ANGRAU	Online	Dec 28, 2020
	Ch Srinivasa Rao	Meeting of the General Body of NASI	Online	Dec 29, 2020
	Ch Srinivasa Rao	Workshop on 'Swachhta Abhiyan' under the chairmanship of Secretary, DARE & DG, ICAR	Online	Dec 30, 2020

6.1 Special Events at NAARM

6.1.1 Virtual Meeting of NAAS Associates of Hyderabad Chapter

A virtual meeting of Associates of NAAS based in Telangana and Andhra Pradesh was held through video conference mode on June 24, 2020 in order to review the ongoing activities of the Regional Chapter of NAAS, Hyderabad and to plan the future activities. Dr. Ch. Srinivasa Rao, Director, NAARM, Fellow of NAAS and NASI and Convener, NAAS-Hyderabad chapter chaired the meeting



and Dr. RM Sundaram, Principal Scientist, ICAR-IIRR, Hyderabad and NAAS Fellow served as the convener. The meeting was attended by a total of four Associates of NAAS belonging to the States of Telangana and Andhra Pradesh. The NAAS Associates who attended the meeting are: Dr. Damodar Reddy, Director, ICAR-CTRI, Rajahmundry, Dr. BM Naveena, Principal Scientist, ICAR-NRCM, Hyderabad, Dr. Manish Pandey, Sr. Scientist, ICRISAT, Hyderabad and Dr. SK Mangrauthia, Sr. Scientist, ICAR-IIRR, Hyderabad.



Fig. 6.1: Snapshots of virtual meeting of NAAS Associates of Hyderabad Chapter

6.1.2 Research Advisory Committee (RAC) meeting

The first meeting of 8th Research Advisory Committee (RAC) of NAARM was conducted virtually on July 24, 2020 under the Chairmanship of Prof. G. Padmanaban, President, NASI and Former Director, IISc, Bangalore. Dr. Ch Srinivasa

Rao, Director, NAARM welcomed and introduced the Chairman and members. He also presented an overview of the overall achievements of the Academy. The RAC reviewed the research and the progress of capacity building, and suggestions were given for further improvements to attain global visibility.



Fig. 6.2: First meeting of 8th Research Advisory Committee under the Chairmanship of Prof. G. Padmanabhan, President, NASI and Former Director, IISc



6.1.3 Academic Committee Meeting

Academic committee, a body of the faculty members meet at the Academy once in a month. In this forum, issues pertaining to the capacity building, research and other academic programmes are discussed and relevant decisions are arrived at. This committee is chaired by the Director and the member Secretary, Dr. A. Dandapani.



Fig. 6.3: Academic Committee Meeting held on September 14, 2020

During the year 2020, a total of twelve meetings were held. Some of the important agenda items taken were periodical review of publications (research papers & policy papers), status of impact studies, regular review of capacity building programmes with progress of previous month and planned programs for the ensuing month, status and progress of PGDMA, PGTMA, and PGETMA programmes. Similarly, the preparations for the special meetings like IRC, and RAC were also discussed in the meeting. The progress the professional society, AMARA

and its journal, JAMRA was also discussed. The flagship programmes such as FOCARS (110th and 111th Batches) were discussed. The conduct of GIS conference and the arrangements have been discussed in detail during the ACM meeting.

6.1.4 23rd Institute Research Council Meeting

The 23rd Institute Research Council (IRC) meeting was held during Jun 2-3, 2020 in virtual mode. The Chairman, IRC and Director appreciated the efforts of the faculty for their contributions in effective implementation of the mandate of the Academy and highlighted the need for considering the recommendations of QRT and RAC while proposing new research projects. Dr. RC Agrawal, Deputy Director General (Education), ICAR gave a virtual inaugural address encouraging the scientists of the Academy for doing all the mandated activities even during such difficult situation. A total of 49 research projects including 19 completed, 20 on-going and 10 new projects were discussed during the seven technical sessions including the plenary, each headed by the Chair and Co-Chair with rapporteur support. Out of 19 completed projects, 8 were in-house and 11 were extra mural while among the on-going projects, 11 were in-house and 9 were extra mural funded. This IRC has extended the project duration of 6 projects in order to accomplish the project objectives



Fig. 6.4: Virtual meeting of 23rd IRC

6.1.5 Virtual Joint Advisory Committee Meeting of NAARM & UoH

The Academy organized a virtual Joint Advisory Committee Meeting of NAARM & UoH on Jul 13, 2020 on Post-Graduate Diploma in Technology Management in Agriculture (PGDTMA) & Post-Graduate Diploma in Education Technology Management (PGDETM) to review and approve the action plan proposed. The meeting was organized under the Chairmanship of Dr. Ch. Srinivasa Rao, Director, NAARM and the JAC members, Prof. S Jeelani, Prof. E Haribabu, Prof. G Bhuvaneswara Lakshmi and Prof. C Raghava Reddy from UoH. Besides, coordinators of both the courses, faculty from NAARM also attended.



Fig. 6.5: Virtual Joint Advisory Committee Meeting of NAARM & UoH

6.1.6 24th Institute Research Council Meeting

The 24th Institute Research Council meeting was held virtually for one day on 16th December 2020.



Chairman of the IRC, Dr. Ch. Srinivasa Rao, Director ICAR-NAARM addressed the members of the IRC and appreciated the faculty for their contribution to various capacity building programmes, academic and research activities as per the mandate of the Academy even during the pandemic situation. He emphasized on improving the Project: Scientist ratio from the current 0.97 to above 1. During this IRC, a total of 20 ongoing in-house; 10 extra-mural and three new in-house projects were discussed in seven technical sessions including the plenary, each headed by the chair and co-chair with rapporteur support. Approved decisions taken or suggestions made during each technical session were presented by the respective session co-chair during the plenary. The project duration of three projects has been extended (two in-house and one extra mural) during the IRC period in order to accomplish the project objectives



Fig. 6.6: 24th Institute Research Council Meeting

6.1.7 Vigilance Awareness Week

Vigilance Awareness Week was celebrated during 27 Oct – 02 Nov, 2020 with the theme “Satark Bharat,

Samridh Bharat” (Vigilant India, Prosperous India) along with Integrity Pledge on 27 October, 2020 through virtual mode. More than 100 employees including FOCARS probationers took the pledge and were also encouraged to take e-pledge. During the week, activities such as display of banners, e-posters display in meeting hall and LED display in hostels and other places, integrity pledge by all employees, essay competition among faculty/officers/staff members, session on vigilance management by Shri. Sandeep Mukherjee, Former Deputy Director (Vigilance), ISTM, DoPT, were carried out as per the CVC guidelines besides elocution competition among PGDMA students. On the closing day, Dr. S. Senthil Vinayagam, Vigilance Officer of the Academy presented the report on activities organized during Vigilance Awareness Week. Dr. SK Soam, Joint Director addressed about importance of vigilance in government organizations and to sensitize every citizen on vigilance. In his presidential address, Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM stressed upon strict adherence of conduct rules, procurement rules, recruitment process and other organizational activities by all the employees of the Academy.



Fig. 6.7: FOCARS Probationers taking pledge during Vigilance Awareness Week

6.1.8 Senior Officers Committee (SOC) Meeting

Senior Officers’ Committee meetings are regularly conducted every month at NAARM. The photo

below exhibits one such meeting organised on Mar 20, 2020 with senior officials of the Academy. The major agenda discussed were budget utilization, civil works and electrical works undertaken and their progress, constraints, farm issues, quarantine facility support systems in the academy, and emergency health care systems of staff and pensioners in health centre. Many meetings of SOC during the year 2020 were conducted virtually due to pandemic.



Fig. 6.8: SOC Meeting at the Academy during Pandemic Period

6.1.9 Institute Joint Staff Council (IJSC) Meeting

Institute Joint Staff Council Meeting was organized on Feb 01, 2020. The action taken report of the previous IJSC meeting and staff welfare issues were discussed during the meeting.



Fig. 6.9: Institute Joint Staff Council (IJSC) Meeting at Academy

6.1.10 Pensioners Meet

Pensioners Meeting was held at the Academy on Sep 29, 2020. During the meeting, Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM said that during the testing times when our whole country is going through a calamity type of situations due to COVID-19, NAARM is continuing its tradition to solve and address the grievances of our senior

citizens, our esteemed pensioners. A total of 36 members participated in the meeting and their grievances related to health cards, medical reimbursements, recognition of private hospitals, revision of pension etc. were addressed.



Fig. 6.10: Pensioners Meet during Pandemic

6.2 Visits and Interactions of Dignitaries at NAARM

6.2.1 Dr. (Smt.) TAMILISAI Soundararajan, Hon'ble Governor, Telangana State Inaugurated the 110th Foundation Course for Agricultural Research Service at NAARM

Dr. (Smt.) TAMILISAI Soundararajan, Hon'ble Governor, Telangana State inaugurated the 110th Foundation Course for Agricultural Research Service (FOCARS) at the Academy on January 07, 2020. A total of 135 ARS Probationers including 56 female probationers representing 28 ARS disciplines and 22 States have undergone the Foundation Course. In her inaugural address, Smt. Soundararajan highlighted the role of research in strengthening agricultural production and efforts made towards achieving the goal of doubling the farmers' income set by the Hon'ble Prime Minister of India by the Year-2022. Highlighting the achievements made in agricultural commodities and the foreign trade of milk and spices, she

stressed upon the innovative approaches that need to be adopted to enhance the livelihoods, food and nutrition security in the context of climate change, emergence of new pests and diseases, etc. She emphasized upon carrying out focused research and extension that can help to solve the issues associated with the country's milk exports by meeting the international standards. The Hon'ble Governor urged the young probationers to take agriculture sector to the new heights and bring second Green Revolution in the country. Smt. Soundararajan appreciated the Academy's efforts taken for building the National Agricultural Research System's capacity.



Fig. 6.11: Dr. (Smt.) TAMILISAI Soundararajan, Hon'ble Governor, Telangana State addressing 110th FOCARS at NAARM

6.2.2 Dr. Trilochan Mohapatra, Secretary DARE and DG ICAR interacts with 110th FOCARS

Dr. T. Mohapatra, Secretary-DARE and DG-ICAR interacted with 110 FOCARS trainees on Feb 7, 2020. During the interaction, he stressed on the need to analyze the state representation of ARS scientists over the past three years in order to promote ARS from all the states. He interacted and motivated young scientists for their dedicated service for the cause of small and marginal farmers of India. He also discussed on how to make ICAR institutions or deemed to be universities and State Agricultural Universities to figure in the top 200 institutions and Universities of the

world. He also discussed on pioneering areas of research in Indian agriculture to be pursued by the young scientists; ways for improving the number of women directors in the ICAR; improving the stature of agricultural research among the common public in comparison with other fields; bringing second green revolution in India, improving the collaborations with research and agri-industry, hungry and poverty issues in the country, foreign exposure to FOCAR trainees, role of ICAR in faculty building in agriculture and allied universities of state. Dr. Ch.Srinivasa Rao, Director, ICAR-NAARM while welcoming highlighted the new interventions included in the foundation course such as research excellence in remote areas, panel discussion with Lal Bahadur Shastri Young Scientist award winners of ICAR, science communication, ethics in research and development, learning positive thinking, interactions with science academies and policy makers besides overall research project management and transforming them into highly productive scientists of ICAR.



Fig. 6.12: Dr. Trilochan Mohapatra, Secretary DARE and DG ICAR with 110th FOCARS Trainees

6.2.3 Shri Sanjay Kumar Singh, Additional Secretary, DARE and Secretary, ICAR interacts with 110th FOCARS probationers and Directors of ICAR Institutes located at Hyderabad

Shri. Sanjay Kumar Singh, IAS (Addl. Secretary DARE and Secretary, ICAR) visited the Academy

on Feb 15, 2020. In his interaction with 110th FOCARS probationers, he discussed the need for a framework to convert our research to field level applications; emphasized on communicating research findings to farmers in regional languages by effectively utilizing visual aids in the form of videos relevant to major farm-related problems and solutions. He underlined the need of pioneer areas such as artificial intelligence, IoT, Machine learning and importance of interdisciplinary research. Besides food security, he advised to concentrate the research on nutrition security issues and climate research and stress management. He urged the young scientists to keep motivated themselves throughout the service by pursuing research in the pioneer areas, benefitting the farming community.



Fig. 6.13: Shri Sanjay Kumar Singh, Additional Secretary, DARE and Secretary, ICAR interacting with 110th FOCARS probationers

Shri Singh had detailed discussions with Directors of ICAR Institutes located at Hyderabad related to research programs, priorities and way forward. He discussed the issues of duplication of research programs among ICAR Institutes, strengthening of field based research, technology-industry linkages and suggested to concentrate on core issues of technology business incubation centres in ICAR. He also suggested to come out with the documentation on the constraints faced by agri Start-ups linking with donor expectations to improve the overall agri entrepreneurship

ecosystems in India. Respective directors of IIRR, IIMR, IIOR, PDP, NRC on Meat, CRIDA, ATARI and NAARM made detailed presentations on the overview of activities. He also interacted with administrative and finance officers of various ICAR institutes located in Hyderabad on budget utilization and manpower vacancies and assured that man power shortage issues will be resolved soon. During his visit, he also attended the ICAR-Pensioners meeting and interacted with ICAR Pensioners at Hyderabad and discussed about the pensioners' issues including monthly pension adalat being organized by various pension-authorizing Institutes in the country.



Figure 6.14: Shri Sanjay Kumar Singh, Additional Secretary, DARE and Secretary, ICAR interacting with Directors of ICAR Institutes located at Hyderabad

6.2.4 Prof. AK Mishra, Chairman, ASRB interacts with 111th FOCARS Probationers

Prof. A.K. Mishra welcomed the scientists to the prestigious Agricultural Research Services and urged them to work towards reducing the market imbalances through crop planning, cluster farming and resource management so as to enhance farm income; and think imaginatively to reduce the time lag in reaching the benefits of research findings to the end users. He also gave an overview about the innovative steps taken by the Board to attract and retain talent for ICAR and suggested the young scientists to stay competitive all through their career. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM gave an

overview of the foundation program, highlighted the steps taken to re-orient the program under the new normal and highlighted the role being played by the Academy in addressing the ever-increasing capacity building needs of NARES and also as a think tank of ICAR.

6.2.5 Dr. AK. Singh, Secretary, NAAS delivers a guest lecture on 'NAAS and Science Excellence in Agriculture'

A.K. Singh, Secretary, NAAS delivered a guest lecture on "NAAS and Science Excellence in Agriculture" on Feb 7, 2020 at the Academy. In his lecture, he covered on various activities of NAAS activities since its inception on June 5, 1990, Mission and Objectives of NAAS, Management bodies of NAAS, Executive council of NAAS, Past Presidents of NAAS, Section wise distribution of NAAS fellowships/Associateships, various Awards instituted by NAAS for recognizing significant contribution of senior, middle and young scientists to promote agricultural research; NAAS Regional Chapters and its activities, Organization of Biennial Agricultural Science Congresses (ASCs), NAAS publications - Policy papers/briefs; Status/Strategy papers of national importance; study reports, NAAS score of science journals. NAAS Fellows and Associates from Hyderabad Chapter also attended this event.



Fig. 6.15: Dr. AK. Singh, Secretary, NAAS felicitated by Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM

6.3 Celebrations

6.3.1 International Women's Day Celebrations

The Academy celebrated International Women's Day on Mar 08, 2020. The main theme of the celebration was "I am Generation Equality: Realizing Women's Rights". On this occasion, Mr. Gauri Shankar Prasad, Branch Manager, SBI, briefed about the various saving schemes of banks for homemakers. He also gave information on tax saving schemes for working women. Ms. Kusumanjali gave information on various post office saving schemes for women. Dr. Surya Rathore, Chairperson Women Cell, NAARM delivered the talk based on theme 'Realizing Women's Rights' and sensitised the women about their rights. She also emphasized that awareness about their legal rights is very important for women. About 45 women including PGDMA girls, staff, workers of NAARM participated in the programme.



Fig. 6.16: Participants of "International Women's Day" celebrations

6.3.2 74th Independence Day Celebrations and felicitation of Corona warriors

The 74th Independence day was celebrated at NAARM in a festive mood with all covid-19 pandemic precautions at hand. Dr. Ch. Srinivasa Rao, Director, NAARM hoisted the national flag and addressed the gathering. He narrated the sacrifices made by our selfless leaders and freedom fighters for the cause of freedom struggle to become an independent nation. He also highlighted the landmark achievements of



Fig. 6.17: 74th Independence Day Celebrations at ICAR-NAARM and Corona warriors' felicitation



the Academy in capacity building, education, research and as a think-tank. The employees who excelled in tough times of pandemic and made extraordinary contributions amidst covid-19 were felicitated during the celebrations.

6.3.3 ICAR-NAARM celebrates 45th Foundation Day

The Academy celebrated 45th Foundation Day on Sep 01, 2020. Padma Vibhushan Dr. RS Paroda, Former Secretary and DG, ICAR was the Chief Guest on the occasion. Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM welcomed all the guests and highlighted the salient achievements of the Academy in the areas of capacity building, research and academics during the preceding year. He acknowledged the guidance and motivation from QRT, RAC and IMC members to the Academy. He particularly recognized the special role played by the staff members of NAARM, who served as frontline warriors of Covid-19 during the pandemic. On this occasion, a publication titled 'ICAR-HRD Units at a glance' was released.

In his Foundation Day lecture, Dr. Paroda congratulated the award winners and overall achievements of the Academy. He highlighted that NAARM, as a Think Tank, has to institutionalize a new paradigm in terms of organizational restructuring, developing a global view, instilling competencies for foresight and visioning, and also in prioritizing the research areas. He emphasized that NAARM should collaborate with other scientific Academies of the country, international research academic institutions and also with the industry. Dr. RS. Paroda called for piloting disruptive innovation within the National Agricultural and Research and Education System (NARES) and suggested that the Academy should strive towards enhancing

accountability of scientists and transparency in research management. He also declared the Annual Awards of NAARM for the best performing employees in scientific, technical, administrative and supporting staff categories. Eleven farmers from across the country, representing different agro-climatic zones of the country were felicitated during the ceremony for their innovative farming practices. The event was also live streamed in Youtube and other social media, in which over 600 dignitaries comprising eminent scientists from India and abroad; former and serving Directors and Vice Chancellors of ICAR/SAUs; scientists and students from various ICAR Institutes. Dr. SK Soam, Joint Director thanked all the dignitaries for honouring the invitation and gracing the occasion.



Fig. 6.18: Padma Vibhushan Dr. RS Paroda, Former Secretary and DG, ICAR addressing through video conferencing

6.3.4 Celebrations of 150th Birth Anniversary of Mahatma Gandhi

The Academy celebrated the 150th Birth Anniversary of Mahatma Gandhi on 2nd October, 2020. Also in compliance with the instructions of ICAR, different activities were organized on the Life and Philosophy of Mahatma Gandhi at the Academy. The activities included Quiz Competition on Life of Mahatma Gandhi separately for PGDMA students and staff of the Academy; Online programme depicting write-



Fig. 6.19: Lecture by Dr. A. Raghu Kumar on the eve of the 150th Birth Anniversary of Mahatma Gandhi

ups/Pictures etc. relating to Gandhian Philosophy / Thoughts among the youngsters. Swachhata Pledge was taken online by Staff Members of the Academy and a lecture on ‘Life & Philosophy of Mahatma Gandhi’ was delivered by Dr. A. Raghu Kumar, Advocate, Telangana State High Court in online mode. The staff and students of the academy participated in the programme and greatly benefited by the lecture.

6.3.5 Celebration of Mahila Kisan Diwas

The ICAR-National Academy of Agricultural Research Management, Hyderabad virtually organized the “Mahila Kisan Diwas” on 15th October, 2020. Emphasizing the importance of this day with the theme: “Empowering Farm Women with Extension and Research interventions”, Dr. Ch. Srinivasa Rao, Director, ICAR – National Academy of Agricultural Management, stressed on the need for quality research on farm women. Dr. B. S. Sontakki, provided inputs regarding various extension strategies for empowering the farm women and emphasizing their role in agriculture. Dr. Surya Rathore talked about the various facets of women empowerment i.e. economic, social, psychological, nutritional, technological and legal and how to achieve them through gender mainstreaming. Along with Dr. Sontakki and Dr. Rathore; Dr. Manju Gerard was also one of the panelists who laid emphasis that airing one’s voice is the only key to empowerment. If women will



not open up and remain in veils, nobody can help them and therefore they will have to come out of their shells. Good key takeaways were delivered by the panel for empowering farm women which included capacity building of women related to entrepreneurship development, legal issues governing farm women and their strict enforcement, introduction of government programmes keeping not only gender equality in view but also gender equity. Due emphasis needs to be given to gender analysis, gender budgeting and gender auditing in organizations and institutions for holistic development of the second gender in agriculture. In all, 69 farm women, scientists and extension personnel were benefitted by NAARM on this eventful day.



Fig. 6.20: Deliberations during Mahila Kisan Diwas

6.3.6 Constitution Week

ICAR-NAARM celebrated ‘संविधान दिवस-Constitution day’ in online mode on 26th November, 2020. The event coincided with the Valedictory Function of Celebration of 70 Years of Adoption of Indian



Fig. 6.21: Glimpses of celebrations of Constitution Week

Constitution organized year-round from 26 November 2019 to 26 November 2020. Dr. Ch Srinivasa Rao, Director, ICAR-NAARM presided over this function and Prof G. Vinod Kumar, Associate Professor & Director, PG Law College, Osmania University, Hyderabad delivered a lecture, who chronicled the evolution of Indian Constitution and highlighted the landmark events in the process. He hailed the immense contributions of Bharat Ratna Dr. B. R. Ambedkar in preparing one of the best and inclusive constitutions in the world. He stressed on core principles of Justice, Equality, Liberty and Fraternity enshrined in the Preamble to Indian Constitution, besides fundamental duties, directive principles and rights.

In his presidential remarks, Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM paid tributes to the Chief Architect of Indian Constitution, Bharat Ratna Dr. B. R. Ambedkar and remembered his contributions to agricultural development. Winners of debate

and quiz competitions organized as part of the celebrations, were announced on this occasion. The Online Programme was attended by around 200 members of the Academy consisting of Faculty, Officers and Staff, 111th FOCARS trainees, PGDM-ABM students, and contractual research staff. Earlier in the day, the members of ICAR-NAARM family participated in the virtual programme of Preamble Reading on the occasion of The Constitution Day.

6.3.7 Celebration of 120th Birth Anniversary of Acharya Shri N.G. Ranga

The Academy and N.G. Ranga Trust virtually celebrated the 120th Birth Anniversary of Shri Acharya N.G. Ranga, Padma Vibhushan Awardee, Freedom Fighter and Farmer's Welfare Worker on November 7, 2020. Shri M. Venkaiah Naidu, Hon'ble Vice-President briefed the outstanding contributions of the respected Freedom Fighter-cum-longest served Parliamentarian (1952-1991),

Shri Acharya N.G. Ranga towards agricultural development in the country during and after independence. The Vice-President highlighted the importance of agriculture in the Indian economy, current challenges and opportunities in the farming sector, viz., climate change, water issues, soil health deterioration, market linkages, retaining youth in agriculture, integrated farming systems, emerging technologies like AI and drones and diversified agriculture and food systems. He also emphasized on the stakeholders' and government's crucial role to drive agricultural production sustainable and profitable through

new technology towards income security of the farming community. Shri Govinda Rajulu Chintala, Chairman, NABARD; Dr. Ancha Srinivasan, Climate Change Advisor, South East Asia, Asian Development Bank; Shri K. Venkata Rosaih, MLA, Ponnuru, Andhra Pradesh; Dr. G. Muniratnam, Founder General Secretary, Rashtriya Seva Samithi, Andhra Pradesh along with other dignitaries also virtually participated in the event. Earlier, Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM, Hyderabad delivered the welcome address. Shri R. Kishore Babu, Chairman, Shri N.G. Ranga Trust proposed the vote of thanks.



Fig. 6.22: Snapshots of Celebration of 120th Birth Anniversary of Acharya Shri N.G. Ranga



6.3.8 World Soil Day Celebrations

The Academy celebrated World Soil Day on December 5, 2020. During the celebrations, a special briefing session and presentation exercise was conducted for the 111th FOCARS scientists-probationers. Mr. Rahul Mishra, ARS Probationer of 111th FOCARS highlighted the importance of soil and its proper management for sustainable farming through a special presentation. An awareness campaign on 'Importance of Soil Health Management and Soil Health Card Scheme of Government of India' was organized at Thurkapally, a village adopted by the Academy under Sub-Plan Scheme. About 50 farmers and farm women and 37 ARS Probationers of 111th FOCARS, faculty members and technical officers of the Academy participated in the programme.

6.3.9 Kisan Diwas (Farmers Day) Celebrations

The ICAR-National Academy of Agricultural Research Management, Hyderabad celebrated the auspicious day, KISAN DIWAS (Farmers' Day), by listening to the perspectives of innovative farmers, besides felicitating them. Speaking on the occasion, Dr. Ch. Srinivasa Rao, Director of the Academy hailed the selfless contributions of farmers of the country in achieving record agricultural production amidst the serious challenges imposed by COVID-19 pandemic and other natural calamities. He specifically called

upon the ARS Probationers currently undergoing 111th FOCARS at the Academy to rededicate their services for the welfare of farming communities. Six innovative farmers namely Shri Ch. Suresh Kumar Reddy, Shrimati Padma, Shri Vijay Kumar, Shri P. Venkatesh, Shrimati K. V. Lavanya and Shri K. V. Ramana Reddy from the adopted villages were felicitated on this occasion. These farmers spoke on the need for improving natural resource management, market linkages, better pricing and other issues that need attention of researchers and policy makers. Elocution, Essay writing and singing competitions on the theme of 'Farmers and Farming' were specially organized for the students and trainees of the Academy.

ARS probationers of Foundation Course responded to the call of the farmers for betterment of the farmers in different states.

6.4 Yoga and Meditation

6.4.1 International Yoga Day-2020

The Academy celebrated the 'International Day of Yoga' on June 21, 2020 with the theme 'Yoga at Home and Yoga with Family', as per the guidelines issued by Ministry of AYUSH, GoI. All the faculty members and officers were guided to perform yoga at home with family and share pictures. The details of warming-up exercises, protocol and guidelines for Yoga as issued by Ministry of AYUSH, GOI were circulated to all the employees



Fig. 6.23: Farmers' felicitation by Director, NAARM during Farmers' day

of ICAR-NAARM. The NAARM staff members practiced the Yoga at home involving their family members during 7.00 - 7.45 AM on June 21, 2020 alongside the Doordarshan live telecast.



Fig. 6.24: Glimpses of the yoga practiced by Director, NAARM and members of NAARM family

6.5 Campus Beautification

6.5.1 NAARM bags Best Landscaping Institute Award

NAARM participated in 6th Garden Festival-2020 conducted by the Department of Horticulture, Government of Telangana. As a part of Selection Process of Awards, various committees visited the Campus and evaluated. The Academy bagged two awards in "Best Landscape Gardens maintained by State/Central Govt. Institutions" Category and



Fig. 6.25: A Panoramic view of lawn maintenance at NAARM campus

"Best Landscape Gardens Maintained by Central Govt. Educational Institutions" Category.

6.5.2 Farm Activities during Lockdown Period

The essential duties in the farm unit continued during the lock down period by following all norms like social distancing and practicing shift system with staggered timing and mandatory use of face mask for all the staff and workers. It continued during post lock down period as well. The main activities carried out during this quarter included watering the avenues, lawns, nursery potted plants, drip irrigation of fruit plants, rose garden, weeding and inter-cultivation, harvesting and selling of mango and sweet lime crops, trimming of lawns and hedge cutting, sowing of vegetable and intercrops etc. Some of the new developments like construction of a rain water harvesting pond in the proposed Agri-biodiversity Park, installation of protective iron frame on the top of the open well of the farm, installation of electrical pump sets for irrigating plantations and gardens and repairing of the gate valves of the main irrigation pipeline leading to water reservoir. The planting beds and surroundings of the water reservoir garden were cleaned to facilitate proposed lawn planting. As part of development and maintenance of the campus, several new initiatives have been taken like beautification of lawn near lake with carpet grass, shade grass





lawn in-front of Faculty Centre, beautification of footpaths from agronomy well to sports complex.



Fig. 6.26: Farm workers on NAARM field during Pandemic period

6.5.3 Technology Demonstration Park

Efforts have been initiated towards setting up an exclusive Technology Demonstration Park in the Academy pursuant to the recommendations of the 8th Research Advisory Committee and commitments made under the XII Plan EFC. The main objective of this initiative is to showcase agri and other technologies of ICAR to the trainees and other visitors to create awareness and interest in diffusing them further.



Fig. 2.27: Polyhouse facility at ICAR-NAARM

With its serene and sprawling campus of about 50 ha, the Academy has provisions for adopting many diverse agri and other green technologies. Realizing this, through extensive site survey and consultations, the academy developed a detailed plan for documenting and cataloguing the technologies, introduce new technologies with emphasis on green technologies, and develop publicity material for showcasing the activities to trainees and other visitors. Currently, 25 technologies have been documented for display and showcasing.

6.6 Visit of Distinguished Guest Faculty

Name of the faculty	Designation and place	Title of the lecture	Date
Dr. Devendrappa S.	Programme Officer, Krishi Community Radio Station, UAS, Dharwad, Karnataka	1) Radio as a tool for Agricultural and Rural development 2) Community Radio Stations: Establishing with a spice of innovation	21/02/2020
Dr. Janardhan Rao Cheeli	Associate Professor (Mass Com) University of Hyderabad	Writing, Recording & Debriefing of Television talk	22/02/2020
Dr. S.N. Meera	Principal Scientist (Agricultural Extension) ICAR - Indian Institute of Rice Research, Hyderabad	1) Expert Systems of Extension in India 2) Rice knowledge portal	24/02/2020
Dr. P.B. Pradeep Kumar	Assistant Professor (Agricultural Extension), Acharya N G Ranga Agricultural University, Vishakapatnam (A.P.)	Mobile App construction	25/02/2020
Dr. B. Vijayalakshmi	Associate Director Centre for Development of Advanced Computing, Hyderabad	Construction of a model Mobile Application	25/02/2020

Name of the faculty	Designation and place	Title of the lecture	Date
Mr. Kiran Yellampalli	Manager, Keansa Solutions, Hyderabad	Online bidding system for Unstructured markets	27/02/2020
Dr. V.P. Sharma	Technical Consultant National Area Rainfed Authority, New Delhi	1.) Overview of ICT Applications in India 2.) ICT enabled drought proof Planning	28/02/2020
Dr. V.K. Jayaraghvendra Rao	Principal Scientist (Agricultural Extension) IIHR, Bangalore	1.) Digital Presentation and its use in Extension 2.) Making a digital presentation	29/02/2020
Mr. Sriram Bandi	Architect, Kalgudi, Hyderabad	Kalgudi – A Linkedin for farmers	06/03/2020
Mr. Sunil Kumar Joshi	Business Manager, DKMA, ICAR, New Delhi	1.) Importance of Online Consortium of Agriculture Knowledge CeRA 2.) Practice Sessions on CeRA	07/03/2020
Mr. A. Prashanth Kumar	Authorised distributor for Telangana & AP, Mobitech Wireless Solutions, Hyderabad	1.) IOT based Irrigation automation 2.) Practical on Irrigation automation	07/03/2020
Mr. Amandeep Panwar	Manager, Bharat Rohan, New Delhi	1.) Use of Drones in Agriculture 2.) Drones: Some successful examples in Agriculture	09/03/2020

6.7 Visits of Students/Trainees/Farmers from various Agricultural Universities and other institutes

Sl. No.	Details of College and University/ Institute	Degree/ Designation	Date of visit	No. of Visitors
1.	College of Agriculture, Vijayapur, UAS, Dharwad	B.Sc. (Ag)	January 2, 2020	63
2.	College of Agriculture, Vijayapur, UAS, Dharwad	B.Sc. (Ag)	January 5, 2020	67
3.	College of Agriculture, Badnapur	B.Sc. (Hons)	January 14-16, 2020	66
4.	Lokmangal College of Agricultural Biotechnology,	B.Sc. (Agri.biotechnology)	January 24, 2020	25
5.	College of Horticulture and Research Station, Rajnandgaon	B.Sc. (Ag)	January 26, 2020	27
6.	College of community science, Central Agricultural University, Tura, Meghalaya	B.Sc. (Community Science)	February 8, 2020	8
7.	Biswarath College of Agriculture, Assam Agricultural University, Assam	B.Sc. (Ag. Hons)	February 9, 2020	42
8.	Dr. C. Narayanaswamy College of Agriculture, Univeristy of Agricultural Sciences, GKVK, Bengaluru	M.Sc. (Agril Exn) and Ph.D	February 17, 2020	21
9.	KRC college of Horticulture, Arabhavi, Karnataka	B.Sc. (Horti)	February 17, 2020	58
10.	College of Agriculture, Dr. PDKV, Akola, Maharashtra	B.Sc. (Ag)	February 24, 2020	125
11.	College of Home Science, Acharya N.G. Ranga Agricultural University, Lam, Guntur, Andhra Pradesh	B.Sc. (Ag. Hons Community Science)	February 25, 2020	71
12.	College of Agriculture, Annamalai University, Tamil Nadu	M.Sc. (Agril Exn)	February 27, 2020	20
13.	Dept of Agricultural Extension, PJTSAU, Hyderabad	M.Sc. (Agril Exn)	February 27, 2020	6
14.	College of Agriculture, Dhule	B.Sc. (Ag)	March 2, 2020	136
15.	College of Agricultural Extension, PRIST university	B.Tech. (Agril Engg)	March 3, 2020	14
16.	College of Agriculture, Osmanabad	B.Sc. (Ag)	Match 4, 2020	66
17.	College of Community Science, VNMKV, Parbhani	B.Sc. (Hons Community Science)	March 5, 2020	17
18.	National Institute of Plant Health Management, Hyderabad	Faculty members & scientists	March 9, 2020	24

The Academy is established in a green serene and sprawling campus of about 50 hectares. It has various facilities like Faculty Center, Academic Block, Administrative Block, Library, Medical and Health Centre, Sports Complex, Children Park, Guest houses and Trainees Hostel, Housing and Residential Quarters.

7.1 Library

The NAARM Library has 31,452 books, 11 International journals, 3 International online journals, and 6 online databases /software, i.e. EBSCO Business Source Premier, Indiastat.com, Commodities, Economic Outlook, iThenticate License, Orbit.com etc. The Library has many of its publications in digital formats like CD, VCD and its own digital repository (EPrints@NAARM). The Academy is also a member of CeRA, and has access to journals and databases through CSIRO, Springer and Open J-Gate etc. Library management has been completely digitized through the open source software “Koha”. The Academy also has a rich repository of online patent search providers, GIS software and other statistical software.



Fig. 7.1: Library at ICAR-NAARM

7.2 Health Centre

NAARM Health center provides preventive and curative treatment for the employees and their

dependents, trainees, students and pensioners and their dependents. On an average 40 patients are seen per day and medicines are provided to the patients. Emergency treatments are provided to the staff members and their dependents of other ICAR institutes. More than 750 ICAR pensioners have registered at NAARM Health center for regular OPD treatment and obtain referral services. Periodical health camps are organized at health center with investigations and specialist doctor's consultation at health center. Many camps were conducted during the year for the benefit of NAARM staff, their family members, trainees and pensioners. Health talks by the reputed specialist doctors of recognized corporate hospitals are conducted at the academy from time to time. A lady medical officer has also been appointed at Health center. Periodical Blood donation camp are organized by health center with the help of Red Cross blood bank. A Cardiologist visits health center once a month for two hours to take care of cardiac patients with free consultation.



Fig. 7.2: Health Center at Academy

7.3 Sports Complex

The sports complex at NAARM has several facilities which includes a stadium for outdoor events like cricket, football, lawn tennis and various athletic events. Indoor complex comprises of

badminton court, table tennis, billiard/snooker and gymnasium. This facility is used by the NAARM staff and trainees and with the use of this facility, NAARM sports contingent have won prizes in zonal, inter-zonal sports meet and inter-institutional sports events.



Fig. 7.3: Sports Complex at NAARM

7.4 Children's Park



Fig. 7.4: Children's park at NAARM

Children park at NAARM was established for the benefit of children of staff who resides in the campus, who spend their leisure time in the open and fresh air of the park. It was inaugurated by Dr. Trilochan Mohapatra, Secretary, DARE& Director, General, ICAR, a year ago. The children's park has several playthings for them to play with and enjoy. The facilities in the park include swings, slides, rods for exercise and many other such items that provide for children the required exercise and enjoyment benefitting both physical and mental health at tender age.

7.5 NAARM Gallery

The Gallery was established keeping in view of visits of large number of college and school

students, faculty, scientists and officials of NARES across the country on day to day basis. Some of the salient information displayed in the gallery include genesis, history and transition of the Academy, focal themes of NAARM, academic programmes offered by the Academy, challenges in agriculture sector etc. Significant emphasis on Centres of Excellence viz. a-IDEA, Technology Business Incubator and Centre for Open and Lifelong Learning in Agriculture (COLLAge) was laid to portray the activities of Academy's contribution to science and agricultural education. Taking a pledge for the welfare of the farming community; emphasizing promotion of good agricultural practices and water conservation, enhancing quality education, ensuring environmental safety, growing more trees, avoiding food wastage, reducing carbon footprint etc. is given importance to the students who visit the gallery.



Fig. 7.5: NAARM Gallery

7.6 Recycling of Solid Wastes

NAARM has adequate facilities for vermicomposting.

Vermicomposting: The leaf litter, grasses, weeds which are collected daily in the campus from the campus roads under the avenue trees, pavements and landscape areas are heaped in the old nursery area. After segregation the farm waste is used for vermicomposting in the old nursery area.

Waste cum compost pits (Wealth from Waste): Ten pits of size (3x2x1.5 m) were dug at different



Fig. 7.6: Vermicomposting at Academy

places in the campus to dispose the additional farm wastes like leaf litter, grasses, weeds etc. These vegetative residues are allowed to decompose and compost is prepared which is utilized for application to plantations.

7.7 Agro- Technology Park

Pursuant to the recommendations of the 8th Research Advisory Committee and commitments made under the XII Plan EFC, efforts have been initiated towards setting up an exclusive Technology Demonstration Park in the Academy. The main objective of this initiative is to showcase agri and other technologies of ICAR to the trainees and other visitors to create awareness and interest in diffusing them further. With its serene and sprawling campus of about 50 ha, the Academy has provisions for adopting many diverse agri and other green technologies. Realizing this,

through extensive site survey and consultations, the Academy developed a detailed plan for documenting and cataloguing the technologies, introduce new technologies with emphasis on green technologies, and develop publicity material for showcasing the activities to trainees and other visitors. Currently, 25 technologies have been documented for display and showcasing. All the technologies demonstrated have respective display boards.

7.8 Herbal Garden

A herbal garden has been established in the campus with 50 species of medicinal and aromatic plants. Display boards depicting the common and scientific names and information regarding usable plant parts and utility are installed for all the sub plots. This garden has demonstration of medicinal and aromatic plants at one place.



Fig. 7.7: Herbal garden in fledgling stage

7.9 Orchard at NAARM

The orchard crops viz. mango, sweet lime, sapota, aamla, custard apple, guava, fig, pomegranate, phalsa, and other miscellaneous fruits are grown in an area of 6.5 hectares with an objective of

demonstrating new agro-technologies to the FOCARS, trainees, PG students, visiting farmers and students. Latest agro-technologies like INM, IPM, mulching practices, micro-irrigation system etc. are practiced and demonstrated.



Fig. 7.8: Sweetlime and Amla crops in fruiting stage

7.10 Organic Farms

A *Prakruti club* has been formed by a group of PG students, faculty and officers of the Academy. The members of the club participate during their

leisure times in different cultural operations and grow vegetable crops in the Organic kitchen garden of the Halls of residence.



Fig. 7.9: Organically grown Vegetables in kitchen garden of hostel and polyhouse

7.11 Floriculture Block

It includes Rose garden, Jasmine plantation, gladiolus, seasonal flower beds cultivated in an area of 0.5 ha. Apart from adding to the beauty of the campus, it depicts the newly released HT

and Indian rose varieties and the latest agro-technologies in growing of the flower crops. It fulfils the requirements of floral arrangement and floral bouquets during various office functions.



Fig. 7.10: Improved Rose varieties grown in the Floriculture block



Fig. 7.11: Newly planted varieties of roses in the floriculture block

During this year the Rose garden extension block was established with 200 bushes of new rose varieties (45 varieties) in the fenced floriculture block. This new rose block is well established.

Ornamental foliage and flowering plants are propagated in the nursery. The potted plants are multiplied in the nursery which are used in the decoration of office buildings during functions.



Fig. 7.12: Ornamental potted plants propagated at the nursery

7.12 Avenue Trees

Several species (about 50 Nos.) of avenue trees are grown in the campus which includes predominantly flowering trees like *Gulmohur*, *Peltoforum*, *Jacaranda*, *Cassia spp.*, *Bahuniapurpurea*, *Tabubaeaspp.Spathodia*, Similarly, shade trees include *Neem*, *Raintree*, *Karanj*, *Pagoda* etc. The predominant foliage avenue trees include *Pterospermum*, *Ashoka*, *Conocarpus*, *Indian almond*, *palm trees* etc. These avenue trees add to the beauty and create an aesthetic ambience in the campus.



Fig. 7.13: Tabebuia avellanedae tree in full bloom in the campus

7.13 Rainwater Harvesting

Initiative was taken up to harvest the maximum quantity of rainwater in the lake situated at the main entrance of the Academy. Clearance of obstructions in the catchment area, diverting the water harvested from roof tops, rainwater drains etc. were implemented and the lake was full to its maximum capacity by the end of monsoon. It helped in charging the nearby bore wells during dry season by enhancing the ground water table. Along with irrigating the landscape gardens around the lake.

The lake water and the surrounding lush green lawns have been adding to the beauty of the campus and created an aesthetic ambience in the Academy. 40,000 fish seedlings of different types of fishes were released into the lake waters during this year. The fish species include *Rohu*, *Katla*, *Grass Carp* and *Common Carp*. Ducks were introduced in the lake waters and it provided ideal habitat for Fishery + Duckery integrated system.



Fig. 7.14: NAARM Lake



Fig. 7.15: Introduction of ducks in NAARM Lake

7.14 Rooftop Rain Water Harvesting

The main objective of rooftop rain water harvesting is to make water available for future use. It was taken up by constructing simple diversion channels and recharge pits near office buildings.

The rainwater collected from the rooftop of office buildings was diverted to the recharge pits which helped in recharging the groundwater aquifers.

Six such units helped in recharging the nearby bore wells from where the water was used for irrigating the landscape gardens.



Fig. 7.16: Rain water harvesting pits near office buildings at NAARM

7.15 Biodiversity Conservation

An Agri-Biodiversity Park (ABD Park) was established in the Academy in an area of about 4 acres with the aim of conserving and preserving native varieties of flora and fauna, including agricultural crops of Deccan Plateau. Important animals seen in the campus include peacock, mynah, sparrow, kingfisher, quail, crane, rabbit, mongoose etc. The theme based plantations with ecological and aesthetic values were planted specially trees, Butterfly Park, Fragrant flower

plantation, shade grass etc. Plantation of various local species like *neem*, *ficus* spp., tamarind, beal, wood apple, and medicinal plants was taken up and these plants are established well. A water harvesting pond is also created in the ABD Park. This park will serve as a show window of the flora and fauna of this area so as to create awareness and interest about the importance of bio-diversity among the farming community, trainees, students and the visitors of the Academy.

8.1 Recognitions

8.1.1 Participation in International Events

Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM participated as Government of India Delegation

Member in UNFCCC June Momentum for Climate Change in virtual mode during June 1-10, 2020.



Fig. 8.1: Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM participated as Government of India Delegation Member in UNFCCC June Momentum for Climate Change in virtual mode during June 1-10, 2020



Fig. 8.2: Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM participated in Virtual "IFPRI South Asia Discussion of the 2020 Global Food Policy Report" held on July 6, 2020



Fig. 8.3: Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM participated in Koronivia Joint Work on Agriculture meeting organized by UNFCCC secretariat in virtual mode on Sep 23, 2020



Fig. 8.4: Dr. Ch. Srinivasa Rao, Director, ICAR-NAARM participated in the UNFCCC Climate Change Dialogues Meetings held in virtual mode during Nov 2, 4, 6, 11, 23, 24, 26, 30 and December 1-4 & 7, 2020

8.1.2 Awards Received by the Faculty and staff of NAARM

Name of Staff	Name of award	Date	Purpose	Awarding body
Ch. Srinivasa Rao	Rafi Ahmed Kidwai Award	July 16, 2020	Outstanding Research in Agricultural Sciences-2019' (for Natural Resource Management & Agricultural Engineering)	ICAR, New Delhi
S. Senthil Vinayagam	Lifetime Achievement Award	July 4-5, 2020	International Scientist Awards on Engineering, Science and Medicine	VDGOOD Professional Association, Tamil Nadu
Bharat S. Sontakki	Swami Sahajanand Saraswati Outstanding Extension Scientist of ICAR-2019	July 16, 2020	Outstanding Contributions in the Area of Agricultural Extension	ICAR, New Delhi
M. Ramesh Naik	Best scientist of the year award-2020	Aug 05, 2020	Noble work done for students, teachers and entire education fraternity.	Glacier Journal Research Foundation, Ahmedabad
P. Supriya	Best Paper Oral Presentation Award	Nov 24-27, 2020	For Oral presentation in International e-conference on 'Advances and Future Outlook in Biotechnology and Crop Improvement for Sustainable Productivity'	University of Horticultural Sciences, Bagalkot



Name of Staff	Name of award	Date	Purpose	Awarding body
N. Vijayalakshmi	ICAR Cash Award	July 16, 2020	Administrative category	ICAR, New Delhi
S.N. Rasool	ICAR Cash Award	July 16, 2020	Skilled Support Staff Category	ICAR, New Delhi

NAARM Foundation Day Awards

Best Performer Awards for the year 2019-20

Under Scientist Category

Dr. Ranjit Kumar, Head, ABM Division
received Best Faculty Award

Dr. P. Krishnan, Principal Scientist, RSM Division
received Best Trainer Award

Dr. S. Senthil Vinayagam, I/C Head, ESM Division
received Best Scientist Award

Under Admin. inc. Finance Category

Officer: Zakir Hussain Khilji, CFAO

Non-Officer: R Chandra Babu, Assistant

Under Technical Category

T1- T5 Grade: ACPRN Nageswara Rao,
Technical Officer

T6 and above: KV Kumar, Chief Technical Officer

Under SSS Category

Male: S. Venkatesham, Xerox Operator

Female: Smt. I. Bharathamma, Farm Section

Under CLTS Category

Male: Shri N. Yadagiri, Farm Section

Female: Smt. M. Lakshamma, Farm Section

Awards for Best Performer of the Decade

Admin. inc. Finance Category:

Smt. T. Vanisri, Private Secretary

Under Technical Category:

Dr. P. Vijender Reddy, Chief Technical Officer

Under SSS Category:

Smt. B. Shanthamma, Library Section

Under CLTS Category:

Shri B. Raju, Farm Section

Special Category Awards

Special Recognition for Energy Conservation-

Shri K. Shivaiah, TO

Special Award for Outstanding Contribution in Institution Building-

Dr. M. Balakrishnan, Principal Scientist, ICM
Division

COVID-19 Warriors-Recognition Awards

Admin category-

Smt. Y Gayatri, UDC, Finance Section

T1- T5 category-

Shri T. Laxman, Technical Officer (Driver)

T5 and above category-

Shri M. Shekhar Reddy
Asst. Chief Technical Officer

SSS category-

Shri C. Venkatesham, CFAO Office

CLTS category-

Shri K. Shamulu, Farm Section

8.1.3 Other Recognitions Received by the Faculty of NAARM

- Dr. Ch. Srinivasa Rao as Member, Executive Board, International Dryland Development Commission (IDDC), Cairo, Egypt.
- Dr. Ch. Srinivasa Rao as Member, Asian Long Term Experimental Network for Agriculture (ALTENA), Tsukuba, Japan.
- Dr. Ch. Srinivasa Rao Awarded Reviewer Certificate for serving as Reviewer for the Journal of Land Degradation and Development by the Wiley.
- Dr. Ch. Srinivasa Rao as Member, Advisory Committee, World Bank funded ICAR-NAHEP Component-2 Project on Investment in ICAR Leadership for Agriculture Higher Education.
- Dr. Ch. Srinivasa Rao as Member, Food and Agriculture Division Council (FADC), Bureau of Indian Standards, Ministry of Consumer Affairs, Food and Public Distribution, Govt of India.
- Dr. Ch. Srinivasa Rao as Member, ICAR Committee on Agricultural Research Service.
- Dr. Ch. Srinivasa Rao as Member, Board of Management, Professor Jayashankar Telangana State Agricultural University (PJTSAU), Hyderabad, Telangana (as ICAR Nominee).
- Dr. Ch. Srinivasa Rao as Member, Academic Council, Professor Jayashankar Telangana State Agricultural University, (PJTSAU), Hyderabad, Telangana.
- Dr. Ch. Srinivasa Rao as Member, Board of Management, Acharya N.G. Ranga Agricultural University, Guntur, Andhra Pradesh (as ICAR Nominee).
- Dr. Ch. Srinivasa Rao as Member, Academic Council, Acharya N.G. Ranga Agricultural University, Guntur, Andhra Pradesh.
- Dr. Ch. Srinivasa Rao as Chairman, ICAR Committee for screening of proposals for new Agribusiness Incubation (ABI) Centres in ICAR Institutes.
- Dr. Ch. Srinivasa Rao as Member, ICT Steering Committee of ICAR.
- Dr. Ch. Srinivasa Rao as Member, NAAS Sectional Committee in Natural Resource Management.
- Dr. Ch. Srinivasa Rao as Member of the Committee to review the Career Advancement Scheme of Scientists of ICAR.
- Dr. Ch. Srinivasa Rao as Member, Selection-cum-Standing Committee for Emeritus Scientist Scheme of ICAR.
- Dr. Ch. Srinivasa Rao as Member of the Committee to revise and explore the possibility of filing APAR online.
- Dr. Ch. Srinivasa Rao as Member, ICAR Training Board, New Delhi
- Dr. Ch. Srinivasa Rao as Member of Chief Advisory Committee for Preparation of Integrated Horticulture Development Perspective Plan, Dept. of Horticulture and Sericulture, Govt. of Telangana.
- Dr. Ch. Srinivasa Rao as Member, Executive Council, National Council for Cooperative Training (NCCT), Department of Agriculture, Cooperation & Farmers Welfare, Ministry of Agriculture & Farmers' Welfare, Govt of India
- Dr. Ch. Srinivasa Rao as Member, Committee to study the payment of honorarium/ remunerations in respect of projects, consultancy and professional activities to make appropriate recommendations, National Council for Cooperative Training (NCCT), Ministry of Agriculture & Farmers' Welfare, Govt. of India.



- Dr. Ch. Srinivasa Rao as Member, Management Committee, Extension Education Institute (Southern Region), Department of Agriculture, Cooperation & Farmers Welfare, Ministry of Agriculture & Farmers' Welfare, Govt. of India.
- Dr. Ch. Srinivasa Rao as Member, Agro and Food Processing Committee of the Federation of Telangana and Andhra Pradesh Chambers of Commerce and Industry (FTAPCCI), Hyderabad
- Dr. Ch. Srinivasa Rao as Member, Indian Chamber of Food and Agriculture, New Delhi
- Dr. Ch. Srinivasa Rao as Member, Regional Advisory Group (RAG), NABARD
- Dr. Ch. Srinivasa Rao as Member, ICAR Steering committee to review and monitor the performance of National Agricultural Innovation Fund (NAIF) Scheme.
- Dr. Ch. Srinivasa Rao as Member, Advisory Committee of the Initiative for Climate Action Transparency Project on Adaptation (ICTA-A) of The Energy and Resources Institute (TERI), New Delhi. (Nominated by ICAR)
- Dr. Ch. Srinivasa Rao as Chief Editor, The Indian Journal of Dryland Agricultural Research and Development.
- Dr. Ch. Srinivasa Rao as Member, ICAR Central Joint Staff Council.
- Dr. S.K. Soam attended Consultative Group Meeting on Geographical Indications for 05 applications on 05-02-2020 at Mumbai for following products; Harnal Chilli (No. 642), Myndoli Banana (No. 680), Chinnor Rice (No. 663), Coconut Vinegar (No. 646) and Goan Khaje (No. 655).
- Dr. S.K. Soam attended Consultative Group Meeting on Geographical Indications for 04 applications on 12-02-2020 at Chennai for following products; Kuttiaattoor Mango (No. 660), Edayur Chilli-Mulaku (No. 675), Kanyakumari Cloves (No. 626), Dharwad Peda (No. 626).
- Dr. S.K. Soam chaired a session on Academia and Industry Linkage at National Conference on Geospatial Technologies in Agriculture Organized by Association for Management of Agricultural Research and Agripreneurship (AMARA) In collaboration with ICAR-National Academy of Agricultural Research Management (NAARM), at ICAR-NAARM during 20-21 February, 2020.
- Dr. S.K. Soam chaired a session on Academia and Industry Linkage at National Conference on Geospatial Technologies in Agriculture Organized by Association for Management of Agricultural Research and Agripreneurship (AMARA) In collaboration with ICAR-National Academy of Agricultural Research Management (NAARM), at ICAR-NAARM during 20-21 February, 2020.
- Dr. S.K. Soam Co-Convener for National Conference on Geospatial Technologies in Agriculture Organized by Association for Management of Agricultural Research and Agripreneurship (AMARA) In collaboration with ICAR-National Academy of Agricultural Research Management (NAARM), at ICAR-NAARM during 20-21 February, 2020.
- Dr. S.K. Soam as an External Member of 'Scrutiny Committee' for recruitment for the Post of Director at National Institute of Agricultural Extension Management (MANAGE), Hyderabad. Participated in the meeting on 9th March, 2020.
- Dr. S.K. Soam nominated as a Nodal Officer (Capacity Building), Joint Working Group of ICAR-NABARD under MoU between ICAR and NABARD.
- Dr. S.K. Soam nominated as a Chairman, TOWLIC Hindi Karyashala at Indian Institute of Oilseeds Research (IIOR), Hyderabad on 27th February, 2020.

- Dr. S.K. Soam as IPR expert member of ITMC of ICAR-IIMR, Hyderabad attended online meeting on 9th June, 2020 and taken decision on several technology licensing matters.
- Dr. S.K. Soam as a Chief Guest for the Online Review Meeting to review the progress of Zonal Technology Management Centre (ZTMC) activities in ICAR Crop Institutes in Southern Indian Zone, organized by ICAR-IIMR, Hyderabad on 15th September, 2020.
- Dr. S.K. Soam as a IPR expert member, provided advice for the 'Development of International linkages for tissue culture protocol development', in addition, the progress on earlier two patents was also discussed during the meeting of Institute Technology Management Committee (ITMC) of Indian Institute of Oil Palm Research (IIOPR), Pedavegi on 18th September, 2020.
- Dr. S.K. Soam provided expert advice during the online meeting conducted by Marketing Department, Govt. of Telangana at Horticulture Research Institute, Sri Konda Laxman Telangana State Horticultural University, Hyderabad for 'Warangal Chapata as Geographical Indication' on 18th July, 2020.
- Dr. S.K. Soam participated in the online ICAR level meeting of 'Third party evaluation of ICAR-Central Scheme' by the ISAP on 12th September, 2020 organized by Education Division of ICAR, and Chaired by Dr. RC Agrawal, DDG (Edn).
- Dr. S.K. Soam Provided expert advice to National Institute of Animal Nutrition and Physiology, Bangalore on the matter of patent decision u/s 15 for application 2954/CHE/2012 in the month of July, 2020.
- Dr. S.K. Soam as a nominated member by ICAR, attended 29th Meeting of Institute Management Committee of National Institute of Agricultural Economics & Policy Research (NIAP), New Delhi on 9th October, 2020 through virtual mode.
- Dr. S.K. Soam as an invited expert contributed in online brainstorming session on 'Agriculture and Food Policy for Five Trillion Dollar Economy', organized by NAAS, New Delhi on 14th October, 2020, Presided by Dr. Ramesh Chand, member, NITI Ayog, and Chaired by Dr. T. Mohapatra, DG, ICAR.
- Dr. S.K. Soam as 'Guest of Honour', inaugural session of the Faculty Development Programme on 'Pedagogy of Online Teaching: Philosophy and Methods' organized by Faculty Development Center, GBPUAT, Pantnagar 23 October, 2020.
- Dr. S.K. Soam participated in the Virtual Meeting on 'Third Party Evaluation of Schemes' held on 2-11-2020, held under the chairmanship of Dr. RC Agrawal, DDG (Edn), ICAR on 02-11-2020.
- Dr. S.K. Soam attended DST-CII India & Portugal Technology Summit on Strengthen Bilateral / Multilateral Trade & Investments, Driven by Science Technology & Innovation through Virtual Mode during 07-09 December, 2020.
- Dr. D. Thammi Raju was elected as Organizing Secretary for AMARA (Association for Management of Agricultural Research and Agripreneurship).
- Dr. D. Thammi Raju participated in TQIMC meeting of Center for Human Resource Development (CHRD), School of Development Studies and Social Justice, National Institute of Rural Development and Panchayat Raj, Hyderabad on 10.7.20 as an external expert.
- Dr. D. Thammi Raju acted as External Examiner for thesis viva voce of OUAT, Bhubaneswar.



- Dr. D. Thammi Raju participated Expert Member of Screening and Interview Committee for the Professor in Vety and AH Extension Education of PVNRTVU, Hyderabad.
- Dr. G. R. Ramakrishna Murthy, ICAR-NAARM is nominated as Academic council member for ANGRAU for 2 years w.e.f. Dec 28, 2020.
- Dr. Alok Kumar, as Member, Organizing Committee, International Extension Education Conference on “Role of NGOs in Extension Services: Opportunities and Challenges” (IECC-BHU 2020) organized by Banaras Hindu University, Varanasi, India during December 27-30, 2020.
- Dr. Alok Kumar, as Co-Chairman, & Convenor, Technical session III on New Dimension of NGO involvement during International Extension Education Conference on “Role of NGOs in Extension Services: Opportunities and Challenges” (IECC-BHU 2020) organized by Banaras Hindu University, Varanasi, India during December 27-30, 2020.
- Dr. P. Krishnan served as the Leader of the Working Group on Socio-economic Vulnerability for developing a Regional Framework for Coastal Vulnerability, during the Central Indian Ocean Leadership Workshop organized at MoES-National Institute of Ocean Technology (NIOT), Chennai during Jan 06-07, 2020.

8.1.4 ICAR-NAARM in Media

Press Coverage of Programmes / Events of ICAR-NAARM	Date of Media Coverage
News coverage about the inauguration of 110th Foundation Course for Agricultural Research Service (110-FOCARS) by Dr. (Smt.) TAMILISAI Soundararajan, Hon’ble Governor of Telangana State at ICAR-NAARM	Jan 8, 2020
Special media coverage was planned and executed to cover NAARM 110 th FOCARS. Arranged interviews with FOCARS trainees, concerned faculty coordinators. ETV Telangana aired this special news, under Jaikisan programme with theme based on Young agricultural scientists heading to agricultural development	Jan 21, 2020
Press coverage on MoU between NAARM and ASCI	Feb 11, 2020
News coverage about the inaugural programme of National Conference on ‘Geospatial Technologies in Agriculture’ held at ICAR-NAARM	Feb 20, 2020
News item about the ICAR Policy Paper on Re-imagining Higher Agricultural Education in India on the Face of Challenge from COVID-19 Pandemic – Strategies for Adapting to the New Normal	June 25, 2020
Press coverage of MoU between NAARM and 6 Agriculture and Veterinary Universities	July 11, 2020
Press Coverage of ICAR Awards to NAARM	July 16 & 17, 2020
News item about Inauguration of Disaster Recovery (DR) Centre in virtual mode by Hon’ble Union Minister of Agriculture & Farmers Welfare and Minister of Rural Development & Panchayati Raj, Shri Narendra Singh Tomar in presence of Hon’ble Ministers of State, Agriculture and Farmers Welfare, Shri Parshottam Rupala, Shri Kailash Choudhary and Dr. Trilochan Mohapatra, Secretary, DARE and Director General, ICAR.	Aug 11, 2020
Press coverage of inauguration of Agri-incubation Centre	Aug 27, 2020
Press coverage for NAARM foundation day	Sept 2, 2020
Press coverage of inauguration of 111 th FOCARS by World Food Prize Winner Dr. Ratan Lal	Oct 6, 2020

Oct 7, 2020

Nov 7, 2020

Nov 9, 2020

► వ్యవసాయ విద్య పరిశోధనలో నార్స్ పాత్ర కీలకం

రాజేంద్రనగర్, న్యూనటుడే: వేళంలో వ్యవసాయ విద్య పరిశోధనలో సమూల మార్పులు తెవవంలో జాతీయ వ్యవసాయ పరిశోధనా నిర్వహణ సంస్థ(నార్సీ) కీలకంగా పనిచేస్తోందని భారత వ్యవసాయ పరిశోధనా మండలి (ఎన్పీఆర్) మాజీ డైరెక్టర్ జనరల్ డాక్టర్ ఆర్ఎస్ పరోడా అన్నారు. నార్సీ 41వ వ్యవస్థాపక డిసీల్నువాన్ని పురస్కరించుకొని రాజేంద్రనగర్లోని సంస్థ ఆవరణలో జరిగిన కార్యక్రమంలో ఆయన పాల్గొన్నారు. ఈ కార్యక్రమంలో నార్సీ డైరెక్టర్ డాక్టర్ సీహెచ్ శ్రీనివాసరావు, శాస్త్రవేత్తలు పాల్గొన్నారు.

हिन्दी गिराण
नार्म ने मनाया 45वाँ स्थापना दिवस

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वार्ता मेरा हैदराबाद

राष्ट्रीय कृषि अनुसंधान प्रणाली में भाग की विशेष पहचान : डॉ. आरएस परोदा



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प्राचीन एवं अति प्राचीन
समय के विषय प्रश्नों का
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महाराष्ट्र के लोग
सिखाए गए हैं और उन्होंने
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लेखों में इसी के अन्वेषण की
एक श्रद्धापूर्वक प्रशंसा की है।
लेखकों ने इसी के अन्वेषण
की प्रशंसा की है।

॥ अथवा विचारों को जो
जहाँ वह पहुँचता है, वहाँ
जाता है। वहाँ वह
अपनी प्रकृति को व्यक्त
करता है। वहाँ वह
अपना, अपना विचार
प्रकट करता, जो ही प्रकृति
को विचार, प्रकृति को
अपनी ही प्रकृति को

వ్యవసాయంలో పాత పద్ధతులకు స్వస్తి చెప్పండి : నాబార్డు ఛైర్మన్

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8.1.5 TV/Radio Talk Delivered

Sl. No.	Name of the Person	Topic of the Talk	Date	Name of the Channel and Program
1.	Ch. Srinivasa Rao	About 89 th Annual Session of the National Academy of Sciences, India (NASI) and Symposium on Science and Technology based Entrepreneurship Development during December 21-23, 2019 at NAARM, Hyderabad	Dec 21, 2019	ETV News Channel (Telugu)
2.	Ch. Srinivasa Rao	110 th Foundation Course for Agricultural Research Service by Dr. (Smt.) TAMILISAI Soundararajan, Hon'ble Governor of Telangana State at ICAR-NAARM on Jan 7, 2020. https://www.youtube.com/watch?v=zOV9X3ceoFY ; https://www.thehansindia.com/news/cities/hyderabad/dr-tamilisai-soundararajan-calls-for-innovative-practices-in-agricultural-sector-595748 ; https://www.youtube.com/watch?v=JpNau-oLz-o	Jan 07, 2020	V6 Telugu News Channel Hans News - ETV Telangana Telugu news channel
3.	Ch. Srinivasa Rao	National Conference on 'Geospatial Technologies in Agriculture' held at ICAR-NAARM on Feb 20, 2020. https://www.youtube.com/watch?v=OtBy5Ilp1XE	Feb 20, 2020	ETV Telangana channel; DD Yadagiri News channel (Detailed coverage telecast on 'Naa Koti Ratnala Maagani' (Agriculture program) on Feb 23, 2020.
4.	Ch. Srinivasa Rao	Interview taken on the sidelines of Mega Event on S&T Interventions for the welfare of Tribal Population, organized by NASI, held at Institute of Life Sciences, Bhubaneshwar, during Feb 24-25, 2020. https://youtu.be/Py05FeOh1O0	Feb 24, 2020	Watch Disha Exclusive Interview With Dr. Ch. Srinivasa Rao YouTube
5.	Ch. Srinivasa Rao	Inauguration of Disaster Recovery (DR) Centre by Hon'ble Union Minister of Agriculture & Farmers Welfare and Minister of Rural Development & Panchayati Raj, Shri Narendra Singh Tomar ji in the presence of Hon'ble Ministers of State, Agriculture and Farmers Welfare, Shri Parshottam Rupala ji, Shri Kailash Choudhary ji and Dr. Trilochan Mohapatra, Secretary, DARE and Director General, ICAR	Sep 11, 2020	ETV Telugu channel on 12.08.2020



Sl. No.	Name of the Person	Topic of the Talk	Date	Name of the Channel and Program
6.	Ch. Srinivasa Rao	Inauguration of 111 th Foundation Course for the Agricultural Research Service (FOCARS) Probationers at ICAR-NAARM by Prof. Rattan Lal, World Food Prize Laureate and Director, Carbon Management and Sequestration Center, The Ohio State University, USA on 05 October 2020 1) https://www.etvbharat.com/telugu/telangana/state/hyderabad/world-food-prizewinner-dr-ratan-lal-on-agriculture/ts20201006092537607 2) https://youtu.be/asQfl6-1v9A	Oct 05, 2020	ETV Telugu channel
7.	Ch. Srinivasa Rao	120 th Birth Celebrations of Shri Acharya N.G. Ranga, Padma Vibhushan Awardee was organized by NAARM, Hyderabad on November 7, 2020 in virtual mode. Hon'ble Vice President of India, Shri Muppavarapu Venkaiah Naidu garu gave Presidential Address on the occasion. Keynote address was given by Shri Govinda Rajulu Chintala, Chairman, NABARD. Dr. Ancha Srinivasan, Climate Change Advisor, South East Asia, Asian Development Bank, Thailand was Guest of Honour. https://www.youtube.com/watch?v=RKDD3a_pyZY	Nov 07, 2020	ETV Telugu channel

8.1.6 Invited Lectures Delivered by the Faculty and Staff

Lecture	Programme/ Organized by
Ch Srinivasa Rao	
Science and Technology Innovations for enhanced livelihoods of Tribal Population in India	NASI-ILS Collaborative Mega Event on 'S&T Interventions for the welfare of tribal population' at Institute of Live Sciences (ILS), Bhubaneswar on Feb 25 & 25, 2020
Restarting Agriculture Post COVID and Gearing for <i>Kharif</i> , 2020: Challenges and Way Forward	Virtual Session on 'Restarting Agriculture Post Covid & Gearing up for <i>Kharif</i> ' organised by Confederation of Indian Industry, Telangana (Online) on May 02, 2020
Science Communication - Myths or any related topic as first expert speaker	Webinar on 'Breaking Myths of Science Communication' under NAHEP, Organized by College of Dairy Science, Amreli, Gujarat (Online) on May 04, 2020
Women-friendly Agriculture Policy Development	Webinar on 'Effective Policy Paper Writing for Women in Agriculture' Organized by ICAR-CIWA, Bhubaneswar (Online) on May 27, 2020
Reforms needed in Agri Higher Education in Post Covid Era	National Webinar on "Challenges, Opportunities and Future of Agri and Allied Research and Education: Post Covid Era" on Topic: " organized by All India Agricultural Students Association (Online) on May 30, 2020
ICTs for Agribusiness Management	National Webinar on "Post Covid-19 Agribusiness: Challenges and Opportunities"-Organised by JAU, Junagadh (Online) on Jun 14, 2020

Lecture	Programme/ Organized by
Biochar in Dryland Agro-ecosystem for Climate Change Mitigation and Sustainability	Webinar on 'Biochar: Potential availability, Usefulness and Limitation for use in Indian Agriculture', organized by ICAR-IISS, Bhopal (Online) on Jun 19, 2020
Opportunities and Challenges in Ag. Education	Online Training Programme on "Teaching Competency Enhancements Programme for Faculty of SKLTSHU, Hyderabad (Online) on Jul 07, 2020
Effect of Climate Change on Agriculture	Webinar on "Climate Resilience Agriculture - Opportunities and potential" organised by Directorate of Extension Services, RVSKVV, Gwalior (Online) on Jul 07, 2020
Resource Conservation Technologies for Sustainability of Agriculture and Environment	National Web Conference on Technological Approaches for Resource Conservation and Management for Environmental Sustainability (TARCMES) being organized by the Academy of Natural Resource Conservation and Management (ANRCM), Lucknow on the occasion of Foundation day of ANRCM (Online) on Aug 16, 2020
Sustainable Agriculture Practices for Food Security and Zero Hunger in India	Training Programme on Achieving Zero Hunger by 2030: Critical role of agriculture and allied sectors by SKUAST-K Srinagar (Online) on Aug 18, 2020
Overview of Agricultural Education	Online Training Program on E-Resource, E-Learning and IoT for faculty of UAS, Raichur (Online) on Aug 25, 2020
Climate Change and mitigation measures	Webinar on "Oceans in the wake of climate change: challenges and solution" being organised by Department of Aquatic Environment Management, FCRI, Thoothukudi (Online) on Aug 26, 2020
Government initiatives and strategies for Nutrisensitive Agriculture	National Webinar on "Nutrisensitive Agriculture: A. way to empower India for a healthier future" organized by College of Community Science, CAU, Tura with CIWA, Bhubaneswar (Online) on Sep 29, 2020
Climate-Resilient Resource Conservation Technologies for Dryland and Rainfed Ecosystems: Challenges and Opportunities	Vashwik Bhartiya Vignanik (VAIBHAV) Summit 2020. - a PMO's initiative to seek possibilities of future collaborations with leading scientists working in India and those of Indian origin placed abroad (Online) on Oct 10, 2020
Innovative Approaches towards Climate-Smart Agriculture	Online Innovative Lecture Series-2020 organised by Parbhani Chapter of Indian Society of Soil Science, Vasantrao Naik Marathwada Krishi Vidyapeeth, Parbhani (Online) on Oct 10, 2020
Innovations in agricultural technology for climate resilient agriculture	Webinar on "Agriculture and Food Technologies and Mechanisms of adaptation to Climate Change" organized by CSIR-National Botanical Research Institute, Lucknow in collaboration with Department for International Cooperation of the Russian Academy of Sciences (Online) on Nov 11, 2020
Protecting Soil and Biodiversity for Sustainable Food Production and Climate Action in India	Webinar on 'Keep soil alive - Protect soil biodiversity' organized by the Department of Soil Science and Agricultural Chemistry, Agricultural College, Bapatla on the theme of on the occasion of World Soil Day on 5th December, 2020 (Online) on Dec 03, 2020
Programmes and Policies for Improving Nutrient Use Efficiency in Indian Agriculture in Session II - Innovation in Fertilizer Management	FAI Annual Seminar 2020 : Fertiliser and Agriculture during Covid-19 - organised by FAI, New Delhi held during December 7-9, 2020 (Online) on Dec 08, 2020



Lecture	Programme/ Organized by
Conservation Agriculture in Rice : Challenges and Opportunities	1 st Indian Rice Congress-2020 - Rice Research and Development for Achieving Sustainable Development Goals organised by Association of Rice Research Workers, Cuttack, Odisha in collaboration with ICAR, ICAR-NRRI, ICAR-IIRR, ICAR-IARI, OUAT, Society for Advancement of Rice Research and IRRI South Asia Regional Centre, Varanasi held during December 8-9, 2020 (Online) on Dec 09, 2020
S K Soam	
Research project formulation and outcome management	Online National Workshop 'Research Methodology: Concepts and Applications' conducted by MPUAT, Udaipur during Jun 26-27 2020
National Workshop on Footprints of Ideal Research Methodology	Nanaji Deshmukh Veterinary Science University, Jabalpur, Madhya Pradesh on Aug 11, 2020
Geographical Indicators and its Implications for Indian Farmers'	Training Programme on 'WTO and its Implications on Indian Agriculture' at MANAGE, Hyderabad on Oct 21, 2020
Agriculture and Trade Related Intellectual Property Rights	Training Programme on 'WTO and its Implications on Indian Agriculture' at MANAGE, Hyderabad on Oct 21, 2020
IPRs and GIs in the Export of Animal Husbandry Products	Training Programme on 'Promotion of Export Management' Extension Education India, Hyderabad on Dec 08, 2020
GI for Community Development	National workshop on Intellectual Property Management in Agriculture, organized by Institute Technology Management Unit of ICAR - Indian Institute of Agricultural Biotechnology, Ranchi, Jharkhand on Nov 28, 2020
A. Dhandapani	
Data visualization using R and Multivariate Analysis	Workshop on statistical analysis for PG and Ph.D. Students, TNAU, Coimbatore during Feb 28-29, 202
Random Models	Ph. D./ P.G. Class for ICAR-CIFE, Mumbai (Online) during Apr. 29-30, 2020
Application of R Software in correlation and regression analysis	Application of R Software for Data Analysis in Agricultural and Life Sciences College of Fisheries, CAU, Imphal (Online) on Aug 24, 2020
Design of Experiments in Fisheries and Aquaculture	Data Analytics in Fisheries Dr. J. Jayalalithaa Fisheries University (TNJFU), Nagapattinam, TN (Online) on Aug 14, 2020
Sampling Techniques	Research Methodology for Social Sciences AAU, Anand under NAHEP (Online) on Sep 02, 2020
Methodological Errors in Agricultural Research	Webinar on "Diagnostics and Remedial Measures for common error in application of statistics, College of Agriculture, Bharuch, Navsari Agricultural University (Online) on Oct 20, 2020
Significance of e-data repositories and their management, policy and publication issues in relevance to Agriculture under e-Governance	Importance of e-data bookkeeping under e-Governance at UAS Dharwad (Online) on Nov 11, 2020
Alok Kumar	
Human Resource Management for Agricultural Transformation	National Webinar on Role of agricultural sector in Resilience of Indian economy amid Covid-19 pandemic at BAU, Sabour (Online) on Sep 25, 2020
B. Ganesh Kumar	
Fisheries Marketing in COVID Period: Innovations and Good Practices	Webinar Series on 'Institutionalizing COVID Period Innovations in Agricultural Marketing by CCS NIAM, Jaipur on May 20, 2020

Lecture	Programme/ Organized by
Supply Chain Management Challenges & Opportunities in Fisheries Sector– Post Pandemic Era	National Level Webinar on ‘Challenges and Opportunities in by Agriculture - A Multidisciplinary Approach’ by Vanavarayar Institute of Agriculture, TNAU, Pollachi on June 14, 2020
Impact of COVID-19 on Fish Marketing	NAHEP Webinar on ‘Fishery Business Eco-system in India’ by AAU, Anand on June 22, 2020
(i) Backward and Forward Linking Farmers to Markets – Harvesting Benefits for Farmers & (ii) Developmental programmes meant for Marketing in Animal Husbandry Sector	Online training programme on ‘Market-led Extension in Animal Husbandry Sector’ by EEI, Hyderabad on Aug 4 & 6, 2020
Integrating Livestock Farmers with Market	MOOCS training programme on ‘Innovative Ideas for Entrepreneurship Development in Livestock Sector’ by MANAGE, Hyderabad on Sep 21, 2020
Value Chain Management Strategies in Livestock and Poultry for Doubling Farmers’ Income	National Webinar on ‘Doubling the Farmers’ Income through Livestock Sector – Prospects and Potential’ by TANUVAS, Chennai on Oct 01, 2020
Livestock based Farming Systems: Adaptation and Mitigation of Climate Change	Online training programme on ‘Climate Resilient Agriculture and Rural Livelihoods’ by NIRD & PR, NERC, Guwahati on Oct 14, 2020
Livestock, Small ruminant and dairy farming	E-training on ‘Community-based Approaches for Sustainable Livelihoods and Climate Resilient Adaptation Practices – Rural Livelihoods’ by NIRD & PR, Hyderabad on Jun 05, Aug 27, Nov 12 & Dec 17, 2020
Marketing Opportunities for Fish in India	Online training programme on ‘Recent Advances in Agriculture Extension and Farm Management’ by SAMETI, Puducherry on Nov 25, 2020
B.S. Sontakki	
Student-centered Teaching and Learning	Foundation Course on Research, Education and Extension for the Newly Recruited Assistant Professors of Kerala Agricultural University, Thrissur on Jan 13, 2020
Student Performance Evaluation	
National Agricultural Extension Policy: Post-Independence Experiences in India	41st Feed the Future - India Triangular Training (FTF-ITT) Program “Agriculture Policy Formulation, Review and Analysis” (21 Jan. to 4 Feb. 2020) MANAGE, Hyderabad on Jan 28, 2020
Student-centered Teaching and Learning	Foundation Course on Research, Education and Extension for the Newly Recruited Assistant Professors at Kerala Agricultural University, Thrissur on Mar 03, 2020
Communication - Key skill in understanding one another	Training Program on “Harmony, Balance & Synergy for Enhancing the performance of the Human Resources University of Agricultural Sciences, Dharwad (Karnataka) on Mar 06, 2020
Organizational climate for fostering effectiveness	Training Program on “Harmony, Balance & Synergy for Enhancing the performance of the Human Resources University of Agricultural Sciences, Dharwad (Karnataka) on Mar 09, 2020
Change management – Keeping up with the times	



Lecture	Programme/ Organized by
Emerging trends in Social Sciences Research with specific reference to research in agricultural and rural extension	International Web series- 2020 on Emerging Trends in Extension & Social Sciences research at Central Agricultural University, Imphal, ICAR - CTRI, Kasaragod & ICAR-NAARM, Hyderabad (Online) Lembucherra - (Tripura) on Jun 06, 2020
Capacity Development of Extension Professionals: Competency-based Approach	International Webinar on Core Competencies for Extension Professionals MANAGE, Hyderabad and USAID Online on July 17, 2020
Scientific writing	Webinar on Technical Writing at Anand Agricultural University, Anand (Online) on Oct 16, 2020
Communication skills for networking and negotiation	Webinar on Effective Networking and Negotiation Skills for Agripreneurs at University of Agricultural Sciences, Dharwad (Karnataka) on Nov 17, 2020
D. Thammi Raju	
Teaching Management Digital interventions in Agricultural Education	Training Workshop on “Contemporary Education Technologies for Agricultural Education” ANGRAU Faculty during June 8-12, 2020.
Soft skills for Dairy Professionals’	Classes for Socially Deprived Students by Eminent Experts, organized by College of Dairy Technology, Kamareddy, Telanagana on Feb 17, 2020
Management of Extension Events of KVKs and Approaches for reaching the Unreached Farmers	Training Programme on Faculty Development Programme for KVKs of Southern States, organized by EEI, Rajendranagar, Hyderabad on Feb 28, 2020
Digital Innovations in Agriculture Education and Extension & Oral Communication	E-Resources, E-Learning and IoT for Students and Staff organized by UAS, Raichur on Feb 26, 2020
Digital Education Initiatives in Education and Extension	Student Programme on Soft skills by SKUAST through online on Aug 22, 2020
Digital Education Initiatives in Education and Extension	Online Training Programme on “E-Resources, E-Learning and IoT” for Faculty of UAS, Raichur on Aug 24, 2020
Teaching Management	Online Training Programme on “E-Resources, E-Learning and IoT” for Faculty of UAS, Raichur on Aug 27, 2020
Digital Education Initiatives in Education and Extension	Training Programme on ‘E-Resources, E-Learning and IoT’ for Students of UAS, Raichur (October 05 – 09, 2020) on Oct 05, 2020
Soft skills for Agri Professionals	Training Programme on “E-Resources, E-Learning and IoT” for Students of UAS, Raichur (October 05-09, 2020) on Oct 10, 2020
G.R. Ramakrishna Murthy	
Innovations in E Learning- MOOCs and OER	Virtual Training Course on E-Learning Ecologies: Tools and Techniques for Enhancing Teaching and Training Competencies’ by College of fisheries, CAU, Imphal on May 28, 2020
Blog creation	Training Programme on Google for Education by College of fisheries, CAU, Imphal on Aug 11, 2020
Cutting edge technologies in fisheries sector	Training Programme on Data Analytics in Fisheries by TNJFU, Thalainayeru on Sep 02, 2020
Digital Agriculture	Training Programme on Design thinking in rural applications conducted by IIITDM, Kurnool on Oct 19, 2020
Overview of conduct of MOOCs	Virtual Training Programme on Innovative Training Methods for Effective Extension Delivery by EEI, Hyderabad on Nov 25, 2020

Lecture	Programme/ Organized by
K.H. Rao	
Learning Strategies and Methods during Covid Period	Indian Agricultural Education System and Entrepreneurship Scope in 21 st century during Aug 05-14, 2020 at Dr. PDKV, Akola (Online) on Aug 5, 2020
M. Balakrishnan	
Emerging trends in data science and its applications	Workshop (Webinar) Department of Physical Sciences, Kasturba Gandhi PG College for Women, Marred Pally Secundrabad on May 23, 2020
Scope of Artificial Intelligence Research in Agriculture	Online HRD training programme for the faculty members of Kerala Agricultural University Central Training Institute (CTI), Kerala Agricultural University, Mannuthy, Thrissur KAU, Kerala On Sep 24, 2020
Manju Gerard	
Health and Environmental Protection during Pandemic Outbreak of COVID-19	Department of Chemistry, SHUATS, Prayagraj, June 25-26, 2020
Identification of New Dimensions for Preparing National/Global Level Database on Women in Agriculture	ICAR-CIWA on Aug 28, 2020
Advances in Agrochemicals and Possible Green Solutions	Lucknow Christian College during July 29-30, 2020
N. Sivaramane	
Farmer Producer Company in India	SAMETI, Puducherry on Recent Advances in Agriculture Extension and Farm Management" on Nov 12, 2020
P D Sreekanth	
Introduction to GIS components, data structures and errors	Training Programme on Geoinformatics and its Applications by Water Technology Centre, PJTSAU, Hyderabad on Jan 31, 2020
Remote Sensing , GIS and GPS with Spatial Focus on Marketing and Insurance in Agriculture and Allied Sectors	Virtual Training Programme on Digital Applications for Promotion of Marketing in Agriculture and Allied Sectors by Extension Education Institute (EEI), Hyderabad on Aug 27, 2020
P. Krishnan	
Research Project Conceptualization and Science Communication	Foundation Course on Research, Education and Extension Management for the newly recruited scientists of KAU at Directorate of Extension, Kerala Agricultural University Thrissur, Kerala (Web Conferencing) on Jan 18, Feb 14, & Mar 04, 2020
Breaking Myths in Science Communication	NAHEP Sponsored 14-days Virtual Training on, "Science Communication (Sci-Com) for Smart Scientists", organized by ICAR-Central Institute of Fisheries Education (CIFE), Mumbai on May 09, 14, & 26, 2020
Science Communication Tips for Social Scientists Plagiarism – An Overview	NAHEP Sponsored Virtual Soft Skill Development Program on Scientific Writing in Social Sciences at University of agricultural Sciences (UAS), Dharwad on Jul 06, 2020
Evidence-based Conservation of Coastal Bio-resources: Recent Trends	Webinar on Conservation and management of marine fisheries resources at Tamil Nadu J. Jayalalitha Fisheries University, Nagapattinam on Jul 28, 2020



Lecture	Programme/ Organized by
Implementation of Nagoya Protocol on ABS in the context of Agricultural Sector	Virtual Training Programme on Access and Benefit Sharing (ABS) in the context of Agriculture and Allied Sectors for the State Biodiversity Boards at National Biodiversity Authority (NBA), Chennai on Aug 21, 2020
Breaking Myths in Science Communication	Online Training Programme on “E-Resources, E-Learning and IoT” for Faculty of UAS, Raichur at ICAR-NAARM, Hyderabad on Aug 27, 2020
Writing Research Papers Breaking Myths & Building Right Perspectives	Online Training Program on Data Analytics in Fisheries at Tamil Nadu J. Jayalalitha Fisheries University, Nagapattinam on Sep 02, 2020
Making a Career Choice: From Confusion to Confidence	Online Training Programme on “E-Resources, E-Learning and IoT” at UAS, Raichur (Online) on Oct 05-09, 2020
Research Publications Breaking Myths & Building Right Perspectives	National e-Training on Research Ethics and Research Paper Writing Skills development at C.S. Azad University of Agriculture and Technology, Kanpur on Nov 27, 2020
Building Right Perspectives for Winning Grants / Doing Research	e - Workshop on Role of TNJFU Women professionals in Fisheries and Aquaculture Development of Tamil Nadu at Tamil Nadu J. Jayalalitha Fisheries University, Nagapattinam on Dec 03, 2020
P. Venkatesan	
FPOs Scope and Opportunity for Market led Extension	Strategies for market led extension and COVID-19 pandemic in Agriculture-for KVK & DAATTCs and officers of State Agriculture Department during Jun 8-11, 2020 at Extension Education Institute Hyderabad on Jun 09, 2020
FPOs Scope and Opportunity for Market led Extension	Virtual Training programme on FPOs for officers of Horticulture Dept. during Jun 23-26, 2020 at Extension Education Institute Hyderabad on Jun 25, 2020
Innovations in Agricultural Extension and Technology Dissemination, Emerging models and approaches in KVKs	Faculty Development Programme for KVKs of Southern Zone during Feb 25-29, 2020 by Extension Education Institute and MANAGE Hyderabad on Feb 28, 2020
Scale development and testing in social sciences	Statistical and Analytical Skills in Social Science and Agribusiness during Jul 3-9, 2020 at Department of Extension Education, PJTSAU Hyderabad on July 06, 2020
Panelist & Co-chairman in experiences of middle & senior level extension professionals in performing EAS during pandemic COVID-19	Dealing Extension Advisory Services during extended COVID-19 pandemic period KVK-Khordha CIFA, OUAT and Reliance Foundation, Bhubaneswar at Bhubaneswar July 24, 2020
Orientation to PRA, Village Transact, Resource Mapping, Agro Ecology Map, Livelihood Analysis & ITK	PRA Tools & Techniques and Micro planning during Sep 3-5, 2020 at Central Academy for State Forest Service, Coimbatore
Social Map, Seasonal Calendar, Venn Diagram, Timeline and Time Trend	PRA Tools & Techniques and Micro planning during Sep 3-5, 2020 at Central Academy for State Forest Service, Coimbatore on Sep 04, 2020
Technology Map, Matrix Ranking, Consequence Diagram, Micro Planning	PRA Tools & Techniques and Micro planning during Sep 3-5, 2020 at Central Academy for State Forest Service, Coimbatore on Sep 05, 2020
Presentation Skills	FDP to meet emerging Academic Demands during Dec 01-21, 2020 at Musiri Institute of Technology, Trichy, on Dec 02, 2020

Lecture	Programme/ Organized by
Effective Training Skills	Training of Trainers during Dec 7-12, 2020 at Central Academy for State Forest Service, Coimbatore on Dec 7, 2020
Training need analysis	Training of Trainers during Dec 7-12, 2020 at Central Academy for State Forest Service, Coimbatore on Dec 9, 2020
Effective Communication Skills	Training of Trainers during Dec 7-12, 2020 at Central Academy for State Forest Service, Coimbatore on Dec 11, 2020
R.V.S. Rao	
Stress management	Health training to institute staff during Covid19 by ICAR institutes, ICAR-DGR, Junagadh (Online) on Sep 07, 2020
Positive Thinking and Motivation	Health training to institute staff during Covid19 by ICAR institutes ICAR-DGR, Junagadh (Online) on Sep 07, 2020
Change Management	General training of KAU Teachers Central Training Institute, Kerala Agricultural University (Online) on Oct 23, 2020
Team Building	21 days programme on Overall Personality Development of Teachers to meet the emerging academic demands Musiri Institute of Technology, Vellalapatti, Trichy District, Tamil Nadu (Online) on Dec 01, 2020
Ranjit Kumar	
e-NAM and Recent Advances in Agricultural Marketing	Webinar under NAHEP-CAAST by KAU, Thrissur on Jul 06, 2020
Essentials of Agripreneurship: Elements, scouting business ideas and identifying customers	Webinar under IDP-NAHEP by CAU, Imphal on Jul 07, 2020
Introduction to Business Plan and Goal Setting	Webinar under NAHEP by Dr. PDKV, Akola on Aug 06, 2020
Essentials of Marketing: STP and 4 Ps	Webinar under IDP-NAHEP by CAU, Imphal on Aug 11, 2020
Developing Entrepreneurial Skill among Agri-Graduates	Webinar Series: Education for Agriculture- Making a Difference by NMIMS-School of Agricultural Sciences & Technology on Aug 21, 2020
Seed Entrepreneurship Development: Small Scale Business Modeling	Webinar under CAAST on Nutritional Crops by CSAUA&T, Kanpur on Aug 26, 2020
Developing Business Plan for Women FPO	National Webinar on Women FPOs by MPUAT, Udaipur on Sep 10, 2020
e-NAM- Potential for Better Price Discovery for Smallholders	International Online Training on "Agriculture Value Chain Analysis for Strengthening Smallholders" by IIM, Jammu on Sep 15, 2020
Agri-Tech Startups for Empowering Smallholders	International Online Training on "Agriculture Value Chain Analysis for Strengthening Smallholders" by IIM, Jammu on Sep 16, 2020
Changing Paradigm of Agricultural Marketing under New Market Reform	National Webinar on Role of Agricultural Sector in Resilience of Indian Economy by BAU, Sabour on Sep 25, 2020
Essentials of Agripreneurship	Webinar under Agri-Business Incubator Project by ICAR Research Complex for Eastern Region, FSRC for Hill & Plateau Region, Ranchi on May 27, 2020
Tackling dairy sector amid COVID times	Lecture Series by a-IDEA, NAARM on May 23, 2020
eNAM: Preparedness, Opportunities and Challenges	Webinar on "eNAM: Challenges and Prospects" under NAHEP-CAAST by Anand Agricultural University, Anand on Jun 10, 2020
Farmer buying behaviour for inputs	Marketing of Agricultural Inputs by MANAGE, Hyderabad on Feb 20, 2020



Lecture	Programme/ Organized by
Writing Research Articles for High Impact Journals	Online National Workshop on Fundamental Concepts and Applications of Research Methodology by Bihar Agricultural University, Sabour (Online) on Oct 06, 2020
Emerging Supply chains – A case of eNAM	Online training program on “Supply Chain Management of Agricultural Produce” by MANAGE, Hyderabad on Dec 11, 2020
Entrepreneurship of Fruits & Agricultural Products	Indian Institute of Packaging (IIP), Hyderabad Centre (Online) on Oct 14, 2020
Opportunities in Agri-Business Management	Rama University, Kanpur (Online) on Dec 18, 2020
S. Ravichandran	
Introduction to big data in fisheries	Online Training Programme on “Data Analytics in Fisheries” organized by TNFU, Nagapattinam Hyderabad on Sep 01, 2020
Chairman of a technical session and delivered a lead lecture on “Covid-19 and its impact on major global Economies”	Global Conference on ‘Emerging Agricultural Research to Endure the Predicament of COVID-19 Pandemic’ on Dec 12-13, 2020 organized by Agricultural Economics and Social Science Research Association (AESSRA), New Delhi.
S. Senthil Vinayagam	
Entrepreneurship & Start Ups	Off campus Training Programme on “E-Resources, E-Learning and IoT” by UAS, Raichur on Feb 2020
Core competencies in research, teaching and technology transfer	Foundation Course on Research, Education and Extension Management for faculty by Kerala Agricultural University Trichur on Mar 06, 2020
Communication Skills	Contemporary Education Technologies for Agricultural Education” for ANGRAU Faculty on Jun 08, 2020
Competency enhancement in digital teaching	Contemporary Education Technologies for Agricultural Education” for ANGRAU Faculty on Jun 08, 2020
Project Appraisal Analysis	National Training Course (NTC-2020) by National Agricultural Development Cooperative limited during June 10, 2020
Communication Skills	Online Training Programme on Teaching Competency Enhancement for Faculty of Sri Konda Laxman Telangana State Horticultural University by SKLTSHU on 07 Jul, 2020
Orientation to Govt. of India Agriculture and Allied Sectors Schemes and Programs	Virtual training programme on Farm Journalism for Effective Transfer of Technology by EEI, Rajendranagar on Aug 18, 2020
Extension Strategies for enhancing market access	Online Training Program on ‘Market Led Extension for Linking Farmers to Market’ by MANAGE, Hyderabad on 20 Aug 2020
Oral Communication	Online Training Programme on “E Resources, E-Learning and IoT” for Staff of UAS, Raichur by UASR on 24 Aug 2020
Sustainable Issues in Agriculture and Food Security	21 days National Training Programme (NTP 2020) by ICAR-IGFRI and National Agriculture Development Cooperative Ltd on Aug 25, 2020
Application of ICTs in Agriculture and Agricultural Extension	Training Program on “New Dimensions in ICTs and Knowledge Management in Agriculture” organized by MANAGE, Hyderabad on Aug 31, 2020
Teaching Learning Strategies of Online Education	One week online programme on Teaching & Learning Strategies for Online Education (TLSOE-2020) organized by Nalla Narasimha Reddy Education Society’s Group of Institutions, Hyderabad on Sep 29, 2020

Lecture	Programme/ Organized by
Application of ICTs in Agriculture and Agricultural Extension	New Dimensions in ICTs and Knowledge Management in Agriculture for UAS Raichur by MANAGE, Hyderabad on Oct 19, 2020.
Innovations in Educational Technology	Online HRD training programme for the faculty members of Kerala Agricultural University by Kerala Agricultural University, Kerala on Oct 22, 2020
Recent Innovations in Educational Technology	Five days online training programme on Next Generation Extension Tools and Techniques for Upcoming Agricultural Professionals under the aegis of NAHEP-IDP by Junagadh Agricultural University, Junagadh on Oct 23, 2020
Competency enhancement for Digital Teaching	Faculty development programme on the focal theme over all Personality Development of Teachers to Meet the Emerging Academic Demands by Musiri Institute of Technology, Trichy on Dec 03, 2020
E-Learning and OER	Faculty development programme on the focal theme over all Personality Development of Teachers to Meet the Emerging Academic Demands by Musiri Institute of Technology, Trichy on Dec 04, 2020
National Education Policy	e-Workshop on "Role of TNJFU Women professionals in Fisheries and Aquaculture Development of Tamil Nadu by TNJFU, Tamil Nadu on Dec 05, 2020
Competency Enhancement for Digital Extension	Training on Extension Management organized by CCSHAU, Hisar on Dec 21, 2020
Sanjiv Kumar	
ICT enabled marketing in Animal Husbandry Sector	Virtual training programme on Market Led Extension in Animal Husbandry Sector by EEI, PJTSAU, Hyderabad on Aug 05, 2020
Financial and Marketing Management aspects of Entrepreneurship Development	National e-Training on Indian Agricultural Education System and Entrepreneurship Scope in 21st Century by PDKV, Akola on Aug 07, 2020
Principles and Strategies of Agribusiness Management	MOOC Training on Innovative Ideas for Entrepreneurship Development in Livestock Sector by MANAGE, Hyderabad on Sep 21, 2020
Entrepreneurship opportunities in Dairy Sector	Webinar under IDP-NAHEP by GADVASU, Ludhiana on Nov 11, 2020
Digitalization and ICT applications in FPOs in Animal Husbandry	Virtual Training Program on Promotion of FPOs for Profitable Farming for the officers of Animal Husbandry Sector by Extension Education Institute, Rajendranagar on Oct 21, 2020
FDA/AGMARK Rules and Regulations	Packaging laws and regulations for Post Graduate Diploma in Packaging by Indian Institute of Packaging, Hyderabad on Nov 19, 2020
Marketing of Agricultural Inputs	Vishwa Vishwani Institute of Systems and Management, Hyderabad during July-Sep, 2020
Surya Rathore	
Building Startup Ecosystem in SAUs	Winter School on Entrepreneurial Skill Development of Rural Youth through Innovative Approaches at Maharana Pratap University of Agriculture & Technology, Udaipur (Rajasthan) on Jan 08, 2020
Innovative teaching methodologies	Winter School on Entrepreneurial Skill Development of Rural Youth through Innovative Approaches at Maharana Pratap University of Agriculture & Technology, Udaipur Udaipur (Rajasthan) on Jan 09, 2020



Lecture	Programme/ Organized by
Soft Skills for Employability generation among Agri-graduates	B. Sc. Agriculture Rural Agricultural Work Experience (RAWE) programme at Govind Ballabh Pant University of Agriculture & Technology, Pantnagar (Uttarakhand) on Jan 25, 2020
Communication Skills for Career Development	Development of Soft Skills for Entrepreneurship among Agri Graduates (NAHEP) of University of Agricultural Sciences, Bangalore (Karnataka) on Feb 12 & 13, 2020
Communication Skills for Career Development	Development of Soft Skills for Entrepreneurship among Agri Graduates (NAHEP) of Professor Jai Shankar Telangana State Agricultural University, Hyderabad (Telangana) on Jan 31, 2020
Impact Evaluation in Extension & Social Sciences	International Web series- 2020 on Emerging Trends in Extension & Social Sciences research during Jun 10-16, 2020 Central Agricultural University, Imphal, ICAR – CTRI, Kasaragod & ICAR – NAARM, Hyderabad (Online) Lembucherra- (Tripura) on Jun 16, 2020
Towards gender mainstreaming: Strategies & Issues	National workshop on Gender Issues & Atmanirbhar Bharat during Oct 15-17, 2020 (IDP - NAHEP) at Central Agricultural University, Imphal, (Online) on Oct 17, 2020
Prospects and Issues of Empowerment among women FPOs	Mahila Kisan Diwas: Webinar on Empowerment of Women in Agriculture: The Roads Ahead at MANAGE, Hyderabad (Online) Hyderabad (Telangana) on Oct 15, 2020
Gender analysis tools	Gender analysis skills for agriculture & allied sectors during Nov 17-18, 2020 (IDP - NAHEP) Central Agricultural University, Imphal, (Online) on Nov 11, 2020
Vocational Scope of agricultural sciences	Celebration of Agriculture education day at Central Agricultural University, Imphal, (Online) College of Community Sciences, Tura (Meghalaya) on Dec 03, 2020

8.2 Publications

8.2.1 Research Papers in Refereed Journals

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- Anand, A., Krishnan, P., Suryavanshi, A.S., Choudhury, S.B., Kantharajan, G., Srinivasarao, Ch., Manjulatha, C., Babu, D.E. 2020. Identification of Suitable Aquaculture Sites Using Large-Scale Land Use and Land Cover Maps Factoring the Prevailing Regulatory Frameworks: A Case Study from India, *Journal of the Indian Society of Remote Sensing*, <https://doi.org/10.1007/s12524-020-01211-7>.
- Balakrishnan M., Venkatesan, P. Soam S.K., and Srinivasa Rao Ch. (2020). Designing of Database for Indigenous Technical Knowledge in Agriculture. *International Journal of Current Research*. 12(7): 12181-12184. DOI: <https://doi.org/10.24941/ijcr.39043.07.2020>.
- Balakrishnan, M and Meena, K. (2020). Coconut yield prediction using back propagation and Split algorithm in bay islands. *CLIO an Annual Interdisciplinary Journal of History*. 06(2): 549-557.
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8.2.9 Copyrights

Title	Registration Number & Date	Authors
Agro-biodiversity Knowledge Management Portal	SW-12734/2019 19 August 2019	M Balakrishnan, SK Soam and PD Sreekanth
Delineating Surface Water Area Using Remote Sensing Imagery	SW-13411/2020 22 May 2020	P D Sreekanth, S K Soam and Ch. Srinivasa Rao
Production and its use in Rainfed Agriculture: Experiences from CRIDA. ICAR-CRIDA	L-94261/2020	Venkatesh G, Gopinath K.A, Sammi Reddy K, Sanjeeva Reddy B, Prasad JVNS, Rajeshwar Rao G, Pratibha G, Srinivasa Rao Ch, Ravindra Chary G, Prabhakar M, Visha Kumari, V, Arun K Shankar, Venkateswarlu B. Biochar

9.1 Budget Allocation and Expenditure during Jan-Dec 2020 (Rs. in lakh)

Sl. No.	Head of Account	Budget	Expenditure
A. Grant in Aid - Capital			
1	Works	63.58	63.58
2	Equipments	73.34	73.34
3	Information Technology	28.77	28.77
4	Library Books & Journals	27.89	27.89
5	Vehicles & Vessels	6.05	6.05
6	Furniture & Fixtures	31.06	31.06
Total Capital		230.69	230.69
Grants in Aid - Salaries (REVENUE)			
1	Establishment Expenses		
2	i. Establishment Charges	2181.36	2181.36
3	ii. Wages	135.39	135.39
4	iii. OTA	0.25	0.25
Total Salaries		2317.00	2317.00
B. Grant in Aid - General (REVENUE)			
1	B. Pension & Other Retirement Benefits	3599.50	3599.50
2	A. Domestic TA / Transfer TA	32.84	32.84
3	Research & Operational Expenses	189.37	189.37
4	Administrative Expenses	1849.84	1849.84
5	A.HRD	27.41	27.41
6	Miscellaneous Expenses	200.16	200.16
Total Grants in Aid - General		5899.12	5899.12
Grand Total (Capital+Salaries+Revenue)		8446.81	8446.81

9.2 Resource Generation

9.2.1 Sponsored Research Projects Budget

(Rs. in Lakhs)

Sl. No.	Name of the Project	Fund Received	Expenditure
1	PJTSAU Food Consumption Project	4.64	4.64
2	Agril.Res.&Dev.Infra.in Select Indian States under NSTMIS Scheme of DST	4.25	0.80
3	IFPRI Sponsored Project - ASTI in Indian NARES- ASTI Project	11.96	10.09
4	Consultancy Project on Adoption of Insect Resistance Mgmt. (Mhyco, Monsanto)	5.73	5.73

Sl. No.	Name of the Project	Fund Received	Expenditure
5	ICAR-NAARM-NIPHM Project "Plant Biosecurity through MOOC Platform"	4.24	1.59
6	ICAR-NAARM-MANAGE Project "Agri-Warehousing Management"	8.37	2.57
7	ICAR-NAARM-PJTSAU Digital Content for MOOCS	6.00	1.45
8	ICAR-NAARM-ITC "Economic Analysis of Eucalyptus and Subabul based Agroforestry Ecosystems in Telangana & A.P	6.29	4.6
9	Profiling the Resources Devoted in Agril.R&D Dev.of National framework for Evidence-based S&T Governance in India Scheme of DST	7.12	7.12
10	Skill Development in Bioinformatics Infrastructure Facility on Agril. Bio Technology under DBT	6.53	1.73
11	Adoption of Natural Farming and its effect on Crop Yield and Farmers Livelihood in India NITI Ayog Project	9.45	3.28
12	ICAR-NAARM-NDRI, Karnal for developing MOOC Courses on Commercial Dairy farming & Milk processing & Value Edition	14.71	10.12
13	ICAR-NAARM-NIPHM Project developing online course on "Rodents & Household Pest Management"	6.80	5.17
14	Preparation of DPR on capacity building/knowledge sharing for Institution & community in karnataka (NCSCM)	13.00	13.00
15	Preparation of DPR activities under ECO smart coastal villages for 5 villages in 5 states (NCSCM)	15.00	15.00
14	IFPRI Sponsored Project - Developing a National Dashboard for Evidence-Based Governance of Agril.Res.India	13.78	0
15	ICAR-NAARM-BCIL, New Delhi "Strengthening Natural Resource Management (SNRM)	11.18	8.34
16	ICAR-NAARM-BCIL, New Delhi "Strengthening Natural Resource Management (SNRM)	26.10	17.78

9.2.2 Off-Campus and Sponsored Programmes from Jan-2020 to Dec 2020

(Rs. in Lakhs)

Sl. No.	Name of the Programme	Total Budget received
1	FDP on Enhancing Efficiency and Effectiveness to Develop Entrepreneurial Ecosystem 11th to 20th February, 2020Feb.	3.44
2	Training Programme on "E resources E-Learning & IOT for students & staff of UAS, Raichur held from 25th to 29th February, 2020	5.04
3	Training Programme on "Entrepreneurial Skill Development" of graduating Students of SVVU, Tirupati	4.72
4	College of Dairy Science, Amereli, Gujarat for conducting 3 Programmes	10.37
5	Traing programme on "Contemporary Education Technologies for Agricultural Education for the Faculty of ANGRAU, Guntur held from 2nd to 6th June, 2020	1.41



Sl. No.	Name of the Programme	Total Budget received
6	Online Training Programme on Teaching Competency Programme for the faculty of SKLTSHU held from 7 th to 9 th July, 2020	0.99
7	Short Certificate Programme through online on "E Resources, E-Learning & IoT" for the students & staff of UAS, Taichur held from 24 th to 28 th August, 2020	1.83
8	Short Certificate Programme through online on "E Resources, E-Learning & IoT" for the students & staff of UAS, Taichur held from 5 th to 9 th October, 2020	1.48
9	Online Training Programme on "Design & Analysis of Experiments for Researchers" for 25 researchers of M/s. Advanta India Ltd.	1.50
10	Virtual training Module on "PRACom-2020:PRA and Communication Strategies for the 2019-21 Batch of IFS Probationary Officers from IGNFA held from 9-11th September, 2020	3.25
	Total	41.01

9.2.3 Resource Generation from Educational Programmes

Sl. No.	Educational Programmes	Receipts (Rs. in Lakhs)
1	PGDMA 2018-20	1.08
2	PGDM-ABM 2019-21	105.68
3	PGDM-ABM 2020-22	106.36
4	PGDTMA 2019 & 2020	8.22
6	PGDETM 2019 & 2020	2.8
	Total	224.14

9.2.4 Resource Generation from other Activities

Sl. No.	Receipts from other Activities	Receipts (Rs. in Lakhs)
1	Farm Produce	1.25
2	Guest House and Quarters	23.87
3	Institute Training Programmes	11.21
4	Miscellaneous Receipts	19.51
	Total	55.84

9.2.5 Net Receipts from Sponsored Research Projects

Sl. No.	Sponsored Research Projects	(Rs. in Lakhs)
1	Adoption of insect resistance management (IRM) practices in BT cotton Cultivation in India	5.6
2	IFPRI sponsored Project Strengthening & Uintegrating ASTI in India NARES	4.13
3	Net Receipts from Educational Programme 2018-20	95.00
4	Net receipts from IRGS Programmes	4.62
5	NIPHM on Plant Bio-Security through MOOC Platform	0.72
6	Development of MOOC courses on Commercial Dairy Farming & Milk Processing and value edition ICAR-NDRA, Karnal	0.1
7	Entrepreneurship Development through farmer led Inovations - NASF	0.84
8	PJTSAU-University Library ICAR-NAHEP Sub Project Digital content for MOOCs	0.9
9	Preparation DPR on capacity Building/Knowledge sharing for Institution & community in Karnataka	2.25
10	Preparation DPR under ECO-Smart coastal villages for 5 villages in 5 states	2.89
11	NAARM-MANAGE Warehousing Management	1.67
	Total	118.72

9.3 Plan Schemes

(Rs. in Lakhs)

Sl. No.	Name of the Scheme	Fund Available / received including Opening balance for 2020-21 (Upto Dec'20)	Expenditure during (Upto Dec'20)
1	National Agril.Innovation Fund (Component-I)-ZTMC	9.00	7.94
2	National Agril.Innovation Fund (Component-II)-ABI	6.25	0.87
3	Technology Enhanced Learning in Agril. Edn - TELAgE Project	2.30	1.19
4	XII Plan EFC Main Scheme KRISHI Project	11.64	8.03
5	Management & Impact Ass. Of Farmer First Project	6.58	5.96
6	ICAR Extra Mural Res.Project "Compete. Analysis among faculty of Agril.Univ.in Tech.Enhanced learning"	1.07	0.76
7	NASF, New Delhi - Entrepreneurship Dev. Through Farmer Led Innovations	8.65	9.29
8	Summer / Winter Short courses	2.01	2.01
9	NAHEP	403.46	115.87

Human Resources at the Academy (As on 31.12.2020)

Category	Sanctioned Strength	In position	Vacant
Scientific - RMP	02	01	01
Scientific - Faculty	60	33	27
Technical	47	28	06
Administrative	49	32	17
Skilled Support Staff	33	30	03
Total	191	124	54

Research Management Positions

1. Dr. Ch. Srinivasa Rao, Director
2. Dr. SK Soam, Incharge Joint Director

Scientific Staff

Agribusiness Management Division

1. Dr. Ranjit Kumar, Head
2. Dr. Tava Srinivas, Principal Scientist
3. Dr. B. Ganesh Kumar, Principal Scientist
4. Dr. N. Sivaramane, Principal Scientist
5. Dr. P.C. Meena, Principal Scientist
6. Dr. Sanjiv Kumar, Scientist

Education Systems Management Division

1. Dr. S. Senthil Vinayagam, Head (I/c)
2. Dr. G.R. Ramakrishna Murthy, Principal Scientist
3. Dr. D. Thammi Raju, Principal Scientist

4. Dr. V.V. Sumanth Kumar, Senior Scientist

Extension Systems Management Division

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3. Dr. Surya Rathore, Principal Scientist
4. Dr. P. Venkatesan, Principal Scientist

Human Resources Management Division

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2. Dr. P. Ramesh, Principal Scientist
3. Dr. K.H. Rao, Principal Scientist
4. Dr. Alok Kumar, Principal Scientist

Information and Communication Management Division

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2. Dr. A. Dhandapani, Principal Scientist
3. Dr. S. Ravichandran, Principal Scientist
4. Dr. M. Balakrishnan, Principal Scientist
5. Dr. N. Srinivasa Rao, Principal Scientist
6. Dr. P.D. Sreekanth, Principal Scientist
7. Dr. B.S. Yashavanth, Scientist
8. Dr. Purru Supriya, Scientist

Research Systems Management Division

1. Dr. I. Sekar, Head
2. Dr. K. Srinivas,

- Principal Scientist
3. Dr. K. Kareemulla,
Principal Scientist
 4. Dr. M.B. Dastagiri,
Principal Scientist
 5. Dr. P. Krishnan,
Principal Scientist
 6. Dr. Manju Gerard,
Principal Scientist
 7. Dr. Umesh Hudedamani,
Scientist
 8. Dr. Ramesh Naik, Scientist

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Joint Director (Admn.) & Registrar

Shri B.D. Phansal

Chief Finance & Accounts Officer

Sh. Z.H. Khilji

Senior Administrative Officer

Vacant

Administrative Officer

Shri Mukul Raj Singh

Asst. Admn. Officer

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2. Mrs. K.K. Rukmani Ammal
3. Dr. K.R. Ghanshyam

Private Secretary

1. Sh. N. Raghunath
2. Mrs. A. Mercy
3. Mrs. T. Vanisri

Junior Accounts Officer

1. Mrs. N. Vijayalakshmi

Assistants

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2. Sh. G. Raj Reddy
3. Sh. C. Phani Raj
4. Mrs. B. Padma Saroja
5. Sh. P. Srinivasu
6. Sh. R. Chandra Babu
7. Sh. C. Julius Samuel
8. Sh. M. Sridhar
9. Mrs. Y. Gayathri

Personal Assistants

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2. Mrs. Y. Anuradha
3. Mrs. S. Sessa Sai
4. Mrs. V. Shailaja
5. Mrs. S. Shanthi

Upper Division Clerks

1. Sh. M.K. Samson
2. Mrs. Rajashri Bokde
3. Sh. K. Suryanarayana
4. Sh. P. Swamy
5. Sh. M. Narsing Rao

Lower Division Clerk

1. Sh. M. Ashok
2. Sh. C. Bikshapathi

Technical Staff

Grade T-9 (Category-III)

1. Dr. Ch. Janardhan Rao,
Chief Technical Officer
(Video cameraman)
2. Dr. M.A. Basith,
Chief Technical Officer
(Jr. Farm Superintendent)
3. Sh. K.V. Kumar,
Chief Technical Officer

(Statistical Assistant)

4. Sh. P. Namdev,
Chief Technical Officer (Arts)
5. Sh. P. Vijender Reddy,
Chief Technical Officer
(Research Assistant)
6. Sh. Sohail Ahmed Khan,
Chief Technical Officer
(Junior Engineer-Civil)

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1. Mrs. G. Aneesa,
Assistant Chief Technical
Officer (Assistant Editor)
2. Sh. M. Shekhar Reddy,
Assistant Chief Tech. Officer
(Dark Room Assistant)
3. Dr. Ahire Laxman Mahar,
Assistant Chief Technical
Officer (Horticultural
Technical Assistant)
4. Sh. M. Ravi,
Sr. Technical Officer
(Photographer-cum-Artist)
5. Sh. Sham Bahadur,
Asst. Chief Technical Officer
(Catering incharge)
6. Sh. P. Mohan Singh,
Asst. Chief Technical Officer
(Computer Assistant)
7. Ms. Savithri Murali,
Asst. Chief Technical Officer
(Catering in-charge)

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cum-Information Officer)

Grade T-5 (Category-II)

1. Sh. N. Naresh Kumar, Technical Officer (VCR-cum-Tele cine Operator)
2. Sh. K. Shivaiah, Technical Officer (Technician)
3. Sh. D. Rajagopal Rao, Technical Officer (Media Operator)
4. Sh. S. Rajukumar, Technical Officer (Electronic Computer Operator)
5. Sh. T. Laxman, Technical Officer (Driver)
6. Sh. A.C.P. Rama Nageswara Rao, Sr. Technical Assistant (Lineman)

Grade T-4 (Category-II)

1. Sh. Pitla Srinivas, Sr. Technical Assistant (Proof Reader)
2. Sh. G. Muthyalu, Sr. Technical Assistant (Driver)
3. Sh. N. Ashok, Sr. Technical Assistant (Driver)
4. Sh. N. Prabhakar Rao, Sr. Technical Assistant (Plumber)
5. Sh. M. Srinivasa Rao, Sr. Technical Assistant (Pump Driver)

6. Sh. B.K. Venkatram, Senior Technical Assistant (Pump Driver)

Grade T-3 (Category-II)

1. Sh. R. Siva Prasad, Technical Assistant (Driver)
2. Sh. Danam Murahari, Technical Assistant (A.V. Operator)

Skilled Support Staff

1. Sirigiri Venkatesham
2. Mrs. S. Shakuntala
3. Sh. Kumba Satyanarayana
4. Sh. K. Pentaiah
5. Sh. P. Yadaiah
6. Sh. G. Pentaiah
7. Sh. J. Chandraiah
8. Sh. R. Sattaiah
9. Ms. Indirapati Bharatamma
10. Sh. C. Chandramouli
11. Sh. L. Satyanarayana
12. Ms. C. Kausalya
13. Sh. M. Ganesh Kumar
14. Sh. Chilumula Venkatesham
15. Sh. Sirigiri Narsimha
16. Sh. Kyasam Satyanarayana
17. Sh. B. Premdas
18. Mrs. A. Saroja
19. Sh. S. Nayab Rasool
20. Sh. B. Ashok
21. Sh. K. Daniel
22. Sh. M. Satyanarayana
23. Sh. D. Srisailam
24. Sh. B. Ratnaiah

25. Sh. D. Babaiah
26. Sh. L. Narasimha
27. Sh. K. Samson

Appointments / New Joinings

1. Shri L. Narsimha, Casual Labourer with Temporary Status (CLTS) has been regularised against the vacant post of SSS w.e.f. 19.03.2020 (FN)
2. Shri K. Samson, Casual Labourer with Temporary Status (CLTS) has been regularised against the vacant post of SSS w.e.f. 19.03.2020 (FN)
3. Dr. Purru Supriya, Scientist (Agricultural Bioinformatics) joined the Academy w.e.f. 04.04.2020 on her posting after completion of 110 FOCARS.
4. Shri B.D. Pansal, joined the Academy as Joint Director (Admn.) & Registrar w.e.f. 01.07.2020 (F.N.)
5. Shri MukulRaj Singh, joined the Academy as Administrative Officer w.e.f. 31.08.2020 (F.N.)

Promotions

1. Mrs. Y. Gayathri, UDC has been promoted under DPC to the grade of Assistant w.e.f. 05.09.2020
2. Shri. P. Swamy, LDC has been promoted under DPC to the grade of UDC w.e.f.

05.09.2020

3. Shri. M. Narsing Rao,
LDChas been promoted
under DPC to the grade of
UDC w.e.f. 05.09.20

Retirements

1. Shri P.G. Kohad,
Asst.Admn.Officer-Retired
on superannuation w.e.f.
29.02.2020 (AN)
2. Shri K. Obulapathi,
Technical Officer
(Technician Electrical)
superannuated w.e.f.
30.04.2020(AN)
3. Shri C. Muralidharan,
Joint Director (Admn.) &
Registrar superannuated

w.e.f. 30.06.2020 (AN)

4. Shri Dasaratha Gajjala,
Skilled Support Staff
superannuated w.e.f.
30.06.2020 (AN)
5. Dr. Sandhya Shenoy,
Principal Scientist retired
on superannuation w.e.f.
31.07.2020
6. Mrs. Mariyamma K.,
Skilled Support Staff retired
on superannuation w.e.f.
31.07.2020
7. Shri K. Veeranasiah,
Sr. Technical Assistant
retired on superannuation
w.e.f. 31.08.2020
8. Mrs. B. Shantamma,

Skilled Support Staff retired
on superannuation w.e.f.
30.09.2020

9. Shri. Jagtanand Lal Das.
Senior Administrative
Officer retired on
superannuation w.e.f
30.11.2020
10. Mrs. Nazia Parveen,
Skilled Support Staff retired
on voluntary retirement
w.e.f. 01.12.2020 (F.N.)

Obituary (while in service)

1. Mrs. Noorjahan,
CLTS Worker has expired on
04.08.2020



List of Committees

Higher Education Committee

- 1 Dr. Ch. Srinivasa Rao, Director - Chairman
- 2 Dr. S.K. Soam, I/c Joint Director - Member
- 3 Dr. Bharat Sontakki,
Head, XSM Division - Member
- 4 Dr. S. Senthil Vinayagam,
I/C Head, ESM Division - Member
- 5 Dr. I Sekar, Head, RSM Division - Member
- 6 Dr. Ranjit Kumar,
Head, ABM Division - Member
- 7 Dr. RVS Rao, Head, HRM Division - Member
- 8 Dr. P Krishnan, Pr. Scientist - Member
- 9 Dr. P Ramesh,
Principal Scientist & Controller of
Examinations (CoE) - Member
- 10 Shri. C. Muralidharan, JD (A) & R - Member
- 11 Coordinators of PGDMA,
PGD-TMA, PGD-ETM-Coopted Members
- 12 Dr. P.C. Meena, Pr. Scientist &
OIC, P.G.S. Unit - Member Secretary

Board of Studies

- 1 Dr. Ranjit Kumar,
Head, ABM Division - Chairman
- 2 Dr. S.K. Soam,
Head, ICM Division - Member
- 3 Dr. Bharat Sontakki,
Head, XSM Division - Member
- 4 Dr. K. H. Rao, Principal Scientist & Advisor,
Placement Cell - Member
- 5 Dr. A. Dhandapani,
Pr. Scientist, ICM Division- Member

- 6 Dr. P Ramesh,
Principal Scientist & COE- Member
- 7 Dr. T. Thammi Raju,
Pr. Scientist, ESM Division-Member
- 8 Dr. B. Ganesh Kumar,
Pr. Scientist, ABM Division - Member
- 9 Shri. BD Phansal, JD (A) & R - Member
- 10 Shri Z.H. Khilji, CFAO - Member
- 11 Coordinators of PGDMA,
PGD-TMA, PGD-ETM - Coopted Members
- 12 Dr. P.C. Meena, Pr. Scientist &
OIC, P.G.S. Unit - Member Secretary

Admission Committee

- 1 Dr. Ranjit Kumar,
Head, ABM Division - Chairman
- 2 Dr. Bharat Sontakki,
Head, XSM Division - Member
- 3 Dr. I Sekar,
Head, RSM Division - Member
- 4 Dr. RVS Rao,
Head (I/C), HRM Division - Member
- 5 Dr. S. Senthil Vinayagam,
I/C Head (I/C), ESM Division - Member
- 6 Dr. A. Dhandapani,
Pr. Scientist, ICM Division - Member
- 7 Dr. Sivaramane,
Principal Scientist - Member
- 8 Shri. BD Phansal, JD (A) & R - Member
- 9 Dr. P.C. Meena, Pr. Scientist &
OIC, P.G.S. Unit - Member Secretary

Works Committee

- 1 Dr. S.K. Soam, I/c JD - Chairman
- 2 Dr. G.R.K. Murthy, Pr. Scientist - Member

- 3 Dr. M. Balakrishnan, Pr. Scientist - Member
- 4 Joint Director (A) & Registrar - Member
- 5 CFAO - Member
- 6 Dr. P. Vijender Reddy, CTO - Member
- 7 Shri P. Mohan Singh, ACTO - Member
- 8 Danam Murahari, Tech Asst - Member
- 9 Shri Sohail Ahmed Khan,
ACTO - Member Secretary

Anti-Ragging Committee

- 1 Joint Director / Dean - Chairman
- 2 Dr. I Sekar, Head, RSM Division - Member
- 3 Joint Director (Admn.) & Registrar - Member
- 4 Academic Coordinators,
PGDM (A) Programme - Members
- 5 Dr. P.C. Meena, Pr. Scientist &
OIC, P.G.S. Unit- Member Secretary

Anti-Ragging Squad

- 1 Dr. (Mrs.) Surya Rathore,
Pr. Scientist - Chairman
- 2 Dr. B Ganesh Kumar,
Pr. Scientist - Member
- 3 Dr. Umesh Hudedamani,
Scientist - Member
- 4 Chief Finance & Accounts Officer - Member
- 5 Officer Incharge, Hostel - Member Secretary

Grievance Committee

- 1 Dr. S.K. Soam, JD I/c - Chairman
- 2 Dr. Bharat Sontakki, HoD - XSM - Member
- 3 Shri Khilji, CFAO - Member
- 4 Shri J.L.N. Das - Member
- 5 Dr. V.V. Sumanth Kumar,
Sr Scientist - Member - Scientific Category

- 6 Shri T. Laxman, Technical Officer -
Member - Technical Category
- 7 Shri P. Swamy,
LDC - Member- Admn Category
- 8 Smt. V. Saroja,
SSS - Member - Supporting Category
- 9 Shri M. Dinesh, AAO - Member Secretary

Swachh Bharat Abhiyan Implementation Committee

- 1 Dr. M.A. Basith, CTO & OIC - Farm - Chairman
- 2 Shri J.N. L. Das, SAO - Member
- 3 Dr. L.M. Ahire, ACTO- Member
- 4 Shri M. Sekhar Reddy, ACTO - Member
- 5 Shri Danam Murahari,
TA - Member Secretary

Farm Advisory Committee

- 1 Dr. Mude Ramesh Naik - Chairman
- 2 Dr. B.S Yashavanth, Scientist - Member
- 3 Dr. P. Vijender Reddy, CTO - Member
- 4 Dr. Ahire Laxman Maharua, ACTO - Member
- 5 Shri P. Mohan Singh, STO - Member
- 6 Shri M. Shekhar Reddy, ACTO - Member
- 7 Shri T. Laxman, Member
- 8 Shri Danam Murahari, TA - Member
- 9 Dr. MA Basith, ACTO - Member Secretary

Library Advisory Committee

- 1 Dr. R.V.S. Rao, Pr. Scientist,
I/C Head, HRM Division - Chairman
- 2 All HoDs - Members
- 3 Dr. P. Krishnan,
Pr. Scientist, RSM Division - Member



- 4 Dr. V.V. Sumanth Kumar,
Scientist, ESM Division - Member
- 5 Joint Director (A) & Registrar - Member
- 6 Chief Finance & Accounts Office - Member
- 7 Dr. N. Srinivasa Rao,
Pr. Scientist - Member Secretary

Newsletter Committee

- 1 Dr. Ranjit Kumar,
Head-ABM Division - Chairman
- 2 Dr. Sanjiv Kumar,
Scientist, ABM Division - Member
- 3 Dr. B.S. Yashavanth,
Scientist, ICM Division-Member Secretary

Annual Report Committee

- 1 Dr. I Sekar,
Head, RSM Division - Chairman
- 2 Dr. Bharat Sontakki,
Head, XSM Division - Member
- 3 Dr. Ranjit Kumar,
Head, ABM Division - Member
- 4 Dr. Tavva Srinivas,
Principal Scientist, ABM Division - Member
- 5 Dr. P. Krishnan,
Pr. Scientist, RSM Division - Member
- 6 Dr. Alok Kumar, Principal Scientist,
HRM Division - Member Secretary
- 7 Dr. Sanjiv Kumar,
Scientist, ABM Division - Member

Campus Residents Welfare Committee

- 1 Dr. S.K. Soam,
I/c Joint Director - Chairman
- 2 Dr. Umesh Hudedemani,
Scientist - Member

- 3 Dr. Mude Ramesh Naik,
Scientist-Member
- 4 Shri Khilji, CFAO - Member
- 5 Dr. B.S. Yashavant,
Scientist - Member Secretary
- 6 Shri Shitantshu Kumar,
SAO, ICAR-IIOR - Outside Member
- 7 Shri Ch. Suresh Kumar,
Assistant, ICAR-IIMR - Outside Member

Examination Unit

- 1 Dr. P. Ramesh,
Pr. Scientist, HRM Division - CoE
- 2 Dr. Manju Gerard,
Principal Scientist, RSM, In - charge, CoE
- 3 Shri JNL Das, Senior Admn. Officer - Member
- 4 Shri Pitla Srinivas,
Senior Technical Assistant - Member
- 5 Mrs. B. Padma Saroja, Assistant - Member
- 6 Shri B Ashok, SSS, Training Unit - Member

Hostel Management Committee

- 1 Dr. M. Balakrishnan, Pr. Scientist - Chairman
- 2 Dr. Mrs. Sandhya Shenoy,
Pr. Scientist - Member
- 3 Dr. Umesh Hudedemani, Scientist - Member
- 4 Joint Director (A) & Registrar - Member
- 5 Chief Finance & Account Officer - Member
- 6 Smt. Savithri Murali,
Sr. Tech. Officer - Member
- 7 Shri A.C.P.R. Nageshwara Rao,
Tech Assistant - Member
- 8 Sh. Sham Bahadur,
Asst. Chief Tech. Officer - Member Secretary

Institute Technology Management Committee

- 1 Dr. Ch Srinivasa Rao, Director - Chairman
- 2 Dr. SK Soam, Jt. Director, I/c - Member
- 3 Dr. S. Senthil Vinayagam,
I/c Head - ESM - Member
- 4 Dr. GRK Murthy, Pr. Scientist - Member
- 5 Dr. K Srinivas, Pr. Scientist - Member
- 6 Dr. V.V. Sumanth Kumar,
Sr. Scientist- Member
- 7 Dr. Umesh Hudedemani, Scientist-Member
- 8 Joint Director (Admin) Registrar - Member
- 9 CFAO - Member
- 10 Dr. RM Sundaram,
Pr. Scientist, IIRR. Hyd.- Member
- 11 Dr. Manju Gerard,
Pr. Scientist, Member Secretary

Sports Complex Maintenance Committee

- 1 Dr. P. Vijender Reddy, ACTO - Chairman
- 2 Dr. M.A Basith, CTO - Member
- 3 Sh. Sohail Ahmd Khan, ACTO - Member
- 4 Dr. Laxman M Ahire, ACTO - Member
- 5 Sh. M. Shekhar Reddy, ACTO - Member
- 6 Sh. P. Mohan Singh, ACTO - Member
- 7 Sh. Sham Bahadur, ACTO - Member
- 8 Sh. M.K. Samson, UDC - Member
- 9 Dr. Debnath, Medical Consultant - Member
- 10 Smt. K.K. Rukmani Ammal,
AAO (S&P) - Member Secretary

Campus Canteen Committee

- 1 Dr. L.M. Ahire, ACTO - Chairman
- 2 Smt. Y. Gayathri, UDC - Member

- 3 Representative from FOCARS - Member
- 4 Two representatives each from
PGDMA I & II - Members
- 5 Secretary, IJSC - Member
- 6 Member, CJSC - Member
- 7 Shri. N Prabhakar, STA - Member Secretary

Internal Complaints Committee

- 1 Dr. Surya Rathore,
Pr. Scientist, XSM Division - Chairperson
- 2 Mrs. G. Aneeja,
Asst. Chief Technical Officer - Member
- 3 Mrs. T. Vanisri, Pvt. Secretary - Member
- 4 Mrs. Vijayalaxmi,
Jr. Account Officer - Member
- 5 Mrs. K.K. Rukmani Ammal,
AAO (S&P) - Member Secretary

Biometric and Performance Monitoring Committee

- 1 Dr. Ch. Srinivasa Rao, Director - Chairman
- 2 Joint Director (A) & Registrar - Member
- 3 CFAO - Member
- 4 Shri Rajukumar,
Nodal Officer - AEBAS - Member Secretary

Academic Committee Meeting

- 1 Dr. Ch. Srinivasa Rao, Director - Chairman
- 2 All Faculty - Members
- 3 Dr. A. Dhandapani,
Pr. Scientist - Member Secretary - 1
- 4 Dr. Alok Kumar,
Pr. Scientist - Member Secretary - 2

Purchase Advisory Committee

- 1 Dr. A. Dhandapani, Pr. Scientist - Chairman
- 2 Dr. P. Ramesh, Pr. Scientist - Member



3. Dr. Umesh Hudedemani, Scientist - Member
4. Joint Director (A) & R - Member
5. CFAO - Member
6. Dr. Vijender Reddy, CTO - Member
7. Shri M. Sridhar, Assistant - Member
8. Smt. K.K. Rukmani Ammal,
AAO (S&P) - Member Secretary

Technical Advisory Committee

1. Dr. G.R.K. Murthy, Pr. Scientist - Chairman
2. Dr. V.V.S. Sumanth Kumar,
Sr. Scientist - Member
3. Shri K.V. Kumar, ACTO-Member Secretary

Placement Committee

1. Dr. K.H. Rao,
Pr. Scientist - Placement Advisor
2. Dr. B. Ganesh Kumar,
Pr. Scientist - Placement Coordinator - 1
3. Dr. N. Sivaramane,
Pr. Scientist - Placement Coordinator - 2

Project Monitoring and Evaluation Committee

1. Dr. T. Srinivas, Pr. Scientist - Head
2. Dr. P. Krishnan, Pr. Scientist -Member
3. Dr. Umesh Hudedamani,
Sr Scientist- Member

Stores Verification Officers

1. Dr. D.Thammi Raju, Pr. Scientist
2. Dr. N. Srinivasa Rao, Pr. Scientist
3. Dr. Vijender Reddy, CTO

(Verification done by the officer listed at S. No. 1 and during his absence it is done by the next nominated officer and so on)

Official Language Committee

1. Dr. Ch. Srinivasa Rao, Director - Chairman
2. Dr. N. Srinivasa Rao,
Pr. Scientist -In-charge – OL
3. Dr. S.K. Soam, I/c Joint Director - Member
4. Dr. R.V.S. Rao, Pr. Scientist - Member
5. JD (A) & Registrar - Member
6. Sr Administrative Officer - Member
7. Shri Pradeep Singh,
AD (OL) - Member Secretary



भाकृअनुप-राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी

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